When dendritic structure helps local credit assignment

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Abstract

Learning requires a neuron to assign a circuit-level error to individual synapses, but the contribution of its dendritic tree to this computation is unknown. We derive exact gradients for conductance-based trees and show that each synaptic update factorizes into local eligibility and a transported compartment error. This factorization yields a credit-operator theory that predicts when a restricted route improves learning: retained task signal must outweigh representation error, gain mismatch and admitted noise. Controlled simulations show that neuron identity supplies most of the useful feedback, whereas subtree addresses and serial dendritic depth help only at intermediate bandwidth or when they match the task hierarchy; depth alone is costly. Reconstructed arbors provide sparse, predominantly coarse addresses, while focal shunting regulates their gain only in permissive electrotonic regimes; measured visual responses provide no evidence that these anatomical routes are aligned with endogenous task credit. Dendritic structure therefore does not generically solve credit assignment. It supplies a conditional routing resource whose value is jointly determined by coordinate precision, address resolution, conductance state and task–morphology alignment.

Introduction

A biological neuron must distribute credit for its output errors across thousands of synapses arrayed on a branching dendritic tree, a problem that the point units of standard neural networks do not face [1–3]. Each synapse samples presynaptic drive, local voltage, reversal potential and input resistance. By increasing local conductance, inhibition can shunt excitation and alter the gain of every dendritic path through that compartment [4, 5].

Backpropagation applies the chain rule from an objective to every parameter [6]. It is therefore the exact optimization reference, not itself a biological mechanism: a literal implementation requires parameter-specific derivatives, coordinated backward Jacobians and suitable forward–feedback relationships. Synapses instead access presynaptic activity, local postsynaptic state and modulatory signals of uncertain spatial precision [7, 8]. Credit assignment in a neuron consequently has two spatial levels: identifying the neuron that should change and distributing that coordinate through its arbor. We therefore ask when a restricted signal can approximate the exact dendritic error field.

Biologically motivated approaches relax different requirements of backpropagation. Three-factor rules combine local eligibility with a modulator [9]; feedback alignment replaces transposed weights with alternative pathways, requiring positive functional alignment rather than exact symmetry [10, 11]; predictive, equilibrium and prospective schemes infer desired neural states [12–14]; and eligibility-propagation methods separate online synaptic traces from neuron-specific learning signals [15]. Dendritic-error and burst models separate forward and teaching signals through compartments, interneurons or temporal codes [16–20]. Most deliver a neuronal coordinate or use a few named compartments, leaving open what a full arbor contributes after credit reaches one neuron.

We first derive the exact gradient required by a conductance-based dendritic network. Each gradient factorizes into *local eligibility* $\times$ *transported error*: eligibility depends on presynaptic activity, driving force and input resistance, whereas a neuronal teaching coordinate is transported through path-specific gains. The factorization turns dendritic credit assignment into a concrete approximation problem: which spatial error fields can a restricted biological signal represent, and when do those fields improve learning?

We formalize a coordinate–address–gain hierarchy inside a common credit-operator theory. Teaching signals provide signed neuronal coordinates; dendritic ancestry provides nested candidate addresses; and conductance determines route gain. The theory predicts a phase boundary: a restricted route helps stochastic learning only when the useful gradient signal it retains compensates for its representation bias, coefficient error and admitted noise. Here, "phase" means an operational regime in alignment–bandwidth–noise space—not a temporal training phase or a thermodynamic phase transition. The theory also distinguishes three possible roles of depth—additional address scales, nonlinear forward composition and the cost of transporting a signal along more variable paths.

The experiments walk this hierarchy in order. Controlled simulations separate coordinate bandwidth, neuron-to-tree assignment and task-dependent routing between sibling subtrees; a hierarchy-depth experiment tests the predicted signal–noise crossover; and an experiment that adds physical dendritic stages at fixed synaptic, parameter and state budgets asks when serial nonlinear stages improve forward learning. MICrONS reconstructions then test route capacity, focal shunting and task alignment; an external six-animal dataset tests for signed neuron-specific coordinates. The evidence is conditional: neuron identity supplies most useful information, dendrites offer sparse addresses, shunting regulates gain only in suitable electrotonic regimes, and anatomy helps only when its routes align with task credit (Fig. 1).

Results

We proceed from the error available to a point neuron to the structured field required by its synapses. A global scalar conveys only error sign or magnitude; a neuron-indexed coordinate identifies the receiving cell; a within-tree address selects a subtree; and route gain sets the strength delivered along that path. Theory defines these levels, controlled networks isolate them, and anatomical and functional data test whether real trees supply and use the predicted routes. Study chronology, numerical gates and exclusions are reported in Methods.

Exact transported compartment error is used only as an information ceiling, never as a proposed biological signal. Every routed advantage reported here is a resource claim rather than a claim of dendrite-exclusive computation: a point model explicitly supplied with the same routing fields and gains matches it by construction, so the question is what anatomy makes available, and at what wiring cost.

> **Core notation.** Indices u, n and i denote a neuron, a compartment and a synapse, respectively. K is the number of communicated feedback channels; H is task-hierarchy depth; D_r is the number of address levels used by a feedback route; and D_p is the number of physical dendritic stages in a trained conductance model. D1, D2 and D3 therefore mean one, two and three serial dendritic stages, not three cell types. Bracketed labels give branching factors from soma to tips: D1 $[8]$, D2 $[2,3]$ and D3 $[2,1,2]$ all contain eight nonsomatic branch units. The symbols δ_u and $\delta_{u,k}$ denote a neuron-level teaching coordinate and its route-k refinement; $\Phi_{u,K}$ is the corresponding K-route dictionary and P_Φ its projector. Bold $\boldsymbol{g} = \nabla\mathscr{L}$ denotes a gradient, whereas scalar g_i denotes a conductance. The Supplementary Information contains the complete symbol table.

Exact dendritic gradients factor into eligibility and transported error

For a point neuron u with drive $s_u = \sum_i w_i x_i$, activity $V_u = f(s_u)$ and error $\delta_u = \partial\mathscr{L}/\partial V_u$, the chain rule gives

$$\frac{\partial \mathscr{L}}{\partial w_i} = \underbrace{x_i f'(s_u)}_{\text{synapse-local eligibility}} \underbrace{\delta_u}_{\text{one coordinate shared by neuron } u} . \tag{1}$$

Backpropagation computes δ_u from the downstream Jacobian, whereas biological models approximate it. A point unit has no subcellular state: all its synapses share this coordinate and differ only through eligibility. We ask how a dendritic tree extends that credit.

The local factorization is not restricted to one feedforward tree. Let the steady state of a compartmental neuron satisfy

$$\boldsymbol{F}(\boldsymbol{V},\boldsymbol{g},\boldsymbol{x})=\boldsymbol{0},\qquad J_V=\frac{\partial \boldsymbol{F}}{\partial \boldsymbol{V}},\qquad J_V^{\mathsf{T}}\boldsymbol{q}=\nabla_{\boldsymbol{V}}\mathscr{L}.\tag{2}$$

This fixed-point adjoint is the steady-state member of the Almeida–Pineda recurrent-backpropagation lineage [21, 22]. Intuitively, q_n measures how much the loss would change if a small current were injected at compartment n. When J_V is nonsingular, so that the steady state varies differentiably with its parameters, implicit differentiation gives the exact parameter gradient

$$\frac{\partial \mathscr{L}}{\partial g_i}=-\boldsymbol{q}^{\mathsf{T}}\frac{\partial \boldsymbol{F}}{\partial g_i}.\tag{3}$$

For a conductance synapse at compartment n, using the residual convention $\boldsymbol{F}=G\boldsymbol{V}-\boldsymbol{b}$, this becomes

$$\frac{\partial \mathscr{L}}{\partial g_i}=\underbrace{x_i(E_i-V_n)}_{\text{synapse-local factor}}\;\underbrace{q_n}_{\text{compartment adjoint}}.\tag{4}$$

The directed path product below is the triangular-tree corollary (an acyclic directed tree whose Jacobian can be ordered triangularly) of Eq. 2. In a reciprocal passive cable, $J_V=G$ is symmetric positive definite and $G^{-\mathsf{T}}=G^{-1}$, giving the Green's-function adjoint used for reconstructed cells. For a general nonsymmetric or recurrent compartmental system the inverse transpose is essential. For the triangular tree system used below, $q_n=R_n^{\text{tot}}\,\partial\mathscr{L}/\partial V_n$, so Eq. 4 and the compartment factorization of Eq. 6 are the same identity with the input-resistance factor moved between the local and transported terms. Thus dendritic credit is always local eligibility multiplied by a compartment-specific adjoint; only the construction of that adjoint changes.

We replace the point unit by a directed, feedforward rooted tree [23] whose compartment n receives synaptic conductances g_i with presynaptic activities x_i and reversal potentials E_i. Let $\mathscr{S}_n$ and $\mathscr{C}_n$ denote its synapses and child compartments, respectively, and let g_n^L and E_L denote leak conductance and reversal potential. At the conductance stage, its steady-state voltage is

$$V_n=R_n^{\text{tot}}\left(\sum_{i\in\mathscr{S}_n}g_ix_iE_i+\sum_{c\in\mathscr{C}_n}g_{c\to n}^{\text{den}}a_c+g_n^LE_L\right),\qquad R_n^{\text{tot}}=(g_n^{\text{tot}})^{-1},\tag{5}$$

where $a_c=f_c(V_c)$ is the activity transmitted by child c. Here $g_n^{\text{tot}}=g_n^L+\sum_{i\in\mathscr{S}_n}g_ix_i+\sum_{c\in\mathscr{C}_n}g_{c\to n}^{\text{den}}$ includes leak, active synaptic conductance, and child coupling. Differentiating with respect to a synaptic conductance gives

$$\frac{\partial \mathscr{L}}{\partial g_i}=\underbrace{x_iR_n^{\text{tot}}(E_i-V_n)}_{\text{synapse-local eligibility}}\;\underbrace{\frac{\partial \mathscr{L}}{\partial V_n}}_{\text{compartment error}}.\tag{6}$$

Presynaptic activity, driving force, and input resistance are local to the synapse and its compartment. All remaining task dependence is contained in $\partial\mathscr{L}/\partial V_n$.

Because a tree has one route from each compartment to the soma, its exact compartment error is

$$\frac{\partial \mathscr{L}}{\partial V_n}=\delta_{0,u}\widetilde{\alpha}_n,\qquad \widetilde{\alpha}_n=\prod_{(i\to k)\in\text{path}(n\to 0)}f_i'(V_i)R_k^{\text{tot}}g_{i\to k}^{\text{den}},\tag{7}$$

where $\delta_{0,u}$ is the somatic error for neuron u. Exact reconstruction from Eqs. 6–7 matched automatic differentiation across 100 diagnostic runs: median relative gradient error was 7.43×10^{-8} and the maximum was 1.88×10^{-7} (Fig. 2a).

Equation 7 applies to the directed artificial network used in the regular-tree experiments. The reconstructed-cell analyses below instead use a reciprocal passive cable matrix grounded in classical dendritic cable theory [24]. In that model the compartment error is an exact Green's-function adjoint rather than this simple product, but the synaptic gradient retains the same separation between local sensitivity and a soma-derived compartment error.

This identity defines what a local learning rule must approximate. Our three-factor rule retains the exact eligibility in Eq. 6 and replaces the compartment error by an available teaching field. Additional four- and five-factor variants use slowly estimated reliability preconditioners, but do not alter the fast local factorization. Exact path transport is used only as an information oracle.

From exact gradients to restricted credit operators. The factorization identifies the object that differs among local rules. Let $\widehat{\boldsymbol{g}} = \boldsymbol{g} + \boldsymbol{\xi}$ be an unbiased stochastic gradient with covariance Σ, and let a fixed credit operator M encode the available coordinate, route assignment and gain. The update is $\boldsymbol{w}^+ = \boldsymbol{w} - \eta M\widehat{\boldsymbol{g}}$. If the population loss has L-Lipschitz gradient, the descent lemma gives

$$\mathbb{E}\mathscr{L}(\boldsymbol{w}^+) \le \mathscr{L}(\boldsymbol{w}) - \eta \boldsymbol{g}^\top M\boldsymbol{g} + \frac{L\eta^2}{2}\left(\|M\boldsymbol{g}\|_2^2 + \mathrm{tr}(M\Sigma M^\top)\right). \tag{8}$$

When $\boldsymbol{g}^\top M\boldsymbol{g} > 0$, optimizing this bound over η yields the phase utility

$$U(M) = \frac{[\boldsymbol{g}^\top M\boldsymbol{g}]^2}{2L\{\|M\boldsymbol{g}\|_2^2 + \mathrm{tr}(M\Sigma M^\top)\}}. \tag{9}$$

Equation 8 is an upper guarantee for a general L-smooth loss and is exact for the corresponding quadratic when its curvature equals L on the update directions; comparisons of $U(M)$ otherwise order guarantees, not necessarily realized losses. Here and throughout, a *phase* means a regime separated by a quantitative crossover—for example, whether rejected noise is larger or smaller than discarded signal. It does not denote a thermodynamic phase transition. Thus more address dimensions are predicted to help only when their retained task signal outweighs finite-step gain and admitted gradient noise. Exact full-batch backpropagation has $M = I$ and $\Sigma = 0$; at a matched infinitesimal update norm no alternative direction can provide greater first-order descent. A projected stochastic gradient can nevertheless outperform an unprojected stochastic gradient because it rejects noise. For an orthogonal projector P on an identity-curvature quadratic at the common step $\eta = 1/L$, the exact crossover is $\mathrm{tr}[(I-P)\Sigma] > \|(I-P)\boldsymbol{g}\|_2^2$: rejected noise must exceed discarded signal (at smaller common steps the rejected-noise threshold rises; Supplementary Information). Backpropagation followed by the same projection is therefore the relevant upper control for any projected local rule.

> **Scope of the credit-operator prediction.** One-step geometry predicts within-family progress and large contrasts caused by retained signal, rejected noise or gross misalignment. It does not predict every advantage accumulated along a training trajectory; in particular, the trained $K = 4$ ancestry advantage is outside the initialization bound's scope.
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> **Capture vocabulary.** *Field capture* is the fraction of exact mean-field energy retained by a dictionary projection; *spectral capture* averages this fraction over a credit covariance; *checkpoint capture* applies it to a trained gradient; and *capture per wiring* divides field capture by feedback-connection density. We use an explicit qualifier whenever the estimand is not field capture.

For a route dictionary Φ, topology controls the span $\mathrm{col}(\Phi)$, whereas learning must also estimate the route coefficients. The total weighted field error separates exactly into the irreducible projection residual and the within-span coefficient-estimation error (Supplementary Information, "Stochastic credit-operator phase theory"). Multiplying fixed route columns by nonzero scalar shunting gains leaves their span and spectral capture unchanged, although it can strongly alter Gram conditioning. By contrast, state- or branch-dependent gain can act as a reliability preconditioner. For disjoint branches with signal energy S_b, noise energy N_b and fixed step $\eta = c/L$, Eq. 8 is optimized over attenuations $0 \le a_b \le 1$ by $a_b^* = \min\{1, S_b/[c(S_b + N_b)]\}$; the reliability factor $S_b/(S_b + N_b)$ is the special case $c = 1$, paralleling the forward reliability weighting derived for conductance-based dendrites [25]. A point model supplied with the identical branch gates must match; the biological question is whether dendritic conductance can make such localized gain available.

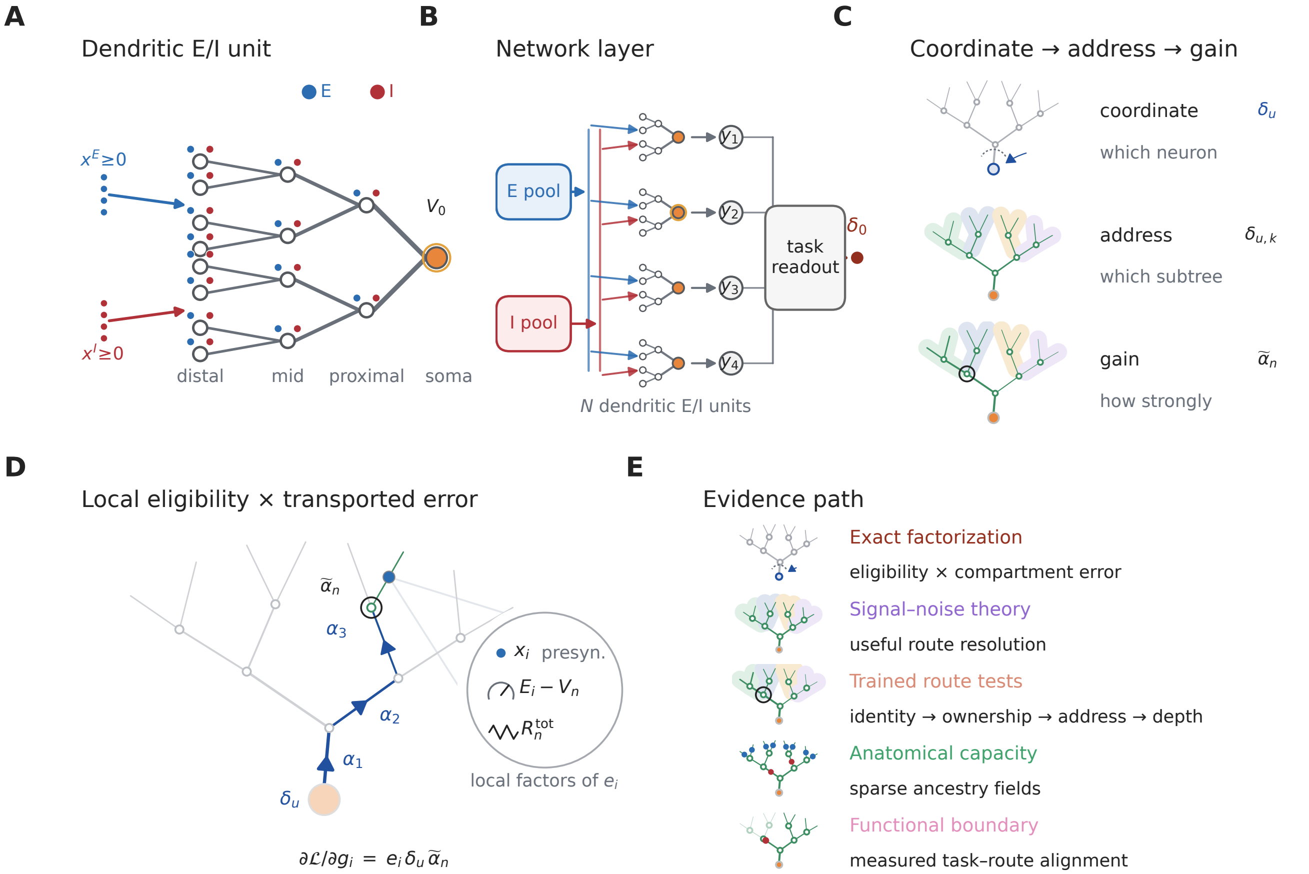

Figure 1: **From neuronal coordinates to dendritic credit addresses.** **A**, Local synaptic states are distributed across a branching tree, whereas the task supplies an error at the neuronal output. **B**, A population layer of dendritic excitatory and inhibitory units supplies a task readout; output error first arrives at each neuronal soma. **C**, Once a neuron-level coordinate δ_u arrives, dendritic subtrees provide distinct addresses and path conductances scale route gain. The following experiments test these levels in that order before asking when additional route resolution is useful. **D**, The central factorization as a teaching cartoon: local synaptic eligibility is multiplied by a neuron-level error transported through a route-specific gain. **E**, Evidence path from factorization and signal–noise theory through trained route tests, anatomical capacity and the functional alignment boundary.

Per-neuron bandwidth dominates restricted feedback

A global scalar is low bandwidth but discards which neuron produced an error. The matched-width/scalar-fallback rule used in our initial experiments retains the somatic vector only at a dendritic stage of identical width and otherwise averages it to one scalar per example. In contrast, neuron-indexed feedback repeats the appropriate somatic coordinate across all compartments belonging to that neuron's tree. It communicates one value per neuron, not one value per compartment.

This scalar-versus-indexed distinction is a feedback-bandwidth effect, not a new scaling principle: global perturbation estimates are known to become noisy as network size grows [10, 26]. Our question is what is added by neuron-to-tree ownership and within-tree address after neuronal identity has been supplied.

We compared these fields in regular $[3,3]$ trees, which contain 13 modeled compartments per neuron. Across 15 paired MNIST seeds, neuron-indexed feedback raised three-factor accuracy from $90.54 \pm 0.54\%$ to $97.16 \pm 0.11\%$ in shunting networks and from $91.76 \pm 0.78\%$ to $97.14 \pm 0.11\%$ in width- and connectivity-matched additive networks. All 15 pairs improved in each architecture. At the same neuron-indexed-trained checkpoints, the cosine between local and exact dendritic gradients rose from 0.012 ± 0.059 to 0.708 ± 0.027 in shunting trees and from -0.001 ± 0.047

to 0.625 ± 0.035 in additive trees (Fig. 2b,c). The increase was positive at all 15 matched checkpoints in each architecture.

Supplying exact transported compartment errors tests the derived local eligibility without feedback approximation. With five paired shunting seeds, three-factor learning with a local decoder reached $97.18 \pm 0.12\%$, compared with $97.03 \pm 0.08\%$ for a same-architecture backpropagation reference. Adding the empirical five-factor preconditioner or a backpropagated decoder provided no material gain under exact transport (Fig. 2d). Thus the poor scalar-feedback result is not a failure of the derived local eligibility. The large scalar-to-neuron-indexed gap results from loss of neuronal identity; the smaller remaining gap to exact transport reflects omitted path information.

Shunting changes the conductance component of path gain. In a separate five-seed diagnostic cohort with five inhibitory synapses per branch, the unweighted path-gain coefficient of variation across compartments within examples was 0.229 for shunting and 1.161 for additive trees, with the same ordering in all five paired seeds. This statistic describes the electrical operating point; it does not establish better gradient direction. Indeed, a 15-seed comparison crossing architecture with analytical or occupancy-calibrated branch-output-nonlinearity initialization showed no general shunting advantage: both architectures reached a mean of 90.92% under analytical initialization (± 0.42 and ± 0.51 s.d. for shunting and additive; paired difference -0.00, $P = 0.982$), whereas occupancy calibration modestly favored the additive model. We therefore interpret conductance as a route-gain mechanism, not as a universal optimizer (Supplementary Fig. S4).

Broader archived controls define the boundary of this conclusion (Supplementary Figs. S1–S4). Under restricted feedback, shunting trees were more stable than matched additive trees in depth and teaching-noise stress tests, but the advantage was task dependent and did not isolate a backward shunting mechanism. Increasing feedback structure consistently improved learning, including on a flattened CIFAR-10 mechanism control, whereas unmatched local errors and forward nonlinearities did not replace the required teaching field. These experiments support a conditional robustness claim rather than a general accuracy advantage.

We prospectively trained 640 models crossing two tasks, two neuron models, one to four dendritic stages, three feedback fields and matched backpropagation, with ten paired seeds per condition; all local conditions used the same factorization-derived eligibility and training protocol. An input-validity audit, defined without reference to outcomes, excluded all 160 synthetic-noise shunting runs, whose signed transfer output is not a valid drive for a positive-conductance shunting denominator (Methods), leaving 480 retained runs comprising both models on MNIST and the additive model on the noise task.

Neuron-indexed feedback increased held-out accuracy over a scalar broadcast by 4.92–6.99 percentage points on MNIST and 9.76–14.16 in the additive noise model, with all 120 retained paired seeds improving (Fig. 2d and Supplementary Fig. S19a,b). Exact transport placed mean three-factor accuracy within 0.14 points of matched backpropagation in all 12 retained task–model–depth conditions (Supplementary Fig. S19c).

A fresh Fashion-MNIST ladder tested whether the dominant identity effect was specific to MNIST (Fig. 2g,h). Without retuning the architecture or learning protocol, neuron-indexed feedback improved over the scalar fallback by 4.46 points in shunting trees (95% paired-seed interval 4.09–4.88) and 3.78 points in additive trees (3.43–4.14), with all ten paired seeds positive in both architectures (exact sign-flip $P = 0.002$). Exact path transport added no reliable benefit beyond neuron identity: -0.11 points in shunting trees (-0.21 to -0.01; 2/10 positive; $P = 0.096$) and 0.13 points in additive trees (-0.08 to 0.35; 8/10 positive; $P = 0.314$). This second dataset strengthens the bandwidth conclusion while leaving the distinct within-tree-address question to the credit-reversal experiments below.

A bandwidth-matched control fixed coordinate count and values but deranged the coordinate-to-tree map. Across the six retained conditions, correct assignment improved accuracy by 0.15–0.70 points and was favored in 58/60 pairs (Fig. 2e); five of six contrasts survived correction. Bandwidth therefore explains most of the scalar-to-neuron-indexed gain, while correct coordinate ownership adds a smaller effect. This comparison does not assign different task-dependent coordinates to different subtrees of one neuron and is not a trained test of within-tree addressing.

We therefore froze a separate two-stream credit-reversal task before its outcomes were inspected. Both sibling streams were active on every example; context selected the stream controlling the somatic output, while the other carried an anticorrelated distractor. Context and label were exactly balanced, so context alone could not predict the target. The same scalar somatic error therefore required different branch updates in paired contexts. The one-neuron field captured 0.509 of the eligibility-weighted exact-gradient energy at initialization, below the prespecified 0.90

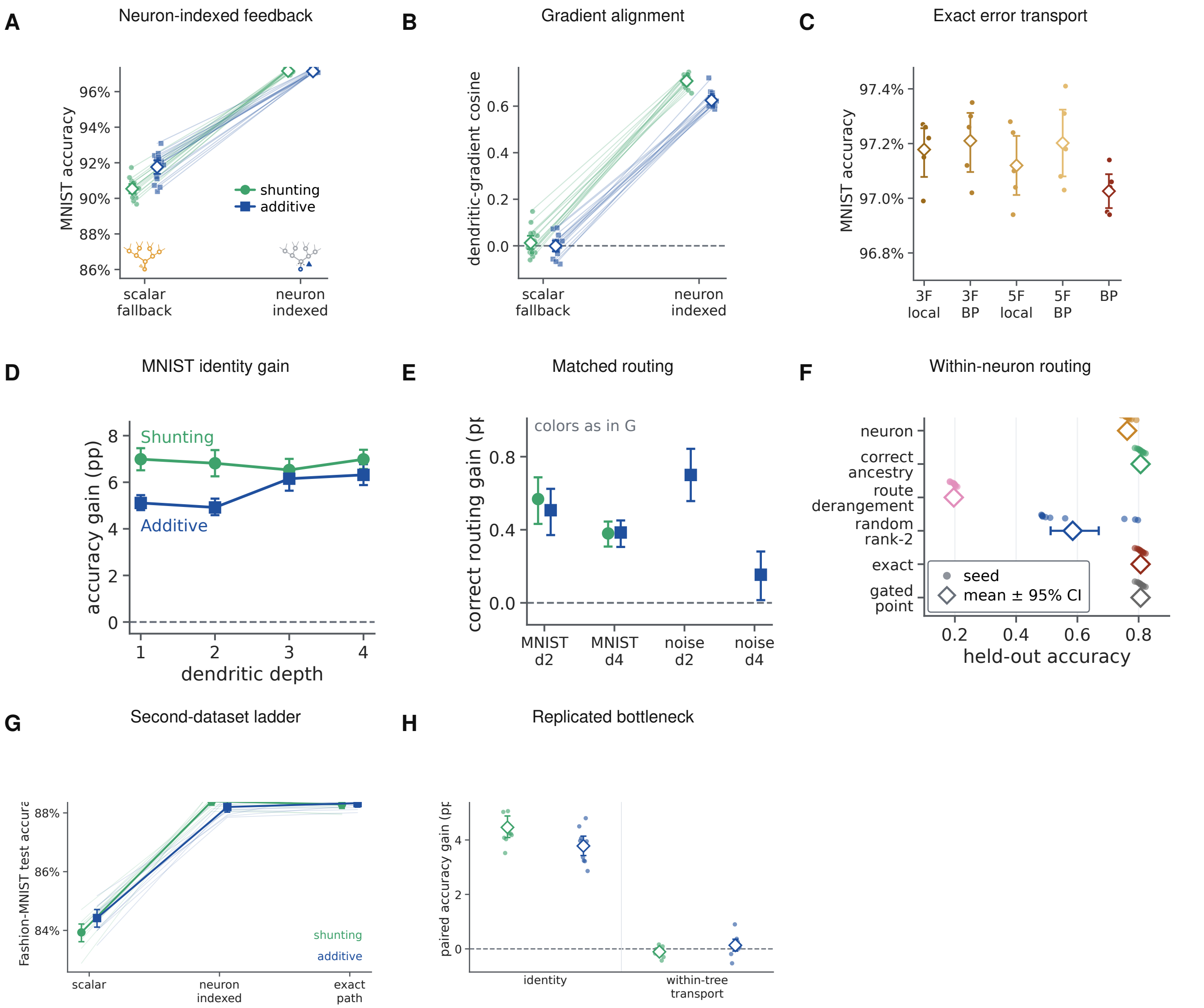

Figure 2: **Neuron identity is the dominant feedback bottleneck, with a smaller cost for incorrect route ownership.** **A**, MNIST accuracy under one scalar feedback signal or a neuron-indexed coordinate. **B**, Exact- gradient cosine at matched checkpoints. **C**, Accuracy under exact path transport, local rules and backpropagation (BP); exact transport is an information oracle. **D**, Neuron-indexed-minus-scalar accuracy across physical depth in the prospective MNIST cohort. **E**, Gain from correct rather than deranged neuron-to-tree ownership at matched bandwidth. **F**, Held-out accuracy in the two-stream credit-reversal task, where the same neuronal error requires different updates in sibling subtrees. **G,H**, Fashion-MNIST replication of the scalar, neuron-indexed and exact- path ladder and their paired contrasts. Faint lines or small points are paired seeds; large symbols and bars are means and 95% seed-bootstrap intervals. Exact transport, gated-point and correct-ancestry values coincide in the shallow representation-matched condition in **F**.

rejection threshold, whereas two correct subtree coordinates captured 1.000.

Across ten paired seeds, correct two-subtree routing reached 80.63% held-out accuracy, compared with 76.24% for one shared neuronal coordinate (difference 4.39 percentage points, 95% paired bootstrap interval 3.33–5.48; 10/10 seeds) and 19.63% after a fixed within-neuron derangement (a permutation assigning every route to the wrong sibling) (difference 61.00 points, 59.93–62.09; 10/10 seeds; Fig. 2f). A dense random rank-two map reached 58.47% and was below correct routing in all ten seeds. Gradient capture tracked norm-matched one-step progress (Supplementary

Fig. S19h). After a single-context adaptation block, the shared rule lost 55.66 points on the previous context, whereas the correct route left the inactive sibling unchanged (difference in forgetting, 55.66 points, 53.84–57.67; Supplementary Fig. S19i).

Correct subtree routing, exact transport and analytic backpropagation were identical on this shallow linear branch model. Crucially, a point-neuron implementation with the same two context-gated parameter groups also matched them exactly. The experiment therefore demonstrates the value of the address, not a unique dendritic computational advantage: the same solution can be paid for by explicit feature gating in a point model. It executes the decisive $K = 2$ support contrast on one forward architecture. We next completed the broader bandwidth and representation factorial.

The same audit excluded the 400-run inhibitory-dose family from inference because its shunting-minus-additive endpoint depends on the invalid signed-shunting cells. An independent clean-source 320-run exact-transport/backpropagation audit instead tested implementation agreement across ten paired seeds, four depths, two tasks and two cores; the synthetic shunting cells used an explicit ReLU transfer before positive conductances were formed (Supplementary Fig. S19e). All 320 runs passed the configuration, log, metric and finite stage-complete checkpoint audit. After averaging depths within seed, exact-minus- backpropagation accuracy was 0.005 percentage points on additive MNIST (95% paired bootstrap interval −0.028–0.041), −0.013 on shunting MNIST (−0.047–0.022), −0.112 on additive noise (−0.465–0.247) and −0.098 on shunting noise (−0.653–0.390). All four intervals included zero. Across all 160 raw method pairs, the mean difference was −0.055 points and the largest absolute pair difference was 5.90 points. These are empirical agreement checks, not formal equivalence tests or pointwise identity claims.

A retained 240-run spatial-map control improved accuracy by 0.36–2.57 points, including under backpropagation and in the additive randomly projected noise model (Supplementary Fig. S7). At 336 contacts, spatial branches covered 336 unique inputs versus 276.2 for random branches, identifying a forward-connectivity rather than task-specific credit-routing effect.

Of 320 fixed-budget runs, the 160 additive cells provide the valid synthetic comparison at 16 terminal branches and 960–968 contacts per soma. Depth four underperformed depth one in all 40 paired seed contrasts: −7.92 points under scalar feedback, −1.06 under neuron-indexed feedback, and −1.37 under both exact transport and backpropagation (Supplementary Fig. S19f; Supplementary Fig. S8). Deeper trees retain more trainable parameters, but scalar feedback clearly amplified depth stress.

At 120 retained backpropagation checkpoints, mean gradient cosine was 0.027 for scalar and 0.586 for neuron-indexed feedback. Norm-matched updates descended at 75/120 and 120/120 checkpoints, respectively, and cosine predicted retained one-step progress ($\rho = 0.893$, 95% checkpoint-bootstrap interval 0.860–0.923; Supplementary Fig. S9). This connects field geometry to loss change without replacing the trained-learning factorial.

Structured subtree addresses help at an intermediate feedback budget

We froze an eight-context hierarchical credit-reversal task and crossed five parameter-matched representations, feedback budgets $K \in \{1,2,4,8\}$, six route families and three references. Twenty paired independent seeds produced all 2,700 planned fits. Context and label were exactly balanced; every route field had unit norm per example; initial weights, data and optimization were paired. The dendritic, explicitly gated point, flat-compartment and grouped-point implementations had 64 forward parameters, 64 active input contacts and one nonlinearity. They received identical routed coefficient fields. A degree- and depth-matched rewired tree preserved resource counts while breaking task–topology alignment.

Correct ancestry routing displayed a non-monotonic advantage. Mean held-out accuracy increased from 0.187 at $K = 1$ to 0.449, 0.801 and 0.810 at $K = 2,4,8$, respectively (Fig. 3b). It strongly exceeded fixed within-neuron route derangement at $K = 2$ (difference 0.264, 95% paired bootstrap interval 0.243–0.285) and $K = 4$ (0.611, 0.601–0.620). Against the best matched non-anatomical control selected separately within each seed, however, correct ancestry lost at $K = 1$ and $K = 2$, won at $K = 4$ by 0.0127 (0.0059–0.0197; 15/20 seeds; two-sided Wilcoxon $P = 0.0048$, Benjamini–Hochberg-adjusted $P = 0.0064$ across the four budgets), and tied at full rank $K = 8$ (Fig. 3c). An unrestricted learned rank-K oracle and random rank-K routes therefore exploited low-dimensional task structure more efficiently at the smallest budgets; nested anatomy became advantageous only when four coordinates were available.

The correct matched tree exceeded the rewired tree by 0.231 (0.210–0.254) at $K = 2$ and 0.050 (0.042–0.060) at $K = 4$, with all 20 seeds positive in both contrasts (Fig. 3d). At $K = 1$ the route is shared, and at $K = 8$ every leaf is addressed, so rewiring had no effect. Exact transport and analytic backpropagation were identical. The four implementation-equivalent dendritic, gated-point, flat and grouped-point controls also matched exactly in every deterministic and paired-stochastic condition (maximum accuracy difference 0; Fig. 3e). Initial exact-gradient capture alone did not determine final accuracy across route families: fields with similar capture but different route signs and task alignment learned differently (Fig. 3f). Thus the result establishes a structured-address resource advantage at an intermediate bandwidth, not an intrinsic superiority of dendritic material: point models reproduce it when explicitly supplied with the same parameter groups and routing fields.

A credit-operator phase theory predicts when restricted routes help

We next froze a controlled experiment directly from Eqs. 8–9. Sixteen coordinates were represented in an orthonormal balanced-tree Haar basis. Fifty independent confirmatory seeds generated 10,800 seed–condition rows crossing route rank, task–tree alignment, task hierarchy, routed model depth, signal and noise retention, and branch-reliability heterogeneity. The experiment holds the forward coordinate budget fixed and isolates stochastic credit geometry; it is not a positive-conductance forward-neuron simulation.

First, the benefit of ancestry routes depended continuously on spectral task–tree alignment $\rho \in [0,1]$, where $\rho = 0$ is the seed-specific unaligned covariance and $\rho = 1$ is tree-Haar-aligned. At full alignment and $K = 4$, ancestry captured 0.552 more task-gradient energy than a random rank-matched route (95% paired bootstrap interval 0.535–0.569; 50/50 seeds; two-sided Wilcoxon $P = 1.78 \times 10^{-15}$; Fig. 4b). The advantage was near zero under random alignment, increased with alignment, never exceeded the Ky–Fan dense principal-subspace upper bound—the maximum covariance capture available to any rank-K orthogonal projector—and vanished at full rank. At full alignment and $K = 4$, ancestry attained the rank-matched upper bound to numerical precision. This attainment is the generic case for hierarchical credit: the K-route ancestry span equals the span of the K coarsest tree-Haar modes, so anatomy attains the rank-K Ky–Fan bound whenever the task covariance is diagonalized by the tree's wavelet basis with a coarse-dominant spectrum—and, in general, exactly when the leading rank-K eigenspace lies inside the laminar span—while its shortfall for a general covariance is bounded by the leading-eigenspace energy outside that span (Supplementary Section S4). A deterministic endpoint reanalysis used the affine covariance mixture to obtain an exact ancestry-versus-random alignment threshold within each seed. Nineteen of 50 random-route draws already favored ancestry at zero alignment; all 50 favored it at full alignment. Counting the former as requiring zero alignment, the median $K = 4$ threshold was 0.0173 and the mean was 0.0523 (95% seed-bootstrap interval 0.0336–0.0730). This is a threshold relative to sampled random routes, not a universal biological constant.

Second, routed depth was optimal when it matched task hierarchy. This match was imposed by the controlled generator—signal occupied levels through H, whereas finer levels carried additional noise—so the test is a quantitative validation of predicted crossover locations, not a discovery that $D_{\mathrm{r}} = H$ in arbitrary tasks. For task depths $H = 1,2,3,4$, the lowest mean final population loss occurred at model depth $D_{\mathrm{r}} = H$ (Fig. 4c,d). Across the 200 seed–task pairs, matched depth reduced loss relative to the best mismatched depth by 0.0957 (95% interval 0.0782–0.1134 after averaging the four task-depth contrasts within each independent seed; 48/50 seeds; two-sided Wilcoxon $P = 3.00 \times 10^{-13}$). Insufficient depth left target components outside the route span; excess depth admitted fine-scale stochastic gradient energy. At $D_{\mathrm{r}} = 4$ the projector is full rank and the update is identical to the common stochastic-backpropagation reference. Depth therefore helps here by matching the address resolution of the task, not by being large in isolation.

Third, independently varying signal and noise retention recovered the predicted boundary. Backpropagation followed by the same route projection had lower mean one-step population loss than unprojected stochastic backpropagation in exactly those of the 25 condition means for which rejected noise exceeded discarded signal (Fig. 4e). The routed local estimator carried an additional specified coefficient-noise cost and never exceeded this projected-backpropagation control. Branch-specific reliability gains then improved progress only when they were aligned with

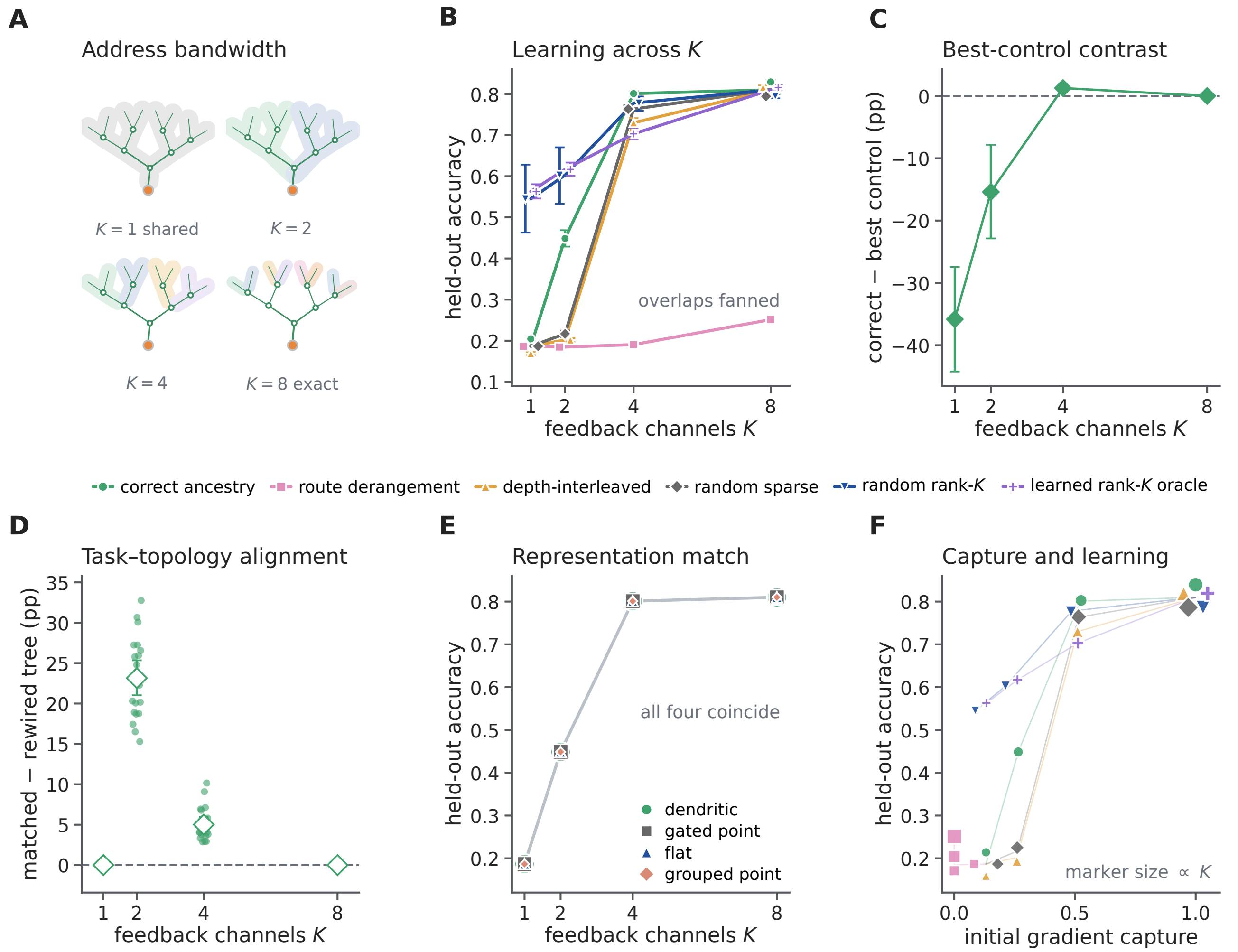

Figure 3: **The benefit of a structured subtree address depends on feedback bandwidth and task–topology alignment.** **A**, Eight leaves are addressed by one shared coordinate or $K = 2,4,8$ nested route fields. **B**, Held-out accuracy across route families in the dendritic-tree implementation. Curves and error bars show means and 95% seed-bootstrap intervals over 20 paired seeds; overlapping points are fanned slightly, with lines meeting at the shared values. The learned rank-K condition is an unrestricted task-derived upper bound. **C**, Correct ancestry minus the best of four matched non-anatomical controls within each seed; positive values favor ancestry. **D**, Correct routing on the task-matched tree minus the same rule on a degree- and depth-matched rewired tree. Small points are paired seed differences. **E**, Correct-route learning is numerically identical for dendritic, explicitly gated point, flat-compartment and grouped-point implementations when each receives the same routed fields. **F**, Initial exact-gradient capture versus final accuracy for each route family and budget; marker size increases with K, and overlapping points are fanned for visibility. Capture alone does not determine learning when route signs and task structure differ. Every condition matched forward parameter count, active contacts, nonlinearity count, initialization, data and optimizer.

branch signal-to-noise ratio. At maximum heterogeneity, reliability-aligned gain exceeded the best single global gain by 1.241 in one-step loss reduction (95% interval 1.116–1.373; 50/50 seeds; $P = 1.78 \times 10^{-15}$), whereas shuffling or reversing the gains removed or reversed the benefit (Fig. 4f). An explicit point gate receiving the same gains matched exactly (maximum difference 0).

We next tested whether a physical conductance could realize this oracle attenuation in a trained branch model. Eight branches received strictly nonnegative input rates and excitatory conductances. A nonnegative shunt to reversal zero reduced branch input resistance, while an oracle compensating current $\kappa_b V_b$ held the forward voltage identical;

this isolates the gain multiplying local eligibility. Fifty fresh confirmatory seeds crossed five reliability-heterogeneity levels and nine matched controls (Supplementary Fig. S13). At maximum heterogeneity, reliability-aligned shunting increased one-step test-loss reduction relative to the best global shunt by 0.001817 (95% paired bootstrap interval 0.001618–0.002020; 49/50 seeds; two-sided Wilcoxon $P = 3.55 \times 10^{-15}$). Its mean one-step advantage over no shunt was 0.001339 (0.000594–0.002096; 33/50 pairs; $P = 0.0030$).

After 40 updates, aligned shunting reduced test loss by 0.02040 relative to the best global shunt (-0.02222 to -0.01871 for aligned minus global; 50/50 seeds favored aligned, $P = 1.78 \times 10^{-15}$), and also outperformed shuffled and anti-aligned shunts. Its final loss differed from the unshunted noisy rule by 0.000425 (-0.002456–0.003071; $P = 0.154$), providing no reliable final-loss advantage in either direction. Exact clean backpropagation remained the lowest-loss reference. A state-matched additive control was identical to no shunt, and the explicit point gate receiving the same sample-dependent factors was computed through an independent update path and agreed with the conductance trajectory to numerical precision. Thus positive conductance approximates the predicted reliability ordering under a state clamp, but the immediate benefit does not establish autonomous shunt learning, dendrite-exclusive computation or uniform training superiority.

Finally, static nonzero gain rescaling changed numerical conditioning without changing the route span: nested indicators, their tree-Haar basis and a gain-scaled version had identical capture to 1.33×10^{-15}, while the condition number ranged from 1 for the Haar basis to 583 for the illustrated scaled parameterization (Fig. 4g,h). We then prospectively trained noisy route coefficients in a separate 50-seed, exactly same-span experiment (Supplementary Fig. S14). The three rank-eight coordinates had condition numbers 1, 15 and 388.5 and zero target address residual. The preregistered unconditional Haar advantage was falsified at effective sample size four: raw nested and scaled nested coordinates reduced final loss by 0.277 and 0.191 relative to Haar because their slow modes filtered update noise. At effective sample size 256 the ordering reversed, with Haar reducing loss by 0.0256 and 0.1709, respectively, in 50/50 paired seeds. An exact finite-time bias–variance calculation predicted median crossovers at effective sample sizes 38.5 and 8.25, and an oracle Gram-preconditioned control made every same-span field trajectory identical to 1.16×10^{-15}. Thus conditioning is neither a pure defect nor new capacity: it trades finite-time bias against estimator variance.

The phase utility was also diagnostic in the already completed 2,700-fit factorial. Reconstructing 64 initialization minibatches for each of 20 seeds gave 540 dendritic-route conditions; $U(M)$ predicted norm-matched one-step progress with Spearman $\rho = 0.937$ (seed-block 95% interval 0.9336–0.9412) and final held-out accuracy with $\rho = 0.916$ (0.905–0.929). Spectral capture alone predicted one-step progress less well ($\rho = 0.877$), with a positive seed-block difference of 0.060 (0.050–0.070; Fig. 4i). This secondary reanalysis explains immediate geometry at the fixed initialization; it is not an independent trained experiment.

The bound also locates optima, with an instructive boundary (Supplementary Fig. S16). Because every tested field is a projector on one nested ladder, utility rises with feedback bandwidth only while each added route scale contributes proportionally more signal than admitted noise, so an interior optimum is predicted whenever the signal spectrum decays faster than the noise (Supplementary Section S4). In the depth experiment, the per-seed argmax of the initialization utility coincided with the trained loss minimum in 135 of 200 seed–task pairs, and the group-mean argmax matched exactly at $H = 1, 2, 3$; at the full-rank boundary $H = 4$ the one-step bound was conservative, favouring $D_\mathrm{r} = 2$. In the factorial, the same utility reproduced the exact ties at $K = 8$, where all spans coincide, but did not predict the trained $K = 4$ ancestry advantage over the best control (6 of 20 seeds): that advantage accrues along the training trajectory and is not visible in one-step initialization geometry, consistent with capture alone not determining learning (Fig. 3f).

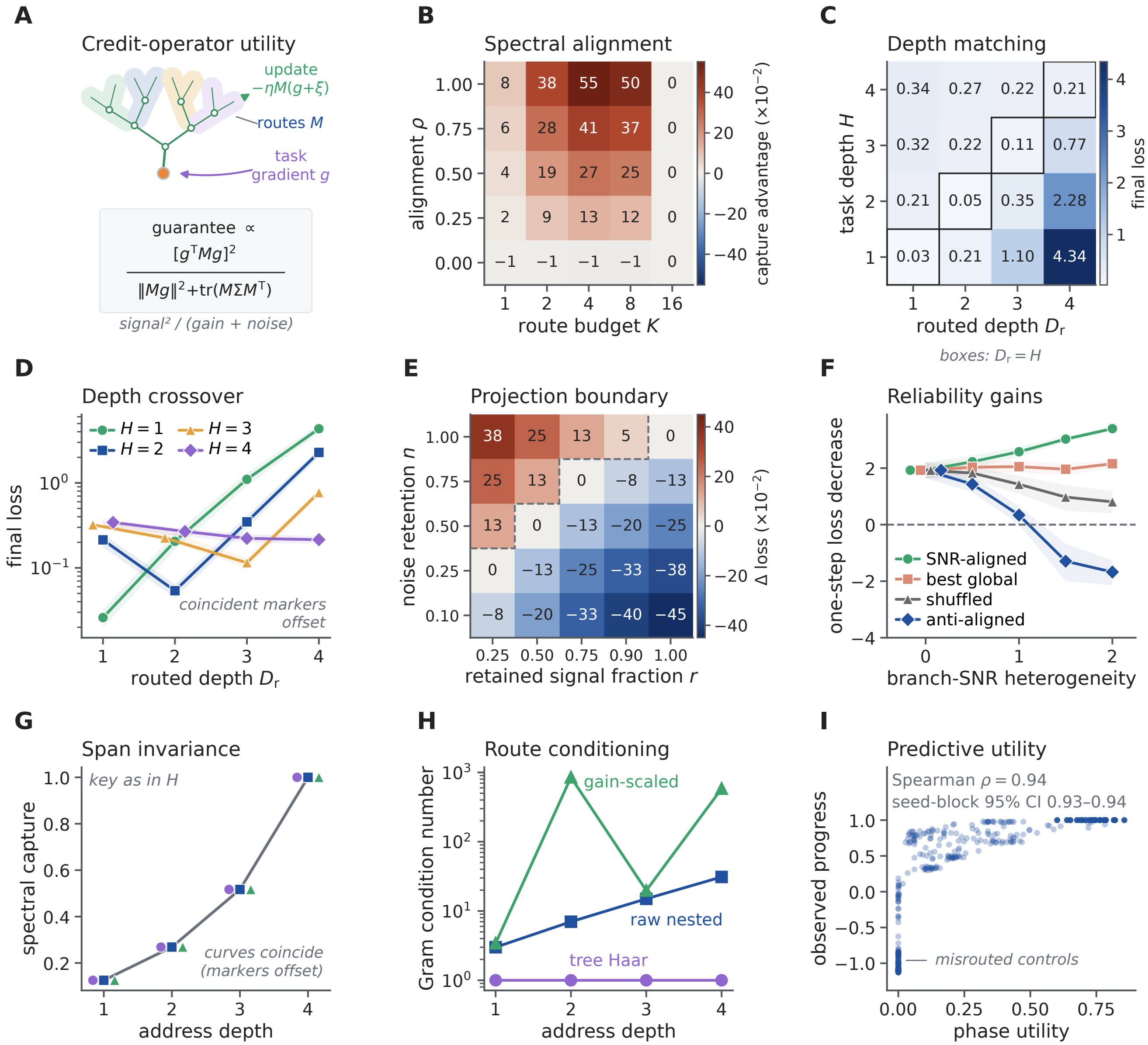

Figure 4: **A signal–noise phase theory predicts when dendritic credit structure is useful.** **A**, A fixed credit operator M maps an unbiased stochastic gradient into a routed local update; its guaranteed utility balances retained signal against finite-step gain and admitted noise. **B**, Mean ancestry-minus-random spectral capture across alignment and rank ($n = 50$ seeds); the advantage grows with alignment and vanishes at full rank. **C,D**, Mean final population loss for crossed task hierarchy H and routed depth D_r. Black outlines mark $D_r = H$; shading in **D** is the 95% seed-bootstrap interval, and coincident markers in **D** are slightly offset. $D_r = 4$ is full rank and equals stochastic backpropagation. **E**, Expected one-step loss of backpropagation plus the route projection minus unprojected stochastic backpropagation. Negative values favor projection; the dashed line is the analytical $n = r$ crossover. **F**, Simulated one-step loss decrease for SNR-aligned, best global, shuffled and anti-aligned gains (means and 95% intervals over 50 seeds). The explicit point gate is identical to the aligned condition, with coincident markers offset at zero heterogeneity. **G,H**, Static nonzero column gains preserve spectral capture but can alter the positive-spectrum Gram condition number. Markers in **G** are horizontally offset to reveal exact overlap. **I**, Prespecified signal–noise–curvature utility versus observed norm-matched one-step progress for 540 dendritic conditions from the 2,700-fit factorial; the interval resamples 20 seed blocks. Panels **B–H** are a fixed-state stochastic quadratic validation, not a positive-conductance forward model or evidence of uniform superiority to exact full-batch backpropagation.

Task-aligned physical depth improves learning at fixed budgets

The fixed-state phase experiment tests address resolution, not whether serial dendritic dynamics contribute to forward computation. We therefore constructed a separate positive-rate task—all inputs are nonnegative rates driving positive conductances—in DendriNet, the conductance-based population-network implementation used throughout (Methods). A distal class signal was modulated by nested fine, coarse and global gains, while separate nonnegative inhibitory streams measured those factors. This is a mechanism-matched existence test built around divisive normalization, not a generic task benchmark [27, 28]. D1 $[8]$, D2 $[2,3]$ and D3 $[2,1,2]$ contained the same eight nonsomatic branch units per soma, 14,336 active contacts, 21,760 candidate slots, 66,178 trainable parameters and 2,944 persistent state scalars. Only the number and placement of serial divisive stages changed (Fig. 5a,b).

With backpropagation and aligned gain sensors, mean test accuracy increased from 0.6096 at D1 to 0.6752 at D2 and 0.9182 at D3. The paired D3-minus-D1 effect was 30.86 percentage points (95% seed-bootstrap interval 30.45–31.30; 10/10 positive pairs; exact sign-flip $P = 0.002$; Fig. 5c,d). The operating point was selected transparently by exploratory signal, coupling and gain ladders and then fixed before these ten inference seeds (Methods and Supplementary Fig. S15); the 31-point contrast must therefore be read as a replicated effect at a calibrated operating point. Physical depth was flat when the sensors were independent (0.02 points) or trial shuffled (-0.01 points), and decreased under exact-resource reversal of fine and global tree placement (-0.43 points, -0.67 to -0.15). The corresponding aligned-minus-control interactions were 30.84, 30.87 and 31.28 points, respectively, all positive in 10/10 paired seeds. Thus the effect requires correspondence between task factors, branch-local sensors and their physical ordering; stage count alone is insufficient.

The nonlinearity also mattered at this operating point. Aligned raw-additive BP decreased from 0.5733 at D1 to 0.5383 at D2 and 0.5190 at D3, whereas the matched shunting series increased (Fig. 5f). This control supports a divisive hierarchical-composition mechanism in this task, not a general impossibility for additive or expanded point networks.

A same-task architecture ladder sharpened that boundary (Fig. 5g,h and Supplementary Fig. S18a). An all-active grouped-star model retained every branch module, mask, parameter, contact and persistent state but removed serial child-to-parent composition. It matched D1 (-0.005 points), remained near 0.610 at every depth, and therefore fell behind the serial tree by 6.54 points at D2 and 30.81 points at D3 (30.41–31.26; 10/10 positive). Reversing fine and global placement reversed the D3 contrast to -0.43 points, giving a 31.24-point alignment interaction (30.65–31.77; 10/10 positive). Thus the large effect requires serial composition in the model, rather than merely possessing the same collection of branch units. It is not an unconstrained capacity advantage: active- and total-parameter-matched point MLPs reached 0.9861 and 0.9902 and exceeded the serial D3 tree by 6.79 and 7.20 points, respectively, in every paired seed (Supplementary Fig. S18b). The tree supplies a task-aligned structured resource, not a computation that a flexible point network cannot learn.

Local learning retained the aligned depth benefit. A single decoder-derived soma coordinate rose from 0.6083 at D1 to 0.7778 at D3 (16.95 points, 16.37–17.44; 10/10 positive), while exact path transport reached 0.8873 at D3 (27.90 points, 27.24–28.53; 10/10 positive; Fig. 5e). Exact-resource reversal removed both effects: the aligned-minus-reversed depth interactions were 17.72 points (17.27–18.18; 10/10 positive) for the shared coordinate and 28.68 points (28.07–29.27; 10/10 positive) for path transport. A separate replay on the 30 aligned BP checkpoints reproduced autograd with exact path transport (minimum gradient cosine 0.9999999999994), whereas mean shared-soma cosine fell from 0.946 at D1 to 0.767 at D3. The local-rule performance and checkpoint diagnostic together show that physical hierarchy can remain accessible to local eligibility when a sufficiently specific error is transported; they do not establish that exact path transport is itself biologically available.

An autograd teaching-coordinate ladder localized the remaining separation (Fig. 5i,j). Replacing each compartment's loss derivative by its owning soma derivative while retaining autograd eligibility and the BP optimizer cost only 0.64 points at D3 (0.15–1.17; 8/10 positive; $P = 0.033$). Because this arm and LocalCA initially used different optimizer groups, a transparently post-outcome diagnostic repeated soma broadcast with the frozen LocalCA optimizer. It then matched shared-soma LocalCA (-0.06 points, -0.23 to 0.09; $P = 0.492$), whereas changing only the broadcast arm's optimizer accounted for 13.47 points (12.89–14.05). Within LocalCA, path transport recovered 10.96 points over the shared coordinate (10.34–11.58); full BP retained a 3.09-point advantage over path LocalCA

(2.48–3.79). In this calibrated task, the dominant BP–local gaps are therefore optimizer regime and within-tree coordinate specificity, not the tested local eligibility expression itself.

An alignment dose response tested whether the depth effect changed before the fully matched endpoint (Supplementary Fig. S18c–e). Intermediate sensor alignments $\alpha \in \{0.25, 0.50, 0.75\}$, with $\alpha = 0$ denoting independent sensors and $\alpha = 1$ matched sensors, were registered after the $\alpha = 0$ and 1 outcomes were known, so this is a prospective interpolation rather than an independent confirmation. The D3-minus-D1 effect increased from 0.02 points at $\alpha = 0$ to 0.21, 0.82 and 3.13 points at the intermediate levels, before rising to 30.86 points at full alignment. Every seed had a positive within-seed linear slope (mean 25.84 points per unit α, 95% interval 25.43–26.26; exact sign-flip $P = 0.002$), but the curve was strongly nonlinear: the $\alpha = 0.75$ minus 0.25 increase was only 2.92 points (2.65–3.17), whereas the final 1.00 minus 0.75 increment was 27.73 points (27.35–28.11). The result therefore localizes a steep alignment boundary near the fully matched task, rather than supporting a generic linear dose law.

We then tightened the point control and changed task hierarchy (Supplementary Fig. S18f–k). The earlier grouped-star control retained serial-shaped child aggregators as independent arms. A literal grouped-point emulation instead replaced each aggregator by a parameter-matched direct projection to the soma: every nonsomatic level kept its local branch module, mask and shunting computation, but received the external streams independently and never inherited another branch state. The replacement copied the corresponding initialized raw conductances, making D1 an exact initialization control. On the existing $H = 3$ task, grouped-point accuracy remained near 0.610 across D1–D3 (aligned D3-minus-D1, -0.01 points, -0.06 to 0.04), while serial-minus-grouped-point reached 30.87 points at aligned D3 (30.48–31.31; 10/10 positive; exact sign-flip $P = 0.002$). The same contrast was -0.44 points after placement reversal (-0.69 to -0.16), giving a 31.31-point alignment interaction (30.73–31.84; 10/10 positive). The earlier grouped star and the literal grouped point differed by only 0.06 points at aligned D3 (0.01–0.13; 7/10 positive; $P = 0.084$), so the conclusion does not depend detectably on which nonserial parameter-matched readout is used.

An independent $H = 2$ hierarchy then changed the sensor partition to two 32-feature tiers, the fixed branch inventory to 4/4 and the serial morphologies to D1 [8] and D2 [4, 1], while retaining the calibrated task family and all other optimizer and stopping settings. Ten fresh paired seeds 10300–10309 were used. Serial BP increased from 0.6588 to 0.9672, a 30.84-point D2-minus-D1 effect (30.46–31.23; 10/10 positive; $P = 0.002$). Reversing the two sensor tiers changed that effect to -1.56 points, yielding a 32.40-point depth-by-placement interaction (32.10–32.70; 10/10 positive). The literal grouped point stayed flat in both placements (aligned, 0.02 points, -0.02 to 0.06; reversed, -0.02 points, -0.06 to 0.02), and serial BP exceeded it at aligned D2 by 30.82 points (30.41–31.23), with a 32.36-point architecture-by-placement interaction. Shared-soma LocalCA retained a 21.20-point aligned depth effect (19.34–23.00) and exact-path LocalCA retained 31.38 points (30.95–31.83); placement reversal made both effects negative, producing 22.93- and 33.36-point interactions, respectively. Every positive interaction was ordered in 10/10 paired seeds. This is an independent hierarchy-depth replication within the same synthetic mechanism-matched task family, not a second dataset or evidence that natural tasks generally favor serial trees.

Together, the new same-task controls separate four explanations that the original depth comparison could not distinguish. The parameter-matched point networks reject a dendrite-exclusive expressivity claim. The serial-minus-star and serial-minus-literal-point interactions identify ordered divisive composition, rather than branch count, parameter count or a particular nonserial readout, as the source of the aligned forward advantage; the $H = 2$ cohort shows that this conclusion is not unique to the original three-stage inventory. The optimizer-matched soma-broadcast comparison finds no detectable penalty from the tested local eligibility expression, whereas path-specific transport recovers 10.96 percentage points over a shared neuronal coordinate; the principal local-credit limitation in this task is therefore the teaching coordinate and its optimization regime. Finally, the alignment interpolation shows that the forward depth advantage appears only near a strongly matched task–sensor hierarchy. The experiments consequently support a conditional role for dendritic structure: it provides a useful structured computation and address system when the task matches the tree, but neither depth alone nor dendritic material guarantees better learning.

Reconstructed arbors provide sparse, mostly coarse route capacity

The evidence so far comes from synthetic trees with hand-specified topology. We next asked whether reconstructed cortical arbors provide such addresses. We constructed ancestry dictionaries from eight layer 2–5 intratelencephalic neurons in MICrONS materialization 1822, mapping 28,669 classified inputs to collapsed cable segments from one mouse. Electrical parameters were normalized passive proxies, and reciprocal axial coupling was solved globally. The analysis complements prior connectomic proposals for anatomical credit assignment and the larger MICrONS morphology map [29, 30].

For neuron u, we define a K-channel feedback dictionary $\Phi_{u,K} \in \mathbb{R}^{N_u \times K}$, with rows for its synaptic or compartment locations and columns for available feedback fields. An *address* is one such column: its support and gain specify which locations can share one task-dependent coefficient. A point neuron has only the shared column $\mathbf{1}$; depth bins supply coarse columns; an ancestry dictionary supplies nested subtree columns; and an unrestricted dense rank-K subspace is a descriptive upper bound. This definition separates neuronal ownership from within-tree address and makes clear that dendrites supply structured, not arbitrary, credit bandwidth.

This distinction can be made quantitative. For a fixed family of K-column dictionaries $\Phi_{u,K}$ and a distribution of exact task-dependent coefficient fields $\boldsymbol{c}_u$, let $P_{\Phi_{u,K}}$ be the W-orthogonal projector onto $\mathrm{col}(\Phi_{u,K})$. For tolerated unexplained field-energy fraction $0 < \varepsilon < 1$, define the ε-address dimension

$$d_{\varepsilon,u} = \min\left\{ K : \frac{\mathbb{E}\|P_{\Phi_{u,K}}\boldsymbol{c}_u\|_W^2}{\mathbb{E}\|\boldsymbol{c}_u\|_W^2} \geq 1 - \varepsilon \right\}. \tag{10}$$

A point neuron with one neuronal teaching coefficient has rank-one task-dependent address space, irrespective of its number of synapses. A K-route dendritic dictionary has address rank at most $\mathrm{rank}(\Phi_{u,K}) \leq K$ and can express K structured coefficient fields without assigning those routes to K different neurons. This is a resource statement, not a claim that dendrites create arbitrary feedback bandwidth: a point-neuron system can emulate the same field if it is supplied with K separately routed teaching coefficients or an equivalent gated feature expansion (Supplementary Information).

For an excitatory site i and candidate inhibitory route k, the direct topological route matrix was

$$\Gamma_{ik} = A_{ik}\beta_k, \tag{11}$$

where A_{ik} marks route ancestry and $\beta_k = g_k^I / g_k^{\mathrm{tot}}$ is normalized inhibitory gain. This is a route-sensitivity proxy, not the exact passive-cable derivative used below. Its nested domains addressed a median 2.67% of mapped excitatory input and had mean participation rank 20.81, versus rank one for a scalar broadcast (Fig. 6a–c).

We measured the fraction of a modeled field retained by K feedback channels. For weighted field $\overline{\boldsymbol{g}} = W^{1/2}\boldsymbol{g}$ and dictionary $\overline{\Phi}_{T,K} = W^{1/2}\Phi_{T,K}$, let $\overline{P}_\Phi$ be the Euclidean projector onto $\mathrm{col}(\overline{\Phi}_{T,K})$. Field capture is

$$\mathscr{C}_K(\boldsymbol{g};\Phi,W) = \frac{\|\overline{P}_\Phi\overline{\boldsymbol{g}}\|_2^2}{\|\overline{\boldsymbol{g}}\|_2^2} = 1 - \frac{\|\overline{\boldsymbol{g}} - \overline{P}_\Phi\overline{\boldsymbol{g}}\|_2^2}{\|\overline{\boldsymbol{g}}\|_2^2}. \tag{12}$$

Here W contains summed mapped excitatory area, so Eq. 12 is first an anatomical field-energy measure. Only if $\overline{\boldsymbol{g}}$ is an exact gradient in explicitly transformed coordinates $\overline{\boldsymbol{w}} = W^{-1/2}\boldsymbol{w}$ and the transformed loss has L_W-Lipschitz gradients does $\mathscr{C}_K$ also equal the ratio of projected and full one-step descent guarantees at step $1/L_W$ (Supplementary Information).

We led with an independent generator: exact state-matched responses to focal shunts in the reciprocal cable model. At eight channels, ancestry routes captured 0.458, versus 0.363 for random paths and 0.249 after row shuffling, with both advantages positive in all eight cells. Differences from depth bins (0.014) and degree- and depth-matched surrogate trees (0.035) were inconclusive: the cell-bootstrap interval for the latter excluded zero, whereas the primary paired Wilcoxon test did not ($P = 0.078$). We therefore infer sparse support beyond random assignment but not a demonstrated advantage of fine ancestry beyond coarse depth and degree (Fig. 6e,f).

As a model-matched ceiling, we then generated fields directly from K.

At eight channels, morphology-selected routes captured 0.487 of the energy in the model-derived route-perturbation fields using 6.92% of the connections in a dense feedback matrix. This was 85.0% of the field energy captured by an unconstrained dense eight-channel oracle. Random paths captured 0.285, depth bins 0.222, and ancestry-shuffled routes 0.198. The morphology advantage was positive in every cell for each control; hierarchical cell-and-stream intervals excluded zero (Fig. 6c,d). Normalizing by wiring makes the resource statement explicit: per unit of feedback wiring, morphology-selected routes captured 14.2 times more field energy than the dense oracle (95% cell-bootstrap interval 10.7–18.2), whereas the ancestry-shuffled control, matched at the identical 6.9% wiring density, reached 5.3-fold; the anatomy-specific advantage at matched density is therefore 2.7-fold, with the remainder contributed by sparsity itself (Fig. 6g,h). The ordering also survived an analysis restricted to directly typed presynaptic E/I inputs: morphology captured 0.696, compared with 0.347 for random paths, 0.447 for depth bins, and 0.308 after ancestry shuffling (Supplementary Fig. S20h).

In 47 additional cells, morphology captured 0.756 with 8.08% of dense wiring, versus 0.302–0.441 for structural controls; every within-cell contrast was positive (Supplementary Fig. S10). The cohort is nucleus-disjoint but from the same mouse and a convenience-derived historical reconstruction.

All comparisons matched channel count. They establish anatomical compression of modeled fields, not a task gradient or route use by the animal.

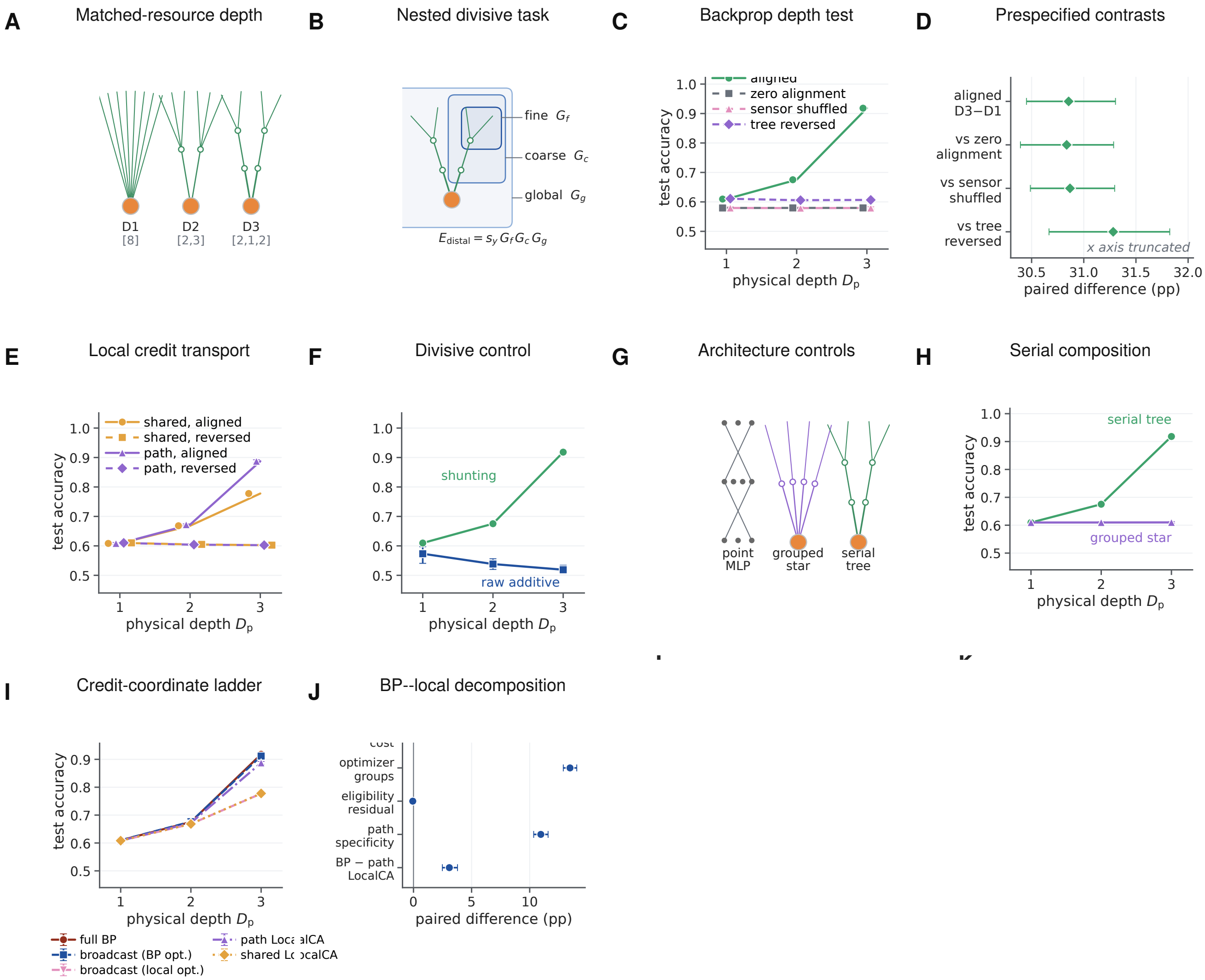

Figure 5: **Matched nonlinear hierarchy makes exact-resource physical dendritic depth useful.** **A**, D1–D3 have $D_\mathrm{p} = 1, 2, 3$ stages. Bracket entries are soma-to-distal branching factors: $[8]$, $[2,3]$ and $[2,1,2]$ give 8, $2+6$ and $2+2+4$ nonsomatic branch units, with identical contact, candidate-slot, parameter and state budgets. **B**, A distal positive-rate signal is multiplied by fine, coarse and global gains; independent sensors, trial-shuffled sensors and reversed physical placement are matched controls. **C**, Test accuracy under shunting backpropagation. Lines in **C,E,F** are ten-seed means and error bars 95% seed-bootstrap intervals; coincident flat-control markers are offset. **D**, Prespecified paired D3-minus-D1 effect and aligned-minus-control interactions; bars are 95% paired-seed bootstrap intervals, all excluding zero. **E**, Aligned and reversed-placement local credit assignment (LocalCA) with one shared-soma coordinate or exact path transport; the coincident reversed curves are drawn with offset markers. **F**, Aligned shunting and raw-additive BP. **G**, Architecture controls that separate a flexible point multilayer perceptron, a nonserial grouped star and the serial tree. **H**, Serial-tree and grouped-star accuracy across physical depth. **I**, Credit-coordinate ladder from full BP through soma-broadcast autograd to shared- and path-transport LocalCA. **J**, Paired D3 decomposition of the BP–local gap into coordinate, optimizer, eligibility and path-specificity contrasts. All inferential runs use paired seeds 10200–10209; lines are seed means and error bars are 95% seed-bootstrap intervals.

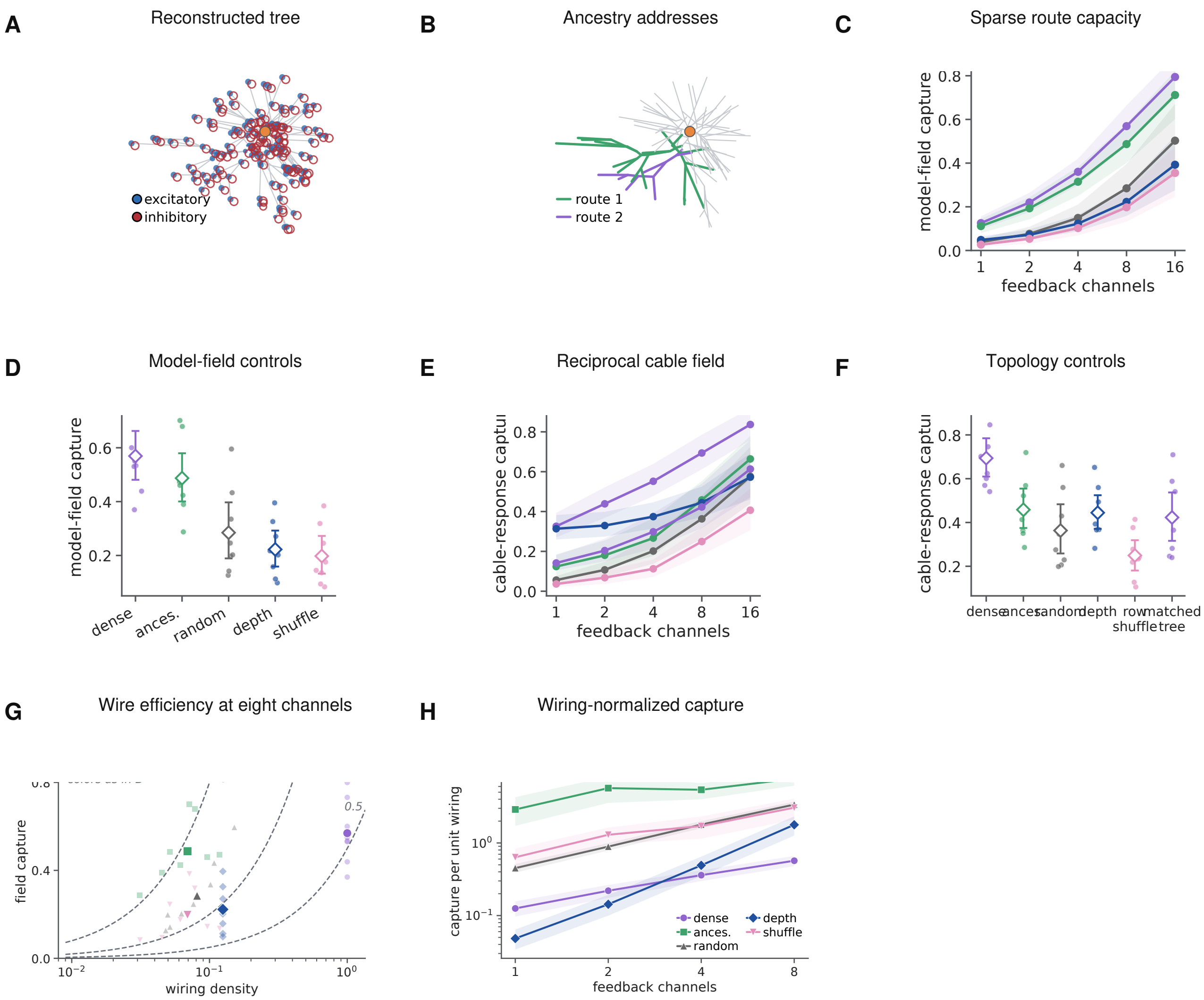

Figure 6: **Reconstructed arbors provide a sparse, mainly coarse routing basis for modeled perturbations. A**, Example reconstructed tree and mapped excitatory and inhibitory contacts. **B**, Two nested ancestry-defined addresses on the same arbor. **C,D**, Capture of the model-derived perturbation field across channel budgets and at eight channels for a dense oracle, ancestry routes and matched structural controls. **E,F**, The same comparison for an independently generated reciprocal- cable response field. **G**, Eight-channel capture versus wiring density; dashed curves mark equal capture per unit wiring. **H**, Capture per unit wiring across budgets. At eight channels the model-matched efficiency is 14.2-fold relative to the dense oracle, whereas the density-matched anatomy-specific factor is 2.7-fold. Lines summarize eight reconstructed cells and bands or intervals are 95% cell-bootstrap intervals. These are modeled fields on measured anatomy, not observed task gradients; wiring counts feedback connections, not cable length or metabolic cost.

The route analysis above used inferred inhibitory distributions. We next asked whether synapse-resolved inhibitory connectivity is organized in a manner compatible with branch-level control. This question builds on physiological and modeling work on tunable dendritic inhibition and inhibitory plasticity [31, 32]. We joined the disjoint reconstruction cohort to the published MICrONS inhibitory census [33]. Twenty of 47 target cells could be joined by stable nucleus identity across releases; five had different root IDs after re-segmentation. The source tables contained 116,401 input locations and 2,789 known contacts from 116 inhibitory cells. Of these, 114,775 inputs and 2,637 contacts from 114 inhibitory cells mapped within 5 μm. This is a structural analysis of one mouse; it does not identify a learning signal, and the packaged census does not provide a per-target proofreading-status field.

For each inhibitory cell–target connection, we retained one representative contact per independent axonal clump. The 402 connections with at least two clumps provided a within-axon test. Repeated contacts from the same inhibitory axon were 12.0 μm closer along the target tree than compartment- and path-distance-matched input locations (95% bootstrap interval -21.5 to -4.2 μm; 15 of 20 target means were negative; one-sided Wilcoxon $p = 0.0032$). They also shared 0.0357 more normalized path (0.0091–0.0766; 15 of 20; $p = 0.0028$) and addressed descendant input domains with 0.0322 greater weighted cosine overlap (0.0060–0.0743; 14 of 20; $p = 0.0053$; Supplementary Fig. S12d–f). These target-level comparisons do not by themselves separate dendritic organization from repeated sampling of the same axons or local tissue geometry. Averaging first within each of 92 presynaptic axons retained the tree-distance effect ($p = 0.0008$), whereas the shared-path and domain-overlap effects were not reliable ($p = 0.104$ and $p = 0.393$). With candidate locations jointly matched for path distance and three-dimensional proximity, none of the three contrasts was reliable ($p = 0.108$, 0.826, and 0.54, respectively). Thus repeated contacts are spatially co-targeted, but these data do not establish geometry-independent organization over descendant credit domains.

Inhibitory classes exposed different spatial scales. After retaining one contact per axonal clump, distal dendrite-targeting contacts addressed a smaller fraction of target-cell input than perisomatic-targeting contacts in all 20 cells. The mean within-target difference of the class medians was -0.0162 (-0.0231 to -0.0105; one-sided $p = 9.5 \times 10^{-7}$; Supplementary Fig. S12g). This class effect should not be confused with a global placement advantage. Across all inhibitory contacts, descendant-domain size did not differ reliably from matched generic input locations (mean difference 0.0016, $p = 0.546$; Supplementary Fig. S12h).

Finally, we measured the internal compressibility of the actual inhibitory route dictionary. At eight sites, its highest-leverage columns captured 0.743 of that same weighted dictionary, close to a dense singular-vector oracle (0.774), compared with 0.379 for random actual sites, 0.397 for depth bins, and 0.369 after depth-preserving ancestry shuffling. Each selected-route contrast was positive in all 20 targets (one-sided $p = 9.5 \times 10^{-7}$ for each; Supplementary Fig. S12i,j). Selection by leverage is an anatomical compression oracle: it measures redundancy within a dictionary using columns of that dictionary, not independent credit routing or a claim that the animal chose those sites. Together, the census establishes class-dependent spatial scales and available nested domains, while the axon-clustered and 3D-matched controls prevent a stronger functional-routing interpretation.

Shunting regulates route gain only in permissive electrotonic regimes

Topology-based reconstruction does not show that conductance can selectively control the routes. At the textbook passive calibration ($R_m = 15{,}000\,\Omega\,\mathrm{cm}^2$), the primary contrast was effectively null (-0.0019, two of eight cells positive); localization emerged only in permissive normalized or higher-conductance regimes. We therefore specified a focal intervention before examining its outcome. This adjoint-credit question extends prior analyses of spatial inhibitory domains in dendrites [4]. The passive conductance matrix of each reconstructed tree was denoted by G, with voltage $\boldsymbol{V} = G^{-1}\boldsymbol{b}$, where $\boldsymbol{b}$ is the current-source vector. Let $\boldsymbol{e}_s$ and $\boldsymbol{e}_k$ be standard basis vectors at the soma and focal compartment k, respectively. For a somatic loss, the adjoint error is $\boldsymbol{q} = G^{-1}\boldsymbol{e}_s\,\partial\mathscr{L}/\partial V_s$. A shunt of conductance κ_k at compartment k changes the transport operator according to

$$\boldsymbol{q}' = \boldsymbol{q} - \frac{\kappa_k q_k}{1+\kappa_k (G^{-1})_{kk}} G^{-1}\boldsymbol{e}_k. \tag{13}$$

Writing $\boldsymbol{c}_k = G^{-1}\boldsymbol{e}_k$ and $r_k = (G^{-1})_{kk}$ gives $\Delta\boldsymbol{q} = -a_k\boldsymbol{c}_k$, where $|a_k| = \kappa_k|q_k|/(1+\kappa_k r_k)$. Thus dose controls the response amplitude, which increases monotonically and saturates at $|q_k|/r_k$, whereas the cable column $\boldsymbol{c}_k$ controls its spatial spread. For descendant set $D(k)$ and a matched comparison set $C(k)$, the transport-only selectivity criterion is

$$S_k = \frac{\mathrm{median}_{i\in D(k)}\,|c_{k,i}|}{\mathrm{median}_{j\in C(k)}\,|c_{k,j}|} > 1. \tag{14}$$

Equation 14 separates electrotonic selectivity from perturbation strength. It is necessary for descendant enrichment of absolute adjoint change under this median contrast, but is not sufficient for the synaptic-gradient statistic, which also depends on baseline adjoints and local driving forces. A first-order-current-matched additive perturbation changes the voltage-driving term but not G^{-1}. This yields a direct test: after a state-matching compensatory current injection restores somatic voltage and the scalar output error, conductance shunting should alter exact synaptic gradients more strongly among descendants than among depth-matched unrelated sites. Reciprocal coupling still changes credit outside the descendant set.

We tested 101 depth-stratified focal sites across the eight cells; the gradient-change profile across topological relations is shown in Fig. 7b. At the unit dose, the descendant-minus-depth-matched-unrelated localization index—descendant minus depth-matched unrelated median log gradient attenuation—was 0.104 for shunting and 0.035 for the matched additive perturbation. The cell-level paired difference was 0.069 log units (95% bootstrap interval 0.053–0.086), positive in all eight cells. The contrast rose monotonically with dose, from 0.018 at 0.25 local-conductance units to 0.128 at two units. True focal relations also exceeded relations sampled from other focal sites, stratified by topological-depth quartile when another site was available, by 0.080 (0.059–0.098), again in all eight cells (Fig. 7b–d).

Because a synaptic gradient is the product of adjoint error and local driving force, we next froze either factor algebraically. Keeping the full post-shunt voltage and driving force but replacing the post-shunt adjoint with its baseline value gave localization 0.026. Restoring the post-shunt adjoint raised it to 0.104, an exact transport-specific difference of 0.079 that was positive in all eight cells (Fig. 7e). This comparison holds the post-shunt driving-force field fixed and therefore isolates the contribution of changed error transport without relying on first-order current matching. The complementary adjoint-only state gave localization 0.130. A two-factor Shapley sensitivity analysis assigned 0.087 of the planned shunt-minus-additive contrast to the adjoint change and -0.017 to the driving-force change, reconciling exactly to the 0.069 net contrast at every site (Supplementary Section S8). These factor substitutions are algebraic controls, not separate biological interventions.

The state-matching residual at the soma was numerically zero, conductance matrices remained positive definite, and adjoint gradients agreed with finite differences to a maximum relative error of 1.03×10^{-6}. The principal contrast also survived changes in normalized conductance scale and inhibitory reversal potential (Supplementary Fig. S21f). Restricting both synaptic weighting and focal sites to directly typed inputs gave a shunt-minus-additive contrast of 0.101 (0.075–0.129), positive in all eight cells (Supplementary Fig. S21g). In the disjoint sensitivity cohort, the same contrast was 0.138 (0.116–0.162) across 235 sites in 45 eligible cells and was positive in every cell (Supplementary Fig. S10f). These results identify a conductance-specific gradient-localization mechanism at fixed somatic state in the normalized proxy model.

We next calibrated axial and leak conductances from reconstructed radii and lengths using explicit specific resistances. With $R_a = 150\,\Omega\,\mathrm{cm}$ and $R_m = 15{,}000\,\Omega\,\mathrm{cm}^2$, the median axial-to-leak ratio was 68.0. The shunt-minus-additive contrast was then -0.0019 in the original cohort (two of eight cells positive) and 0.0024 in the v661 cohort (45 of 45 positive). Lower effective membrane resistance, representing a progressively higher-conductance state, restored a larger effect: at $R_m = 1{,}000\,\Omega\,\mathrm{cm}^2$ the original eight-cell contrast was 0.0164, and at $R_m = 300\,\Omega\,\mathrm{cm}^2$ it was 0.0674 (eight of eight positive in both conditions; Fig. 7f). Low input resistance and conductance-driven gain modulation are established features of in-vivo-like high-conductance states [34, 35]. These values use an explicit relative calibration of synapse size because MICrONS synaptic area is not a conductance measurement. The focal signature is therefore an electrotonic-regime-dependent model prediction, not a universal property of the reconstructed arbors or evidence that inhibitory synapses carried endogenous teaching signals in the animal.

We then froze a passive sensitivity matrix that removes the relative-dose confound. Across the same 101 sites and eight cells, it crossed three membrane resistances, three levels of distributed leak-like background conductance, and relative-local, fixed-absolute and input-conductance-normalized doses (16,362 site–condition–intervention rows). At a fixed 0.5-nS shunt, the shunt-minus-additive localization contrast was 0.0104 at $R_m = 300\,\Omega\,\mathrm{cm}^2$, 0.0073 at

$R_m = 1{,}000\,\Omega\,\text{cm}^2$, and -0.0039 at $R_m = 15{,}000\,\Omega\,\text{cm}^2$; the first two were positive in all eight cells, whereas only two of eight were positive in the last regime. At the high-resistance setting, adding one and four times the baseline leak as distributed background conductance raised the input-conductance-normalized contrast from -0.0004 to 0.1210 and 0.2587, respectively. Every tested descendant gradient was attenuated in magnitude; none was enhanced or changed sign. Thus passive shunting supplies graded attenuation, not signed reversal, and its conductance-specific spatial advantage requires both sufficient dose and a selective electrotonic column (Supplementary Fig. S11).

Finally, we asked whether the same localization survives weak active conductances rather than a purely passive operator. The ensemble was evaluated at the permissive electrotonic operating point identified above ($R_m = 1{,}000\,\Omega\,\text{cm}^2$ with one baseline-leak unit of distributed background conductance; Methods); it extends the localization result within that regime rather than restoring it at the high-resistance calibration. For each of the eight cells, we solved 64 channel-density draws containing voltage- dependent Na, K, Ca, HCN and NMDA currents, then linearized the exact nonlinear steady-state residual to obtain its local Jacobian. All 512 equilibria were accepted; every Jacobian remained positive definite and the maximum equilibrium residual was 9.6×10^{-11}. Draws were nested sensitivities, not independent cells. At the unit input-conductance-normalized dose, focal shunting produced descendant localization 0.529 (95% cell-bootstrap interval 0.513–0.541), attenuated every modeled descendant gradient, produced no sign flips and reduced descendant gradient energy to 0.330 of baseline (Fig. 7g–j). Its localization exceeded the state-matched additive control by 0.382 (0.356–0.409), positive in all eight cells (two-sided Wilcoxon $P = 0.0078$). The effect rose monotonically across the three doses. This active steady-state ensemble strengthens the local-Jacobian prediction but is not a kinetic-channel or spiking simulation.

Measured responses reveal a boundary for morphology-specific alignment

Structural capacity used ancestry-kernel perturbations and the focal result used modeled adjoints. We next tested whether the routes align with measured functional inputs and response-derived task gradients. A MICrONS/DANDI join identified seven reconstructed targets and 136 repeated visual conditions per scan. The original one-scan-per-target cohort contained 69 connected, imaged excitatory partners. Functional similarity did not increase consistently with shared ancestry after controlling for Euclidean separation and depth: the mean target-level partial rank effect was -0.069 (95% target-bootstrap interval -0.255 to 0.108), positive in three of seven cells (Fig. 8a,b). This estimand asks whether within-tree position adds organization beyond geometry and complements the broader like-to-like wiring principle reported in the same MICrONS volume [36]. Using the deterministic rule "all automatic-conservative scans with at least five connected imaged partners" expanded the analysis to 13 scan observations. After averaging scans within target, the partial shared-path effect was -0.048 (-0.224 to 0.118), positive in three of seven cells. The scan-complete analysis therefore removed the selection ambiguity without changing the null conclusion.

We trained a branch surrogate for each postsynaptic response and projected its exact error into four-channel dictionaries. Each partner mapped through its dominant contact to one nonlinear state on the first major branch below the soma. Sites on that branch therefore shared one task-dependent error. With only 3–8 occupied major branches per target, this cannot resolve within-branch credit or strongly test fine topology. Stimulus identities were held out. Mean held-out field capture was 0.999 for a training-only dense oracle, 0.475 for morphology paths, 0.564 for nonempty random paths, 0.621 for depth bins, 0.446 for ancestry-shuffled paths, and 0.222 for a scalar broadcast (Fig. 8c). Ancestry routes minus ancestry-shuffled capture was 0.028 (95% interval -0.146 to 0.206), with three of seven target cells positive (Fig. 8c).

Learning outcomes gave the same boundary. Mean normalized held-out error was 0.778 for exact backpropagation, 0.781 for the dense oracle, 0.791 for depth bins, 0.801 for scalar broadcast, 0.834 for random paths, 0.837 for morphology paths, and 0.842 after ancestry shuffling. Thus morphology was close to its ancestry-shuffled control but did not outperform the simpler depth-bin or scalar dictionaries on this task. The paired morphology improvement was 0.004 normalized-error units (95% interval -0.021 to 0.027), positive in four of seven targets (Fig. 8d). A pooled capture–performance association vanished after centering within target ($\rho = -0.150$; label-permutation $P = 0.419$; Supplementary Fig. S22g). Higher within-target capture predicted lower error at one channel ($\rho = -0.621$, $P = 0.0006$) and eight channels ($\rho = -0.480$, $P = 0.0104$), but not two or four (Source Data). Channel-wise morphology-minus-shuffle contrasts (Supplementary Fig. S22h), reliability thresholds and a matched

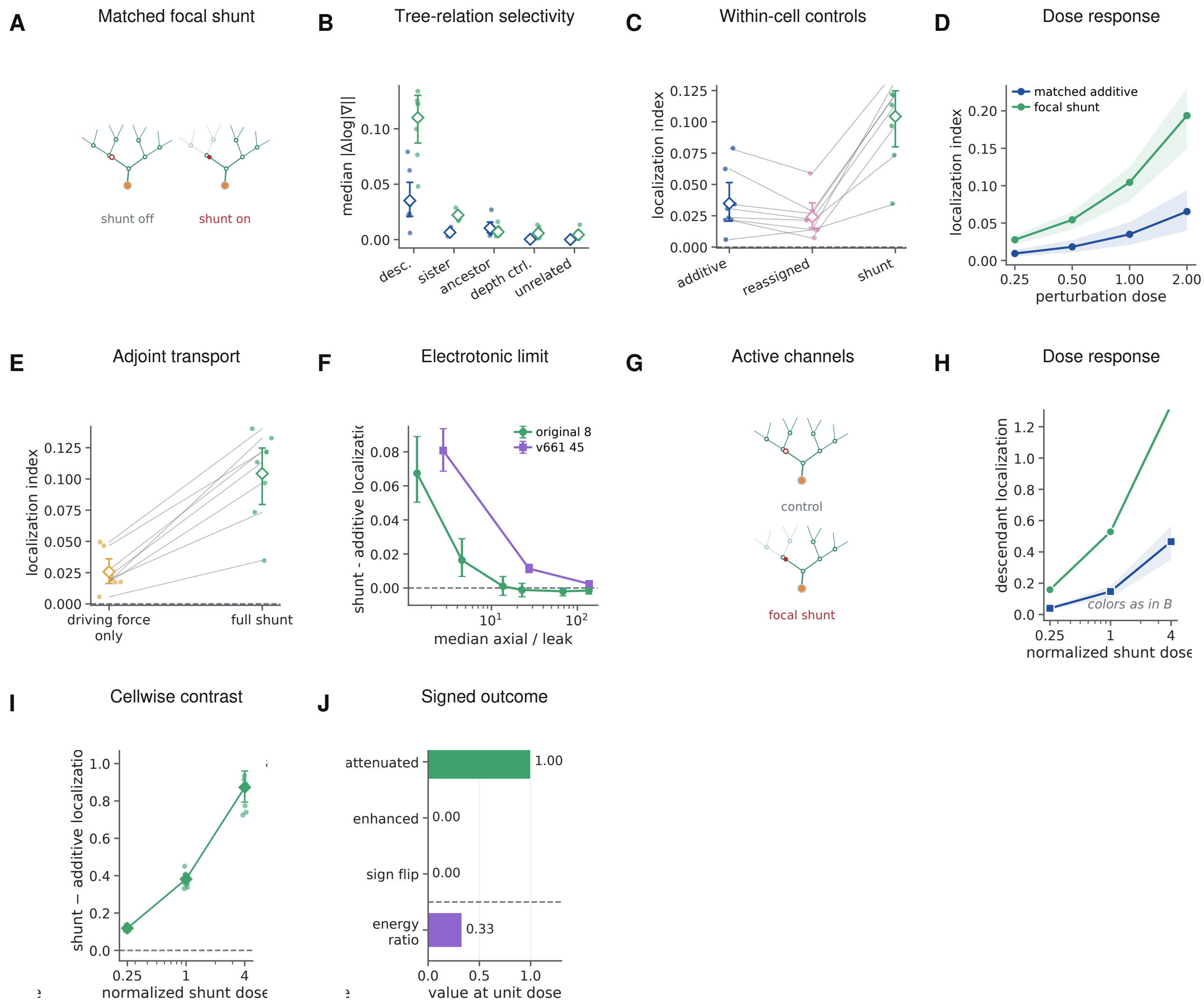

Figure 7: **Focal shunting produces a descendant-enriched modeled-credit signature in defined electrotonic regimes.** **A**, Focal intervention with state-matching somatic current, shown as a shunt-off/shunt-on tree pair; the faded descendant subtree marks where the shunt attenuates transported credit. **B**, Absolute gradient change across tree relations. **C**, Within-cell localization for the matched additive perturbation, route reassignment and focal shunting. **D**, Dose response. **E**, Exact factor-freeze comparison isolating changed adjoint transport. **F**, Shunt-minus-additive localization across the cable-calibrated axial-to-leak ratio; the standard high-resistance calibration is effectively null. **G**, Active-channel steady-state design. **H,I**, Active- model dose response and paired shunt-minus-additive localization. **J**, Descendant outcomes at unit dose: attenuation, enhancement, sign reversal and residual gradient energy. Points are reconstructed-cell means after averaging sites and channel draws; lines, bands and intervals are 95% cell-bootstrap summaries. The reconstructed cell is the inferential unit. Active validation uses the permissive electrotonic regime and does not restore the effect at the standard passive calibration.

subset did not establish morphology-specific benefit.

We then replaced the coarse branch surrogate with an exact passive solve on each complete compressed reconstructed tree. Every imaged partner supplied a trainable non-negative conductance at its mapped segment; the exact local rule used the trial-specific soma-to-segment adjoint. Restricted rules used only scalar output error, local eligibility and a fixed four-channel feedback vector calibrated from topology before learning. All 13 eligible scans and ten held-out-stimulus splits per scan completed without exclusion, giving 520 method fits. Split replicates were averaged within scan, then scans within each of seven target cells.

Exact compartment error achieved mean normalized held-out MSE 0.788. The topology-matched, random-route and site-shuffled rules reached 0.832, 0.833 and 0.839, respectively (Fig. 8g). Topology improved MSE over site shuffling by 0.0070 units (95% target-bootstrap interval -0.0207 to 0.0321) and over random routes by 0.0013 (-0.0090 to 0.0113), neither a reliable advantage. At the common exact checkpoint, topology captured 0.232 of the eligibility-weighted update field, below site-shuffled routes (0.439) and random routes (0.358; Fig. 8h). Restricted dictionaries had identical effective rank within every scan–split run, and exact conductance gradients passed finite differences with maximum relative error 3.1×10^{-7}.

Thus route availability did not imply route use in either the coarse or segment-resolved task. Four dense dimensions nearly matched exact learning in the coarse model, but anatomical routes did not align reliably with measured- response credit on the full tree. The dense-oracle capture curve indicates why the coarse-model null is expected under the credit-operator account: a single channel already captured 0.965 of this response family's credit energy (effective rank 1.07), so a four-channel budget exceeds the task's credit rank and every equal-rank dictionary spans it—the regime in which the theory predicts the observed tie (Fig. 8k). Oracle calibration makes each restricted method a capacity ceiling. The null does not exclude other response windows, higher input coverage or electrophysiologically fitted conductances.

Measured responses showed no reliable morphology-specific alignment; the conditional claim, however, predicts that the same dictionary should become useful exactly when task credit rotates into its span. To test the conditional statement directly, we generated alignment-controlled quadratic objectives on each reconstructed tree. Exact-gradient fields were rotated from the span of the morphology dictionary into its orthogonal complement while channel count, dictionary, gradient energy, and loss curvature were held fixed. The same fields were evaluated with morphology, random-path, depth-bin and ancestry-shuffled dictionaries before norm-matched or iterative projected descent (Fig. 8e,f and Supplementary Fig. S5).

The morphology dictionary captured exactly the imposed aligned energy: from 0 at complete orthogonality to 1 at complete alignment. At 40% alignment it captured 0.400 and supported 0.615 of the loss reduction from a norm-matched full-gradient step. Its capture advantage over the three controls ranged from 0.203 to 0.228 and was positive in all eight cells. At complete orthogonality, morphology lost to every control in all eight cells; at complete alignment, it exceeded the controls by 0.710–0.799 and won in all eight. Thus the same fixed anatomy changes from an unhelpful to a useful feedback basis as only its alignment with task credit changes. Across the three control dictionaries and alignment levels, initial capture tracked 20-step progress within each cell (median Spearman $\rho = 0.990$, range 0.969–0.996). The one-step relation is shown in Fig. 8f and the 20-step relation in Supplementary Fig. S5.

This constructive sufficiency test imposes alignment, fits oracle projection coefficients and uses a quadratic objective. It links field capture to optimization under controlled conditions but does not show that MICrONS cells used these gradients; independently measured responses showed no reliable morphology-specific benefit.

The theory assumes that a neuron-specific teaching coordinate is available. We tested that assumption using published source data from a six-animal brain–computer-interface experiment in which the causal signs of two selected neurons, P+ and P-, were fixed by the experimental mapping [37]. We defined one contrast before inspecting its sign: standardized dendritic population residual during error reduction minus the residual during error increase. If the dendritic signal follows causal credit rather than one common scalar error, the contrast should be positive around P+ and negative around P-.

The P+ contrast was positive in five of six animals (mean 0.0753; 95% bootstrap interval 0.0295–0.1214; exact one-sided random-sign permutation $P = 0.0313$). The P- contrast was negative in all six (mean -0.1498; -0.2054 to -0.0866; $P = 0.0156$). Their signed separation was positive in all six animals (mean 0.2251; 0.1416–0.3108; $P = 0.0156$; Fig. 8i). Decomposing the six animal contrast vectors showed that 83.7% of pooled squared projection energy lay in the signed P+/P- mode, compared with 16.3% in a common scalar mode (animal-bootstrap 95% interval for signed-mode energy, 60.6–95.7%; leave-one-animal-out range, 76.5–90.2%) (Fig. 8j); descriptive neuron-level residual distributions from the published source table are shown in Supplementary Fig. S17d. With six animals, these exact permutation tests sit at or one step above their resolution floor. The analysis independently supports the first level of the theory—signed neuron identity—but does not determine how the signal is transported within a dendritic tree or uniquely implicate shunting.

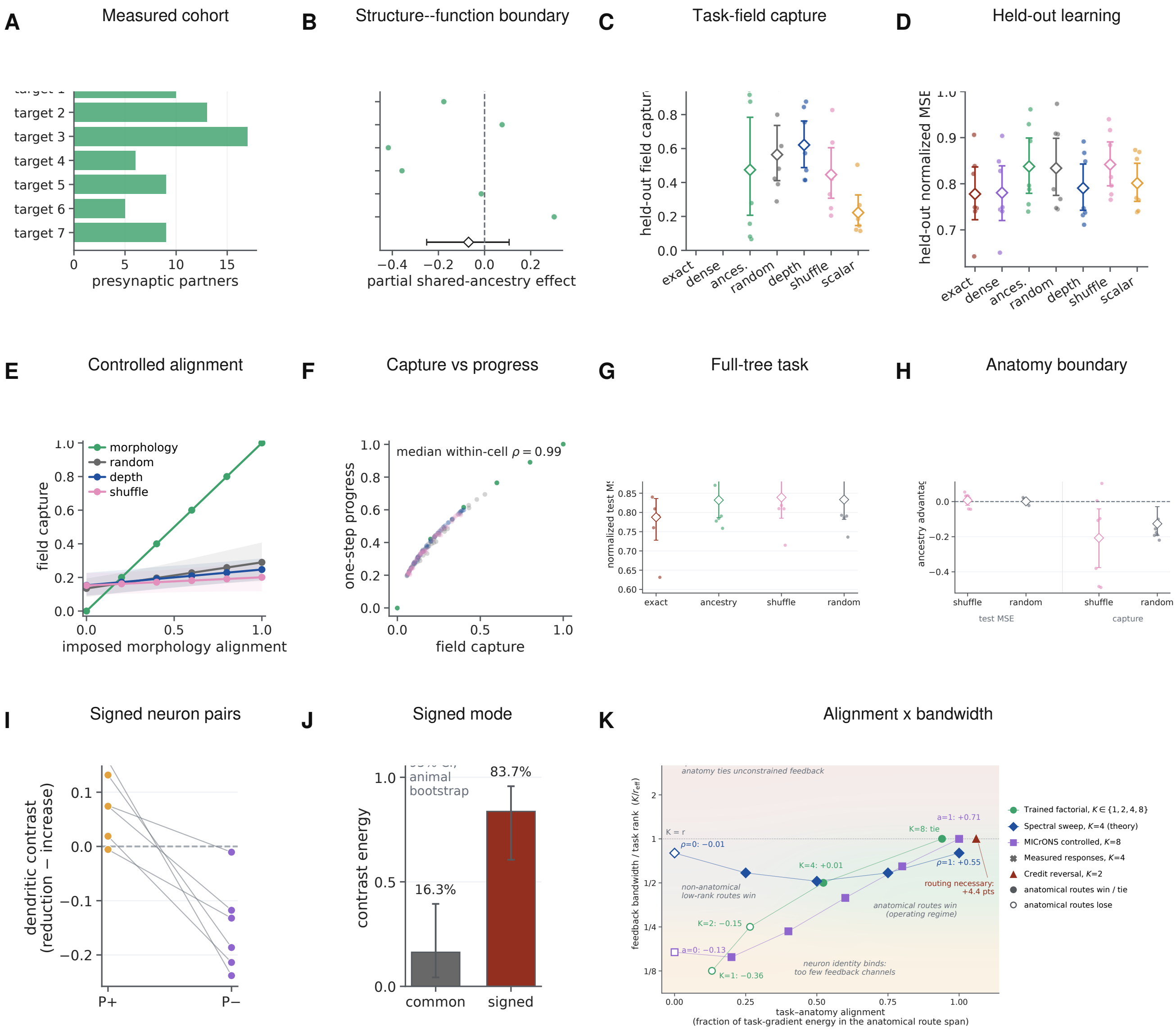

Figure 8: **Measured responses define the biological boundary, while imposed alignment validates the conditional prediction. A,B**, Seven- cell measured-response cohort and the geometry-controlled structure–function effect. **C,D**, Held-out task-field capture and learning for exact, dense, ancestry and matched control dictionaries; projection coefficients are oracle ceilings. **E,F**, Rotating task credit into the fixed ancestry span increases its capture, and capture predicts one-step progress. **G,H**, Complete-tree learning and ancestry-minus-control contrasts across all eligible scans; neither shows a morphology-specific benefit. **I**, Signed neuron-coordinate contrast in six animals: P+ and P- are defined by the causal experimental mapping. **J**, Most contrast energy lies in the signed P+/P- mode rather than a common scalar mode. **K**, Alignment-by-bandwidth synthesis across trained and controlled experiments. Filled markers denote an ancestry win or tie; open markers denote a loss. Shaded regions summarize the credit-operator regimes and are not fitted to the displayed outcomes. Target cells or animals are the inferential units; nested splits and simulations are averaged within them.

693 Together, the experiments occupy a common alignment-by-bandwidth plane (Fig. 8k). Low bandwidth loses neuronal identity, whereas high bandwidth can make equal-rank spans coincide. Between these limits, ancestry routes help only when task credit falls sufficiently inside their span. The trained $K = 4$ interior optimum, the spectral and controlled- alignment sweeps, the credit-reversal task and both measured-response nulls all fall on the corresponding side of this boundary. The shaded regions are theory-derived regimes, not a fitted classifier of the experiments.

Discussion

Each exact synaptic gradient separates into locally available eligibility—presynaptic activity, driving force and input resistance—and a transported compartment error. Biological learning in a dendrite is therefore a well-posed approximation problem: recover a defined credit field from restricted signals. Point-neuron theories ask how an error reaches a neuron or gates eligibility; a dendritic theory must additionally ask which subtree receives that coordinate and how strongly it is transported. Learning can fail by losing neuron identity, addressing the wrong tree or descendants, or omitting route gain.

The prospective experiments separate these failures quantitatively. One coordinate per neuron closes most of the scalar-feedback gap, while deranging the neuron-to-tree map at matched bandwidth produces a smaller but reproducible cost. A 2,700-fit credit-reversal factorial then shows that the value of within-neuron structure is bandwidth dependent. Correct ancestry is inferior to unrestricted low-rank routes at one and two coordinates, acquires a small but reliable advantage over the best matched non-anatomical control at four coordinates, and ties at full rank. It also outperforms a degree- and depth-matched rewired tree at the intermediate budgets. Dendritic, flat, grouped-point and explicitly gated point implementations are identical when given the same routed fields, locating the benefit in structured address resources rather than dendritic material per se. Exact transported feedback makes the derived local eligibility sufficient to approach matched backpropagation, and the detached clean audit found no material exact-transport–backpropagation difference, bounding the claim to empirical agreement rather than formal equivalence. Gradient alignment predicts immediate descent.

The credit-operator bound now unifies these observations. Route quality is not capture alone: it is retained task signal divided by finite-step gain and admitted stochastic noise. This utility predicted the existing factorial better than spectral capture alone. In the independent phase experiment, ancestry became useful as task covariance rotated into its span; routed depth was best at the task's hierarchy depth; and a projection improved a noisy gradient only below the rejected-noise/discarded-signal boundary. These are not routes by which a local rule can beat exact full-batch backpropagation on the same objective. They explain how a restricted stochastic rule can improve relative to another noisy or bandwidth-limited rule, with backpropagation plus the same projection remaining the relevant upper control. The interior-optimum reanalysis sharpens this division of labour: the bound's optimum located the trained depth optimum wherever one existed, whereas the trained $K = 4$ anatomical advantage lies beyond one-step geometry (Supplementary Fig. S16). The results therefore order into four comparisons. A restricted rule never beats exact full-batch backpropagation at matched step norm; it beats stochastic backpropagation exactly when rejected noise exceeds discarded signal; it beats a bandwidth-matched non-anatomical rule only when task covariance aligns with the anatomical span; and any within-tree address beats scalar or neuron-only feedback whenever exact credit varies across the tree in an addressable way. Reliability-weighted route gains improve each comparison when branch signal-to-noise is heterogeneous. Figure 8k places every bandwidth-varied experiment of this study in the resulting alignment-by-bandwidth plane, including the measured-response null, which falls where the theory predicts a null. In operator terms, topology sets the span of addressable fields, conductance sets the metric that weights their use, and the task determines whether either matters.

Depth can affect learning in at least three distinct ways. First, nested branch points reuse route coordinates across exponentially many leaves and provide progressively finer addresses; the matched-depth phase result isolates this role. Second, serial dendritic stages can compose nonlinear forward operations, a role deliberately absent from the quadratic phase experiment. The exact-resource positive-rate experiment now isolates one such case: aligned divisive composition improved accuracy by roughly 31 percentage points in both $H = 2$ and $H = 3$ variants, whereas literal grouped-point models stayed flat and reversed placement made serial depth harmful. Independent and shuffled-sensor controls removed the H3 gain, and the raw-additive series worsened with depth. This is evidence for matched hierarchical shunting within the calibrated task family, not a general law that deeper trees are better. Third, a deeper physical route transports a shared coordinate through more differentiated branch states. Consistent with that cost, shared-soma gradient cosine fell from 0.946 at D1 to 0.767 at D3, whereas exact path transport reproduced autograd at every checkpoint. The earlier additive fixed-contact control likewise showed no intrinsic depth benefit when terminal branches and contacts were fixed. Thus the dominant requirement remains a signed coordinate for the correct neuron, followed conditionally by enough task-aligned address and computational depth—not depth in isolation. The new controls make the point–dendrite boundary explicit: the serial tree beat a resource-identical nonserial star only under the aligned hierarchy; it also beat the stricter literal grouped-point model at both H2 and H3, yet ordinary parameter-matched point MLPs beat the tree by about seven points. Dendritic depth is therefore a structured inductive and wiring resource in this task, not an absolute expressive separation. The credit ladder draws the complementary BP–local boundary: soma-broadcast autograd nearly matched BP under a common optimizer, optimizer choice accounted for the largest raw separation from shared LocalCA, and path specificity recovered most of the remaining local-rule deficit. Stated as an accounting at matched resources, the tree's contribution separates into a forward-composition term isolated by the backpropagation depth contrast, an address-resolution term isolated by the bandwidth factorial, and a transport cost isolated by the falling shared-coordinate cosine, with the local-rule retention measuring their interaction.

Real dendrites then constrain how that coordinate could be distributed. Reconstructed arbors compress model-matched and independently generated cable fields into sparse, ancestry-defined supports using a small fraction of dense feedback wiring—in wiring-normalized terms, an order-of-magnitude advantage in captured energy per feedback connection over a dense oracle, of which a 2.7-fold factor is anatomy-specific at matched density (Fig. 6g,h). A dendritic tree, on this account, multiplexes K addressable fields within one neuron's arbor. An equivalent point implementation requires the same K separately addressable teaching coordinates; its wiring cost depends on how those coordinates are routed. Yet fine morphology does not reliably outperform controls matched for depth and parent degree. The synapse-resolved inhibitory census similarly supports coarse co-targeting and class-specific spatial scales, while shared-path effects do not survive axon-level aggregation and joint geometric matching. These one-mouse measurements establish available addresses, not their use for endogenous learning.

Shunting has a precise and bounded role within this hierarchy. Unlike models in which inhibitory circuits compute an error [38–40], focal conductance here changes the sensitivity of routes crossing a compartment. Exact factor freezing assigns the positive localization effect in the normalized proxy to altered adjoint transport, partly opposed by the change in synaptic driving force. Physical calibration shows that the effect can become very small in passive regimes with high axial-to-leak coupling and re-emerge in higher-conductance states. This spatial boundary is determined by the focal Green's-function column and cable length constant: strong axial coupling spreads the perturbation and reduces the descendant-versus-matched off-path contrast. The passive inverse-operator corollary is used only as a global magnitude bound, $\|\Delta G^{-1}\|_2 \leq \kappa \|G^{-1}\|_2^2/[1+\kappa/\lambda_{\max}(G)]$; it does not predict spatial selectivity. Shunting is therefore a state-dependent route-gain mechanism, not a universal optimizer and not evidence that inhibition itself carries a teaching signal. The active-channel ensemble extends this conclusion beyond a passive operator: exact linearization at 512 accepted Na/K/Ca/HCN/NMDA steady states preserved descendant-enriched attenuation without gradient sign reversal. It remains a local steady-state result, not a claim about channel kinetics or spiking. The reliability phase identifies a possible computational use for that attenuation: fixed-step branch gains $\min\{1, S_b/[c(S_b+N_b)]\}$ suppress unreliable credit and outperform one global gain when reliability differs across branches. The state-matched positive-conductance experiment joins that prediction to a physical input-resistance mechanism in a trained branch model: aligned shunts improved the immediate credit step and final loss over a global shunt, but not final loss over unshunted noisy learning. The independently computed point-gate control agrees to numerical precision and proves that the operation is not mathematically exclusive to dendrites. This differs from the sibling gain–load study, which asks when branch-local inhibition improves forward signal robustness; here the forward state is matched and conductance weights backward credit [41]. Moreover, branch reliabilities and shunt placement were supplied by an oracle, and the compensating current removed forward state changes. The experiment therefore validates a conductance-based, state-dependent approximation to a reliability factor, not autonomous inference of that factor or a completed biological causal chain.

Anatomical capacity is useful only when it matches the task. Neither all 13 eligible visual-response scans nor 520 learning fits on the complete compressed segment trees reveal a reliable morphology-specific advantage for the derived objective. By contrast, rotating exact task credit into or out of the same anatomical span reverses its learning benefit. The six-animal reanalysis supports the availability of signed neuron-specific coordinates, but does not identify their transport within a tree. Route overlap and off-route leakage then predict one-step interference (Supplementary Fig. S6). Together, these results replace the claim that more dendritic structure is inherently better with a conditional statement: route capacity matters when the task requires credit fields that the arbor can express.

The alignment requirement has a population-level analogue: motor learning is faster for perturbations within an existing neural manifold than outside it [42]. Our claim is the subcellular counterpart, not an identification of the population mechanism: a pre-existing ancestry span helps only when the task-gradient field lies sufficiently within it.

These results relate to, and delimit, three strands of prior work. Studies of dendritic nonlinearity establish that branch-local processing expands a single neuron's forward capacity [43–46]; here forward resources were deliberately matched or budgeted, and the tree's contribution was measured instead in feedback bandwidth, so the two accounts are complementary rather than competing. Dendritic-error and segregated-compartment learning models deliver a teaching signal to one or a few named compartments [17–19, 47]; the present framework asks how far such a signal can usefully be distributed within a full arbor, and answers with an explicit bandwidth–alignment condition rather than an architectural preference. Observations of branch-specific plasticity and structured synaptic placement [48–50] demonstrate that learning is spatially organized in vivo; our analysis specifies what that organization must deliver to constitute credit routing rather than forward gain control—nested addresses whose span aligns with task-gradient covariance—and provides the perturbation signature that distinguishes the two.

Bio-inspired machine-learning systems likewise use explicit context gates, active dendritic segments or localized dendritic updates to allocate learning capacity [51–53]. Those algorithms establish useful engineered routing schemes; the present theory instead predicts when a restricted biological route should help or fail at a fixed bandwidth.

The same accounting suggests why credit might be routed through trees at all. With a fixed budget of routed teaching coefficients, a circuit can spend them as many neurons carrying one coordinate each or as fewer neurons with finer within-tree addresses; the implementations are mathematically interchangeable, but their wiring is not. Per-neuron coordinates require long-range feedback axons, whereas per-branch routes reuse local inhibitory synapses on existing arbors, so the wiring-efficient allocation depends on whether task credit varies mostly across neurons or hierarchically within them. On this view the tree is the inexpensive way to buy feedback bandwidth—a question of circuit economics rather than computational exclusivity.

The framework makes a branch-resolved experimental prediction. After matching somatic voltage, output and global error, focal dendrite-targeting inhibition during learning should alter subsequent plasticity more strongly for spines below the perturbed branch than for equidistant spines on sister subtrees. The contrast should scale with local input resistance and perturbation conductance, and a current-matched additive perturbation should not reproduce it. Simultaneous dendritic recording, focal inhibition and longitudinal spine measurements could distinguish credit routing from forward gain control [37, 48].

The theory assumes steady-state, non-spiking trees with unique directed paths; the reconstructed-cell analyses are sensitivity models rather than cell-specific physiological fits. The primary, sensitivity and inhibitory cohorts come from one mouse, the functional cohort contains seven targets, projection and transport coefficients are oracles, and the external animal analysis is retrospective. The trained within-neuron factorial uses one synthetic eight-context hierarchy, analytically controlled route fields and implementation-equivalent linear parameterizations. The active extension linearizes steady states and does not simulate channel kinetics. Temporal eligibility, spikes, recurrent paths and long-term network adaptation remain outside the current model. In particular, branch plateau potentials and behavioral-timescale synaptic plasticity provide documented dendritic teaching signals beyond this steady-state theory [54, 55]. The hierarchy-depth phase uses a fixed-state quadratic family and prescribed task covariance. The nonlinear physical-depth bridge now tests $H = 2$ and $H = 3$, two fixed sensor inventories and one calibrated operating-point family; it does not cross the full $D_r \times H \times K \times \rho$ design, infer sensor alignment or establish prevalence beyond this synthetic task. The positive-conductance reliability experiment trains excitatory branch weights but fixes oracle shunts and uses a compensating current clamp; it does not demonstrate autonomous estimation or learning of shunt placement. We establish no memory, latency or benchmark advantage over backpropagation. Within these limits, the result is an information hierarchy for cellular learning: coordinates identify neurons, ancestry partitions synapses, conductance regulates routes and task alignment determines utility.

Methods

Directed conductance-tree network

Each compartment combines nonnegative excitatory and inhibitory synaptic conductances, leak, and child-to-parent dendritic coupling. Excitatory and inhibitory reversal potentials were fixed within an experiment. Raw trainable parameters were transformed to nonnegative conductances. A compartment could apply a monotone branch output nonlinearity $a_n = f_n(V_n)$ before transmitting activity to its parent; this mechanism is named "reactivation" in the implementation, and f_n was the identity when it was disabled. The full forward equations, raw-parameter derivatives, and temporal order of the local update are provided in Supplementary Section S1.

Writing $p(n)$ for the parent of compartment n, the compartment-error recursion follows from the chain rule on the tree:

$$\frac{\partial \mathscr{L}}{\partial V_n} = \frac{\partial \mathscr{L}}{\partial V_{p(n)}} f_n'(V_n) R^{\mathrm{tot}}_{p(n)} g^{\mathrm{den}}_{n\to p(n)}. \tag{15}$$

Multiplying this recursion along the unique path yields Eq. 7. The local derivative of Eq. 5 with respect to g_i yields Eq. 6. Analytical gradients were checked against PyTorch automatic differentiation on fixed batches and checkpoints. Parameter tensors, rather than individual elements, were the units of the reconstruction diagnostic.

Local learning and feedback fields

The three-factor update was the batch average of

$$\widehat{\nabla}_{g_i} \mathscr{L} = x_i R^{\mathrm{tot}}_n (E_i - V_n) \widehat{\delta}_n. \tag{16}$$

The tested $\widehat{\delta}_n$ fields included a global scalar, the matched-width/scalar-fallback implementation, neuron-indexed sharing, low-rank random feedback, selected paths, and exact path transport. The neuron-indexed field repeats one somatic coordinate over all descendants of that soma. It does not reveal an independent error at each compartment. When $\widehat{\delta}_n = \partial \mathscr{L} / \partial V_n$, the three-factor rule equals the exact conductance gradient; the exact-transport experiment therefore isolates feedback approximation from local eligibility. Four- and five-factor variants multiply the three-factor update by bounded, slowly estimated stage-level scalar reliability terms. They were treated as preconditioners rather than theoretically required teaching signals.

Regular-tree feedback comparisons used MNIST classification with matched $[3,3]$ morphologies, optimizers, decoders, schedules, and paired initializations within architecture. The feedback-definition experiment used 15 independently initialized seeds per architecture and condition. Exact transport used five matched shunting seeds per rule and decoder-update condition. Test data remained hidden during checkpoint selection. Full hyperparameters and provenance are reported in Supplementary Section S5 and the accompanying machine-readable configurations. This feedback-definition cohort is a re-execution under the released implementation with the width- and connectivity-matched additive architecture; the original conference cohort gave $90.75 \pm 1.01 \to 97.15 \pm 0.12$ (shunting) and $91.58 \pm 0.61 \to 97.14 \pm 0.10$ (additive), consistent with the values reported here [23].

The Fashion-MNIST replication changed only the dataset and used the same regular $[3,3]$ architecture, 128 somata, contact counts, three-factor rule, local decoder, architecture-specific initialization, optimizer and validation-based checkpoint selection. Images were flattened to 784 inputs without normalization or augmentation. Within each fresh seed 10200–10209, the 60,000-image training set was split into 48,000 training and 12,000 validation examples; the official 10,000-image test set was used only after selection. Sixty fits crossed two architectures with scalar fallback, neuron-indexed feedback and exact path transport. The primary replication criterion required the paired neuron-indexed-minus-scalar interval to exclude zero and at least 8/10 positive seed signs in each architecture. Intervals used 50,000 paired-seed bootstrap draws and exact two-sided sign-flip tests enumerated all 2^{10} assignments. The artifact gate required complete finite results and checkpoints, the registered seeds and no fallback warning; all gates passed. A frozen-source guard divided each architecture across two source manifests after a concurrent edit. The exact diff only added an optional rule override that no resolved configuration enabled, and the audit retains both manifests.

Regular-tree regime and one-step analyses

The task, inhibition-dose, depth, broadcast-noise, rule-family, error-source, mechanism-control, feedback-ladder, and CIFAR-10 analyses use the frozen source tables and configurations from the original regular-tree study. Unless stated otherwise, each condition contains five independent training seeds. The three-factor, four-factor, five-factor comparison and the local-mismatch control contain three seeds and are treated as descriptive. The context-gating and noise-resilience tasks are synthetic classification tests described in Supplementary Section S5. CIFAR-10 images were flattened; no convolutional front-end, augmentation pipeline, or large-scale benchmark claim was used.

The prospective depth-by-feedback cohort was frozen before outcome inspection. It crossed two tasks, two neuron models, four dendritic depths, and three local feedback fields at ten seeds, together with one matched backpropagation model per task–core–depth–seed combination (480 local-learning and 160 backpropagation runs). The local conditions used the factorization-derived three-factor rule. Configurations fixed the decoder, optimizer, training horizon, reactivation initialization policy, 40 excitatory and 20 inhibitory contacts per distal branch, and one population of 128 modeled neurons. The single-population design prevented equal neuron indices in different feedforward populations from sharing a teaching coordinate. Nominal depth changed compartment and parameter count; it was therefore treated as a depth stress test rather than a capacity-matched comparison.

Each frozen run had to contain its resolved configuration, seed record, checkpoint, complete learning curves, resource record, and finite held-out metrics. The artifact audit also required every expected dendritic stage and excluded critical log errors, non-finite parameters, degenerate coupling stages, and transport-shape fallbacks. A subsequent outcome-independent input- validity rule excluded the 160 synthetic-noise shunting runs because a signed transfer output cannot parameterize positive conductances. This left 480 publication-facing runs. Condition means used the ten paired seeds. Intervals for paired contrasts used 20,000 bootstrap resamples, and exact two-sided Wilcoxon tests were computed from the ten seed differences.

The bandwidth-matched routing cohort was frozen separately and crossed the two tasks and neuron models with depths two and four, correct neuron-indexed routing or a fixed derangement, and seeds 42–51. The derangement maps every somatic coordinate to another neuron and has no fixed point. It preserves the number and per-example distribution of teaching coordinates, the forward network, and all training settings. The same validity rule removed 40 synthetic-noise shunting runs, leaving 120 runs and six task–model–depth conditions. Correct-minus-deranged accuracy was paired by seed. We report paired bootstrap intervals, exact two-sided Wilcoxon tests, and Benjamini–Hochberg correction across the six retained contrasts.

Trained within-neuron subtree-address experiment

The two-stream support-task configuration, task parameters, ten confirmatory seeds and artifact gates were frozen before the canary outcome was inspected. Each example contained two 12-dimensional sibling-stream inputs, a binary context and a balanced binary label. For the context-selected stream, the input mean was the label sign times a seed-specific unit teacher vector; the other stream had the opposite sign at 0.75 times the amplitude. Independent Gaussian noise with standard deviation 1.15 was then added. Every train and test split contained identical counts for all four context–label combinations. Thus context had zero label correlation, both streams were active, and the inactive stream was a structured distractor rather than a zero mask.

For branch weights $\boldsymbol{w}_A, \boldsymbol{w}_B$, the somatic logit was

$$z = \mathbb{1}[c=0]\boldsymbol{w}_A^\top \boldsymbol{x}_A + \mathbb{1}[c=1]\boldsymbol{w}_B^\top \boldsymbol{x}_B. \tag{17}$$

All conditions used the same paired initialization, mini-batches and forward model. The one-neuron and depth-shared fields applied the logistic output error to both branches. Correct $K=2$ feedback applied it only to the selected sibling, while fixed derangement applied it to the other sibling. The random rank-two control rotated the two context coordinates by a seed-fixed orthogonal matrix. Exact transport and analytic backpropagation used the selected-branch derivative. The gated point emulation used the same two parameter groups and context-gated input eligibility; it therefore represents the explicit point-model resource that reproduces the dendritic address.

Each seed used 2,048 training and 2,048 independent test examples, mini-batch SGD with batch size 64 and learning rate 0.3 for 60 epochs, and Gaussian weight initialization with standard deviation 0.03. Switch interference was measured after a further 12 epochs on 1,024 balanced-label examples from context 1 only, without resetting optimizer or weights. Gradient cosine and scaled capture used the concatenated per-example parameter gradients. For one-step progress, each candidate batch gradient was norm matched to the exact gradient before applying a fixed step of 0.25. Canary seed 4099 was used only to enforce balance, nondegenerate capture, minimum exact-route accuracy and exact–BP equivalence. The unchanged code and configuration then ran confirmatory seeds 4100–4109. The independent seed was the paired inferential unit; intervals used 20,000 paired bootstrap resamples and exact two-sided Wilcoxon tests are reported in Source Data. This experiment tests $K = 2$ on one shallow matched forward architecture and served as the support check for the larger factorial.

The full subtree-address protocol used eight contexts assigned to the leaves of a balanced binary hierarchy and eight eight-dimensional input streams. Every context–label cell was exactly balanced. The selected stream carried unit label signal; non-selected streams carried an opposite-sign distractor whose amplitude increased with tree distance (0.15 for a sibling, 0.45 within the same half and 0.75 across halves), followed by independent Gaussian noise of standard deviation 1.1. Each seed supplied 2,048 train, 2,048 test and 1,024 switch examples. Full-batch gradient descent used learning rate 0.3 for 80 epochs and 20 switch epochs. Confirmatory seeds 4200–4219 were fixed before outcome inspection; canary seed 4199 was excluded from inference.

Five representations were crossed with $K = 1,2,4,8$: the task-matched dendritic tree; an explicitly context-gated point model; a flat compartment model; grouped point subunits; and a degree- and depth-matched tree with a seed-fixed leaf permutation. The first four implementations deliberately received identical routed coefficient fields and are computationally equivalent controls. Every representation contained 64 trainable forward weights, 64 active input contacts, one output nonlinearity and one scalar decoder. Route fields were normalized to unit Euclidean norm for each context. The six families were correct contiguous ancestry subtrees, a fixed within-neuron route derangement, depth-interleaved bins, random sparse groups, an unrestricted random rank-K projector and a training-derived rank-K PCA upper bound. Shared neuronal feedback, exact compartment transport and analytic backpropagation supplied references. Random controls were keyed only by seed, family and bandwidth, ensuring identical stochastic realizations across implementation-equivalent representations.

The frozen canary required exact context–label balance, zero context–label correlation, neuron-shared capture below 0.20, exact capture above 0.999999999, exact–backprop endpoint difference below 10^{-12}, exact implementation equivalence below 10^{-12} and zero fallbacks. All gates passed before the 2,700 confirmatory fits ran with unchanged script and configuration hashes. Outcomes were paired by seed. Intervals used 20,000 paired bootstrap draws; two-sided Wilcoxon tests used the 20 seed differences. The per-seed best matched non-anatomical control was prespecified across route derangement, depth-interleaved, random-sparse and random-rank conditions; the learned PCA route remained an explicitly labeled upper bound.

Credit-operator phase experiment and frozen-factorial reanalysis

The phase experiment used 16 Euclidean coordinates and the orthonormal Haar basis of a balanced binary tree, ordered from the global mode (level 0) to single sibling contrasts (level 4). Its configuration, 50 confirmatory seeds 7000–7049, two artifact-only canary seeds and numerical tolerances were frozen before confirmatory outcomes. The canary required basis orthogonality, projector idempotence, static-gain span invariance and exact agreement of the explicit point gate. Confirmatory execution required unchanged SHA-256 hashes of the configuration, runner and numerical reference. Every expected row was present: 3,750 spectral, 1,800 depth, 3,750 projection and 1,500 reliability rows.

For the spectral screen, task covariance assigned total energy $(0.15, 0.35, 0.30, 0.20, 0)$ to tree levels 0–4. An alignment coefficient $\rho \in \{0, 0.25, 0.5, 0.75, 1\}$ mixed this covariance with an independently rotated covariance having the same eigenvalues. Ancestry, random rank-matched and dense principal-eigenspace projectors were compared at $K \in \{1,2,4,8,16\}$. The dense value was also evaluated as the Ky–Fan sum of the leading K eigenvalues divided by total covariance energy. Because the two endpoint covariances had equal trace, ancestry-minus-random capture was affine in alignment; endpoint differences supplied the exact per-seed crossover, with already favorable zero-alignment seeds assigned threshold zero. Capture was $\mathrm{tr}(P\Sigma_g)/\mathrm{tr}(\Sigma_g)$. The same-span diagnostic compared raw nested indicators, their tree-Haar basis and indicators multiplied by fixed nonzero lognormal column gains; rank, coherence, positive-spectrum Gram condition number and capture were computed directly.

For the depth screen, a unit-norm target was sampled from modes at levels 0 through H, with $H \in \{1,2,3,4\}$. Gradient-noise standard deviation in level ℓ was $0.65[1+1.5\max(\ell-H,0)]$. Starting from zero, 80 common-noise updates of size 0.12 used the projector onto levels 0 through $D_{\mathrm{r}} \in \{1,2,3,4\}$; a fixed leaf permutation supplied the rewired control and the identity supplied stochastic backpropagation. The projection screen independently set retained signal r and retained noise n to $\{0.25, 0.5, 0.75, 0.9, 1\}$ and compared an unprojected stochastic gradient, backpropagation plus the same projection and a routed estimator with an added 0.05 total coefficient-noise fraction. At the unit step and identity-curvature quadratic, their expected losses were calculated analytically and stochastic losses were recorded from paired draws.

For the reliability screen, eight disjoint blocks had signal and noise energies proportional to $\exp(hx_b)$ and $\exp(-hx_b)$, respectively, with equally spaced $x_b \in [-1,1]$ and heterogeneity $h \in \{0, 0.5, 1, 1.5, 2\}$. We compared no attenuation, the best common gain, $S_b/(S_b+N_b)$, seed-shuffled and reversed gains, and a point gate receiving the identical branch factors. Intervals resampled the 50 paired seeds with 20,000 draws; paired tests were two-sided Wilcoxon signed-rank tests.

The secondary analysis reconstructed the frozen training data, initialization and route matrices for all 2,700 previously completed subtree-factorial fits. For each of 20 seeds, 64 independently sampled minibatches of size 64 estimated gradient-noise energy at initialization. The exact full-data gradient supplied signal, the largest logistic Hessian eigenvalue supplied local smoothness, and Eq. 9 supplied utility. Correlations were restricted to the 540 dendritic-tree route conditions and uncertainty resampled whole seed blocks. This is an initialization diagnostic on an existing experiment, not a second confirmatory training cohort.

Positive-rate nonlinear physical-depth experiment

To connect the fixed-state hierarchy result to the original rate-based network, we used the registered `hierarchical_gain_load` generator and the production population-network DendriNet. Inputs were ordered as nonnegative excitatory and inhibitory rate streams. For binary label y, the distal excitatory signal had the schematic form

$$E_j = (e_0 + y\Delta + \varepsilon_j)\exp\left\{\sum_{\ell=1}^{3} \sigma z_{\ell,a_\ell(j)} - \frac{1}{2}\sum_{\ell=1}^{3}\sigma^2\right\}, \tag{18}$$

where $a_\ell(j)$ indexes the fine, coarse or global latent factor containing coordinate j. Three disjoint inhibitory blocks separately measured those factors. With sensor alignment α, their log gains were $\sigma[\alpha z_{\ell,a_\ell(j)} + \sqrt{1-\alpha^2} z'_{\ell,a_\ell(j)}]$. Thus $\alpha = 1$ supplied matched branch-local nuisance sensors, whereas $\alpha = 0$ preserved the complete factor marginals with independent sensors. A trial-shuffled control cyclically permuted the sensor field after class balancing, preserving its marginal distribution while removing example-wise pairing.

At every physical stage the production shunting equation was

$$V_n = \frac{E_n + C_n}{1 + E_n + I_n + G_n + \varepsilon}, \tag{19}$$

where C_n and G_n are child current and child conductance. All rate and conductance terms were nonnegative. Reactivation was the identity, so the divisive branch equation was the only intermediate nonlinearity. D1 [8], D2 $[2,3]$ and D3 $[2,1,2]$ had one, two and three nonsomatic stages, respectively, but each retained eight nonsomatic branch units per soma and the same 4/2/2 fine/coarse/proximal sensor inventory. Every model contained 64 somata, 16 excitatory and 12 inhibitory contacts per branch, 14,336 active synapses, 21,760 candidate slots, 66,178 trainable parameters and 2,944 persistent state scalars per sample. Reversing the fine and global feature ranges provided the same-resource tree-placement control.

The operating point used signal contrast $\Delta = 0.80$, matched per-factor train/test log-gain standard deviation 0.25, child conductance 16 and mechanism-neutral initialization. It was selected transparently after an original severe-shift canary failed at every depth and two-seed exploratory gain, coupling and signal ladders located an accessible non-ceiling boundary (Supplementary Fig. S15). No pilot seed entered inference. The fresh confirmatory seeds 10200–10209 crossed aligned, zero-alignment, trial-shuffled and reversed-placement shunting BP; aligned raw additive BP; and aligned or reversed-placement shunting LocalCA with shared-soma or exact path transport, for 270 fits. Adam ran for at most 180 epochs with validation-based early stopping. Learning rate was 0.003 for BP and 0.0008 for LocalCA; all comparisons were paired by seed.

The prespecified primary effect was aligned D3 minus D1 accuracy. Topology specificity required the aligned depth effect minus the corresponding zero-alignment, shuffled-sensor or rewired-tree depth effect. A positive local claim separately required its depth effect and aligned-minus-rewired interaction. Intervals used 50,000 seed-paired bootstrap draws; exact two-sided sign-flip tests enumerated all 2^{10} paired sign assignments. The artifact gate required all 270 finite results, no LocalCA fallback message, exact resource equality, at least nine of ten aligned BP seeds above 0.55 at every depth and condition means below 0.98. A dedicated checkpoint replay measured branch voltage, total conductance, input resistance, activation derivative, path-gain dispersion and local-versus-autograd conductance-gradient cosine on a fixed held-out batch.

Point–dendrite and BP–local-credit controls

The same frozen generator, operating point, stopping rule and paired seeds were used for a 140-fit reviewer-requested extension. The all-active grouped-star model retained the hierarchical model's complete state dictionary and resource counts but evaluated every nonsomatic level independently and pooled the resulting child aggregators only once at the soma. D1 is therefore an equality control; D2 and D3 remove serial composition without removing branch modules. The aligned and reversed-placement regimes were crossed with D1–D3. Two ordinary point MLPs were trained on the aligned task: one matched the active parameter count (15,055), and the nearest integer-width model matched the total count (66,200 versus 66,178 in the tree). Neither MLP was supplied the task hierarchy or branch grouping.

For the teaching-coordinate control, the forward pass and synaptic autograd derivatives were unchanged, but a backward hook replaced every nonsomatic output derivative by the derivative at its owning soma. This "soma-broadcast autograd" condition was first run with the standard BP optimizer, which makes its contrast with full BP coordinate-specific. A configuration audit performed after those outcomes revealed that it could not be compared mechanistically with LocalCA, which used split parameter groups and learning rates selected for its stability. Before viewing the amendment outcomes, we therefore froze 60 additional aligned and reversed D1–D3 fits that changed only soma-broadcast to the LocalCA optimizer groups (global rate 0.0008, branch-weight rate 0.0007 and decoder rate 0.0015). This amendment is identified as a post-outcome diagnostic, not a prospective rescue of the primary analysis.

The architecture, alignment-interaction and credit contrasts were computed within seed. Intervals used 50,000 paired-seed bootstrap draws, and exact two-sided sign-flip tests enumerated all 2^{10} assignments. The artifact gate required 200 finite new fits, no fallback pathway, seeds 10200–10209 and exact equality of trainable parameters, active contacts, candidate slots and persistent state between serial and star models. The implementation tests also required identical D1 forward logits and exact equality of forward logits between full and soma-broadcast autograd modes before training. All gates passed. Configurations, the dated primary contract and its amendment, seed-level outcomes and the machine-readable audit are included with the source data.

Physical-depth alignment dose response

The interpolation extension retained the same generator, shunting-BP model, operating point, early-stopping rule, resources and paired seeds as the physical-depth endpoint experiment. It crossed $\alpha \in \{0.25, 0.50, 0.75\}$ with D1–D3 for 90 new fits and combined them with the unchanged $\alpha = 0$ and 1 endpoint outcomes. The dated contract was frozen after the endpoint outcomes were available but before any intermediate run was inspected. Its primary summary was the ordinary least-squares slope, within seed, of D3-minus-D1 accuracy on alignment; the $\alpha = 0.75$ minus 0.25 change and adjacent-level changes diagnosed linearity. Intervals used 50,000 paired-seed bootstrap draws, and exact two-sided sign-flip tests enumerated all 2^{10} assignments.

A concurrent source-tree edit interrupted 26 of the first 90 scheduled jobs through the frozen-source guard.

Those jobs were rerun under the new source hash; the 64 completed pre-edit jobs were retained because the only code difference introduced an optional rule override that no resolved configuration enabled. A machine-readable two-manifest audit and the dated source-equivalence record accompany the source data. The final gate required exactly 90 unique, finite intermediate outcomes, no fallback pathway, the expected seeds and exactly equal resources across depth; all conditions passed.

Literal grouped-point and second-hierarchy experiments

After the original $H = 3$ endpoints, nonserial-star controls and alignment-dose outcomes were known, we froze a final 220-fit matrix before opening any new outcome. The literal grouped-point model retained each nonsomatic branch module, structured input mask, local shunting equation and trainable scalar, but replaced the serial child aggregator at every level by a direct parameter-matched projection to its owning soma. Each replacement contained the same number and initialized value of raw conductance parameters as the aggregator it replaced. Every nonsomatic level therefore operated on the raw external streams without receiving another branch state. Unit tests required finite trainable gradients, copied conductances and exact D1 logits against the serial model.

The H3 arm crossed aligned and reversed placement, D1–D3, BP and the existing paired seeds 10200–10209 for 60 new fits; the already observed serial cohort was retained unchanged as its paired reference. The H2 arm used two 32-feature sensor tiers, fixed inventory counts 4/4 and exact-resource morphologies D1 [8] and D2 $[4,1]$. It crossed aligned and reversed placement with serial BP, literal grouped-point BP and serial LocalCA using shared-soma or exact-path transport over ten fresh paired seeds 10300–10309, for 160 fits. Task-family, signal, gain, coupling, learning-rate, early-stopping and all other selected operating-point settings were inherited without retuning from H3.

Primary estimands were the H3 aligned serial-minus-grouped-point contrast at D3 and its placement interaction; the H2 serial-BP D2-minus-D1 effect and its placement interaction; the H2 serial-minus-grouped-point contrast at D2 and its placement interaction; and the H2 shared- and path-LocalCA depth effects and placement interactions. Intervals used 50,000 paired-seed bootstrap draws, and exact two-sided sign-flip tests enumerated all 2^{10} assignments. A directional positive claim required an interval excluding zero and at least eight of ten positive differences. All 220 fits completed with finite metrics, no fallback or nonfinite alert, the expected seeds and exact resource equality. Frozen manifests, run-level outcomes, every paired contrast and the machine-readable audit are distributed with Source Data.

State-matched positive-conductance reliability experiment

The locked mechanism experiment used eight branches with four nonnegative rate inputs per branch, 512 training examples, 2,048 test examples and one learned positive excitatory conductance per input. Branch voltages were $V_b = g_b^E/(g_b^L + g_b^E)$. A nonnegative shunt κ_b with reversal zero was paired with the oracle current $\kappa_b V_b$, so the post-intervention voltage was exactly $(g_b^E + \kappa_b V_b)/(g_b^L + g_b^E + \kappa_b) = V_b$, while the local eligibility was attenuated by $(g_b^L + g_b^E)/(g_b^L + g_b^E + \kappa_b)$. This state clamp separated physical input-resistance attenuation from forward feature suppression.

Initial full-data gradients defined branch signal energy S_b. Independent coefficient noise was scaled to give $N_b = S_b \exp(-2hx_b)$ for eight equally spaced $x_b \in [-1,1]$ and $h \in \{0, 0.5, 1, 1.5, 2\}$. With the actual step $\eta = c/L$ and $c = 0.5$, shunts were fixed from the initial oracle attenuation $\min\{1, S_b/[c(S_b + N_b)]\}$ or its step-consistent best global, shuffled or reversed control; they were not learned. Other controls were clean exact backpropagation, noisy credit without a shunt, a state-matched additive current, an explicit point gate receiving the identical factors, and aligned attenuation with clean credit. Forty common-noise full-data updates used one half of the numerically estimated initial smoothness limit. This correction was frozen after identifying that the preceding pilot had paired this half step with the $c = 1$ gain. Two new artifact-only canary seeds preceded 50 fresh paired confirmatory seeds 8100–8149. The frozen gates required nonnegative inputs and shunts, voltage matching and independently computed point/additive equivalence within 10^{-12}, and relative finite-difference errors below 10^{-5} for both the unshunted objective and shunted frozen-clamp surrogate. Observed maxima were 2.22×10^{-16} for state mismatch, 4.00×10^{-15} and 1.31×10^{-14} for the equivalence gates, and 3.60×10^{-8} for the physical finite-difference error.

Intervals resampled paired seeds with 20,000 draws; paired tests were two-sided Wilcoxon signed-rank tests. Configuration, runner hashes, all 2,250 seed–condition rows and the prespecified contract are supplied with Source Data.

Same-span noisy coefficient-learning experiment

The rate-based coefficient experiment fixed a 16-coordinate balanced-tree space and the rank-eight span through depth three. Three parameterizations shared this projector: eight orthonormal Haar columns, 15 redundant raw nested indicators and the same indicators multiplied by fixed positive gains spanning $\exp(-3)$ to $\exp(3)$. Their positive Gram condition numbers were 1, 15 and 388.52. Each seed supplied a unit-norm target in the common span and 80 common noisy target estimates. Observation standard deviation was $2/\sqrt{n_{\mathrm{eff}}}$ for $n_{\mathrm{eff}} \in \{4, 16, 64, 256\}$. Vanilla coefficient descent used 0.2 of each dictionary's stability limit. A separate oracle control multiplied the coefficient gradient by the pseudoinverse Gram matrix, producing the same projected field step in every parameterization.

The design and predictions were frozen before canary seeds 9098–9099 and 50 paired confirmatory seeds 9100–9149. All 4,800 checkpoint rows completed. Projectors agreed to 9.44×10^{-16}, target address residual was at most 1.45×10^{-30}, and Gram-preconditioned trajectories agreed to 1.16×10^{-15}. The unconditional orthonormal-coordinate prediction failed in the low-data regime. We therefore report both the prespecified contrast and the exact post-result bias–variance decomposition rather than relabeling the outcome as confirmatory. Intervals resampled paired seeds with 20,000 draws and tests were two-sided Wilcoxon signed-rank tests.

The historical inhibitory-dose cohort used the noise-resilience task and crossed two neuron models, five inhibitory-contact counts per branch (0, 5, 10, 20 and 40), backpropagation or one of the three local feedback fields, and ten paired seeds, for 400 runs. Its prespecified endpoint was a shunting-minus-additive contrast. Because every shunting cell used the invalid signed-input convention, the complete family is retained in the run-validity ledger but excluded from figures and inference.

The replacement exact-transport/backpropagation audit crossed the two tasks, two cores, depths one to four and paired seeds 42–51, giving 320 runs and 160 method pairs. Training used detached tracked-clean source commit `74792ca` and nested synthetic-data commit `4f3612a`. The noise- task additive core retained signed inputs; the positive-conductance shunting core used an explicit ReLU transfer before conductances were formed. Acceptance required resolved configurations and seeds, complete learning summaries, finite held-out metrics, finite stage-complete checkpoints, current scheduler logs without a fatal error or transport-shape fallback, and clean source identity. Exact-minus-backpropagation effects were paired within task, core, depth and seed; the plotted task–core contrasts first averaged four depths within seed. Incomplete setup and resource attempts supplied no endpoint and were excluded independently of outcome.

The fixed-topology cohort contained 320 runs at depth four and crossed the two tasks, neuron models, backpropagation or three local feedback fields, random or spatially partitioned sparse input indices, and ten paired seeds. Both maps had 21 contacts per distal branch. The spatial map recursively divided the 28×28 feature plane into 16 disjoint rectangles per neuron; the random map sampled each branch independently from all 784 coordinates. A configuration-level audit quantified unique coordinate coverage and cross-branch overlap for the same ten topology seeds. The validity rule removed the 80 synthetic-noise shunting runs, leaving 240 comparisons. Because any spatial effect shared by backpropagation is a forward-connectivity effect, this cohort was not used to claim topology-specific credit transport.

The fixed-budget depth cohort used the synthetic task and contained 320 runs. The retained 160 additive runs crossed four depths, three local feedback fields and matched backpropagation at ten paired seeds while fixing 16 terminal branches and 960–968 active contacts per soma. This matches terminal branches and input contacts, not compartment or trainable-parameter count.

For the prospective fixed-checkpoint diagnostic, scalar, neuron-indexed, and exact teaching fields were evaluated at the 120 input-valid backpropagation checkpoints on one frozen test batch of 128 examples. Only non-somatic dendritic stages and their trainable branch parameters entered the field and gradient vectors. Scaled capture was one minus the squared residual after optimally rescaling a candidate vector to the exact vector. For the learning endpoint, each candidate gradient was normalized to the exact-gradient norm and applied without further training. Step size was $r\|\theta\|/\|\nabla\mathscr{L}\|$ for $r \in \{10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}\}$; 10^{-5} was the primary plotted operating point and all four are supplied as source data. Relative progress was the candidate loss decrease divided by the loss decrease from the norm-matched exact update. The association interval resampled checkpoints while retaining the paired scalar and neuron-indexed fields.

MICrONS morphology processing

Cells were drawn from the functionally co-registered MICrONS cohort and resolved from stable nucleus identifiers to roots in materialization 1822 of `minnie65_public`. Skeletons were downloaded through the MICrONS skeleton service. Incoming synapses came from `synapses_pni_2`. The original eight cells were a convenience pilot selected under the earlier analysis contract, not a random or representative sample; every anatomical claim from this cohort is therefore explicitly single-animal and descriptive. Degree-two dendritic chains were collapsed into branch-to-branch segments. Synapses were assigned to the nearest dendritic cable node and retained within $5\,\mu$m. Presynaptic coarse type supplied E/I identity where available; otherwise, a high-confidence target-structure prediction supplied a secondary proxy from the in-development `synapse_target_predictions_ssa_v2` table. This table does not currently have a stable public release identifier and is not required for the direct-type-only or disjoint-v661 analyses. The direct-type-only analysis discarded all proxy-classified contacts.

Raw axial coupling was proportional to radius squared divided by cable length; it was normalized by its nonzero cell median and clipped to 0.05–20. Raw leak was proportional to radius times the larger of cable length and radius; it was normalized by its nonzero median, and was clipped to 0.1–10. Excitatory and inhibitory synaptic areas were normalized by the median nonzero total synaptic area within cell. These quantities define sensitivity weights, not fitted physical conductances. The soma was the root of every compressed tree. Ancestry matrices were computed by explicit parent traversal and checked for acyclicity and a unique path to the soma.

For the disjoint sensitivity cohort, we froze a 55-cell eligibility universe before inspecting outcomes, mapped all candidates to the public MICrONS v661 static release, and excluded the original eight cells by stable nucleus identifier. All 47 remaining cells passed the structural criteria. Only direct presynaptic E/I annotations were used. Direct labels were available for 11,024 of 234,301 incoming synapses before spatial mapping (4.71%); 10,516 contacts mapped to retained cables (median 212 directly typed synapses per cell). The cohort was not a population-random sample and contained 34 L2 IT, 2 L3 IT, 10 L4 IT, and 1 L5 ET cell. The same structural-capacity procedure was applied without pooling the two cohorts.

MICrONS inhibitory-census analysis

The inhibitory analysis joined the 47-cell v661 cohort to the published v795 census by stable nucleus identifier, before evaluating any endpoint. All 20 overlapping targets were retained. We used the published cell-type table, inhibitory-synapse table, complete input-synapse table, and target SWCs [33]. Raw all-input coordinates were mapped into the published slanted micrometre frame with an affine transformation fit from 80,114 synapse locations represented in both coordinate systems. The maximum transform residual was $1.64 \times 10^{-12}\,\mu$m. Input and known inhibitory locations were mapped to the nearest target skeleton node and retained within $5\,\mu$m. Inputs without a known presynaptic inhibitory cell are called generic rather than excitatory because the census does not label every incoming contact.

For a multisynaptic inhibitory connection, one largest contact was retained within each published axonal clump. Every retained contact was matched to 64 generic input segments in the same compartment with nearest log path distance to the soma. Five hundred independent matched assignments estimated the null within each connection. Pairwise outcomes were cable-tree distance, normalized shared path to the lowest common ancestor, and cosine overlap between the two input-size-weighted binary descendant domains. Outcomes were first averaged over inhibitory connections within each postsynaptic target. The target, not a contact or connection, was the inferential unit.

Two sensitivity analyses tested dependence and spatial matching. First, connection-level contrasts were averaged within each of 92 presynaptic axons before inference. Second, each candidate location was ranked jointly by its log path-distance and three-dimensional Euclidean proximity to the observed contact, using the sum of the two within-pool percentile ranks; the same 500 matched assignments were then repeated. These analyses ask whether a target-level contrast persists after repeated connections from one axon are clustered and after local tissue proximity is included in the null.

For route capacity, rows were segments with mapped input and columns were segments bearing at least one known inhibitory contact. The binary ancestry matrix was weighted by the square root of mapped input size by row and known inhibitory size by column. Actual sites were ranked by column norm. At each budget, we reconstructed the full weighted actual-route matrix from the selected columns. Controls were 200 random subsets of actual sites, depth bins, depth-preserving within-column ancestry shuffles, and an unconstrained SVD oracle. The leverage selection is a capacity diagnostic, not a biological learning rule. Because the selected columns and reconstruction target come from the same actual-route matrix, this endpoint measures internal dictionary compressibility rather than generalization to an independently generated credit field.

Credit dictionaries and capture

The weighted direct route matrix was $\Gamma_{ik} = A_{ik}\beta_k$. It was a normalized route-sensitivity proxy, not the exact derivative of the reciprocal passive cable solve. Perturbation fields were sampled by multiplying independent route amplitudes by K. At each channel budget, ancestry routes were ranked by β_k times the fraction of total mapped excitatory synaptic area among their descendants. Controls were an unconstrained principal-component basis, random nonempty real paths, bins of cable depth, and column-wise ancestry shuffles. Channel count and the number of evaluated examples were matched. Feedback wiring density was the fraction of nonzero entries in the site-by-channel dictionary.

For each dictionary, weighted least squares supplied the optimal projection after mapping both the field and dictionary into the synaptic-area-weighted coordinates of Eq. 12. Twenty independent Monte Carlo streams, each containing 384 perturbation fields and 64 random or shuffled dictionaries per cell and channel count, measured numerical sensitivity. Streams were not treated as biological replicates. Primary comparisons were averaged within cell before paired testing.

For the independent-generator control, the reconstruction target was not sampled from K. We added a small focal shunt at every inhibitory-bearing segment of the reciprocal passive-cable system, restored the baseline somatic voltage with an independently solved state-matching compensatory current, and finite-differenced the log magnitude of the exact excitatory-conductance gradient at every excitatory-bearing segment. The derivative dose was 10^{-3} times the focal segment's membrane-plus-synaptic conductance. This produced a weighted response operator H whose columns were exact focal responses. Expected field-energy capture for a dictionary Φ was evaluated without Monte Carlo sampling as

$$\frac{\|P_{W^{1/2}\Phi} W^{1/2} H\|_F^2}{\|W^{1/2} H\|_F^2}, \tag{20}$$

where W contained mapped excitatory synaptic area and P denoted the orthogonal projector onto the stated column space. In addition to the primary controls, 200 surrogate trees per cell reassigned child subtrees among parent slots at the preceding depth. This preserved every node and edge attribute, every node's topological depth, and every parent's out-degree while changing ancestral membership. A separate row shuffle preserved the selected route values and sparsity while breaking their assignment to excitatory segments.

Focal conductance perturbations

Before running the focal analysis, we fixed the site-selection rule, primary dose, comparison groups, and localization endpoint. Sites were selected by a depth-stratified rule and required at least three excitatory descendants and three comparison sites; at most 16 sites were sampled per cell across topological-depth quartiles. Primary excitatory and inhibitory synaptic scales were both 0.35, with normalized reversal potentials $E_E = 1$ and $E_I = -0.2$. Conductance doses were 0.25, 0.5, 1, and 2 times the local membrane-plus-synaptic conductance. The additive control injected the first-order current $\kappa_k(E_I - V_k)$ at the same node without changing G. An independently solved state-matching compensatory current injection restored baseline somatic voltage after either intervention. This is not the derivative of a continuously adjusted voltage-clamp controller.

The objective was squared somatic error, with the target voltage set one normalized unit below baseline. Excitatory-conductance gradients were obtained by an adjoint solve, treating the solved compensatory current as fixed during the post-intervention derivative, and were checked by finite differences. If $d_i = E_E - V_i$ and $d_i' = E_E - V_i'$, the post-shunt gradient $q_i' d_i'$ was decomposed algebraically into an adjoint-only substitution $q_i' d_i$ and a driving-force-only substitution $q_i d_i'$, alongside the full-shunt and matched-additive gradients. Two-factor Shapley values allocated the nonlinear full-shunt-minus-additive localization payoff between the adjoint and driving-force changes and reconciled exactly to the observed contrast at each site. The matched-additive reference is not a literal no-factor physical intervention, so an unperturbed-reference allocation was also computed as a sensitivity analysis. These substitutions are not independent biological interventions. Synapses were classified as descendants, ancestors, sister-subtree sites, or unrelated sites. Unrelated synapses within each template were matched to descendant cable depth. For baseline gradient g_i and perturbed gradient g_i', we defined $m_i = |\log[(|g_i'| + \varepsilon)/(|g_i| + \varepsilon)]|$, where $\varepsilon = 10^{-15} \max(1, \max_i |g_i|)$. The frozen site-level localization index was the median m_i among descendants minus the median among depth-matched unrelated synapses. Site effects were averaged within cell; the primary paired contrast used the eight cell means. A topology control sampled 200 relation templates from other focal sites, stratified by focal-site topological-depth quartile when another site was available; otherwise it used another site in the same cell. Axial and leak normalization and clipping are specified in Supplementary Section S6.

We separately calibrated the passive cable in physical units. Non-root segments were treated as cylinders with axial conductance $\pi r^2/(R_a \ell)$ and leak conductance $2\pi r \ell / R_m$; the root used the spherical membrane area $4\pi r^2$. Conductances were expressed in nS. Synaptic areas are not conductance measurements, so excitatory and inhibitory synaptic conductances retained the prespecified relative scale and were referenced to the cell-median physical leak conductance. We evaluated $R_a \in \{100, 150, 250\}\,\Omega\,\mathrm{cm}$ at $R_m = 15{,}000\,\Omega\,\mathrm{cm}^2$ and, at $R_a = 150\,\Omega\,\mathrm{cm}$, $R_m \in \{300, 1000, 3000, 5000, 15{,}000, 30{,}000\}\,\Omega\,\mathrm{cm}^2$. The same sites, state-matching current injection, adjoint gradient, unit relative dose, and additive comparison were used. Two zero-length non-root segments in the v661 source skeletons used their radius as the cable length; this fallback and all electrotonic ratios are recorded in source data.

The frozen passive focal-sensitivity matrix used the original eight cells at $R_a = 150\,\Omega\,\mathrm{cm}$. We crossed membrane resistivities of 300, 1000 and $15{,}000\,\Omega\,\mathrm{cm}^2$ with distributed leak-like background conductance equal to 0, 1 or 4 times each segment's baseline leak. At every selected site we evaluated three dose schemes: 0.25, 1 and 2 times local membrane-plus-synaptic conductance; fixed doses 0.05, 0.5 and 5 nS; and 0.25, 1 and 4 times local input conductance $[(G^{-1})_{kk}]^{-1}$. Rank-one voltage and adjoint updates were computed exactly with Sherman–Morrison, followed by the same state-matching somatic current. In addition to absolute localization, the frozen outputs included signed log-magnitude change, attenuated and enhanced fractions, sign-flip fraction, descendant gradient-energy ratio, local input resistance and S_k from Eq. 14. Site values were averaged within cell before eight-cell intervals and shunt-minus-additive contrasts. A numerical canary required positive-definite matrices and somatic residual below 10^{-10} before the full matrix ran with unchanged code and configuration.

For the active steady-state sensitivity, we added voltage-dependent Na, K, Ca, HCN and NMDA currents to the physical passive system at $R_a = 150\,\Omega\,\mathrm{cm}$, $R_m = 1{,}000\,\Omega\,\mathrm{cm}^2$ and one baseline-leak unit of distributed background conductance. Channel activation followed sigmoidal steady-state gates with channel-specific half-activation, slope, power and reversal values fixed in the machine-readable protocol. Global and segment-level maximal-conductance multipliers were independent log-normal draws. Newton iterations with residual-reducing line search solved the full nonlinear equilibrium. Draws were accepted only when the voltage remained within the prespecified range, the residual was below 10^{-10} and the symmetrized local Jacobian was positive definite above 10^{-8}. With purely local voltage-dependent gates, the off-diagonal Jacobian entries are the reciprocal axial couplings, so the steady-state Jacobian is symmetric and the symmetrized gate and the plain inverse used below are exact rather than approximations.

For every accepted equilibrium, the adjoint used the exact inverse of that local Jacobian. Focal rank-one shunts and first-order-current-matched additive inputs were evaluated at 0.25, 1 and 4 times local input conductance, followed by the same state-matching compensatory somatic current. We required 64 accepted draws in every cell, up to 256 attempts; all 512 first attempts were accepted. The 64 channel draws and up to 16 focal sites were averaged within each reconstructed cell before bootstrap intervals and paired tests. The active ensemble is a steady-state local-linearization sensitivity, not a simulation of channel kinetics or spikes.

The focal analysis was also repeated after removing all proxy-classified contacts. In the disjoint v661 sensitivity cohort, 45 of 47 cells supplied 235 eligible focal sites for the primary shunt–additive comparison. Forty cells with at least two eligible sites supplied the reassigned-relation control. Cells, rather than focal sites, remained the units of inference.

Measured-response analyses

Functionally imaged presynaptic partners were linked to their anatomical contacts through the MICrONS/CAVE and DANDI mappings. The original frozen one-scan cohort included 69 partners across seven targets. The scan-complete analysis deterministically retained every automatic-conservative target–session–scan cohort with at least five unique connected imaged presynaptic roots, giving 13 scan observations nested within the same seven targets. Each scan contained 464 trials, 280 stimulus identities and 136 repeated identities. Multiple contacts from one presynaptic partner were pooled; its analysis site was the segment with greatest summed contact area. Functional similarity was the Pearson correlation of condition-mean responses over repeated identities. For each target, the partial-rank analysis rank-transformed similarity and shared ancestry, separately residualized both against log Euclidean separation and path-depth difference, and correlated the residuals. Scan-level statistics were averaged within target before the target bootstrap and signed-rank test. The target neuron, not a scan or synaptic pair, was the inferential unit.

For task-derived credit, a small branch model predicted each postsynaptic response from its connected presynaptic responses. Complete visual-stimulus identities were assigned to held-out splits. Exact branch errors were computed on training trials, and dictionary projectors were fit without held-out identities. Matched models were then trained from the same initialization with exact or projected branch errors. Ten fixed split/initialization replicates were averaged within each of seven target cells. Because projection coefficients used exact errors, these analyses estimate the capacity of a feedback family rather than a biological encoder. Only input-weight errors were projected; branch-readout weights and bias received exact gradients. Normalized MSE was held-out MSE divided by the held-out error of the training-mean predictor. Complete model and optimizer specifications are in Supplementary Section S9.

For the segment-resolved analysis, each complete compressed reconstruction was instantiated as a reciprocal passive conductance system. A trainable non-negative excitatory conductance was placed at the dominant mapped contact of every imaged partner. Exact gradients of somatic squared prediction error used the trial-specific inverse conductance operator and factored into $x_i(E_E - V_i)$ times the segment adjoint. Restricted methods first projected the baseline soma-to-site transfer vector into a four-channel conductance-weighted ancestry dictionary, a column-value-preserving site shuffle or an equal-rank random set of nonempty anatomical routes. This vector was calibrated once without target responses and frozen during learning; each restricted update then used only scalar output error, the fixed site vector and local eligibility. Exact readout and bias gradients were used in all methods. Ten seed-fixed splits held out complete stimulus identities within every scan. Replicates were averaged within scan and all eligible scans within target before seven-cell inference. Finite differences audited one exact conductance coordinate per run, input hashes and runner hash were archived, and any absent route, rank mismatch, non-finite value or incomplete scan caused exclusion; none occurred.

The capture–learning association excluded the exact and dense oracles. Both variables were centered within target before computing Spearman's ρ. Its interval used 20,000 target-level bootstrap draws, and its two-sided null used 20,000 independent reassignments of learning outcomes among method labels within each target.

Alignment-controlled optimization

For each reconstructed cell, we rebuilt the conductance-weighted ancestry kernel and selected eight routes by inhibitory fraction times descendant excitatory synaptic-area fraction, before generating gradient fields. Segment coordinates were weighted by the square root of summed mapped excitatory synaptic area. If Q is an orthonormal basis for the selected routes, each unit-energy exact gradient was constructed as

$$\boldsymbol{g}(a) = \sqrt{a}\,\boldsymbol{u}_{\parallel} + \sqrt{1-a}\,\boldsymbol{u}_{\perp}, \tag{21}$$

where $\boldsymbol{u}_{\parallel} \in \operatorname{span}(Q)$, $\boldsymbol{u}_{\perp} \perp \operatorname{span}(Q)$, and $a \in \{0, 0.2, 0.4, 0.6, 0.8, 1\}$. Directions were paired across alignment levels. Eight random real paths, eight cable-depth bins, and row-permuted ancestry routes supplied controls. The permutation preserved column values and nonzero count while breaking ancestry.

Each cell contributed 40 nested Monte Carlo streams with 256 gradients per stream. The objective was a positive diagonal quadratic with an anatomy-independent curvature spectrum from 0.5 to 1.5, permuted within stream and fixed across methods and alignment levels. One-step updates had norm 0.1 times the exact-gradient norm. Iterative projected descent used learning rate 0.25 for 20 steps. Both outcomes were normalized by the corresponding full-gradient progress. Streams and gradient fields measured numerical sensitivity; cell-level means were the inferential units.

External animal-data reanalysis

Francioni et al. source data were obtained from the public Nature record associated with DOI 10.1038/s41586-026-10190-7. The archived workbook was verified by SHA-256 before analysis. Extended Data sheet 13E supplied one P+ and one P- contrast per animal, defined by the source study as the change in soma–dendrite residual between error-reduction and error-increase periods. We retained the source signs. Signed separation was P+ minus P- and was checked against the value reported in the workbook. The common basis was $(1,1)/\sqrt{2}$ and the signed basis was $(1,-1)/\sqrt{2}$; squared projection norms were divided by total two-coordinate energy. Sheet 5E supplied neuron-level distributions for visualization only. Directional animal-level hypotheses followed the known causal mapping. We report exact random-sign and Wilcoxon tests together with 50,000-draw animal bootstrap intervals.

Static task-interference calculation

The task-interference identity follows from exact quadratic expansion around a Task-A optimum, where the linear term vanishes. For its equal-route illustration, each task occupied 1,000 coordinates, route overlap varied from zero to one, and the Task-B target opposed Task A on shared coordinates. Off-route leakage supplied an update of relative magnitude λ on Task-A-only coordinates. We evaluated a 101×101 grid over overlap and leakage at step size 0.2. Explicit vector updates were compared with the closed form at every grid point. This analysis contains no temporal or spiking neural dynamics and uses the published motor-learning study only for qualitative, cross-study interpretation.

Interior-optimum and wiring-normalized reanalyses

Two deterministic secondary analyses of frozen outputs support Supplementary Fig. S16 and Fig. 6g,h; neither alters or reruns any training outcome. For the interior-optimum analysis, the per-seed initialization utility of Eq. 9 was taken from the archived operator metrics of the 2,700-fit factorial and from the frozen depth-phase tables. The predicted optimum is the argmax of that utility over routed depth, or of the ancestry-minus-best-control utility contrast over budget; the observed optimum is the argmin of final population loss, or the argmax of the corresponding held-out accuracy contrast. Before any new quantity was reported, the pipeline reproduced the published $K = 4$ ancestry-minus-best-control accuracy contrast (0.0127; 15 of 20 seeds) from the frozen seed outcomes. For the wiring-normalized analysis, capture per unit wiring is weighted field-energy capture divided by feedback wiring density, computed per cell from the frozen routing-capacity curves after averaging Monte Carlo streams within cell; the published eight-channel capture and wiring anchors were reproduced before any ratio was formed. Intervals are 95% seed- or cell-bootstrap intervals with 20,000 resamples. The laminar-capture proposition and the marginal utility condition (Supplementary Section S4) are verified numerically in the released test suite.

Statistics and reproducibility

All tests were two sided unless a directional hypothesis had been frozen in advance. Effect sizes and 95% intervals accompany central comparisons. Reconstructed-cell intervals resampled cells first and, where appropriate, a Monte Carlo stream within cell. Textual focal-attribution intervals used a hierarchical bootstrap over cells and sites; Fig. 7h instead shows cell-bootstrap intervals after averaging sites within each cell. Measured-response intervals resampled target cells. Artificial-network means and standard deviations used independent initialization seeds. Exact Wilcoxon signed-rank tests were reported for small paired cell cohorts; no site, trial, epoch, or Monte Carlo stream was counted as an independent biological replicate. Null results are described as a lack of reliable evidence rather than proof of equality. Bootstrap intervals used 20,000 draws. The four-budget primary subtree scan used Benjamini–Hochberg correction; other parameter sweeps and sensitivity analyses are labeled secondary and are not treated as independent confirmation.

Seed counts followed the evidential role of each cohort rather than one pooled power calculation: three to five seeds were reserved for numerical or mechanism diagnostics; ten for fresh paired confirmatory replications with interval and sign-agreement gates; 15–20 for inherited trained factorials; and 50 for phase boundaries and crossover distributions. Effect sizes were not known before the new simulation families, so no prospective parametric power calculation was used. Positive confirmatory claims required a paired 95% interval excluding zero and the prespecified fraction of concordant seed signs; diagnostic cohorts were not promoted to inferential evidence. A machine-readable table in Source Data maps every headline endpoint to its independent unit and identifies nested or descriptive observations.

Analysis code records configuration files, seeds, software versions, source tables, and machine-readable summaries. Numerical source data are organized by figure and panel in the accompanying package. Tests cover tree construction, projection identities, and exact-gradient checks.

Reporting of computational assistance

OpenAI Codex was used to assist code development, statistical and consistency audits, figure preparation, literature checks, and language editing. It was not listed as an author and did not determine the scientific conclusions. The authors retain responsibility for the study design, interpretation, accuracy of the code and numerical results, citations, and final manuscript.

Ethics and data reuse

No new experiments involving animals or human participants were performed. All biological analyses reused public MICrONS, DANDI, and published animal source data; experimental collection and institutional approvals are described in the source studies [33, 37, 56].

Data availability

MICrONS anatomy and connectivity are available through the public CAVE datastack `minnie65_public`. The primary cohort used materialization 1822, skeleton-service reconstructions, and the `synapses_pni_2` table. The disjoint sensitivity cohort used the public v661 static release. The secondary target-structure labels used only in the full-contact analysis came from the in-development `synapse_target_predictions_ssa_v2` table and are therefore distributed only as derived labels subject to the source-data access terms; all principal structural findings were repeated without those labels. Volumetric imagery is archived in BossDB under DOI 10.60533/BOSS-2021-T0SY, and co-registered functional data are available as DANDI:000402; exact asset and table identifiers are recorded in the supplied manifest. The source publication describes the full data release [56]. Public benchmark datasets are available from their original repositories. The inhibitory-census tables and SWCs are available with the source record for Schneider-Mizell et al. [33]. The Francioni et al. workbook is available from the Source Data link associated with the Nature article [37]; its checksum and used sheet names are recorded in the manifest. Derived cell and exclusion manifests, numerical source data underlying every figure, and non-restricted summaries are supplied with this manuscript. Authentication tokens and externally hosted raw caches are not redistributed.

Code availability

The accompanying reviewer software archive contains the complete training implementation used here, exact-gradient and perturbation analyses, frozen configurations, tests, figure scripts and byte-identified snapshots of the critical training components under the MIT License. A separate Source Data archive contains the numerical data underlying every display item. Submission is gated on replacing this sentence with the public repository URL and versioned archival DOI; no public identifier is asserted before that record is created.

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Acknowledgements

We thank the MICrONS Consortium and the maintainers of the associated CAVE and DANDI resources. We also thank the authors of the published animal-learning studies for making their source data available. This work was supported in part by a gift from the Chan Zuckerberg Initiative Foundation to establish the Kempner Institute for the Study of Natural and Artificial Intelligence at Harvard University.

Author contributions

H.S. developed the computational framework, performed the analyses, and wrote the initial manuscript. M.R. contributed to the local-credit-assignment theory, implementation, and interpretation. B.L.S. supervised the work and contributed to the biological framing, study design, and manuscript. All authors reviewed and revised the manuscript.

Competing interests

The authors declare no competing interests.

Supplementary Information for
When dendritic structure helps local credit assignment

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Scope of the Supplementary Information

This document records the derivations, implementation conventions, analysis contracts, sensitivity analyses and numerical boundaries underlying the main article. We distinguish the evidential levels throughout. First, exact path-product identities and regular-tree learning concern the directed feedforward conductance network. Second, the structural MICrONS analysis uses an ancestry-based route-sensitivity proxy on measured morphology. Third, focal perturbations use an exact adjoint of a reciprocal passive cable model on the same morphology. Neither MICrONS model measures learning in the animal. Fourth, the active-conductance ensemble tests the exact local Jacobian around nonlinear steady states; channel-density draws are nested sensitivity samples, not animals or kinetic simulations. Fifth, measured-response analyses use all eligible scans and independently observed visual responses to test task alignment, not to infer an endogenous teaching signal. Sixth, the alignment-controlled quadratic experiment imposes exact credit and tests a conditional optimization prediction; it is not independent biological evidence. Seventh, the 50-seed credit-phase experiment tests spectral alignment, hierarchy-depth matching, projection denoising and oracle branch reliability in a fixed-state quadratic family; it is not a trained positive-conductance dendrite. Eighth, a separate 50-seed experiment trains positive excitatory conductances while fixed nonnegative shunts realize oracle branch-reliability factors under an exact state clamp; it does not learn the shunts or show a dendrite-exclusive operation. Ninth, the inhibitory census is a synapse-resolved structural association in one mouse. Tenth, the task-interference calculation is an exact static consequence. Eleventh, the six-animal brain–computer-interface analysis is an external retrospective consistency test of signed neuron identity. Twelfth, the nonlinear physical-depth experiment trains the production positive-rate population model on a selected hierarchical task after transparently reported accessibility pilots; it is not a full $D_{\mathrm{r}} \times H \times K \times \rho$ phase grid or an impossibility result for grouped point models. Exact transported errors and fitted projection coefficients are information oracles wherever they appear.

The primary analysis unit is the independent training seed for artificial networks, the reconstructed cell for anatomy and perturbation analyses, the postsynaptic target cell for measured-response and inhibitory-census analyses, and the animal for the signed-credit reanalysis. Sites, channel draws, scans, splits, synapses, stimulus trials, epochs and Monte Carlo streams are nested observations. All reconstructed cells come from one animal and are not independent biological replicates at the animal level.

S1 Exact gradients in a directed conductance tree

1.1 Steady-state compartment voltage

Let a rooted dendritic tree have soma node 0. A non-somatic compartment n receives synaptic conductances g_i^{syn} with presynaptic activity x_i and reversal potential E_i, child-to-parent coupling conductances $g^{\mathrm{den}}_{c\to n}$, and leak g_n^L with reversal E_L. A child transmits $a_c = f_c(V_c)$, where f_c is the branch output nonlinearity (named "reactivation" in the implementation) and is the identity when that mechanism is disabled. The steady-state voltage is

$$g_n^{\mathrm{tot}} = g_n^L + \sum_{i \in \mathscr{S}_n} g_i^{\mathrm{syn}} x_i + \sum_{c \in \mathscr{C}_n} g_{c \to n}^{\mathrm{den}}, \tag{1}$$

$$V_n = R_n^{\mathrm{tot}} \left(g_n^L E_L + \sum_{i \in \mathscr{S}_n} g_i^{\mathrm{syn}} x_i E_i + \sum_{c \in \mathscr{C}_n} g_{c \to n}^{\mathrm{den}} a_c \right), \qquad R_n^{\mathrm{tot}} = (g_n^{\mathrm{tot}})^{-1}. \tag{2}$$

For nonnegative conductances and activities, Eq. 2 is a conductance-weighted average of leak, synaptic reversal potentials and child activities. This boundedness is useful for numerical stability, but is not required for the chain-rule factorization below.

1.2 Local sensitivities and synaptic eligibility

Differentiating the numerator and denominator of Eq. 2 gives

$$\frac{\partial V_n}{\partial g_i^{\mathrm{syn}}} = x_i R_n^{\mathrm{tot}} (E_i - V_n), \tag{3}$$

$$\frac{\partial V_n}{\partial V_c} = f_c'(V_c) g_{c \to n}^{\mathrm{den}} R_n^{\mathrm{tot}}, \tag{4}$$

$$\frac{\partial V_n}{\partial g_{c \to n}^{\mathrm{den}}} = R_n^{\mathrm{tot}} (a_c - V_n). \tag{5}$$

Thus the exact gradient of a synaptic conductance is

$$\frac{\partial \mathscr{L}}{\partial g_i^{\mathrm{syn}}} = \underbrace{x_i R_n^{\mathrm{tot}} (E_i - V_n)}_{\text{local eligibility}} \quad \underbrace{\frac{\partial \mathscr{L}}{\partial V_n}}_{\text{compartment error}} \quad . \tag{6}$$

The eligibility contains quantities available at the synapse or its host compartment. All remaining task dependence lies in the compartment error.

1.3 Transport of the somatic error

Let $p(n)$ be the unique parent of non-somatic node n, and let $\delta_0^V = \partial \mathscr{L} / \partial V_0$. The chain rule gives

$$\frac{\partial \mathscr{L}}{\partial V_n} = \frac{\partial \mathscr{L}}{\partial V_{p(n)}} f_n'(V_n) R_{p(n)}^{\mathrm{tot}} g_{n \to p(n)}^{\mathrm{den}}. \tag{7}$$

Because the graph is a tree, iterating Eq. 7 along the unique path from n to the soma yields

$$\widetilde{\alpha}_n = \prod_{(i \to k) \in \mathrm{path}(n \to 0)} f_i'(V_i) R_k^{\mathrm{tot}} g_{i \to k}^{\mathrm{den}}, \tag{8}$$

$$\frac{\partial \mathscr{L}}{\partial V_n} = \widetilde{\alpha}_n \delta_0^V. \tag{9}$$

The effective path gain $\widetilde{\alpha}_n$ includes every local transfer derivative. With identity reactivation, it reduces to the conductance-stage gain α_n^{cond} obtained by removing the f_i' factors. Equation 9 is an exact derivative, not a separately fitted model parameter.

1.4 Raw-parameter transforms

Trainable raw parameters θ are transformed to nonnegative conductances by a monotone map $g = \psi(\theta)$. The optimizer receives

$$\frac{\partial \mathscr{L}}{\partial \theta} = \frac{\partial \mathscr{L}}{\partial g}\psi'(\theta). \tag{10}$$

This positive transform preserves the sign relation between the conductance gradient and the local rule. Active synapse masks are applied after forming the conductance-space gradient. The exact-gradient diagnostics compare gradients in the same parameter space and include the transform derivative.

1.5 General implicit-adjoint form

The preceding path product is a corollary of a more general steady-state identity. Suppose $\boldsymbol{F}(\boldsymbol{V}, \boldsymbol{g}, \boldsymbol{x}) = \boldsymbol{0}$ and the compartment Jacobian $J_V = \partial \boldsymbol{F}/\partial \boldsymbol{V}$ is nonsingular. Define the adjoint $\boldsymbol{q}$ by

$$J_V^{\mathsf{T}} \boldsymbol{q} = \nabla_{\boldsymbol{V}} \mathscr{L}. \tag{11}$$

The implicit-function theorem gives

$$\frac{\partial \mathscr{L}}{\partial g_i} = -\boldsymbol{q}^{\mathsf{T}} \frac{\partial \boldsymbol{F}}{\partial g_i}. \tag{12}$$

For the residual convention $\boldsymbol{F} = G\boldsymbol{V} - \boldsymbol{b}$, a conductance synapse at compartment n has $\partial \boldsymbol{F}/\partial g_i = x_i(V_n - E_i)\boldsymbol{e}_n$, and therefore

$$\frac{\partial \mathscr{L}}{\partial g_i} = x_i(E_i - V_n)q_n. \tag{13}$$

Equations 11–13 recover the directed path product when J_V is triangular. For the reciprocal passive cable, $J_V = G$ is symmetric positive definite, so $G^{-\mathsf{T}} = G^{-1}$. A general nonsymmetric or recurrent operator instead requires the inverse transpose. Active steady states enter through their Jacobian whenever the required differentiability and nonsingularity conditions hold; no passive symmetry is assumed by the general identity.

1.6 Conductance-specific control of path gain

Holding coupling conductances and the forward state fixed, add inhibitory conductance $\Delta G_k^I \geq 0$ at compartment k. For a compartment n whose path crosses k,

$$\frac{\alpha_n^{\text{cond}}(\Delta G^I)}{\alpha_n^{\text{cond}}(0)} = \prod_{k \in \mathscr{A}(n)} \frac{g_k^{\text{tot}}}{g_k^{\text{tot}} + \Delta G_k^I}, \qquad \frac{\partial \log \alpha_n^{\text{cond}}}{\partial G_k^I} = \begin{cases} -R_k^{\text{tot}}, & k \in \mathscr{A}(n), \\ 0, & k \notin \mathscr{A}(n), \end{cases} \tag{14}$$

where $\mathscr{A}(n)$ contains the compartments through which credit from n passes to the soma. Equation 14 proves path-selective attenuation. It does not prove that attenuation improves optimization.

For $z_n = \log \alpha_n^{\text{cond}}$ and path-dependent attenuation β_n, the perturbed log gain is $z_n' = z_n - \beta_n$, so

$$\operatorname{Var}(z') = \operatorname{Var}(z) + \operatorname{Var}(\beta) - 2\operatorname{Cov}(z, \beta). \tag{15}$$

Shunting narrows this distribution only if $\operatorname{Cov}(z, \beta) > \operatorname{Var}(\beta)/2$. Path-gain compression is therefore an operating-point-dependent empirical outcome rather than a theorem.

S2 Local rules and feedback objects

2.1 Three-factor update

The three-factor rule retains the exact local eligibility and substitutes an available voltage-space teaching field $\widehat{\delta}_n$ for the exact compartment error:

$$\widehat{\nabla}_{g_i^{\mathrm{syn}}}\mathscr{L} = \left\langle x_i R_n^{\mathrm{tot}}(E_i - V_n)\widehat{\delta}_n \right\rangle_B, \qquad \widehat{\nabla}_{g^{\mathrm{den}}_{c\to n}}\mathscr{L} = \left\langle R_n^{\mathrm{tot}}(a_c - V_n)\widehat{\delta}_n \right\rangle_B, \tag{16}$$

where $\langle\cdot\rangle_B$ denotes a mini-batch average. Excitatory and inhibitory contacts use the same expression; their reversal potentials set the sign and magnitude of the driving force.

The four- and five-factor rules multiply Eq. 16 by slowly varying, bounded stage-level scalar reliability estimates. For stage ℓ, mini-batch t, and example b, the implementation averages the recorded layer voltage and corresponding somatic population activity over their feature axes, giving $\bar{V}^{(t)}_{\ell,b}$ and $\bar{V}^{(t)}_{0,b}$. The archived runs used the uncentered dot-product mode,

$$s_{\ell,t} = \frac{1}{B}\sum_{b=1}^{B} \bar{V}^{(t)}_{\ell,b}\bar{V}^{(t)}_{0,b}, \tag{17}$$

$$m_{\ell,1} = s_{\ell,1}, \qquad m_{\ell,t} = 0.9 m_{\ell,t-1} + 0.1 s_{\ell,t} \quad (t > 1), \tag{18}$$

$$r^{4\mathrm{F}}_{\ell,t} = \mathrm{clip}_{[0.1,2.0]}(m_{\ell,t}). \tag{19}$$

Thus $r^{4\mathrm{F}}_{\ell,t}$ is one scalar for a recorded stage, not a separate branch-wise signal, and is not a centered covariance or a Pearson correlation. Recorded voltages are detached from automatic differentiation, and the same mini-batch supplies $s_{n,t}$ and the local update. The positive clamp prevents a negative or very small dot product from reversing or eliminating the update.

The five-factor rule retains this multiplier and adds a conditional variance-ratio factor. Let $Y^{(t)}_\ell$ be the recorded stage voltage and $X^{(t)}_\ell$ its source-resolved parent proxy, each centered across mini-batch t. With $\mathscr{E}_{0.1}[\cdot]$ denoting an EMA with update rate 0.1, define

$$v_{Y,\ell,t} = \mathscr{E}_{0.1}[\langle (Y^{(t)}_\ell)^2\rangle], \qquad v_{X,\ell,t} = \mathscr{E}_{0.1}[\langle (X^{(t)}_\ell)^2\rangle], \qquad c_{YX,\ell,t} = \mathscr{E}_{0.1}[\langle Y^{(t)}_\ell X^{(t)}_\ell\rangle], \tag{20}$$

$$\beta_{\ell,t} = \frac{c_{YX,\ell,t}}{v_{X,\ell,t} + 10^{-3}}, \qquad v_{\mathrm{res},\ell,t} = \max\{v_{Y,\ell,t} - \beta_{\ell,t} c_{YX,\ell,t}, 10^{-12}\}, \tag{21}$$

$$\phi_{\ell,t} = \mathrm{clip}_{[0.25,4.0]}\left(\frac{v_{Y,\ell,t}}{v_{\mathrm{res},\ell,t}}\right), \qquad \widehat{\nabla}^{5\mathrm{F}} = r^{4\mathrm{F}}_{\ell,t}\phi_{\ell,t}\widehat{\nabla}^{3\mathrm{F}}. \tag{22}$$

If $v_{X,\ell,t} \le 10^{-12}$, the implementation sets $\phi_{\ell,t} = 1$. These factors are empirical preconditioners. They are neither needed for the exact factorization nor extra task-error channels. Under exact transported error, the three-factor rule recovers the exact dendritic parameter gradient in the stated steady-state model. This separates feedback approximation from the validity of the local eligibility term; agreement of trained endpoints with backpropagation is tested empirically rather than assumed from the identity.

2.2 Feedback fields

Table S13 defines the feedback objects without conflating bandwidth with routing. The matched-width/scalar-fallback field reuses a somatic vector only at a stage of the same width; at wider stages it uses the same robust scalar reduction as the global scalar condition. For a two-dimensional somatic error vector $\boldsymbol{\delta}_b$ in example b, the code first computes $\mu_b = N^{-1}\sum_u \delta_{b,u}$. With $\varepsilon = 10^{-6}$,

$$s_b = \begin{cases} \mu_b, & |\mu_b| > \varepsilon, \\ \delta_{b,u^*}, \quad u^* = \arg\max_u |\delta_{b,u}|, & |\mu_b| \le \varepsilon. \end{cases} \tag{23}$$

For a one-dimensional input the coordinate is retained. The replacement prevents cancellation from silently producing a zero teaching signal. The historical jobs did not record the replacement count, so its cohort-specific activation rate is an explicit output of the clean rerun rather than inferred post hoc. The matched-width field does not repeat each neuron's coordinate over that neuron's descendants. The ancestry-shared field does exactly that, using one communicated coordinate per neuron and the tree as a fixed ownership map. Exact transport supplies the entire path derivative and is an information upper bound.

2.3 Conditional gradient alignment

For one tree and one example, suppose the correct somatic error is available but path gain is omitted. Write local eligibility as e_i and let $n(i)$ be the host compartment. Then

$$g_i^{\text{exact}} = e_i \widetilde{\alpha}_{n(i)} \delta_0^V, \qquad g_i^{\text{shared}} = e_i \delta_0^V. \tag{24}$$

For nonnegative path gains and nonzero gradients,

$$\cos(g^{\text{shared}}, g^{\text{exact}}) = \frac{1}{\sqrt{1+\mathrm{CV}_w(\widetilde{\alpha})^2}}, \qquad w_i = (e_i \delta_0^V)^2. \tag{25}$$

This identity explains why path-gain dispersion can matter after the correct somatic coordinate has been supplied. It does not solve the separate problem of supplying the correct coordinate across network layers.

S3 Topology-matched credit capacity

3.1 Ancestry dictionary

Let A_T be the synapse-by-route ancestry matrix of tree T, and let $\boldsymbol{\beta}$ contain state-dependent route gains. A low-bandwidth teaching vector $\boldsymbol{z} \in \mathbb{R}^K$ generates a synaptic field

$$\widehat{\boldsymbol{g}} = \Phi_{T,K} \boldsymbol{z}, \qquad \Phi_{T,K} = W_e A_T \operatorname{diag}(\boldsymbol{\beta}) S_K, \tag{26}$$

where W_e holds local eligibility or anatomical weights and S_K assigns the K communicated channels to route sites. Topology fixes support. Conductance changes gain and conditioning but does not create an ancestry relation that is absent from A_T.

3.2 Address dimension and the point-neuron resource comparison

Separate the synapse-local eligibility from the task-dependent coefficient field and let $\boldsymbol{c}_u \in \mathbb{R}^{N_u}$ denote the latter for neuron u. For a prespecified nested family of dictionaries $\Phi_{u,K}$, define

$$d_{\varepsilon,u} = \min \left\{ K : \frac{\mathbb{E}\|P_{\Phi_{u,K}} \boldsymbol{c}_u\|_W^2}{\mathbb{E}\|\boldsymbol{c}_u\|_W^2} \geq 1-\varepsilon \right\}. \tag{27}$$

This definition concerns the task-dependent coefficient after eligibility has been factored out; heterogeneous presynaptic activity alone is therefore not counted as an extra teaching coordinate.

For any fixed $\Phi \in \mathbb{R}^{N_u \times K}$, all fields generated as $\Phi \boldsymbol{z}$ lie in $\operatorname{col}(\Phi)$ and hence occupy at most $\operatorname{rank}(\Phi) \leq K$ dimensions. Conversely, the least-squares field in that space is $P_\Phi \boldsymbol{c}_u$, and its expected weighted residual energy is

$$\mathbb{E}\|(I-P_\Phi)\boldsymbol{c}_u\|_W^2 = \operatorname{tr}\left[(I-\overline{P}_\Phi)\overline{C}_c\right], \qquad \overline{C}_c = \mathbb{E}[W^{1/2} \boldsymbol{c}_u \boldsymbol{c}_u^\top W^{1/2}]. \tag{28}$$

Equations 27–28 prove the address-rank statement used in the main text. The ordinary point-neuron local rule has $\Phi = \mathbf{1}$ and therefore one task-dependent coefficient per neuron. A K-route dendritic dictionary can use up to K structured fields inside that neuron. A point-neuron network is not mathematically forbidden from matching them, but doing so requires K separately routed teaching coefficients, K gated parameter groups, or an equivalent expansion into additional units. Accordingly, the comparison is about where address resources are represented, not about an absolute expressive separation between all possible dendritic and point-neuron networks.

3.3 Field capture

Let W be a positive diagonal matrix that defines the coordinate weighting. Set $\overline{\boldsymbol{g}} = W^{1/2}\boldsymbol{g}$, $\overline{\Phi}_{T,K} = W^{1/2}\Phi_{T,K}$, and let $\overline{P}_\Phi$ be the ordinary Euclidean orthogonal projector onto the columns of $\overline{\Phi}_{T,K}$. Field capture in this coordinate system is

$$\mathscr{C}_K(\boldsymbol{g};\Phi,W) = \frac{\|\overline{P}_\Phi\overline{\boldsymbol{g}}\|_2^2}{\|\overline{\boldsymbol{g}}\|_2^2} = 1 - \frac{\|\overline{\boldsymbol{g}} - \overline{P}_\Phi\overline{\boldsymbol{g}}\|_2^2}{\|\overline{\boldsymbol{g}}\|_2^2}. \tag{29}$$

The unweighted case has $W = I$. The MICrONS structural analysis uses summed mapped excitatory synaptic area on the diagonal of W. For transformed gradient covariance $\overline{C}_g = \mathbb{E}[\overline{\boldsymbol{g}}\overline{\boldsymbol{g}}^T]$, expected capture for a fixed projector is

$$\mathscr{C}_K(\overline{C}_g;\Phi,W) = \frac{\operatorname{tr}(\overline{P}_\Phi\overline{C}_g)}{\operatorname{tr}(\overline{C}_g)}. \tag{30}$$

Equation 30 states the topology–task matching condition: a sparse anatomical basis is useful only to the extent that task credit lies in its span.

Proposition 1 (Ky–Fan credit-capacity bound)**.** *Let the eigenvalues of $\overline{C}_g \succeq 0$ satisfy $\lambda_1 \geq \cdots \geq \lambda_p \geq 0$. For every rank-$K$ orthogonal projector P,*

$$\frac{\operatorname{tr}(P\overline{C}_g)}{\operatorname{tr}(\overline{C}_g)} \leq \frac{\sum_{j=1}^{K}\lambda_j}{\sum_{j=1}^{p}\lambda_j}. \tag{31}$$

Equality is attained by the projector onto the leading K eigendirections. The difference between the right side and anatomical capture is the rank-matched spectral regret.

Proof. Ky–Fan's maximum principle gives $\max_{\operatorname{rank}(P)=K}\operatorname{tr}(P\overline{C}_g) = \sum_{j=1}^{K}\lambda_j$. Division by the fixed total spectral energy proves Eq. 31. □

For the spectral screen, $C(\rho) = (1-\rho)C_0 + \rho C_1$ and $\operatorname{tr}C_0 = \operatorname{tr}C_1$. Hence for two fixed projectors P and Q, the normalized capture difference has the sign of

$$\Delta(\rho) = \operatorname{tr}[(P-Q)C_0] + \rho\operatorname{tr}[(P-Q)(C_1-C_0)], \tag{32}$$

so an interior crossover is determined exactly by its two endpoints. This affine result distinguishes alignment from rank: increasing K raises the oracle bound, whereas increasing ρ determines whether the anatomical subspace approaches it.

3.4 One-step descent bound

The weighted statement is a Euclidean statement in explicitly transformed parameter coordinates. Define

$$\overline{\boldsymbol{w}} = W^{-1/2}\boldsymbol{w}, \qquad \boldsymbol{w} = W^{1/2}\overline{\boldsymbol{w}}, \tag{33}$$

and define $\overline{\mathscr{L}}(\overline{\boldsymbol{w}}) = \mathscr{L}(W^{1/2}\overline{\boldsymbol{w}})$. Then $\nabla_{\overline{\boldsymbol{w}}}\overline{\mathscr{L}} = W^{1/2}\boldsymbol{g} = \overline{\boldsymbol{g}}$. If $\overline{\mathscr{L}}$ has L_W-Lipschitz gradients, the descent lemma gives

$$\overline{\mathscr{L}}(\overline{\boldsymbol{w}} - \eta\overline{P}_\Phi\overline{\boldsymbol{g}}) \leq \overline{\mathscr{L}}(\overline{\boldsymbol{w}}) - \eta\overline{\boldsymbol{g}}^T\overline{P}_\Phi\overline{\boldsymbol{g}} + \frac{L_W\eta^2}{2}\|\overline{P}_\Phi\overline{\boldsymbol{g}}\|_2^2. \tag{34}$$

Orthogonal projection implies $\overline{\boldsymbol{g}}^T\overline{P}_\Phi\overline{\boldsymbol{g}} = \|\overline{P}_\Phi\overline{\boldsymbol{g}}\|_2^2$, hence

$$\overline{\mathscr{L}}(\overline{\boldsymbol{w}} - \eta\overline{P}_\Phi\overline{\boldsymbol{g}}) \leq \overline{\mathscr{L}}(\overline{\boldsymbol{w}}) - \eta\left(1 - \frac{L_W\eta}{2}\right)\|\overline{P}_\Phi\overline{\boldsymbol{g}}\|_2^2. \tag{35}$$

At $\eta = 1/L_W$, the ratio between the projected- and full-gradient guarantees is Eq. 29. In the original coordinates, the projected step is

$$\boldsymbol{w}^+ = \boldsymbol{w} - \eta W^{1/2}\overline{P}_\Phi W^{1/2}\boldsymbol{g}, \tag{36}$$

which is a particular preconditioned update rather than an ordinary Euclidean gradient step in $\boldsymbol{w}$. The implementation was checked on 10,000 random positive-definite quadratic losses: there were no bound violations, minimum numerical slack was 2.67×10^{-7}, and the capture identity agreed to 1.89×10^{-15}. That audit checks the Euclidean projector algebra, which also applies after an explicit coordinate transformation; it does not validate synaptic-area weighting as a physical parameter metric. The result is a one-step guarantee only when the tested field is an exact gradient in the stated coordinates. It is not a convergence or generalization theorem and does not show that a biological circuit can infer the oracle coefficients.

S4 Stochastic credit-operator phase theory

4.1 Expected descent with signal, gain and noise

Let $\mathscr{L} : \mathbb{R}^p \to \mathbb{R}$ have L-Lipschitz gradient at the state under study. Write $\boldsymbol{g} = \nabla \mathscr{L}(\boldsymbol{w})$ and let $\widehat{\boldsymbol{g}} = \boldsymbol{g} + \boldsymbol{\xi}$, where $\mathbb{E}\boldsymbol{\xi} = 0$ and $\mathbb{E}\boldsymbol{\xi}\boldsymbol{\xi}^\top = \Sigma$. A fixed matrix M collects every restriction imposed between the population gradient and the parameter update: coordinate ownership, route span, coefficient gain and any linearized local decoder. The update is

$$\boldsymbol{w}^+ = \boldsymbol{w} - \eta M \widehat{\boldsymbol{g}}. \tag{37}$$

Proposition 2 (Stochastic credit-operator bound)**.** *For Eq. 37,*

$$\mathbb{E}\mathscr{L}(\boldsymbol{w}^+) \le \mathscr{L}(\boldsymbol{w}) - \eta \boldsymbol{g}^\top M \boldsymbol{g} + \frac{L\eta^2}{2}\left[\|M\boldsymbol{g}\|_2^2 + \operatorname{tr}(M\Sigma M^\top)\right]. \tag{38}$$

If $A = \boldsymbol{g}^\top M \boldsymbol{g} > 0$ and $B = \|M\boldsymbol{g}\|_2^2 + \operatorname{tr}(M\Sigma M^\top) > 0$, the right side is minimized by $\eta^* = A/(LB)$ and guarantees decrease

$$U(M) = \frac{A^2}{2LB}. \tag{39}$$

If $A \le 0$, the bound does not certify a positive step.

Proof. The descent lemma gives $\mathscr{L}(\boldsymbol{w}+\boldsymbol{d}) \le \mathscr{L}(\boldsymbol{w}) + \boldsymbol{g}^\top \boldsymbol{d} + (L/2)\|\boldsymbol{d}\|_2^2$. Substitute $\boldsymbol{d} = -\eta M(\boldsymbol{g}+\boldsymbol{\xi})$, take expectations, use $\mathbb{E}\boldsymbol{\xi} = 0$ and $\mathbb{E}\|M\boldsymbol{\xi}\|_2^2 = \operatorname{tr}(M\Sigma M^\top)$, then minimize the resulting scalar quadratic in η. ☐

At fixed infinitesimal update norm, $-\boldsymbol{g}$ is the steepest Euclidean descent direction by Cauchy–Schwarz. Consequently, no routed local operator provides a uniform first-order improvement over exact full-batch backpropagation on the same parameters and objective. Equation 39 instead compares restricted or noisy estimators and exposes when rejecting stochastic energy compensates for discarded signal.

Corollary 1 (Projected stochastic crossover)**.** *Consider the identity-curvature quadratic $\mathscr{L}(\boldsymbol{w}) = \frac{1}{2}\|\boldsymbol{w} - \boldsymbol{w}^*\|_2^2$, initialize at $\boldsymbol{w} = 0$, set $\boldsymbol{g} = -\boldsymbol{w}^*$, use unit step size, and let P be an orthogonal projector. The expected post-update losses of the unprojected and projected unbiased gradients are*

$$\mathbb{E}\mathscr{L}[-(\boldsymbol{g}+\boldsymbol{\xi})] = \tfrac{1}{2}\operatorname{tr}(\Sigma), \tag{40}$$

$$\mathbb{E}\mathscr{L}[-P(\boldsymbol{g}+\boldsymbol{\xi})] = \tfrac{1}{2}\left[\|(I-P)\boldsymbol{g}\|_2^2 + \operatorname{tr}(P\Sigma P)\right]. \tag{41}$$

The projected estimator is better precisely when

$$\operatorname{tr}[(I-P)\Sigma] > \|(I-P)\boldsymbol{g}\|_2^2, \tag{42}$$

that is, rejected noise exceeds discarded signal. A local estimator using the same P but additional coefficient noise cannot exceed backpropagation plus P in expectation.

For completeness, at a common step $0 < \eta < 2$ the unprojected-minus-projected expected loss is

$$\frac{\eta}{2}\left\{\eta \operatorname{tr}[(I-P)\Sigma] - (2-\eta)\|(I-P)\boldsymbol{g}\|_2^2\right\}. \tag{43}$$

Hence projection is better exactly when $\operatorname{tr}[(I-P)\Sigma] > [(2-\eta)/\eta]\|(I-P)\boldsymbol{g}\|_2^2$; the unit-step result above is the special case $\eta = 1$. Smaller common steps therefore require proportionally more rejected noise to compensate for the discarded signal.

4.2 Interior optimum along the nested route ladder

Every feedback field studied in this work is, up to coefficient noise, a projector on one nested ladder, $P_{\text{scalar}} \subset P_{\text{neuron}} \subset P_{\text{ancestry},K} \subset I$. Along such a ladder the capture of Eq. 29 is monotone by construction, but the guaranteed utility is not. For an orthogonal projector with retained signal energy $S = \boldsymbol{g}^\top P \boldsymbol{g} = \|P\boldsymbol{g}\|_2^2$ and admitted noise $N = \operatorname{tr}(P\Sigma P)$, the optimal-step utility of the credit-operator bound is

$$U(S,N) = \frac{S^2}{2L(S+N)}. \tag{44}$$

Let one rung up the ladder add marginal signal $\Delta S \geq 0$ and marginal noise $\Delta N \geq 0$. Substituting into Eq. 44 and clearing denominators gives the exact marginal condition

$$U(S+\Delta S, N+\Delta N) > U(S,N) \iff \Delta S\left[(S+N)(2S+\Delta S) - S^2\right] > S^2\,\Delta N. \tag{45}$$

The bracket is positive, so for small marginal additions the condition reduces to a threshold on the marginal signal-to-noise ratio, $\Delta S/\Delta N > S/(S+2N)$: the utility keeps rising while each added route scale contributes proportionally more signal than the running threshold, and falls once the added scales admit mostly noise. When the signal spectrum decays faster than the admitted noise across ladder scales, U therefore attains an interior maximum at a finite budget rather than at full rank. Equations 44 and 45, together with the interior-maximum construction, are verified numerically in the released test suite (`tests/test_laminar_optimality.py`).

Two frozen experiments locate this optimum empirically (Supplementary Fig. S16; analysis code is included in the release). In the hierarchy-depth phase experiment the initialization utility of the aligned tree, computed per seed and task depth, placed its argmax at the trained loss minimum in 135 of 200 seed–task pairs, and the group-mean argmax matched the trained optimum exactly at $H = 1,2,3$; at the full-rank boundary $H = 4$ the one-step bound was conservative and favoured $D_{\text{r}} = 2$, reflecting its single-step horizon. In the trained subtree factorial the same initialization utility reproduced the exact ties at $K = 8$, where every span coincides, but did not predict the trained $K = 4$ ancestry advantage over the best matched control (6 of 20 seeds; the mean utility contrast at $K = 4$ is negative against both the published four-control set and its three implementable, non-oracle members). The trained $K = 4$ advantage is therefore a training-dynamics effect that accrues along the trajectory rather than a property of one-step initialization geometry, consistent with the main-text observation that initial capture alone does not determine learning.

4.3 Laminar route families and hierarchical task covariance

The distinction between a point neuron and a dendritic tree can be stated as a constraint on the feasible credit operators at fixed rank. A point neuron addresses its synapses with the single shared column $\mathbf{1}$: a rank-one operator with uniform support. An ancestry dictionary is a rank-K operator whose columns form a *laminar* family: any two supports are nested or disjoint. An unconstrained rank-K feedback matrix is the comparison class. For the balanced binary tree used in the factorial, the span of the K-route ancestry dictionary equals the span of the K coarsest vectors of the orthonormal tree-Haar basis $W = [\boldsymbol{h}_1, \ldots, \boldsymbol{h}_n]$ (constant, then sibling-subtree differences, coarse to fine) for every admissible budget $K \in \{1,2,4,\ldots,n\}$.

Proposition 3 (Laminar capture of hierarchical covariance)**.** *Let the task-credit covariance be $C = W\Lambda W^\top$ with $\Lambda = \operatorname{diag}(\lambda_1,\ldots,\lambda_n)$ ordered so that $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0$ along the coarse-to-fine Haar order. Then for every admissible budget K the ancestry dictionary attains the Ky–Fan bound,*

$$\frac{\operatorname{tr}(P_{A,K}C)}{\operatorname{tr}(C)} = \frac{\sum_{i\leq K}\lambda_i}{\operatorname{tr}(C)} = \max_{\operatorname{rank}(P)=K} \frac{\operatorname{tr}(PC)}{\operatorname{tr}(C)}. \tag{46}$$

For a general covariance C with eigenpairs $(\lambda_i, \boldsymbol{v}_i)$ ordered by decreasing eigenvalue, the shortfall from the Ky–Fan bound decomposes exactly as

$$\sum_{i\leq K}\lambda_i - \operatorname{tr}(P_{A,K}C) = \sum_{i\leq K}\lambda_i\|(I-P_{A,K})\boldsymbol{v}_i\|_2^2 - \sum_{j>K}\lambda_j\|P_{A,K}\boldsymbol{v}_j\|_2^2, \tag{47}$$

the leading-eigenspace energy outside the laminar span minus the trailing energy that the span retains. It is therefore bounded above by the former term, and is zero exactly when the span of the top-K eigenvectors of C is contained in the ancestry span.

Proof. The ancestry span equals the span of the K coarsest Haar vectors, which under the stated ordering are the top-K eigenvectors of C; a rank-K orthogonal projector maximizes $\mathrm{tr}(PC)$ exactly on a top-K eigenspace (Ky–Fan). For general C, expanding $\mathrm{tr}(P_{A,K}C) = \sum_j \lambda_j \|P_{A,K}\boldsymbol{v}_j\|_2^2$ and $\|P_{A,K}\boldsymbol{v}_i\|_2^2 = 1 - \|(I - P_{A,K})\boldsymbol{v}_i\|_2^2$ for $i \le K$ gives the stated decomposition; it vanishes iff $P_{A,K}$ projects onto a top-K eigenspace. □

Thus anatomy is an optimal rank-K feedback basis precisely for hierarchically organized credit: a node offers one coefficient per neuron, a tree offers a wavelet hierarchy, and the hierarchy is the right basis exactly when task-gradient covariance is organized on the tree. The spectral phase experiment realizes both regimes: at full alignment and $K = 4$ ancestry attained the rank-matched Ky–Fan bound to numerical precision, whereas at random alignment its advantage over sampled random routes vanished (main text). The attainment, the shortfall identity and the failure case with fine-scale-dominant covariance are verified numerically in `tests/test_laminar_optimality.py`.

4.4 Point-neuron partition residual and route-coefficient error

Let c_i be the exact task-dependent coefficient remaining after local eligibility has been factored out, let $w_i > 0$ be coordinate weights, and let $\pi(i)$ assign synapse i to a group that must share one teaching coefficient. The best allowed field minimizes $\sum_i w_i[c_i - z_{\pi(i)}]^2$.

Proposition 4 (Shared-coefficient residual)**.** *For every group G, the optimum is its weighted mean*

$$\bar{c}_G = \frac{\sum_{i\in G} w_i c_i}{\sum_{i\in G} w_i}, \tag{48}$$

and the irreducible residual is

$$\mathscr{R}(\pi) = \sum_G \sum_{i\in G} w_i (c_i - \bar{c}_G)^2. \tag{49}$$

Refining a partition cannot increase $\mathscr{R}$; it decreases the residual only when an old group contains coefficient variation resolved by the new groups.

Proof. Differentiate the separable weighted least-squares objective with respect to each z_G. The refinement statement follows because the old solution remains feasible when two or more refined coefficients are constrained to be equal. □

The ordinary point-neuron rule has one group per neuron. Dendritic ancestry offers a particular nested sequence of refinements, but is helpful only if the task coefficient varies across those subtrees. An explicitly grouped point model can implement the same partition, which is why the resource-matched point controls in the main article coincide with the dendritic implementation.

For a general dictionary Φ and positive diagonal W, let $\boldsymbol{z}^* = \arg\min_{\boldsymbol{z}} \|\boldsymbol{c} - \Phi\boldsymbol{z}\|_W^2$ and let $\widehat{\boldsymbol{z}}$ be the coefficients actually supplied by a local circuit. Weighted projection orthogonality yields the exact decomposition

$$\|\boldsymbol{c} - \Phi\widehat{\boldsymbol{z}}\|_W^2 = \underbrace{\|\boldsymbol{c} - \Phi\boldsymbol{z}^*\|_W^2}_{\text{address / representation error}} + \underbrace{\|\Phi(\widehat{\boldsymbol{z}} - \boldsymbol{z}^*)\|_W^2}_{\text{coefficient-estimation error}}. \tag{50}$$

If the coefficient error is unbiased with covariance Ω_z, the expected second term is $\mathrm{tr}(\Phi^\mathsf{T} W \Phi \Omega_z)$. The Gram matrix $G_\Phi = \Phi^\mathsf{T} W \Phi$ therefore controls how coefficient error is expressed. Small positive eigenvalues amplify the inverse problem used to estimate $\boldsymbol{z}^*$, while column coherence makes individual route coefficients poorly identifiable even when their combined field is stable. Address rank alone is not a complete learning metric.

4.5 Static route gains preserve span but can change conditioning

Proposition 5 (Route-wise static-gain invariance)**.** *Let B be an invertible diagonal matrix of fixed nonzero route gains. Then*

$$\mathrm{col}(\Phi B) = \mathrm{col}(\Phi), \qquad P_{\Phi B,W} = P_{\Phi,W}. \tag{51}$$

Thus route-wise static scalar gains cannot change address rank, projection residual or spectral capture. They can change the Gram matrix from $\Phi^\top W\Phi$ to $B^\top\Phi^\top W\Phi B$ and hence alter conditioning and coefficient dynamics.

This proposition concerns nonzero scalar rescaling of existing route columns. Spatially varying row gains, state-dependent gating and gains that reach zero can change the effective span or active support; those cases are not covered by Eq. 51.

4.6 Finite-time coefficient learning has a bias–variance crossover

Equal span does not imply equal coefficient dynamics. Let the realized field be $\boldsymbol{f}_t = \Phi \boldsymbol{z}_t$, normalize vanilla coefficient descent by $\lambda_{\max}(\Phi\Phi^\top)$, and define $A = \Phi\Phi^\top/\lambda_{\max}$. For noisy rate-based target estimates $\boldsymbol{g}+\boldsymbol{\xi}_t$, with $\mathbb{E}\boldsymbol{\xi}_t = 0$ and $\mathrm{Cov}(\boldsymbol{\xi}_t) = \sigma^2 I/n$, the field iteration is

$$\boldsymbol{f}_{t+1} = \boldsymbol{f}_t - cA(\boldsymbol{f}_t - \boldsymbol{g} - \boldsymbol{\xi}_t). \tag{52}$$

Proposition 6 (Exact finite-time coefficient risk)**.** *Assume $\boldsymbol{f}_0 = 0$, $\boldsymbol{g} \in \mathrm{col}(\Phi)$ and let $(\mu_j, \boldsymbol{u}_j)$ be the positive eigenpairs of A, with $\theta_j = \boldsymbol{u}_j^\top \boldsymbol{g}$. For independent isotropic noise and a stable step, the expected population loss after T updates is*

$$\mathbb{E}\frac{\|\boldsymbol{f}_T - \boldsymbol{g}\|_2^2}{2} = \frac{1}{2}\sum_j \left[(1-c\mu_j)^{2T}\theta_j^2 + \frac{\sigma^2}{n}\frac{(c\mu_j)^2[1-(1-c\mu_j)^{2T}]}{1-(1-c\mu_j)^2}\right]. \tag{53}$$

For Gram-preconditioned descent, $A = P_\Phi$ and all positive $\mu_j = 1$; therefore every dictionary with the same projector has the same field trajectory under common observations.

Proof. Project Eq. 52 onto each eigenvector. The deterministic error contracts by $1-c\mu_j$ per step. Independent innovations have variance $(c\mu_j)^2\sigma^2/n$ and accumulate as a finite geometric series. Summing half the modal second moments gives Eq. 53. Gram preconditioning replaces $\Phi^\top\Phi$ by its pseudoinverse on the positive subspace, so the induced field operator is P_Φ. □

Equation 53 makes the low-data boundary explicit. Small μ_j retain more finite-time bias but admit less estimator variance. For two same-span operators, write their risk difference as $\Delta B + (\sigma^2/n)\Delta V$. If $\Delta B > 0$ and $\Delta V < 0$, the ordering reverses at $n^* = -\sigma^2\Delta V/\Delta B$: slow coordinates can help below n^* by implicit noise filtering, whereas well-conditioned coordinates help above it by reducing bias. This is a finite-horizon regularization effect, not new address capacity and not an unconditional advantage of ill-conditioning.

4.7 Branch reliability shrinkage

Partition parameter coordinates into disjoint branch blocks b. Let $S_b = \|\boldsymbol{g}_b\|_2^2$ and $N_b = \mathrm{tr}(\Sigma_b)$. For $M = \mathrm{blockdiag}(a_b I_b)$ and fixed step $\eta = c/L$, Eq. 38 guarantees a loss decrease of

$$\mathscr{G}_c(\boldsymbol{a}) = \frac{c}{L}\sum_b \left[a_b S_b - \frac{c a_b^2}{2}(S_b + N_b)\right]. \tag{54}$$

Proposition 7 (Fixed-step reliability-optimal attenuation)**.** *Equation 54 is maximized independently at*

$$a_b^* = \min\left\{1, \frac{S_b}{c(S_b+N_b)}\right\}, \tag{55}$$

for a physical attenuator constrained to $0 \le a_b \le 1$*. The best common gain is* $a_{\mathrm{g}}^* = \min\{1, \sum_b S_b / [c \sum_b (S_b + N_b)]\}$*, and the branch-specific guarantee is no smaller because its feasible set contains every common gain. At* $c = 1$*, Eq. 55 reduces to the reliability factor* $S_b/(S_b+N_b)$*; in that case* $\mathscr{G}_{\mathrm{branch}} = (2L)^{-1} \sum_b S_b^2/(S_b+N_b)$ *and* $\mathscr{G}_{\mathrm{global}} = (2L)^{-1}(\sum_b S_b)^2 / \sum_b (S_b+N_b)$*, with equality when reliability is constant across blocks.*

Proof. Differentiate each concave quadratic in Eq. 54 and project its unconstrained maximizer onto $[0,1]$. Restricting all coordinates to a common value gives a_{g}^*. The branchwise optimum cannot be worse than this restricted solution. For $c = 1$, the closed-form comparison also follows from Cauchy–Schwarz applied to $S_b/\sqrt{S_b+N_b}$ and $\sqrt{S_b+N_b}$. □

The result is an oracle prescription, not a claim that a neuron knows S_b or N_b. It also has no dendrite-exclusive content: any point implementation provided with the same branch partition and multiplicative factors is identical. Its biological relevance depends on whether local conductance can estimate or realize those factors.

4.8 Passive focal shunts attenuate a positive adjoint

Proposition 8 (No sign reversal in a passive grounded tree)**.** *Let G be a symmetric positive-definite nonsingular M-matrix, let* $\boldsymbol{s} \ge 0$*, and define the passive adjoint* $\boldsymbol{q} = G^{-1}\boldsymbol{s}$*. Adding a nonnegative shunt* κ *at compartment k gives* $G' = G + \kappa \boldsymbol{e}_k \boldsymbol{e}_k^\top$ *and*

$$\boldsymbol{q}' = \boldsymbol{q} - \frac{\kappa q_k}{1 + \kappa (G^{-1})_{kk}} G^{-1} \boldsymbol{e}_k. \tag{56}$$

Both G^{-1} *and* $(G')^{-1}$ *are elementwise nonnegative, hence* $0 \le \boldsymbol{q}' \le \boldsymbol{q}$ *componentwise. A focal passive shunt attenuates but cannot reverse a positive adjoint field.*

Proof. Equation 56 is the Sherman–Morrison identity. A nonsingular M-matrix has a nonnegative inverse; adding a nonnegative diagonal term preserves the M-matrix property. The subtracted vector is nonnegative, and $\boldsymbol{q}' = (G')^{-1}\boldsymbol{s} \ge 0$. □

This ordering does not by itself guarantee descendant selectivity relative to a matched current injection; that spatial comparison depends on the Green's function, focal site and electrotonic regime and is evaluated numerically in the focal experiments.

Corollary 2 (Spectral size of a focal passive perturbation)**.** *Under the preceding assumptions, let* $H = G^{-1}$ *and* $H' = (G + \kappa \boldsymbol{e}_k \boldsymbol{e}_k^\top)^{-1}$*. The inverse-operator change is rank one and*

$$\|H' - H\|_2 = \frac{\kappa \|H\boldsymbol{e}_k\|_2^2}{1 + \kappa H_{kk}} \le \frac{\kappa \|H\|_2^2}{1 + \kappa/\lambda_{\max}(G)}. \tag{57}$$

Consequently, for any adjoint source $\boldsymbol{s}$*,* $\|(H'-H)\boldsymbol{s}\|_2 \le \|H'-H\|_2 \|\boldsymbol{s}\|_2$*.*

Proof. Sherman–Morrison gives $H' - H = -\kappa (H\boldsymbol{e}_k)(H\boldsymbol{e}_k)^\top / (1 + \kappa H_{kk})$. The norm of this symmetric rank-one matrix is its scalar coefficient times $\|H\boldsymbol{e}_k\|_2^2$. Use $\|H\boldsymbol{e}_k\|_2 \le \|H\|_2$ and $H_{kk} \ge \lambda_{\min}(H) = 1/\lambda_{\max}(G)$ for the bound. □

Equation 57 makes the electrotonic boundary explicit: a well-conditioned, strongly grounded cable has small $\|G^{-1}\|_2$ and therefore cannot show a large global adjoint change at fixed shunt dose. The bound is global and conservative; the descendant-enriched spatial contrast remains a Green's-function question.

4.9 Why depth can help or hurt

Let $P_1 \preceq P_2 \preceq \cdots \preceq P_{D_{\mathrm{r}}}$ be nested tree-route projectors. Increasing depth reduces the address bias $\|(I - P_{D_{\mathrm{r}}})\boldsymbol{g}\|^2$ but increases admitted noise $\mathrm{tr}(P_{D_{\mathrm{r}}} \Sigma P_{D_{\mathrm{r}}})$ whenever the new modes carry stochastic energy. Equation 42 therefore predicts an interior optimum when task signal is concentrated above one hierarchy level and noise continues into finer levels. This is the controlled role isolated by the hierarchy-depth phase experiment. It is distinct from nonlinear forward composition, which can make deeper dendritic subunits representationally useful even when the backward address span is unchanged.

Depth can also impose a transport cost. In an idealized ensemble with equal local eligibility, let the route gain be $A = \prod_{\ell=1}^{D_p} G_\ell$ and assume independent $\log G_\ell$ with variances σ_ℓ^2. The cosine between a shared coefficient and the exact gain-weighted field in the large-route limit is

$$\frac{\mathbb{E}A}{\sqrt{\mathbb{E}A^2}} = \exp\left(-\frac{1}{2}\sum_{\ell=1}^{D_p}\sigma_\ell^2\right). \tag{58}$$

Additional levels can therefore add address resolution while simultaneously making uncompensated shared feedback less aligned. The balance depends on task hierarchy, gain variability, noise and the available coefficient estimator; "more depth" has no unconditional sign.

4.10 Controlled phase experiment

The experiment used the 16-column orthonormal Haar basis of a balanced binary tree. The spectral screen mixed a tree-aligned covariance with an independent rotation having the same eigenvalues and compared ancestry, random rank and dense principal-eigenspace projectors. The depth screen sampled targets from tree levels no finer than H and assigned increasing noise to levels finer than H; aligned and leaf-rewired projectors were updated with identical noise draws for 80 steps. The projection screen directly crossed signal retention and noise retention. The reliability screen crossed eight branch blocks with five degrees of oppositely ordered signal and noise energy. Full parameters, seeds, gates and row counts are frozen in `configs/credit_phase_theory/confirmatory.json`; seed-level tables, summaries and contrasts are in `source_data/credit_phase_theory/`.

The two canary seeds tested numerical artifacts only. Fifty untouched confirmatory seeds supplied the inferential unit; intervals used 20,000 paired seed-bootstrap draws and central paired contrasts used two-sided Wilcoxon tests. All full-rank projectors agreed to 4.44×10^{-16}, route-wise static gain rescaling preserved the projector to 1.33×10^{-15}, and the explicit point gate matched reliability-aligned attenuation exactly. This fixed-state quadratic experiment validates the phase logic but does not join oracle reliability to learned positive conductances.

4.11 State-matched positive-conductance reliability experiment

We next tested the physical realization of Eq. 55 in a separate trained branch model. Every input rate x_{bid} was nonnegative and every trainable excitatory conductance was $g_{bd} = \text{softplus}(q_{bd})$. For branch b, define total unshunted conductance and voltage

$$G_b = g^L + \sum_d x_{bd} g_{bd}, \qquad V_b = \frac{\sum_d x_{bd} g_{bd}}{G_b}. \tag{59}$$

A fixed shunt $\kappa_b \geq 0$ had reversal zero. To isolate its effect on input resistance from its effect on the forward feature, an oracle current $\kappa_b V_b$ was added on each example. The perturbed state is therefore

$$V_b' = \frac{\sum_d x_{bd} g_{bd} + \kappa_b V_b}{G_b + \kappa_b} = V_b, \qquad \frac{\partial V_b'}{\partial q_{bd}} = \frac{x_{bd}(1 - V_b)}{G_b + \kappa_b}\,\text{sigmoid}(q_{bd}). \tag{60}$$

Relative to the unshunted eligibility, the physical input-resistance factor is $G_b/(G_b + \kappa_b)$. The compensating current was recomputed from the pre-shunt state and treated as a clamp, not differentiated as an independently learned pathway. Consequently this manipulation deliberately removes forward feature suppression while retaining conductance attenuation in the local update.

The binary task used eight branches, four inputs per branch, 512 training examples and 2,048 held-out examples. Input rates were generated from a positive baseline plus a class-dependent branch pattern and clipped at zero. Initial full-data gradients supplied $S_b = \|\boldsymbol{g}_b\|_2^2$. Coefficient-noise variance was then scaled so that

$$N_b = S_b \exp(-2hx_b), \qquad x_b \in [-1, 1], \tag{61}$$

for eight evenly spaced branch indices and $h \in \{0, 0.5, 1, 1.5, 2\}$. The statistical reliability was $r_b = S_b/(S_b + N_b)$, lower-bounded at 0.03 for numerical stability. Because the common step was $\eta = c/L$ with $c = 0.5$, the corrected fixed-step target was $a_b^* = \min\{1, r_b/c\}$ and the initial shunt was $\kappa_b = \overline{G}_b[(a_b^*)^{-1} - 1]$. Shunts remained fixed during the 40 updates; the excitatory conductance parameters were trained, and the same coefficient-noise realizations were used across matched methods within seed.

Nine conditions separated the relevant alternatives: exact clean backpropagation, noisy credit without a shunt, a state-matched additive control, the step-consistent best single global shunt, reliability-aligned, seed-shuffled and anti-aligned shunts, an explicit point gate receiving the identical branch factors, and aligned shunting with clean credit. The point gate is a resource control, not a biological assertion: because it receives the same branch partition and factors, its update must equal Eq. 60. The clean-backpropagation condition retains the unattenuated exact gradient and is the optimization reference.

The preceding pilot is retained as a historical artifact because it used r_b despite taking $c = 0.5$. After identifying that inconsistency, the corrected configuration and predictions were frozen before outcomes for two new artifact-only canary seeds. Confirmatory execution then used 50 fresh paired seeds 8100–8149 with unchanged configuration and runner hashes. All 2,250 expected seed–condition rows completed. The minimum input rate was zero, maximum state mismatch was 2.22×10^{-16}, and independently computed point-gate and additive-control trajectory differences were 4.00×10^{-15} and 1.31×10^{-14}. A new finite-difference gate held the compensating current fixed while perturbing an excitatory parameter; its maximum relative error was 3.60×10^{-8}. All prespecified gates passed.

At $h = 2$, reliability-aligned minus best-global one-step test-loss reduction was 0.001817 (95% paired seed-bootstrap interval 0.001618–0.002020), positive in 49/50 seeds (two-sided Wilcoxon $P = 3.55 \times 10^{-15}$). Aligned minus unshunted reduction averaged 0.001339 (0.000594–0.002096), positive in 33/50 seeds ($P = 0.0030$). After 40 updates, aligned-minus-global test loss was -0.020404 (-0.022220 to -0.018705), favoring aligned in 50/50 seeds ($P = 1.78 \times 10^{-15}$). Aligned shunting also had lower final loss than shuffled and anti-aligned shunts by 0.009204 and 0.013781, respectively. However, aligned-minus-unshunted final loss was 0.000425 (-0.002456–0.003071; $P = 0.154$), so the immediate improvement did not establish a sustained loss difference from no shunt. Exact clean backpropagation retained the lowest mean final loss. These results validate a positive-conductance approximation and ordering of fixed-step oracle reliability factors under a state clamp; they do not establish autonomous shunt learning, dendrite-exclusive computation or general superiority to unshunted learning.

4.12 Same-span noisy coefficient-learning experiment

We tested the distinction between address and coefficient errors with a new rate-based factorial. The 16-leaf balanced-tree Haar basis supplied the common rank-eight span through depth three. The tested dictionaries were the eight orthonormal Haar modes, all 15 nested indicators through depth three, and the same nested indicators scaled by positive gains evenly spaced on the log scale from $\exp(-3)$ to $\exp(3)$. The dictionaries' projectors agreed to 9.44×10^{-16} and their positive-spectrum Gram condition numbers were 1, 15 and 388.52.

Each independent seed drew a unit-norm target in the common span. At every update the estimator observed the target plus isotropic Gaussian noise with standard deviation $2/\sqrt{n_{\text{eff}}}$, where $n_{\text{eff}} \in \{4, 16, 64, 256\}$. The same 80 observation vectors were used for every matched parameterization. Vanilla coefficient descent used $0.2/\lambda_{\max}(\Phi^\top\Phi)$; the oracle coordinate control used $0.2(\Phi^\top\Phi)^+$. Population field loss was recorded after 1, 5, 20 and 80 updates. The target address residual was below 1.45×10^{-30} in every condition, so all reported loss is within-span coefficient-estimation error.

The predictions were frozen before two artifact-only canary seeds and 50 paired seeds 9100–9149. The prediction that Haar coordinates would always reduce finite-time error was false. At $n_{\text{eff}} = 4$, raw-minus-Haar final loss was -0.2774 (95% paired interval -0.3260 to -0.2313) and scaled-minus-Haar was -0.1906 (-0.2500 to -0.1335): slower modes filtered high estimator noise. At $n_{\text{eff}} = 256$, the effects reversed to 0.02555 (0.02171–0.02952) and 0.17091 (0.14576–0.19686), positive in 50/50 seeds for both contrasts. Equation 53, added to explain this falsification, predicted median crossovers at effective sample sizes 38.53 and 8.25. The Gram-preconditioned field trajectories agreed across dictionaries to 1.16×10^{-15}. Thus static parameterization can alter finite-data learning without altering the attainable credit field.

4.13 Nonlinear physical depth in the production rate model

The hierarchy-depth screen above changes the resolution of a linear route projector. We therefore added a separate bridge to the original nonlinear population implementation. The registered `hierarchical_gain_load` generator emits 64 strictly positive excitatory rates followed by 64 strictly positive inhibitory rates. Three independent zero-mean Gaussian node fields define fine, coarse and global lognormal gain factors. Their product multiplies signal-bearing excitatory coordinates in the distal block; the three inhibitory blocks separately measure the corresponding factors. Lognormal means are corrected by $-\sigma^2/2$ at every factor, so changing hierarchy or sensor alignment does not silently change first moments.

The matched condition uses the same latent node in excitatory signal and its inhibitory sensor. In the zero-alignment condition, each sensor instead uses an independent node field with the same scale and grouping. In the shuffled condition, a nonzero cyclic trial shift is applied to the complete inhibitory field after class balancing. The tree-placement condition leaves every data tensor unchanged and reverses the fine and global feature ranges across the fine/coarse/proximal inventory tiers. These controls remove task–tree alignment in different ways: latent correspondence, example-wise access and physical placement, respectively. The final placement control reverses the fine and global ranges and preserves their 21/22/21 widths. An initial three-cycle was invalidated by the prospective resource gate because it changed candidate mask slots; that complete execution remains in the audit archive and does not enter inference.

Branch factors are ordered soma-to-distal by the production layout planner. For architecture $[b_1,\ldots,b_{D_\mathrm{p}}]$, the number of nonsomatic units is $\sum_{j=1}^{D_\mathrm{p}}\prod_{r=1}^{j} b_r$. Thus D1 $[8]$ has eight distal units per soma; D2 $[2,3]$ has two proximal and six distal units; and D3 $[2,1,2]$ has two proximal, two middle and four distal units. The unary middle expansion is deliberate: all depths have eight nonsomatic units. The production structured-mask builder assigns all 4/2/2 tiers to the single D1 stage, fine+coarse tiers to the D2 distal stage and the global tier to its proximal stage, and one fine, coarse or global tier to each D3 stage. Construction-time resource records additionally verified equal trainable parameters (66,178), active synapses (14,336), candidate slots (21,760) and persistent state scalars per sample (2,944). The physical branch law was the same positive-conductance division used elsewhere in the paper, with identity reactivation and learned softplus-positive synaptic weights.

The original canary used signal contrast 0.24, training gain SD 0.25 and an unseen test gain SD 1.2. Deeper networks fit training better, but BP and both LocalCA transports remained at chance on the shifted test distribution (Supplementary Fig. S15A). This failed the accessibility gate and is retained as a negative result. A two-seed test gain ladder showed monotone erosion under increasing shift. A coupling ladder at matched train/test gain identified a seed-consistent interaction: from child conductance 1 to 64, the D3-minus-D1 accuracy contrast increased by 0.1236 on average, while D1 changed minimally. A signal ladder at child conductance 16 then located the non-ceiling signal-contrast boundary. Contrast 0.72 narrowly failed the frozen D1 accessibility threshold (mean 0.597), and the separately run 0.80 boundary passed with D1, D2 and D3 means 0.6086, 0.6737 and 0.9227. These exploratory seeds 10100–10141 were excluded from inference.

Before observing any seed 10200 outcome, we froze a 270-fit replication at signal contrast 0.80, per-factor train/test log-gain SD 0.25 and child conductance 16. The seven cohorts were aligned shunting BP; zero-alignment, shuffled-sensor and reversed-placement shunting BP; aligned raw-additive BP; and aligned or reversed-placement shunting LocalCA with shared-soma and path transport. Every cohort crossed D1, D2 and D3 over the same ten paired seeds 10200–10209. The primary contrasts, resource/accessibility gates, bootstrap procedure and claim boundaries were fixed in the dated experiment contract. In particular, aligned D3–D1 alone tests forward capacity. Task–tree specificity requires an aligned-minus-control depth interaction, and a local-credit claim requires the corresponding LocalCA contrast. Raw additive is a mechanism control; a point model with sufficient flexible capacity can in principle emulate the hierarchy, a boundary tested directly in the same-task controls below.

All 270 frozen fits passed the completeness, finite-metric, no-fallback and exact-resource gates. With aligned shunting BP, mean accuracy was 0.6096, 0.6752 and 0.9182 at D1–D3, giving a paired D3-minus-D1 effect of 30.86 percentage points (95% seed-bootstrap interval 30.45–31.30; 10/10 positive; exact sign-flip $P = 0.001953$).

The raw depth effects were 0.02 points for independent sensors, −0.01 for shuffled trials and −0.43 for reversed placement. Their aligned-minus-control depth interactions were 30.84, 30.87 and 31.28 points, respectively, all positive in 10/10 paired seeds. Aligned raw-additive BP instead fell from 0.5733 to 0.5190. Under LocalCA, aligned D3-minus-D1 was 16.95 points (16.37–17.44) for shared-soma feedback and 27.90 points (27.24–28.53) for path transport. The corresponding aligned-minus-reversed interactions were 17.72 points (17.27–18.18) and 28.68 points (28.07–29.27), both positive in 10/10 seeds. These interactions, rather than the aligned depth effects alone, support task–tree specificity.

As a separate post-training mechanism diagnostic, we replayed one fixed held-out batch through each of the 30 aligned-shunting BP checkpoints. Exact path LocalCA reproduced the selected autograd conductance-gradient direction at every checkpoint (minimum cosine 0.9999999999994; no missing gradient tensors). Mean shared-soma cosine was 0.9457 at D1, 0.7990 at D2 and 0.7674 at D3. Its paired D3-minus-D1 change was −0.1784 (95% seed-bootstrap interval −0.2082 to −0.1490; 0/10 positive pairs). This replay validates the implemented path recursion and shows that a common neuronal error becomes a poorer directional proxy as additional physical stages differentiate branch states; it was not a separately preregistered performance endpoint.

4.14 Same-task point, nonserial and credit-coordinate controls

We added three controls without changing the calibrated task, data, seeds, initialization policy or stopping rule. First, an all-active grouped-star model used exactly the same branch layers, structured masks and parameters as the serial tree, but each nonsomatic layer read the external input independently; the retained child aggregators were summed only once at the soma. State-dictionary values and resource counts were identical at initialization. D1 logits were numerically identical, as required. At D1 the serial-minus-star contrast was −0.005 percentage points (−0.067 to 0.054; 6/10 positive; $P = 0.844$). At D2 and D3 it was 6.54 (6.20–6.88) and 30.81 points (30.41–31.26), respectively, both positive in 10/10 seeds ($P = 0.001953$). Under reversed placement the D3 serial-minus-star contrast was −0.43 points (−0.68 to −0.16; 1/10 positive; $P = 0.0137$), and the paired alignment interaction was 31.24 points (30.65–31.77; 10/10 positive; $P = 0.001953$). Serial divisive composition, rather than the inventory of branch units, is therefore required for the aligned effect in this implementation.

Second, point MLP widths were chosen before fitting to match either 15,055 active parameters or, to the nearest realizable integer width, 66,200 total parameters (the tree has 66,178). Their mean test accuracies were 0.9861 and 0.9902. They exceeded serial D3 by 6.79 points (6.19–7.34) and 7.20 points (6.75–7.68), respectively, in 10/10 paired seeds ($P = 0.001953$ for each). Neither point model was given the hierarchy. This rejects an absolute expressivity advantage for the tree while leaving a matched structured-resource interpretation intact.

Third, soma-broadcast autograd retained the standard synaptic derivatives but replaced each nonsomatic loss derivative by the loss derivative at its owning soma. Forward values and logits were identical to full BP before training. With the standard BP optimizer, full BP minus soma broadcast at D3 was 0.64 points (0.15–1.17; 8/10 positive; $P = 0.0332$). Only after this 140-fit primary extension completed did a configuration audit reveal that the frozen LocalCA reference used a different, split optimizer. A dated amendment therefore froze 60 additional broadcast fits using the LocalCA optimizer before their outcomes were viewed. This diagnostic broadcast reached 0.7771 at aligned D3, compared with 0.7778 for shared LocalCA: broadcast minus LocalCA was −0.06 points (−0.23 to 0.09; 5/10 positive; $P = 0.492$). Changing only the broadcast optimizer cost 13.47 points (12.89–14.05; 10/10 positive). Within LocalCA, path transport exceeded the shared coordinate by 10.96 points (10.34–11.58; 10/10 positive), while full BP remained 3.09 points above path LocalCA (2.48–3.79; 10/10 positive). Thus, for this task, the tested eligibility expression adds no detectable penalty after matching teaching coordinate and optimizer; path specificity and optimizer regime account for the larger separations. The amendment is diagnostic rather than prospectively confirmatory, and no comparison turns autograd into a biological mechanism.

Across the primary and amended controls, all 200 new fits were finite, contained no fallback warning and used exactly seeds 10200–10209. Serial and star models had 66,178 trainable parameters, 14,336 active contacts, 21,760 candidate slots and 2,944 persistent state scalars. Seed-level outcomes, paired contrasts, configuration manifests and a machine-readable audit are distributed in `source_data/point_dendrite_credit_controls/`.

4.15 Alignment-dose interpolation of the physical-depth effect

After the $\alpha = 0$ and 1 endpoint results were available, we froze an interpolation at $\alpha \in \{0.25, 0.50, 0.75\}$ without changing any other generator, architecture, optimizer, stopping or seed setting. The 90 new fits crossed three physical depths with ten paired seeds. Combined with the frozen endpoints, paired D3-minus-D1 effects were 0.019, 0.207, 0.820, 3.126 and 30.855 percentage points at $\alpha = 0, 0.25, 0.50, 0.75, 1$, respectively. The corresponding positive-pair counts were 6, 8, 10, 10 and 10 of ten. Exact two-sided sign-flip tests gave $P = 0.844$ at $\alpha = 0$, 0.0703 at 0.25 and 0.00195 at every higher level.

The frozen primary summary fit an ordinary least-squares line to the five paired depth effects within each seed. Its mean slope was 25.84 percentage points per unit α (95% paired-seed bootstrap interval 25.43–26.26; 10/10 positive; exact sign-flip $P = 0.001953$). This positive summary does not make the response linear. The change from $\alpha = 0.25$ to 0.75 was only 2.92 points (2.65–3.17), whereas the final 0.75 to 1 step contributed 27.73 points (27.35–28.11). The interpolation therefore supports a steep boundary near complete alignment in this calibrated task. It does not supply a second task family, hierarchy depth or biological replication.

A frozen-source guard stopped 26 scheduled jobs when an unrelated source edit occurred during the first array. The completed pre-edit jobs were retained and the stopped jobs rerun because the exact source diff only introduced an optional rule override that none of these resolved configurations enabled. Both manifest hashes, the condition-level deduplication and the equivalence argument are recorded in the dated mid-array source-equivalence report. All 90 unique intermediate fits passed the final finite-metric, no-fallback, seed and equal-resource gates. Machine-readable outcomes and paired contrasts are in `source_data/physical_alignment_dose/`.

4.16 Literal grouped-point control and independent hierarchy depth

The all-active grouped star above retains serial-shaped child aggregators as independent arms. We therefore implemented a stricter grouped-point control in which each nonsomatic level receives the raw external excitatory and inhibitory streams independently and projects directly to its owning soma. Every local branch module, structured mask and shunting equation is retained, but no level inherits another branch state. Each direct projection has the same scalar count as the serial child aggregator it replaces and copies its initialized raw conductances. Tests required copied projection parameters, finite gradients and exact D1 logits against the serial hierarchy.

After the preceding $H = 3$ endpoints were known, a dated contract froze 60 literal grouped-point fits crossing aligned or reversed placement, D1–D3 and the same paired seeds 10200–10209. The frozen serial outcomes were retained unchanged as the reference. Under alignment, grouped-point means were 0.6096, 0.6100 and 0.6095 at D1–D3. Serial-minus-grouped-point was 30.87 percentage points at D3 (95% paired-seed bootstrap interval 30.48–31.31; 10/10 positive; exact sign-flip $P = 0.001953$), but -0.44 points after placement reversal (-0.69 to -0.16). The resulting alignment interaction was 31.31 points (30.73–31.84; 10/10 positive). The earlier grouped star minus the literal grouped point was 0.06 points at aligned D3 (0.01–0.13; 7/10 positive; $P = 0.0840$), which did not pass the frozen directional claim gate.

We separately changed hierarchy depth before viewing any new outcome. The $H = 2$ generator used two 32-feature sensor tiers, fixed inventory counts 4/4 and exact-resource D1 [8] and D2 [4, 1] morphologies. All remaining task, signal, gain, axial-coupling, optimizer and stopping settings were inherited without retuning. Aligned and reversed placement were crossed with serial BP, literal grouped-point BP and serial LocalCA with shared-soma or exact-path transport over ten fresh paired seeds 10300–10309, yielding 160 fits.

Aligned serial BP rose from 0.6588 at D1 to 0.9672 at D2. The 30.84-point depth effect (30.46–31.23; 10/10 positive; $P = 0.001953$) became -1.56 points under reversed placement, giving a 32.40-point interaction (32.10–32.70). Grouped-point depth effects were 0.02 points under alignment (-0.02 to 0.06) and -0.02 under reversal (-0.06 to 0.02); their alignment interaction was 0.04 points (-0.03 to 0.11). Serial-minus-grouped at aligned D2 was 30.82 points (30.41–31.23), and its architecture-by- placement interaction was 32.36 points (32.02–32.71). Shared-soma LocalCA retained a 21.20-point aligned depth effect (19.34–23.00) and a 22.93-point placement interaction (21.02–24.77). Exact-path LocalCA retained 31.38 points (30.95–31.83) and a 33.36-point interaction (33.00–33.75). Every positive interaction was ordered in 10/10 seeds with exact sign-flip $P = 0.001953$.

All 220 new fits were finite and stage complete, with no fallback or nonfinite alert, the expected seeds and exact within-hierarchy resource equality. The second hierarchy is a replication within the same synthetic mechanism-matched task family, not an independent dataset. Configurations, frozen manifests, run-level metrics, source hashes and every paired contrast are in `source_data/remaining_physical_experiments/`.

S5 Artificial-tree experiments

5.1 Architecture and input sparsity

MNIST images were converted to tensors, flattened to 784 inputs and not normalized or augmented. The 60,000-image training set was split by seed into 48,000 training and 12,000 validation images; the official 10,000-image test set was evaluated only after training. The core contained 128 modeled excitatory neurons and no separate inhibitory-neuron population. Each neuron had morphology $[3,3]$: three proximal compartments, three distal children per proximal compartment and one soma, for 13 compartments per neuron. There were no somatic input synapses.

At each of the 12 dendritic compartments, separate TopK parameter arrays selected 40 excitatory and 20 inhibitory contacts from the same 784-pixel input. The model therefore contained 92,160 active input contacts among 2,408,448 candidate synaptic parameters. Inactive local weights were not updated. Child-to-parent couplings and all active conductances were learned after a softplus transform. A linear 128×10 decoder produced the class scores. The archived model-resource record contained 2,414,602 trainable raw parameters in total, including candidate synaptic arrays, couplings, reactivation parameters and decoder.

5.2 Raw additive control

The configuration alias `dendritic_additive` set shunting and additive normalization to false and used the raw additive mode. For compartment n, its source implementation can be written

$$E_n^{\text{add}} = \sum_{i \in \mathscr{S}_n^E} g_i^E x_i, \qquad I_n^{\text{add}} = \sum_{i \in \mathscr{S}_n^I} g_i^I x_i, \tag{62}$$

$$C_n^{\text{add}} = \sum_{c \in \mathscr{C}_n} g_{c \to n}^{\text{den}} a_c, \qquad V_n^{\text{add}} = E_n^{\text{add}} - I_n^{\text{add}} + C_n^{\text{add}}, \qquad a_n = f_n(V_n^{\text{add}}). \tag{63}$$

Consequently,

$$\frac{\partial V_n^{\text{add}}}{\partial g_i^E} = x_i, \quad \frac{\partial V_n^{\text{add}}}{\partial g_i^I} = -x_i, \quad \frac{\partial V_n^{\text{add}}}{\partial g_{c \to n}^{\text{den}}} = a_c, \quad \delta_c^V = \delta_n^V g_{c \to n}^{\text{den}} f_c'(V_c). \tag{64}$$

Thus this control retained subtractive E/I input, the same branching graph and reactivation, but omitted the conductance denominator, reversal-potential driving force and parent-compartment resistance. Raw-parameter gradients in both architectures also included the softplus derivative.

Every compartment used the learned reactivation

$$f_n(V) = \frac{\tanh[m_n(V - b_n)] + 1}{2}, \qquad f_n'(V) = \frac{m_n}{2}\left\{1 - \tanh^2[m_n(V - b_n)]\right\}. \tag{65}$$

The shunting architecture used its analytical initialization policy. Because the raw additive voltage scale differs, the additive architecture used its predefined occupancy-quantile policy: the empirical 0.1 and 0.9 voltage quantiles were mapped to reactivation outputs 0.1 and 0.9, with minimum quantile width 10^{-3}, maximum slope 50 and reversion on an invalid fit. These were architecture-specific, prespecified gate policies rather than identical initial gate values.

5.3 Training protocols and archive boundary

The primary feedback-definition experiment used the three-factor rule, a locally updated decoder, mini-batches of 256 and 180 epochs. Adam used $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-8}$ and no weight decay. Raw TopK parameters, child couplings, reactivation parameters and decoder used learning rates 1.5×10^{-3}, 7×10^{-4}, 5×10^{-4} and 1.5×10^{-3}, respectively. Local gradients were clipped elementwise at 5. The best validation checkpoint was restored; early stopping and mixed precision were disabled. Runs used deterministic PyTorch settings, disabled TF32 and cuDNN benchmarking, and used the highest float32 matrix-multiplication precision.

The reported feedback comparison contains 15 seeds (42–56) per architecture and feedback condition. It is a clean current-code cohort run after freezing the design and replacement policy. Complete resolved configurations, checkpoints, logs, source and output hashes, and job records are retained for all 60 runs. The clean cohort replaced the earlier mixed archive irrespective of the observed effect; the superseded rows remain in the audit archive but are not used in the article.

The Fashion-MNIST replication retained this architecture, contact pattern, three-factor local rule, local decoder, optimizer, 180-epoch limit and architecture-specific initialization without retuning. Images were flattened to 784 inputs without normalization or augmentation. For each fresh seed 10200–10209, 48,000 of the 60,000 training images were used for fitting and 12,000 for validation; the official 10,000-image test set was used only after checkpoint selection. Sixty fits crossed shunting and additive trees with scalar fallback, neuron-indexed feedback and exact path transport.

Mean accuracies for scalar, neuron-indexed and exact path fields were 0.8393, 0.8839 and 0.8829 in shunting trees and 0.8442, 0.8820 and 0.8833 in additive trees. Neuron identity improved over scalar fallback by 4.46 percentage points (95% paired-seed interval 4.09–4.88) in shunting trees and 3.78 points (3.43–4.14) in additive trees; every seed difference was positive in both architectures (exact two-sided sign-flip $P = 0.001953$). Exact path minus neuron identity was -0.11 points (-0.21 to -0.01; 2/10 positive; $P = 0.0957$) and 0.13 points (-0.08 to 0.35; 8/10 positive; $P = 0.314$), respectively. Thus the dominant scalar-to-identity bottleneck replicated, while within-tree transport added no reliable benefit on this dataset.

The preregistered replication gate required at least 8/10 positive identity differences and an interval above zero in each architecture. The artifact gate required all 60 finite endpoints, complete checkpoints, exact registered seeds and no fallback warning; all gates passed. A frozen-source guard split each architecture across two source manifests after a concurrent edit. The exact diff added an optional rule override that no resolved configuration enabled; the audit retains both manifests. Seed-level outcomes, contrasts and source identities are in `source_data/fashion_feedback_ladder/`.

The exact-transport factorial was a separate five-seed protocol (seeds 42–46): it used 200 epochs, a ReLU core output, and crossed three- versus five-factor eligibility with local versus backpropagated decoder updates. TopK, coupling, reactivation and decoder learning rates were 1.5×10^{-3}, 7×10^{-4}, 5×10^{-4} and 1.5×10^{-3}. Its same-architecture backpropagation reference used the same seeds, 200-epoch budget, ReLU output, and validation-based checkpoint selection, with method-specific learning rates of 10^{-3} for TopK and decoder and 5×10^{-4} for coupling and reactivation. Test data were not used for checkpoint selection. Tables S9– S11 provide the frozen specifications and archive status; Table S14 reports the principal outcomes.

The result is not that shunting universally improves learning. Preserving neuron identity recovers most of the scalar-feedback deficit in both tested architectures. The separate exact-transport experiment establishes that the local three-factor eligibility is sufficient when the remaining compartment error is supplied accurately.

Exact reconstruction of the analytic gradient against automatic differentiation used 100 diagnostic runs. Median relative error was 7.43×10^{-8} and the maximum was 1.88×10^{-7}. Parameter tensors, rather than scalar tensor elements, were the diagnostic units.

5.4 Prospective depth-by-feedback experiment and checkpoint diagnostic

The prospective experiment crossed two tasks, two neuron models, one to four dendritic stages, and three local teaching fields at ten paired seeds. A matched backpropagation condition was included for each task–model–depth–seed combination, giving 480 local-learning and 160 backpropagation runs. The local conditions all used the theorem-derived three-factor eligibility. One modeled population of 128 neurons was used so that equal neuron indices in different feedforward populations could not share a teaching coordinate. The design, configurations, seeds, audit criteria and analysis code were frozen before confirmatory outcomes were inspected.

Every run was required to contain a resolved configuration, checkpoint, complete learning curves, resource record and finite held-out metrics. The audit also required every expected dendritic stage, nondegenerate learned couplings and the absence of transport-shape fallbacks or critical log errors. All 640 runs passed those artifact checks. A later outcome-independent input- validity audit excluded the 160 synthetic-noise shunting runs because their signed transfer output is not a valid positive-conductance drive. The 480 retained runs contain both neuron models on MNIST and the additive model on the noise task. Nominal depth changed compartment and parameter count and is therefore interpreted as a depth stress rather than a capacity-matched comparison. Table S15 summarizes the retained feedback contrasts.

To separate feedback dimension from assignment, an additional frozen cohort compared correct neuron-indexed sharing with a fixed derangement at depths two and four. The derangement preserves the number and per-example distribution of teaching coordinates, the forward network, and the training path; only the coordinate-to-tree address changes. Applying the same validity rule removed the 40 synthetic-noise shunting runs. Table S17 reports the six retained paired contrasts from 120 runs.

A second frozen cohort tested inhibitory dose on the noise-resilience task. It crossed two neuron models, inhibitory-contact counts of 0, 5, 10, 20 and 40 per branch, backpropagation or three local feedback fields, and ten paired seeds, for 400 runs. Its primary endpoint was the change from dose 0 to dose 40 in the shunting-minus-additive accuracy difference. Because all 200 shunting cells used the invalid signed-input convention, the full family is excluded from figures and inference; its executions remain in the public validity ledger.

5.5 Detached clean exact-transport/backpropagation audit

The replacement implementation audit crossed exact path transport or standard backpropagation, MNIST or the synthetic noise-resilience task, additive or positive-conductance shunting cores, depths one through four and seeds 42–51. It therefore contained 320 training runs and 160 method pairs. All accepted runs executed from detached tracked-clean source commit `74792ca91724066148f36c758191744a315d62b3`; the nested synthetic-data repository was fixed at commit `4f3612a52756c3691e5bd269c6465c02f240c1e8`. The additive synthetic model retained its signed transfer output. The positive-conductance shunting model used an explicit ReLU output transfer for both learning methods before conductances entered the denominator. This is a corrected input-valid audit, not a byte-identical replication of the historical signed-shunting cells.

Acceptance required a resolved configuration, seed record, learning summary, finite final metrics, finite stage-complete checkpoint, checkpoint and configuration hashes, and the newest scheduler and training logs without a fatal error, non-finite training event or transport-shape fallback. If device memory was insufficient only for full-dataset metric materialization, the runtime logged the event and recomputed the same final metric in streaming batches; the collector recorded this path separately and still required a finite endpoint. Initial attempts that lacked the ignored nested data repository, requested insufficient host or GPU memory, or reached the signed- shunting input guard produced no accepted endpoint. Missing configurations were rerun unchanged on H200 accelerators. Inference paired exact transport and backpropagation by seed within task, core and depth. The four plotted task–core effects first averaged the four depths within seed and then used a 20,000-draw paired-seed bootstrap interval; two-sided Wilcoxon tests are supplied as descriptive implementation checks rather than equivalence tests. All 320 runs passed these gates. Forty-two used the explicitly logged streaming recomputation for at least one terminal metric; every accepted final metric was finite. Table S16 reports the four depth-averaged comparisons. Across all 160 depth-specific method pairs, mean exact-minus-backpropagation accuracy was -0.0545 percentage points and the largest absolute individual pair difference was 5.90 points. The latter precludes interpreting the audit as pointwise equality even though each depth-averaged task–core interval included zero.

A third cohort compared fixed spatial and random sparse input maps at depth four. The 320 runs crossed two tasks, two neuron models, backpropagation or three local feedback fields, two maps, and ten paired seeds. Each distal branch had 21 contacts. Spatial branches recursively partitioned the 28×28 feature plane into 16 disjoint regions per neuron, whereas random branches sampled independently from all 784 coordinates. Supplementary Fig. S7 reports the 240 validity-qualified learning contrasts together with a configuration-level coverage audit. MNIST retains both cores; the noise task retains the additive core. Because the effect persists under backpropagation and on the randomly projected task, it is treated as a forward sparse-connectivity prior rather than credit-routing evidence.

A fourth frozen cohort varied depth while holding 16 terminal branches and approximately fixed input contact budget. The four depth conditions contained 960, 960, 968 and 960 active contacts per soma, respectively. It crossed two neuron models, backpropagation or three local feedback fields, four depths and ten paired seeds, giving 320 runs. The input-valid synthetic comparison retains the 160 additive runs. This control matches terminal branches and contacts, not trainable parameters: deeper networks still contain more compartments and more coupling and reactivation parameters. Supplementary Fig. S8 reports their complete depth profiles and the prespecified depth-four-minus-depth-one contrasts.

We then evaluated scalar, neuron-indexed and exact teaching fields at the 120 input-valid backpropagation checkpoints on one frozen test batch. Only non-somatic stages and their trainable branch parameters entered the field and gradient vectors. For each candidate field we computed scaled error-field capture, eligibility-weighted gradient capture and cosine, and the loss change after matching its update norm to the exact gradient. The primary relative step was 10^{-5}; 10^{-6}, 10^{-4} and 10^{-3} are retained as sensitivity values. The association interval resampled checkpoints while keeping paired feedback fields together. The diagnostic did not update or select checkpoints. Table S20 reports the primary operating point. Supplementary Fig. S9 displays the complete diagnostic.

5.6 Frozen phase-1 trained subtree-address experiment

This experiment implements the primary within-neuron support contrast from the trained-address contract on a deliberately minimal two-sibling model. It does not implement the complete architecture-by-K factorial. The phase-1 JSON configuration and executable were hashed before canary seed 4099 was run. The canary checked artifact integrity and nondegeneracy only: neuron-shared capture was 0.506, correct-route capture was 1.000, correct-route held-out accuracy was 0.814, and analytic exact transport agreed identically with analytic backpropagation. Because all frozen gates passed, the same unchanged files were used for confirmatory seeds 4100–4109.

For seed-specific unit vectors $\boldsymbol{u}_A$ and $\boldsymbol{u}_B$, binary label sign $y \in \{-1, 1\}$ and context $c \in \{0, 1\}$, the two input means were

$$\mathbb{E}[\boldsymbol{x}_A \mid y, c] = y\,[\mathbb{1}(c=0) - 0.75\mathbb{1}(c=1)]\boldsymbol{u}_A, \tag{66}$$

$$\mathbb{E}[\boldsymbol{x}_B \mid y, c] = y\,[\mathbb{1}(c=1) - 0.75\mathbb{1}(c=0)]\boldsymbol{u}_B. \tag{67}$$

Each stream had 12 coordinates and independent Gaussian noise with standard deviation 1.15. The 2,048-example train and test splits contained exactly 512 examples in each context–label cell. The somatic logit selected branch A when $c = 0$ and branch B when $c = 1$. Both inputs nevertheless remained active, preventing the one-coordinate rule from acquiring branch specificity through a zero eligibility on the distractor.

Let $\delta = \sigma(z) - \mathbb{1}(y=1)$ and $\boldsymbol{r}(c) = (\mathbb{1}[c=0], \mathbb{1}[c=1])$. Exact and correct-subtree coefficients were $\delta\boldsymbol{r}(c)$; within-neuron derangement used $\delta\boldsymbol{r}(c)_{::-1}$; the shared neuronal and same-depth controls used $\delta(1,1)$; and the random rank-two control used $\delta\boldsymbol{r}(c)R_s^{\top}$ for a seed-fixed orthogonal rotation R_s. The gated point emulation used one somatic coefficient together with the context-gated features $(\mathbb{1}[c=0]\boldsymbol{x}_A, \mathbb{1}[c=1]\boldsymbol{x}_B)$, making its alternative address resource explicit. All conditions had 24 trainable weights and the same paired initialization and batch order. The random dense control had four routing nonzeros; the sparse two-route conditions had two.

Training used mini-batch SGD, learning rate 0.3, batch size 64 and 60 epochs. The switch endpoint applied another 12 epochs to 1,024 context-1 examples with balanced labels and measured the change in context-0 accuracy. For gradient geometry, exact and candidate per-example gradients were concatenated over both branches. Scaled capture was squared cosine after optimal scalar rescaling. The one-step candidate was normalized to the exact batch-gradient norm and evaluated at step 0.25. Table S1 reports every confirmatory condition; no failed condition was removed.

The very poor deranged loss and the variable random-map loss were retained as prespecified failures. The equality of correct subtree routing, exact transport, backpropagation and gated point emulation is equally important: in this shallow model the exact path coefficient is only the binary branch gate. The result therefore isolates address

Table S1: Frozen phase-1 trained within-neuron address results. Values are means across ten independent paired confirmatory seeds; intervals for the principal differences are reported in the main text and Source Data.

Condition	Channels	Test accuracy	Initial capture	Switch forgetting
Neuron shared	1	0.7624	0.5095	0.5566
Same-depth shared	1	0.7624	0.5095	0.5566
Correct sibling subtrees	2	0.8063	1.0000	0.0000
Within-neuron derangement	2	0.1963	0.0000	0.0006
Random dense rank two	2	0.5847	0.2791	−0.0014
Exact transport	2	0.8063	1.0000	0.0000
Analytic backpropagation	2	0.8063	1.0000	0.0000
Gated point emulation	1 + gated eligibility	0.8063	1.0000	0.0000

alignment but does not establish a dendrite-specific advantage over a point system supplied with equivalent gated parameter groups.

5.7 Confirmatory subtree-address bandwidth and architecture factorial

The complete factorial replaced the two-sibling task with eight context- selected streams arranged in a balanced binary hierarchy. Context and binary label were exactly balanced in train and test sets, every stream remained active on every example, and the sign of the task signal reversed outside the selected context. This prevents a shared coefficient from recovering the required address through silent eligibility. The prespecified feedback budgets were $K \in \{1,2,4,8\}$ nested routes. Twenty paired confirmatory seeds (4200–4219) crossed five representations and 27 method–budget combinations, giving 135 fits per seed and 2,700 fits in total.

The dendritic tree, explicitly gated point model, flat-compartment model and grouped-point model were implementation-equivalent controls: each contained 64 trainable weights, 64 active input contacts and one output nonlinearity, and each received the same routed coefficient field. The fifth representation was a degree- and depth-matched rewired tree. Its resource counts were unchanged, but its ancestry groups were permuted relative to the task hierarchy. Feedback families comprised correct ancestry, fixed within-neuron derangement, depth-interleaved groups, random sparse groups, a seed-fixed random rank-K map, and a task-fitted rank-K upper bound. Neuron-shared, exact-transport and analytic-backpropagation references completed the design. Route fields were unit-normalized per example. Data, initialization, mini-batch order and every stochastic control map were keyed to seed, method family and K, so matched representations saw identical stochastic realizations.

A preconfirmatory canary required exact context balance, zero context–label correlation, nonzero activity in all streams, neuron-shared gradient capture below 0.2, exact capture of one, numerical equality of analytic exact transport and backpropagation, zero fallback counts, and exact agreement among the four implementation-equivalent representations. The first large execution was invalidated after a post-run audit detected that stochastic control maps were not paired across representation labels. Those rows were quarantined before scientific interpretation. The random streams were corrected, the implementation-equivalence check was promoted to a canary gate, and the entire 2,700-fit cohort was rerun. Only this corrected cohort is reported.

Table S2 gives the primary bandwidth contrasts. Correct ancestry strongly exceeded route derangement at $K=2$ and $K=4$, but its comparison with the best matched non-anatomical control was deliberately more stringent: ancestry lost at $K=1$ and $K=2$, won by 0.0127 accuracy at $K=4$ (95% paired seed-bootstrap interval 0.0059–0.0197; 15/20 seeds; two-sided Wilcoxon $P=0.0048$), and tied at full rank. Correct routing on the task-matched tree exceeded the rewired tree at $K=2$ and $K=4$ in all 20 seeds. The four matched representations were exactly equal in every condition (maximum held-out-accuracy difference zero). The experiment therefore supports an intermediate-bandwidth advantage for a task-aligned nested address, while showing that an explicitly routed point implementation can reproduce it.

5.8 Inherited regular-tree figures and expanded regime archive

The three figures below are reproduced without redrawing from the final NeurIPS/arXiv revision of the regular-tree study. Their PDF assets, panel definitions, and frozen generator scripts are preserved byte for byte in the journal repository. We retain them here because they show the original mechanistic and learning controls in exactly the form in which they were first reported. The following expanded archive reorganizes the same validated regular-tree evidence alongside additional task and stress controls; it is an extension, not a replacement for these inherited displays.

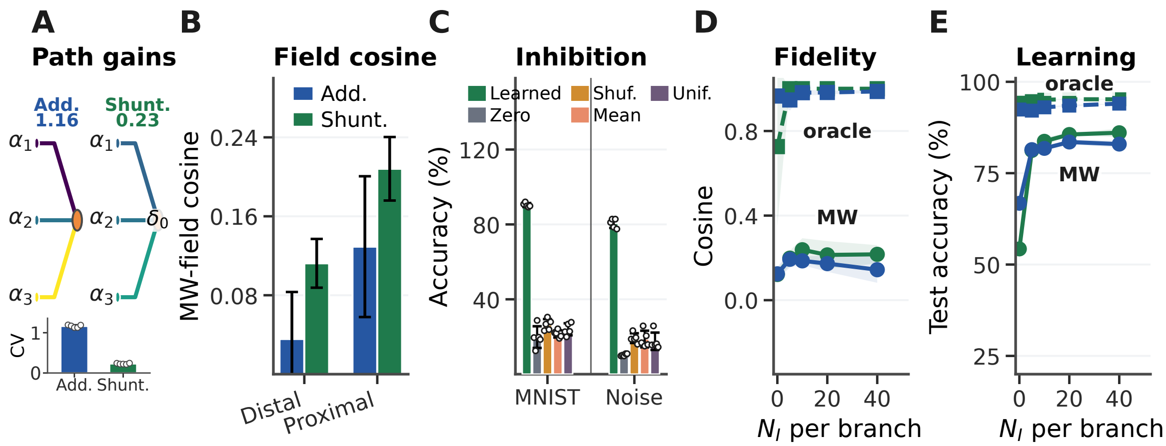

Figure S1: **Mechanistic chain from path gains to learning, reproduced from the original regular-tree study.** **A**, Conductance-stage path-gain field at five inhibitory synapses per branch; the inset shows paired-seed coefficients of variation. **B**, Stage-resolved cosine between the matched-width/scalar-fallback field and the exact MNIST error on distal and proximal compartments. The width-matched somatic stage is excluded because its cosine is one by construction. **C**, Post-training intervention in the learned shunting model: inhibition is zeroed, sample shuffled, replaced by a per-branch batch mean, or replaced by one uniform matched mean. **D,E**, Feedback fidelity and learning on the noise-resilience task; solid lines denote matched-width/scalar-fallback feedback and dashed lines exact transported error. Error bars are ±1 s.d. over checkpoints. This figure is the unchanged NeurIPS/arXiv asset generated by the archived original script. Its cross-architecture comparisons define operating regimes and do not establish a general backward shunting advantage. Noise-task shunting cells in panels **D,E** use the signed-input convention that the journal input-validity rule excludes for fresh cohorts; the corrected non-negative-transfer audit in the main text supersedes them for inference.

The journal synthesis retains the validated task, inhibition-dose, depth, broadcast-noise, rule-family, error-source, feedback-rank and flattened CIFAR-10 controls from the original study. These analyses use the historical frozen configurations rather than a newly tuned joint sweep. Most conditions contain five independent seeds; the rule-family and error-source comparisons contain three and are descriptive. Broadcast noise is zero-mean Gaussian noise added to the available teaching signal, with σ measured relative to the unperturbed signal scale. The depth sweep changes the number of dendritic stages at fixed somatic population width. CIFAR-10 images are flattened and use no convolutional front-end or competitive vision recipe. These experiments define operating regimes and feedback limits; cross-architecture differences do not isolate a backward shunting mechanism. Selected endpoints are reported in Supplementary Figs. S1, S2, S3, and S4, and Table S21.

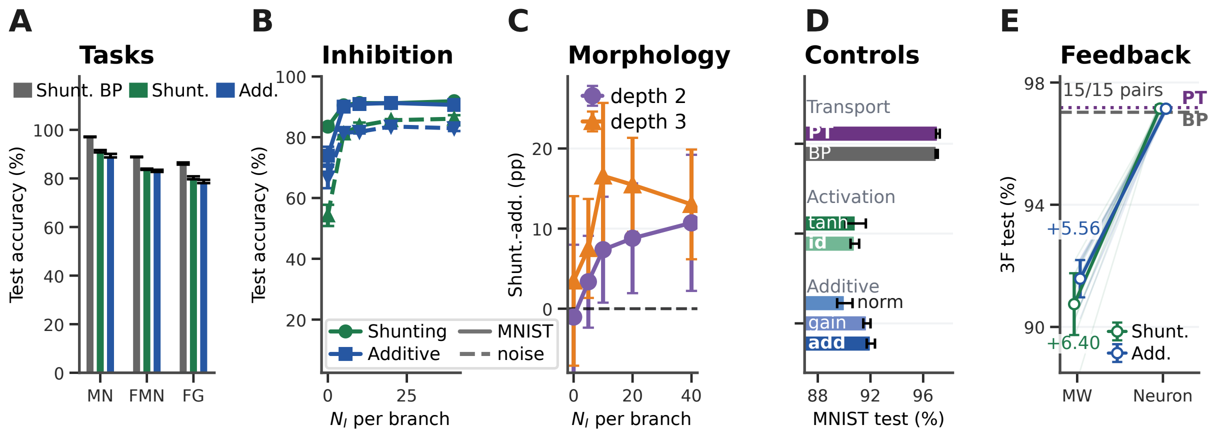

Figure S2: **Matched-capacity performance, regime dependence, and feedback definition, reproduced from the original regular-tree study.** **A**, Matched shunting backpropagation and five-factor local learning on MNIST, Fashion-MNIST, and figure-ground MNIST. **B**, Inhibitory dose-response. **C**, Descriptive morphology-by-inhibition performance. **D**, Exact-transport, reactivation, normalized-eligibility, gain-matched-additive, and additive-core controls on MNIST. **E**, Fifteen-seed comparison showing the gain from preserving one soma teaching coordinate for every compartment of the corresponding modeled neuron. Error bars are ± 1 s.d. This is the unchanged NeurIPS/arXiv figure asset; the prospective journal experiments extend its feedback result across depth, input-valid task–core conditions, and routing controls.

S6 MICrONS morphology reconstruction

6.1 Cohort and data sources

The main analysis used eight layer 2–5 intratelencephalic cells from the functionally co-registered MICrONS cohort [1]. Stable nucleus identifiers were resolved to root IDs in materialization 1822 of `minnie65_public`. Dendritic skeletons were downloaded through the MICrONS skeleton service and incoming synapses came from `synapses_pni_2`. Table S22 reports the frozen cell-level inputs to the structural analysis.

Degree-two chains were collapsed into branch-to-branch cable segments. A synapse was assigned to the nearest dendritic cable node and retained if its distance was at most $5\,\mu$m. Across the eight cells, 28,669 classified incoming synapses passed the mapping and E/I classification criteria. The mean mapping pass fraction was 0.980. Presynaptic coarse cell type supplied E/I identity when available. Otherwise, high-confidence target structure was used as a secondary proxy from the in-development `synapse_target_predictions_ssa_v2` table: spine targets were classified as excitatory and shaft or soma targets as inhibitory. This table does not currently have a stable public release identifier. Proxy and direct presynaptic labels agreed for 73.0% of 2,012 jointly labeled contacts. The principal capacity ordering was therefore repeated on the 2,077 directly typed contacts only.

6.2 Normalized electrical proxy

Raw axial coupling was proportional to radius squared divided by cable length; it was normalized by the median positive value within cell and clipped to 0.05–20. Raw leak was proportional to radius times the larger of cable length and radius; it was normalized by its median positive value and clipped to 0.1–10. Excitatory and inhibitory synaptic areas were normalized by the median nonzero total synaptic area. The resulting quantities are passive sensitivity weights, not fitted membrane conductances. The soma was the root of each compressed tree. Parent traversal verified acyclicity and one path from every retained segment to the soma. The compressed cable model used reciprocal axial coupling and a symmetric passive conductance matrix. It is therefore distinct from the directed

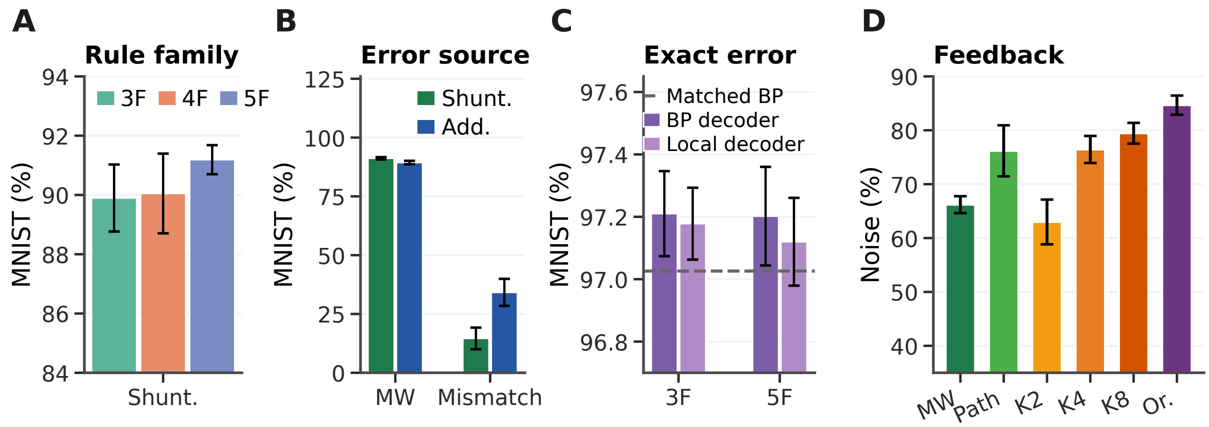

Figure S3: **Rule and feedback controls identify the principal learning bottleneck, reproduced from the original regular-tree study.** **A**, Within-shunting comparison of the three-, four-, and five-factor rules in an archived three-seed cohort. **B**, Somatic teaching error versus an unmatched compartment-local quantity. **C**, Five-seed exact-transport factorial showing that the theorem-derived three-factor rule approaches matched backpropagation when the required compartment error is supplied. **D**, Noise-resilience feedback ladder spanning scalar, path-propagated, random low-rank, and transported-oracle fields. Error bars are ± 1 s.d. This is the unchanged NeurIPS/arXiv figure asset and generator; exact transport is an information oracle. Noise-task shunting cells in panel **D** use the signed-input convention that the journal input-validity rule excludes for fresh cohorts; the corrected non-negative-transfer audit in the main text supersedes them for inference.

child-to-parent artificial network of Supplementary Note 1. The two models share a local synaptic sensitivity multiplied by a soma-derived compartment adjoint, but only the directed model has the path product in Eq. 9.

For excitatory segment i and candidate inhibitory route k, the direct model route matrix was

$$\Gamma_{ik} = A_{ik}\beta_k, \tag{68}$$

where A_{ik} indicates that k lies on the path from i to the soma and β_k is its normalized inhibitory-conductance fraction. This is a normalized ancestry-based route-sensitivity proxy. It is not the exact Green's-function path derivative of the reciprocal passive model used for the focal intervention. The participation rank of the excitatory-area-weighted kernel was $(\sum_j s_j^2)^2/\sum_j s_j^4$, where s_j are singular values.

S7 Structural credit-capacity analysis

Perturbation fields were generated by multiplying independent route amplitudes by the kernel in Eq. 68. For each cell and channel count, the tested dictionaries were: a dense principal-component oracle; selected real ancestry paths; random nonempty paths from the same tree; bins of cable depth; and a column-wise ancestry shuffle that preserved dictionary dimensions and sparsity. Ancestry routes were ranked by β_k times the fraction of total mapped excitatory synaptic area among their descendants.

Both fields and dictionaries were multiplied by the square root of summed mapped excitatory synaptic area before ordinary least-squares projection, as defined in Eq. 29. Twenty independent Monte Carlo streams were used, each containing 384 perturbation fields and 64 random or shuffled dictionaries per cell and channel count. Streams measure numerical sensitivity and were averaged within cell before inference. The eight reconstructed cells were the units of analysis; they came from one animal.

At eight channels, morphology-selected paths captured 0.487 of the weighted energy in the model-derived route-perturbation fields (hierarchical 95% interval 0.400–0.581) at 6.92% dense-feedback wiring. This was 0.850

of the dense oracle's capture. Table S23 reports all matched controls. On the directly typed subset, weighted capture was 0.782 for the dense oracle, 0.696 for morphology, 0.347 for random paths, 0.447 for depth bins and 0.308 for ancestry shuffling. Ancestry-route capture exceeded every structural control in all eight cells. The direct-type analysis addresses sensitivity to E/I labels, but its smaller input set is not a population estimate. These fields are not observed task gradients. Their descent interpretation is conditional on treating them as exact gradients in the transformed coordinate system; the alignment-controlled analysis supplies an explicit optimization test.

We also froze a 55-cell sensitivity universe before computing outcomes, mapped it to the public MICrONS v661 static release, and excluded the original eight cells by stable nucleus identifier. Twenty independently generated perturbation streams were averaged within cell. At eight channels, morphology paths in the 47 remaining cells captured 0.756 of modeled field energy with 8.08% of dense wiring, compared with 0.839 for the dense oracle, 0.302 for random paths, 0.441 for depth bins and 0.330 for ancestry shuffles (Supplementary Fig. S10b–d). The morphology advantage over every structural control was positive in 47/47 cells. The cohort is from the same mouse, is layer-imbalanced, descends from a convenience export, and uses an earlier reconstruction release. Direct presynaptic E/I labels were available for 11,024 incoming synapses before mapping (4.71%); 10,516 mapped contacts entered the analysis. It is therefore a disjoint-cell robustness analysis, not an independent-animal or population replication. The frozen candidate and endpoint-exclusion manifests accompany the source data.

To remove the direct generator–dictionary correspondence, we formed an independent response operator on the original eight trees. A focal shunt of 10^{-3} times local leak-plus-synaptic conductance was added at every inhibitory-bearing segment of the reciprocal passive model. After an independently solved state-matching compensatory current restored the baseline somatic voltage, we finite-differenced the log magnitude of the exact excitatory-conductance gradient at every excitatory-bearing segment. The resulting response operator was full rank in all eight cells and placed 17.6% of its weighted energy, on average, outside direct ancestry support. For a dictionary Φ, expected capture was evaluated exactly as the squared Frobenius norm after projection of this operator onto $W^{1/2}\Phi$, divided by its total weighted Frobenius norm.

We added 200 surrogate trees per cell. At each topological depth, child nodes were randomly assigned among the original parent slots, preserving every node's depth and every parent's out-degree. All edge and node attributes remained attached to their original child nodes. The same leverage rule was then applied to the surrogate dictionary. Row-shuffled selected paths preserved each selected column's values and nonzero count. At eight channels, morphology exceeded random real paths and row-shuffled paths in every cell, but did not reliably exceed depth bins or degree- and depth-matched surrogate trees (Table S24). This control supports sparse ancestry structure relative to random assignment while bounding the component specific to fine topology beyond depth and degree.

S8 Focal shunting perturbations

8.1 Adjoint formulation and matched additive control

For the passive conductance matrix G, voltage was $\boldsymbol{V} = G^{-1}\boldsymbol{b}$. For a loss depending on somatic voltage, the adjoint error was

$$\boldsymbol{q} = G^{-1}\boldsymbol{e}_s \frac{\partial \mathscr{L}}{\partial V_s}, \qquad \frac{\partial \mathscr{L}}{\partial w_i} = q_i(E_E - V_i). \tag{69}$$

A focal shunt η_k gives $G' = G + \eta_k \boldsymbol{e}_k \boldsymbol{e}_k^T$. The Sherman–Morrison identity yields

$$\boldsymbol{q}' = \boldsymbol{q} - \frac{\eta_k q_k}{1+\eta_k (G^{-1})_{kk}} G^{-1}\boldsymbol{e}_k. \tag{70}$$

Let $\boldsymbol{c}_k = G^{-1}\boldsymbol{e}_k$ be the kth Green's-function column and let $r_k = c_{k,k} = (G^{-1})_{kk} > 0$. Then

$$\Delta\boldsymbol{q} = -a_k \boldsymbol{c}_k, \qquad a_k = \frac{\eta_k q_k}{1+\eta_k r_k}, \qquad \frac{\partial |a_k|}{\partial \eta_k} = \frac{|q_k|}{(1+\eta_k r_k)^2} > 0. \tag{71}$$

The perturbation amplitude saturates at $|q_k|/r_k$, while its normalized spatial profile is fixed by $\boldsymbol{c}_k$ in this linear passive state. For a descendant set $D(k)$ and a prespecified matched comparison set $C(k)$, define

$$S_k(D,C) = \frac{\mathrm{median}_{i\in D(k)}\,|c_{k,i}|}{\mathrm{median}_{j\in C(k)}\,|c_{k,j}|}. \tag{72}$$

Because the common amplitude $|a_k|$ cancels, $S_k > 1$ is necessary and sufficient for a positive descendant-minus-comparison contrast in absolute adjoint change at fixed k. It is only a transport-level criterion. Relative or logarithmic changes additionally divide by the baseline q_i, and the full synaptic gradient also contains the post-perturbation driving force. This is why axial coupling, leak, background conductance and local input resistance must be varied explicitly rather than inferred from topology alone. The first-order-current-matched additive control injected $\eta_k(E_I - V_k)$ at the same node without changing G. An independently solved state-matching compensatory current injection restored baseline somatic voltage after both interventions. This is not a continuously adjusted voltage clamp. Therefore the somatic scalar error was fixed, while shunting changed adjoint transport and the additive control did not. Dendritic voltages were not clamped, so both interventions could also change local driving forces.

8.2 Frozen endpoint and validation

Before running the analysis, we fixed the site-selection rule, primary dose, comparison groups, and localization endpoint. Focal sites were selected by a depth-stratified rule without access to gradient outcomes. A site required at least three excitatory descendants and three comparison sites. At most 16 eligible sites were sampled per cell across quartiles of topological depth. The primary model used excitatory and inhibitory synaptic scales of 0.35 and normalized reversal potentials $E_E = 1$ and $E_I = -0.2$. Synapses were classified as descendants, ancestors, sister-subtree sites or unrelated sites. Unrelated synapses within each template were matched to descendant cable depth. For baseline gradient g_i and perturbed gradient g_i', we defined $m_i = |\log[(|g_i'| + \varepsilon)/(|g_i| + \varepsilon)]|$, where $\varepsilon = 10^{-15}\max(1, \max_i |g_i|)$. For each site, the localization index was the median m_i among descendants minus the median among depth-matched unrelated synapses. Site effects were first averaged within cell.

The somatic loss was one-half the squared difference between somatic voltage and a target one normalized unit below its baseline value. Conductance doses were 0.25, 0.5, 1, and 2 times the local leak-plus-synaptic conductance. The state-matching current was solved after each perturbation and then treated as fixed when evaluating the post-intervention conductance gradient. The topology control sampled 200 relation templates from other focal sites, stratified by focal-site topological-depth quartile when another site was available; otherwise it sampled another site in the same cell.

The primary unit-dose test included 101 sites in eight cells. Mean localization was 0.104 under shunting and 0.035 under matched additive perturbation. The paired cell-level difference was 0.069 log units (95% bootstrap interval 0.053–0.086), positive in all eight cells. A relation control sampled 200 templates with the stratified-with-fallback procedure above. True focal relations exceeded this control by 0.080 (0.059–0.098), again in all eight cells. The dose contrast increased from 0.018 at 0.25 local-conductance units to 0.128 at two units.

We separated the two factors in the exact gradient algebraically. Writing $d_i = E_E - V_i$ and $d_i' = E_E - V_i'$, full shunting uses $q_i' d_i'$, adjoint-only substitution uses $q_i' d_i$, and driving-force-only substitution uses $q_i d_i'$. At unit dose, their mean localization indices were 0.104, 0.130 and 0.026, respectively, compared with 0.035 for the matched additive control. Adjoint-only minus driving-force-only localization was 0.104 (95% interval 0.080–0.125), positive in all eight cells.

More directly, full shunting and the driving-force-only state share the same post-shunt voltage field and differ only in whether the adjoint transport field is updated. Their localization difference was 0.079 and was positive in all eight cells. This is the exact factor-freeze contrast plotted in main focal-perturbation figure, panel h.

Because the gradient factors and the localization statistic are nonlinear, these four payoffs are not an additive decomposition. We therefore allocated the full-shunt-minus-additive contrast with two-factor Shapley values. For

v_{00} equal to matched-additive localization, v_{10} adjoint-only, v_{01} driving-force-only and v_{11} full-shunt localization,

$$\phi_q = \tfrac{1}{2}[(v_{10} - v_{00}) + (v_{11} - v_{01})], \tag{73}$$

$$\phi_d = \tfrac{1}{2}[(v_{01} - v_{00}) + (v_{11} - v_{10})]. \tag{74}$$

The identity $\phi_q + \phi_d = v_{11} - v_{00}$ held at every site to a maximum absolute residual of 5.6×10^{-17}. Averaging sites within cells, the adjoint allocation was 0.087 (hierarchical 95% interval 0.063–0.111), while the driving-force allocation was -0.017 (-0.029 to -0.009). The former was positive and the latter negative in all eight cells. Within this algebraic allocation, the adjoint change accounted for more than the net positive contrast, while the driving-force change partially cancelled it. The matched additive reference is the relevant control but is not a literal no-factor corner of a physical 2×2 intervention. Using the unperturbed field as that corner assigned 0.104 (0.074–0.136) to the adjoint and approximately zero to the driving force. Both analyses allocate a nonlinear modeled payoff; the factor substitutions are not independent biological interventions.

All conductance matrices in the primary analysis were positive definite. The somatic state-matching residual was numerically zero. Analytic adjoint gradients agreed with finite differences to a maximum relative error of 1.03×10^{-6}. Table S25 reports prespecified conductance-scale and reversal-potential sensitivity runs. The direction of the shunting-minus-additive effect was positive in every condition, with seven or eight positive cells. These results establish a mechanism in the passive model on measured trees. They do not establish endogenous inhibitory teaching signals in vivo.

With proxy-classified contacts removed, the primary shunt-minus-additive contrast was 0.101 (0.075–0.129) across 55 sites and was positive in all eight cells. In the disjoint v661 cohort, 45 cells supplied 235 eligible sites. The same contrast was 0.138 (0.116–0.162), positive in 45/45 cells. Forty cells with at least two eligible sites supplied the reassigned-relation control; the true relation exceeded this control by 0.118 (0.094–0.144), with 39 positive cells and one numerical tie (main v661 robustness figure, panels f,g). Cells, rather than sites or relation draws, were the units of inference.

We also computed axial and leak conductances in physical units from the same compressed geometry. A cylindrical non-root segment had axial conductance $\pi r^2/(R_a \ell)$ and membrane leak $2\pi r \ell / R_m$; the root was assigned spherical area $4\pi r^2$. All values were converted to nS. Because MICRONS synapse size is not an electrical conductance, synaptic sizes retained the prespecified relative scaling with the median physical leak as reference. We varied R_a and R_m without changing site selection, relative dose, the state-matching somatic current, gradient definition or additive control. Table S26 shows the resulting electrotonic dependence. The large normalized-proxy contrast was recovered only in low-R_m, high-conductance settings; at $R_a = 150\,\Omega\,\mathrm{cm}$ and $R_m = 15{,}000\,\Omega\,\mathrm{cm}^2$ it was negligible or small. These are sensitivity settings, not fits to the recorded cells, and the model remains passive and steady-state.

We then froze a phase-1 matrix before its numerical canary. It fixed $R_a = 150\,\Omega\,\mathrm{cm}$ and crossed $R_m \in \{300, 1000, 15{,}000\}\,\Omega\,\mathrm{cm}^2$, distributed leak-like background conductance of 0, 1 or 4 times baseline leak, and three dose definitions. Relative-local doses were 0.25, 1 and 2 times membrane-plus- synaptic conductance; fixed doses were 0.05, 0.5 and 5 nS; input-normalized doses were 0.25, 1 and 4 times $[(G^{-1})_{kk}]^{-1}$. The same 101 sites in eight cells supplied 16,362 site–condition–intervention rows. Every conductance matrix was positive definite (minimum eigenvalue 0.107 nS) and the maximum somatic-state residual was 3.33×10^{-16}.

At fixed absolute dose, the conductance-specific contrast did not follow the raw focal-shunt localization alone. At 0.5 nS, shunt-minus-additive localization was 0.0104 (0.0080–0.0126), 0.0073 (0.0044–0.0104), and -0.0039 (-0.0075 to -0.0004) at $R_m = 300$, 1,000 and 15,000, respectively. At 5 nS the corresponding values were 0.0974, 0.0741 and 0.0061; the last interval included zero. Distributed background conductance restored the input-normalized effect in the high-resistance condition: at unit input-conductance dose, the contrast changed from -0.0004 with no background to 0.1210 and 0.2587 at background multipliers one and four. Across all 8,181 focal-shunt rows, every descendant gradient magnitude was attenuated, none was enhanced and no sign flip occurred. The passive mechanism is therefore a selective attenuator; signed reversal would require dynamics or active conductances absent from this phase (Supplementary Fig. S11).

8.3 Active steady-state local-Jacobian ensemble

We next tested whether the focal result survives voltage-dependent currents. The physical reconstructed-tree system used $R_a = 150\,\Omega\,\mathrm{cm}$, $R_m = 1{,}000\,\Omega\,\mathrm{cm}^2$ and distributed background conductance equal to baseline leak. Steady-state Na, K, Ca, HCN and NMDA current families had fixed reversal potentials, half-activation voltages, slopes and gate powers specified in the frozen JSON protocol. Global and segment-level maximal conductances were drawn independently from prespecified log-normal distributions. These draws vary plausible operating regimes; they are not fits to cell-specific channel measurements.

For each draw, a residual-reducing Newton solve found the nonlinear steady state. Acceptance required all voltages to lie in the frozen range, maximum absolute state residual below 10^{-10}, and minimum eigenvalue of the symmetrized local Jacobian above 10^{-8}. The implicit-adjoint calculation then used the exact inverse of this Jacobian. Focal shunting and a first-order-current-matched additive input were evaluated at 0.25, 1 and 4 times local input conductance; an independently solved somatic current restored the baseline somatic voltage. Up to 256 attempts were permitted to obtain 64 accepted draws per cell. All first 64 attempts were accepted in every cell, giving 512 equilibria, 101 focal sites and 38,784 site–draw–condition rows. All values were finite, the largest equilibrium residual was 9.51×10^{-11} and the smallest Jacobian eigenvalue was 1.084.

Channel draws and focal sites were averaged within each of the eight cells before inference. At doses 0.25, 1 and 4, mean shunt localization was 0.159, 0.529 and 1.339, respectively (Table S6). At unit dose, shunting exceeded the additive control by 0.382 (95% cell-bootstrap interval 0.356–0.409), positive in 8/8 cells (two-sided Wilcoxon $P = 0.0078$). Every modeled descendant gradient was attenuated, none was enhanced or changed sign, and descendant gradient energy fell to 0.330 of baseline. This extends the local sensitivity result beyond a passive matrix, but remains a steady-state linearization rather than a kinetic-channel or spiking simulation.

S9 Functional topology and task-derived credit

9.1 Direct anatomy–function join

Functionally imaged excitatory partners were linked to anatomical contacts through MICrONS/CAVE and DANDI mappings. The analysis contained 69 connected presynaptic partners across seven postsynaptic target cells. Automatic matches came from `coregistration_auto_phase3_fwd_apl_vess_combined_v2` in materialization 1822. We retained residual at most 10 and score at least zero, then deduplicated each presynaptic root by ascending residual and descending score. All mapped contacts from a retained presynaptic root contributed to its partner-level contact summary. For the one-site-per-partner model, contacts were first grouped by postsynaptic segment. The partner was assigned to the segment with greatest summed contact size, breaking ties by synapse count, and its major branch was the first segment below the soma on that segment's ancestry chain. Thus a partner with contacts on several segments supplied one activity feature and one modeled learning site; its total contact size and synapse count still included all retained contacts.

The target universe was the eight-cell morphology pilot. The prospective specification included every target in that universe with a DANDI scan and directly connected, functionally imaged excitatory partners; seven targets qualified. Operational checks required exactly one postsynaptic target in an extract, agreement between partner and contact counts, at least five mapped partners for the topology screen, at least two sites after any requested task filter, and at least one nonempty inhibitory ancestry route. The primary task analysis applied neither a manual-match restriction nor a repeat-reliability threshold. The exact target/session/scan pairs are frozen in Table S27. The archived specification does not record a deterministic rule for choosing among multiple eligible scans of the same target, so these scan choices are treated as fixed pilot assets rather than as the output of a coverage-ranking procedure. The omitted structural target, root 864691136043380566 (nucleus 292648), had one archived trial-manifest row, for session 8/scan 4, but `has_dandi_asset=False` and no DANDI asset identifier or path. No additional exclusion reason is inferred.

To remove ambiguity from selecting one scan per target, the extension applied one deterministic rule to the complete extracted scan universe: retain every automatic-conservative target–session–scan cohort with at least five unique connected imaged presynaptic roots. Thirteen scans across the same seven targets qualified. Pairwise statistics were computed within scan, scan values were averaged within target, and the seven targets remained the units of inference. Mean shared-path Spearman correlation was -0.0245 (95% target-bootstrap interval -0.214 to 0.147); the partial shared-path correlation after geometry and depth control was -0.0485 (-0.224 to 0.118), positive in three of seven targets. Negative tree-distance correlation and the same-major-branch contrast were also centered near zero (Table S7). Thus the deterministic scan-complete cohort preserves the original null boundary.

Each target had 464 trials and 280 unique condition hashes, of which 136 were repeated. Trial families comprised 384 Clip, 40 Monet2 and 40 Trippy presentations. For each trial and region of interest, the response was the mean raw `RoiResponseSeries` fluorescence over samples whose timestamps fell from trial start through trial stop, including both endpoints. Start and stop offsets were zero. No trials were dropped; every response used here was finite. Table S27 gives target-level asset identifiers and sample counts.

Functional similarity between two partners was the Pearson correlation of their vectors of mean raw fluorescence over the 136 condition identities with at least two trials. Repeat reliability was computed separately for each region of interest: occurrences of every repeated condition were divided into alternating first and second halves, responses were averaged within each half, and the two vectors of condition means were compared by Spearman correlation. No reliability correction was applied, and the primary cohort retained all partners regardless of this diagnostic; thresholds of 0.1 and 0.2 were used only in sensitivity analyses.

The controlled shared-ancestry statistic was computed separately within each target. Functional similarity and shared-path fraction were rank transformed. The two controls were $\log(1+d_{\text{Euclidean}})$ and $\log(1+|d_{\text{soma},i}-d_{\text{soma},j}|)$, where the second term is the absolute difference in soma path length; each control was also rank transformed. We regressed each ranked variable on an intercept and the two ranked controls by ordinary least squares, then took the Pearson correlation between the two residual vectors. The mean target-level partial-rank effect was -0.069 with a target-bootstrap 95% interval of -0.255 to 0.108; three of seven targets had positive effects. Pairwise partner comparisons were not treated as independent replicates.

The functional cohort is deliberately interpreted narrowly. MICrONS calcium imaging measured excitatory neurons, observed partners cover only a small fraction of each target's complete inputs, reliability is modest, the volume contains one animal, and the recordings were not made during learning.

9.2 Task-derived branch credit

A branch model predicted the standardized postsynaptic response from the standardized responses of its connected presynaptic partners. Let $b(i)$ be the dominant major branch receiving partner/site i. The fitted model was

$$d_b = \sum_{i:b(i)=b} x_i w_i, \qquad h_b = \tanh(d_b), \qquad \widehat{y} = \sum_b v_b h_b + c. \tag{75}$$

The objective was

$$\frac{1}{2N}\sum_{t=1}^{N}(\widehat{y}_t - y_t)^2 + \frac{\lambda}{2}(\|\boldsymbol{w}\|_2^2 + \|\boldsymbol{v}\|_2^2), \qquad \lambda = 10^{-3}, \tag{76}$$

with no penalty on c. The exact voltage-space error at site i was $(\widehat{y}-y)v_{b(i)}[1-\tanh^2(d_{b(i)})]$.

Whole condition-hash stimulus identities, not trials of the same identity, were split: 20% of the 280 identities and all of their trials were held out. Input columns and the target were standardized separately using training-set means and sample standard deviations (ddof $=1$); invalid or zero scales were replaced by one and remaining nonfinite standardized values by zero. A stable `SeedSequence` combined base seed 20260721, target root, stream and replicate. There were ten fixed split and initialization replicates per target.

Input weights were initialized as $\mathcal{N}(0, 0.12^2/n_{\text{site}})$, branch readout weights as $\mathcal{N}(0, 0.25^2/n_{\text{branch}})$, and $c=0$. Each method used 1,200 full-batch Adam steps with learning rate 0.015, $(\beta_1,\beta_2) = (0.9, 0.999)$ and $\varepsilon = 10^{-8}$. The exact model was trained first. Its exact site-error fields on training trials defined the dictionaries and its held-out exact field defined reported capture. All projected-learning methods then restarted from the identical initial state. Only the input-weight error was projected; $\boldsymbol{v}$ and c always received their exact gradients.

The primary budget requested four channels, reduced only when a target had fewer sites or nonempty candidate routes. Ancestry candidates were columns of ancestry multiplied by local inhibitory fraction g_i/g_{tot}; routes were ranked by this fraction times their descendant-site fraction. Random paths sampled nonempty candidates uniformly without replacement. Each ancestry-shuffle column was permuted independently over sites. Depth controls were path-length quantile bins, scalar feedback was an all-ones column, and the dense oracle used the leading right singular vectors of the exact training error. Projectors used relative singular-value tolerance 10^{-10}. No held-out response was used to fit a dictionary.

Held-out normalized MSE divided a method's held-out MSE by the MSE of predicting the training mean response. Replicates were averaged within target before inference; the seven target cells were the units. Intervals used 20,000 target bootstrap draws. Paired method comparisons used two-sided Wilcoxon tests, and the capture–utility analysis used method-label permutations stratified within target.

Table S28 reports the four-channel primary analysis. The dense four-dimensional oracle nearly matched exact learning, establishing that feedback dimension alone was not limiting in this surrogate. Ancestry-route did not reliably outperform ancestry shuffling in capture or learning. The within-target association between capture and normalized error was $\rho = -0.150$ with a target-stratified method-label permutation $P = 0.419$. The target-bootstrap 95% interval for ρ was -0.449 to 0.144. The coefficients are least-squares oracles and therefore upper bounds on the capacity of each feedback family, not biologically implemented encoders.

Sensitivity analyses at one, two and eight channels, with direct manual matches, with repeat-reliability thresholds, and with a linear branch model did not establish a reliable morphology-specific learning benefit. Some subsets improved morphology-specific capture without improving learning. This separation is the empirical reason the article distinguishes anatomical route capacity from task alignment.

9.3 Complete reconstructed-tree learning across all eligible scans

The segment-resolved extension eliminated the major-branch state reduction. Every complete compressed reconstruction was instantiated as a reciprocal passive conductance system. Each imaged partner supplied a trainable non-negative excitatory conductance at its dominant mapped segment. For trial-specific soma prediction error, the exact input gradient factored into the local term $x_i(E_E - V_i)$ and the exact soma-to-segment adjoint from the inverse conductance operator. Readout and bias parameters received exact gradients under every method.

Restricted rules received scalar output error, local eligibility and one fixed four-channel site vector. The vector was calibrated before training from a conductance-weighted topology dictionary, an equal-rank random selection of nonempty anatomical routes, or a column-value-preserving site shuffle. No target response entered this calibration. Ten seed-fixed splits per scan held out complete stimulus identities. Runs were averaged within scan and then within target. The resulting design contained four methods, 13 scans and ten splits, for 520 fits. A run was excluded for missing routes, rank mismatch, nonfinite values, incomplete scan coverage or failed finite differences; no run was excluded. One exact conductance coordinate per run agreed with finite differences to maximum relative error 3.05×10^{-7}.

Mean normalized held-out MSE was 0.788 for exact compartment error, 0.832 for topology, 0.833 for random routes and 0.839 for site shuffling (Table S8). The topology MSE improvement over shuffling was 0.0070 (95% target-bootstrap interval -0.0207 to 0.0321) and over random routes 0.0013 (-0.0090 to 0.0113). At the common exact checkpoint, topology captured 0.232 of the eligibility-weighted update field, compared with 0.358 for random routes and 0.439 for site shuffling. Thus a full segment tree and deterministic scan inclusion do not rescue anatomy alignment for this measured-response objective. This is a defined null result, not evidence against topology–function alignment under other behavioral states, response windows, input coverage or cell-specific biophysical calibration.

S10 Alignment-controlled optimization

The alignment-controlled experiment was specified as a constructive positive control for the conditional claim. For each reconstructed cell, we rebuilt the conductance-weighted ancestry kernel and selected eight ancestry routes by inhibitory fraction times descendant excitatory synaptic-area fraction before drawing any gradient fields. Coordinates were weighted by the square root of summed mapped excitatory synaptic area. Let Q be an orthonormal basis for those selected routes. Paired unit vectors $\boldsymbol{u}_{\parallel} \in \mathrm{span}(Q)$ and $\boldsymbol{u}_{\perp} \perp \mathrm{span}(Q)$ generated

$$\boldsymbol{g}(a) = \sqrt{a}\,\boldsymbol{u}_{\parallel} + \sqrt{1-a}\,\boldsymbol{u}_{\perp}, \qquad a \in \{0, 0.2, 0.4, 0.6, 0.8, 1\}. \tag{77}$$

Every field therefore had unit energy and exactly fraction a in the morphology-selected span. The parallel and orthogonal directions were reused across alignment levels.

Controls were eight random paths from the same tree, eight path-distance quantile bins, and an independent row permutation of each selected morphology route. The last control preserved each column's values and nonzero count while breaking ancestry. Random and shuffled dictionaries were redrawn within each nested stream and held fixed across alignment levels in that stream. Each of the eight cells contributed 40 streams with 256 fields per stream.

Optimization used a positive diagonal quadratic whose curvature spectrum ran from 0.5 to 1.5. The spectrum was independently permuted within each stream and then fixed across methods and alignment levels. One-step comparisons matched update norm at 0.1 times exact-gradient norm. Iterative projected descent used learning rate 0.25 for 20 steps. Progress was normalized by the corresponding full-gradient progress. Supplementary Fig. S5 and Table S33 report the principal values. Across the three control dictionaries and alignment levels, the median within-cell Spearman association between initial capture and 20-step progress was $\rho = 0.990$ (cell range 0.969–0.996). This analysis establishes conditional sufficiency and links capture to loss reduction under a controlled objective. Because alignment is imposed and coefficients are fitted from the exact gradient, it is neither an independent biological result nor a model of how a teaching circuit learns its coefficients.

S11 Synapse-resolved inhibitory route analysis

The inhibitory-census endpoints and controls were frozen in a prespecified analysis contract before the data join. The frozen cohort was the complete stable-nucleus-ID intersection between the 47-cell v661 sensitivity cohort and the v795 inhibitory census: 20 targets, with no selection by an outcome. Because the two releases use different coordinate representations, the raw-to-slanted affine map was estimated from paired published inhibitory-synapse locations and required to reproduce all paired locations to less than $10^{-6}\,\mu$m before any target was analyzed. Five targets had different root IDs across releases and were joined by stable nucleus identity. The source tables contained 116,401 input locations and 2,789 known contacts from 116 inhibitory cells before spatial mapping; 114,775 inputs and 2,637 known contacts from 114 inhibitory cells mapped within $5\,\mu$m. The packaged census table has no per-target proofreading-status field, so no such status is inferred.

The connection analysis retained one largest synapse per published axonal clump. This avoided counting immediately adjacent contacts from the same presynaptic axonal event as independent control sites. A matched candidate pool was constructed separately for every retained contact. Matching preserved target cell and compartment and minimized difference in log distance to the soma. All connection outcomes were averaged within target before testing. There were 402 inhibitory cell–target connections with at least two retained clumps, nested within 20 targets.

For descendant-domain overlap, let r_k be a binary vector indicating mapped input locations below candidate site k, multiplied coordinate-wise by the square root of synapse size and normalized to unit length. Two control sites therefore have overlap $r_k^{\mathsf{T}} r_l$. This value is one for identical weighted descendant domains and zero for disjoint domains. It is an anatomical co-control statistic, not a measured correlation of inhibitory activity. Table S29 reports the frozen target-level endpoints.

Two sensitivity controls followed the primary target-level analysis. First, connection contrasts were averaged within each of 92 presynaptic inhibitory axons before testing, which accounts for one axon contacting multiple targets. Second, candidate sites were ranked jointly by proximity in log path distance and three-dimensional Euclidean distance, and 500 matched assignments were repeated within target. The coarse tree-distance contrast survived presynaptic-axon aggregation. Shared-path and descendant-domain overlap did not; no endpoint survived joint 3D and path-distance matching (Table S30). Thus the primary result establishes spatial co-targeting, not geometry-independent organization over credit domains.

The route-capacity matrix used all actual inhibitory-bearing segments as columns, rather than one column per presynaptic inhibitory cell. Row weights were square roots of mapped input size and column weights were square roots of known inhibitory synapse size. Highest column norm defined the selected-site order. Random-site and shuffled controls were independently generated 200 times per cell and budget. This diagnostic asks whether the set of observed locations contains an efficient ancestry basis; it does not show that the highest-leverage subset was selected or used in vivo. Because the target is the same actual-route matrix from which the selected columns are drawn, this is an internal compressibility analysis rather than an independent routing test.

S12 External signed-credit analysis and task-interference identity

The Francioni et al. workbook was downloaded from the source-data link of the published article and frozen by SHA-256 `be1b87481d0f1d1bffc85d8a4cabfd37b7055d640306afab5b3622de0ced8264`. The animal-level contrasts in sheet "Ext 13E" were used directly. The source table's signed-separation column was required to equal P+ minus P- to numerical precision. The orthonormal common and signed projections sum exactly to total two-coordinate energy. Sheet "5E" supplied the four neuron-level distributions in the display but was not used to increase the sample size of the animal-level test.

For $n = 6$, the one-sided exact random-sign test enumerated all 2^6 sign assignments. Bootstrap intervals sampled animals with replacement. The directions were fixed by the experimental mapping: P+ positive, P- negative, and P+ minus P- positive. This is a retrospective test because the prediction was formulated after publication of the experiment, although it was evaluated from the numerical source table rather than the displayed figure.

12.1 Route overlap and task interference

Branch-specific plasticity during motor learning suggests that different tasks can recruit different dendritic regions [2]. Let $\boldsymbol{w}_A$ be an optimum of a quadratic Task-A objective with Hessian H_A, and let an approximate Task-B teaching field produce update $-\eta M_B \boldsymbol{g}_B$. Expansion about the Task-A optimum removes the first-order term, leaving the exact loss increase

$$\mathscr{L}_A(\boldsymbol{w}_A - \eta M_B \boldsymbol{g}_B) - \mathscr{L}_A(\boldsymbol{w}_A) = \frac{\eta^2}{2} \boldsymbol{g}_B^\top M_B^\top H_A M_B \boldsymbol{g}_B. \tag{78}$$

In the equal-route example, opposing updates on the shared fraction ω have magnitude two, whereas leakage onto the Task-A-only fraction has magnitude λ. Dividing by route size gives $\eta^2[4\omega + \lambda^2(1-\omega)]/2$. Numerical vector updates matched this identity to 4.2×10^{-17} absolute error (Supplementary Fig. S6). Separated routes therefore protect a learned task only when both anatomical overlap and teaching-field leakage are small. The published SST perturbation is consistent with the resulting prediction that loss of branch-selective inhibition increases effective leakage and cross-task interference [2], but it does not uniquely identify the credit-routing mechanism. This static identity does not model dendritic calcium, spikes, eligibility timing, or the SST manipulation in the source study.

S13 Statistics, reproducibility and claim controls

Central comparisons report paired effect sizes, 95% intervals and the number of positive cell or seed pairs. Tests were two sided unless a direction had been frozen before analysis. Exact Wilcoxon signed-rank tests were used for small paired cell cohorts. Reconstructed-cell intervals resampled cells first and, where relevant, one Monte Carlo stream within cell. Measured-response intervals resampled target cells. Artificial-network summaries used independently initialized training runs. Inhibitory-census outcomes were first averaged within postsynaptic target. Signed-credit intervals and exact sign tests used animals; neuron-level source rows were descriptive only. Bootstrap intervals used 20,000 draws. The four-budget primary subtree scan used Benjamini–Hochberg correction across

$K = 1, 2, 4, 8$; other parameter sweeps and sensitivity analyses received no multiplicity correction, are labeled secondary, and are not interpreted as independent confirmation.

The following measures are not used to support cross-architecture conclusions: pooled error-field metrics that mix a somatic stage whose reconstruction is exact by construction with dendritic stages; unweighted path-gain dispersion as evidence of gradient direction; and post-hoc comparisons between unmatched training cohorts. Table S32 records the supported interpretation and boundary of each evidence class.

Every main panel is accompanied by a machine-readable source table containing the plotted values, biological or training unit, nesting variables, seed, condition names, and analysis version. The release audit verifies that figure scripts read these tables rather than duplicating values in plotting code. Authentication tokens, restricted caches, and raw data already hosted by MICrONS or DANDI are not redistributed.

The release also includes the three archived regular-tree sweep files, representative fully resolved configurations, the surviving frozen manifest, the seed-level result tables, and an executable NumPy reference for the raw additive forward model and derivatives. Exact source copies of the morphology, routing-capacity, focal-perturbation and projection-bound analyses are provided with SHA-256 provenance. The measured-response directory contains the CAVE join, DANDI/NWB extraction and task-derived learning scripts together with a machine-readable four-channel specification and target manifest. The expanded release also contains the stable-ID inhibitory-census join, coordinate-transform audit, target-level topology tables, the verified Francioni source checksum and extracted contrasts, and the static interference verification grid. Private filesystem paths in portable configurations were replaced by paths relative to the article directory; numerical settings were not changed. The provenance manifest records each origin and every portability edit.

An exact package lock was not archived for the June exact-transport or backpropagation jobs, nor for the measured-response analysis. These versions are therefore not guessed from the current environment. The complete July feedback extension froze Python 3.10.13, its Git identity and selected source hashes, but recorded a dirty tracked worktree. Table S11 and the accompanying reproducibility README distinguish exact, partial and unverifiable provenance fields.

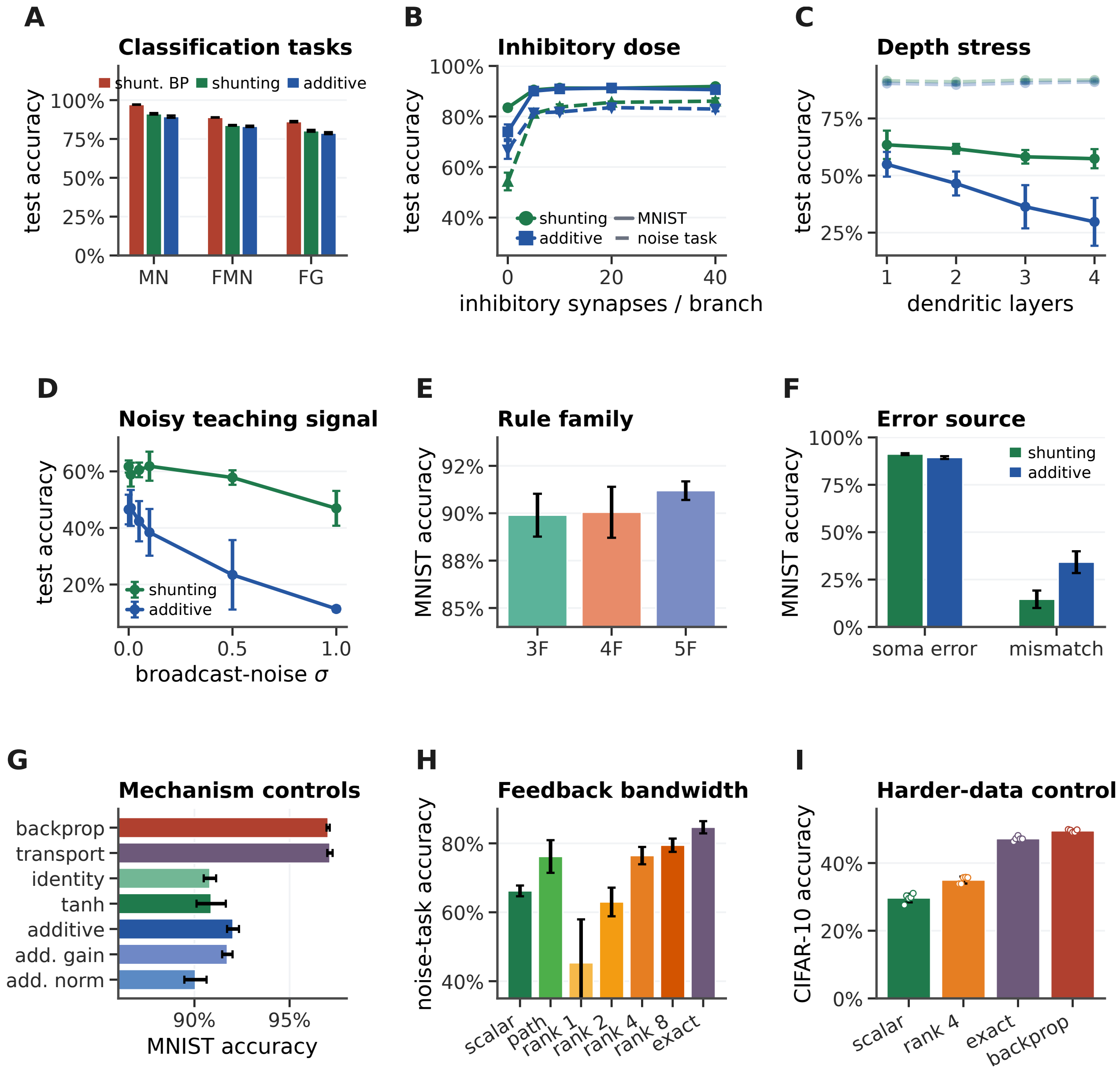

Figure S4: **Regular-tree learning across tasks, operating regimes, and feedback constraints. A**, Test accuracy on MNIST (MN), Fashion-MNIST (FMN), and figure-ground MNIST (FG) under shunting backpropagation and low-bandwidth local learning in shunting and additive trees. **B**, Local-learning accuracy as inhibitory synapses per branch vary on MNIST (solid) and the noise-resilience task (dashed). **C**, Accuracy as nominal dendritic depth increases. Solid lines are local learning and faint dashed lines are matched backpropagation; colour denotes shunting or additive architecture. **D**, Accuracy after adding zero-mean Gaussian noise with standard deviation σ to the broadcast teaching signal. **E**, Three-, four-, and five-factor rules within the same shunting, scalar-feedback cohort. **F**, Accuracy when the teaching factor is the somatic error or an unmatched local quantity. **G**, Separate five-seed controls for exact transport, reactivation, and additive gain or normalization, shown with a matched backpropagation reference. **H**, Noise-task accuracy across scalar, path-propagated, random low-rank, and exact-transport feedback. **I**, Flattened CIFAR-10 control in the shunting architecture. Bars and lines show means; error bars show sample standard deviations across independent training seeds ($n = 5$ unless noted). Panels **E,F** use the archived three-seed rule and error-source cohorts.

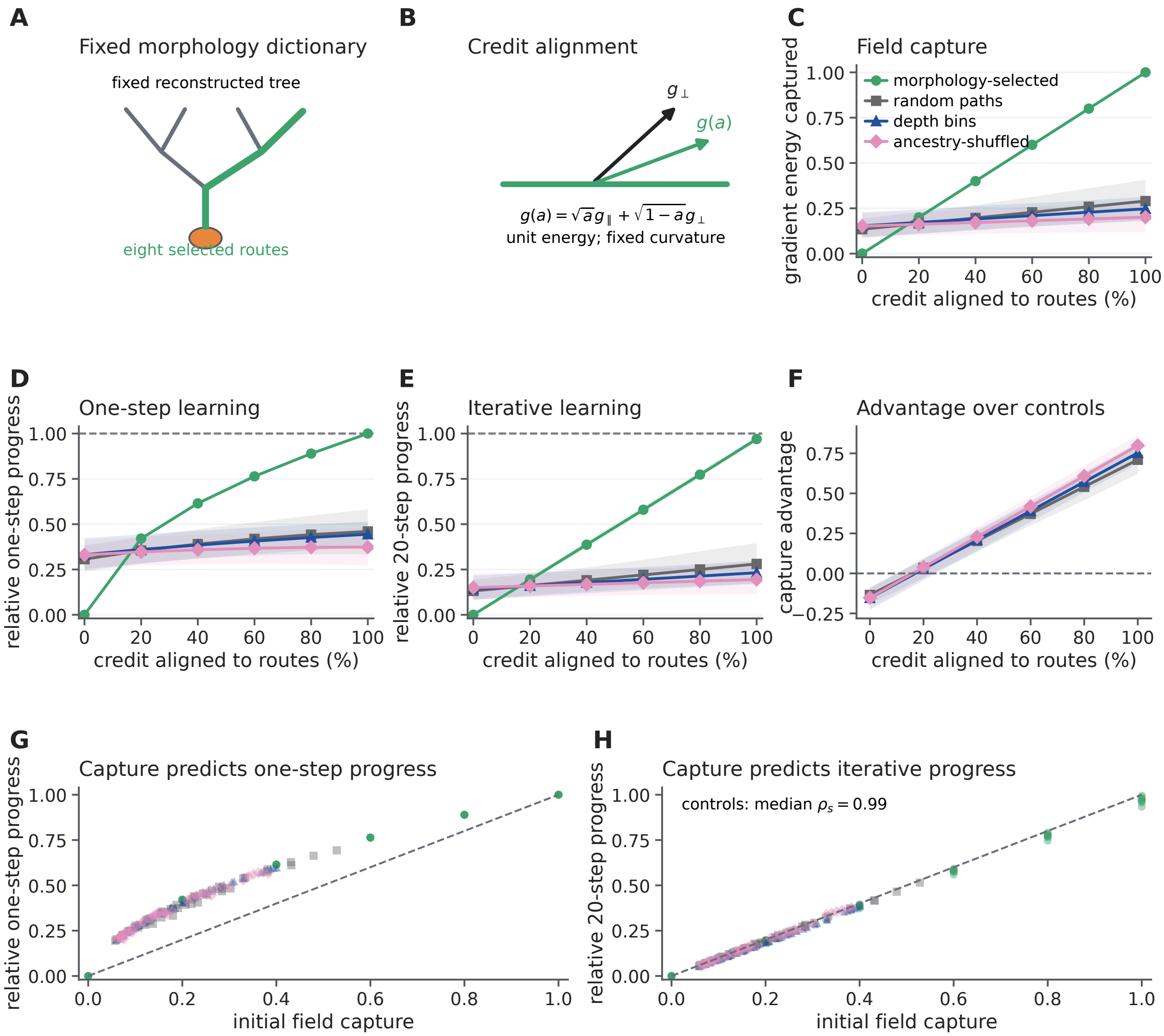

Figure S5: **Constructed route alignment controls projected learning.** **A**, A fixed eight-route dictionary on a reconstructed tree. **B**, Unit-energy credit fields are rotated between its span and the orthogonal complement. **C**, Capture across imposed alignment levels. **D**, Loss reduction from one norm-matched projected-gradient step, relative to the full-gradient step. **E**, Relative loss reduction after 20 projected steps. **F**, Ancestry-route capture relative to random paths, depth bins and ancestry shuffling. **G,H**, Initial capture versus one- and 20-step progress. Means in **C–F** are across eight cells with hierarchical 95% intervals. Ancestry-route capture equals imposed alignment by construction; coefficients are oracle projections. This is a conditional positive control, not independent evidence about the recorded neurons.

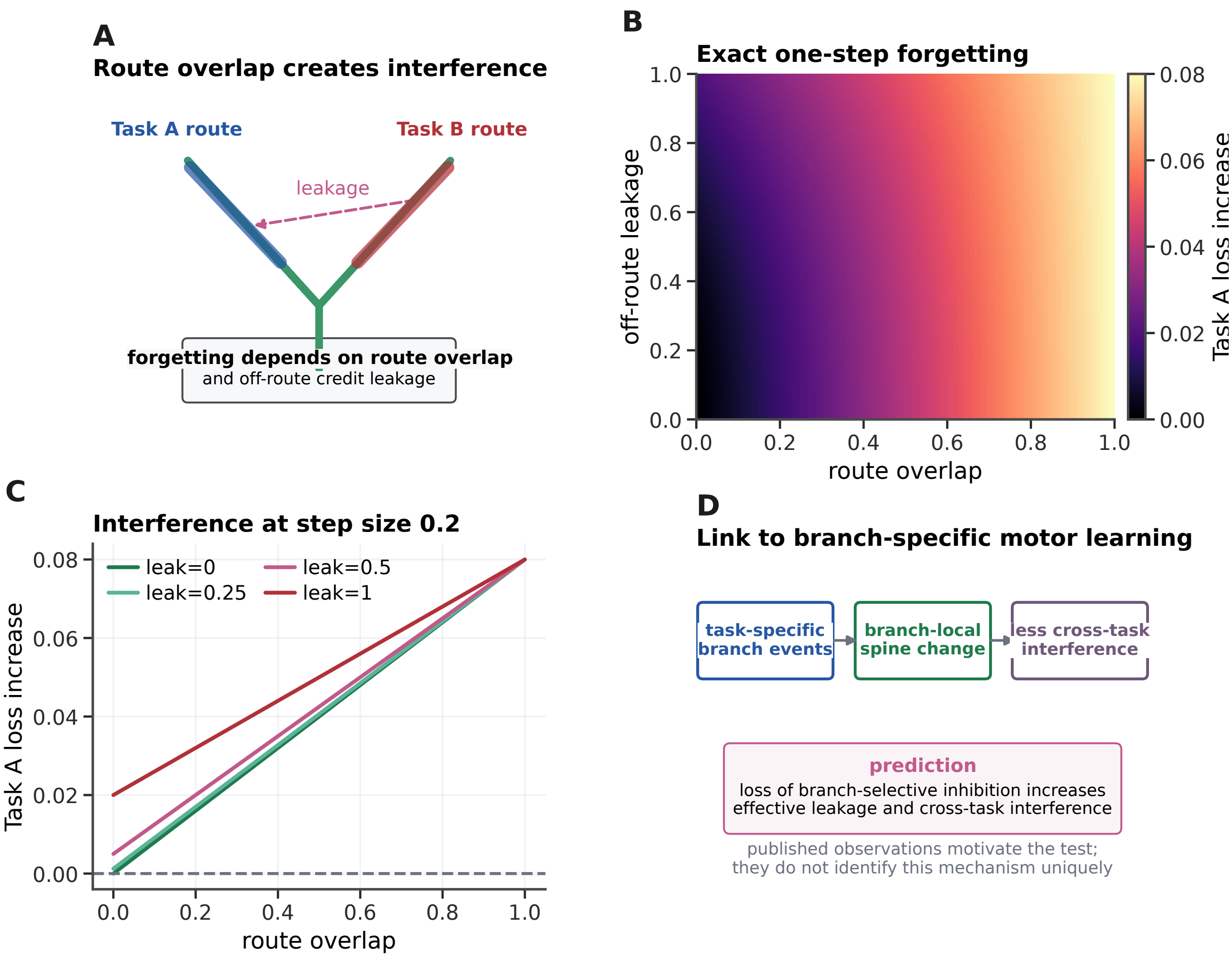

Figure S6: **Route overlap and off-route leakage determine one-step task interference.** **A**, Two tasks address partly separated dendritic routes; an approximate teaching field can also leak onto the other route. **B**, Exact Task-A loss increase after one Task-B update across route overlap and off-route leakage. **C**, Slices of the exact surface at step size 0.2. **D**, Relation to published branch-specific motor-learning observations and the resulting prediction. Panels **A–C** are an exact static quadratic consequence and numerical verification, not a temporal neural model. Panel **D** is a cross-study interpretation rather than a reanalysis of the original motor-learning data.

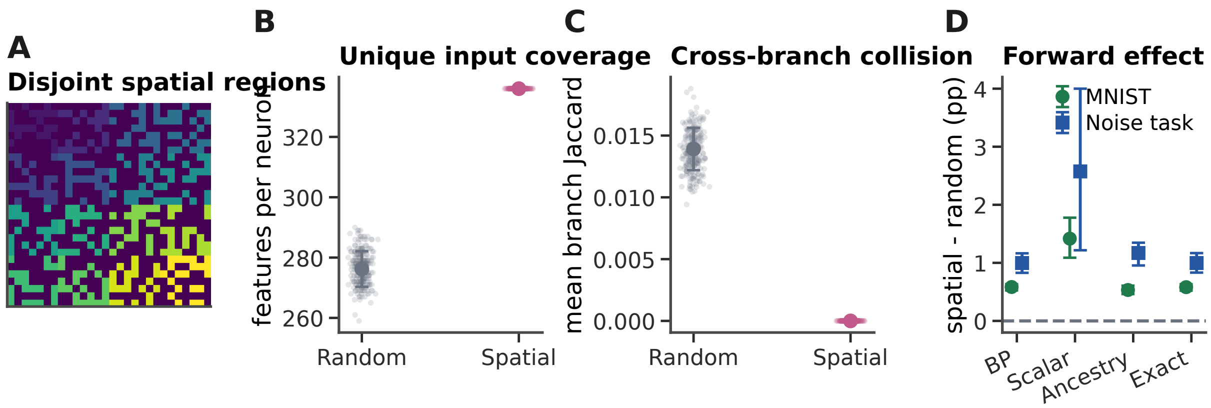

Figure S7: **The fixed spatial map changes forward input coverage. A**, Example contacts from one neuron. The 16 distal branches occupy disjoint image regions. **B**, Unique input coordinates reached by each neuron under independently sampled random branches and the spatial map. **C**, Mean pairwise Jaccard overlap between branch input sets. **D**, Spatial-minus-random accuracy under backpropagation (BP), scalar, neuron-indexed ancestry and exact feedback. MNIST averages shunting and additive models within seed; the randomly projected noise task uses the input-valid additive model only. Points and intervals show means and 95% paired-seed bootstrap intervals. Each learning condition contains ten paired seeds. The map has the same number of contacts in both conditions but greater unique coverage and no cross-branch collisions in the spatial condition; its performance effect is therefore not specific to local credit assignment.

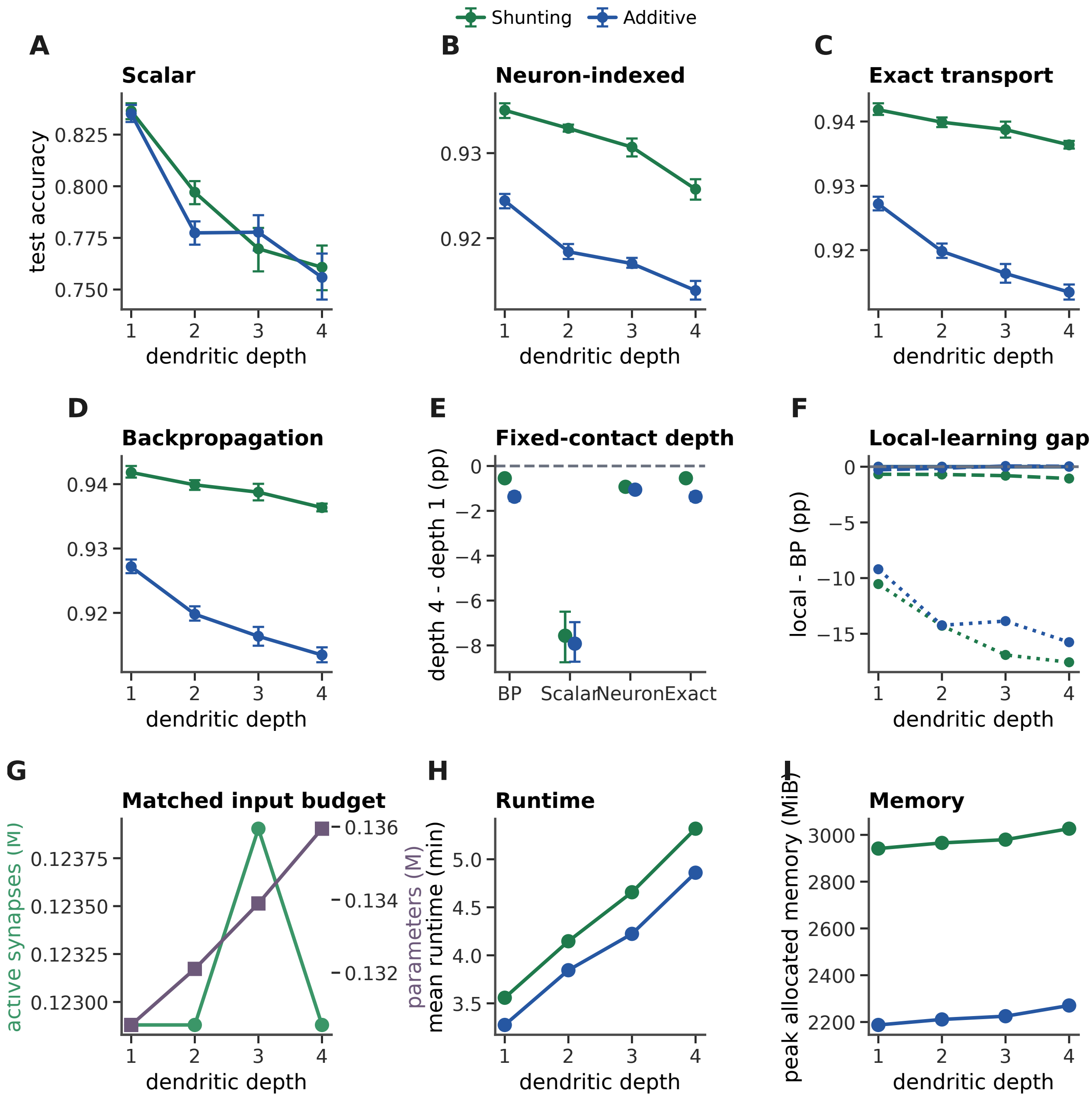

Figure S8: **Fixed terminal-branch and input-contact budgets do not make greater dendritic depth intrinsically beneficial.** **A–C**, Held-out accuracy across depths one to four under scalar, neuron-indexed ancestry and exact feedback for the input-valid additive network. **D**, Matched backpropagation. **E**, Prespecified paired depth-four-minus-depth-one accuracy contrasts. **F**, The gap between local and matched backpropagated learning. **G**, Active contacts and trainable-parameter counts. **H,I**, Runtime and peak memory. Each condition contains ten paired seeds; points and intervals are means and 95% paired-seed bootstrap intervals. All four retained depth-four-minus-depth-one contrasts were negative in 10/10 pairs (Wilcoxon $P = 0.001953$). The design fixes 16 terminal branches and 960–968 active contacts per soma, but does not match compartment or trainable-parameter count.

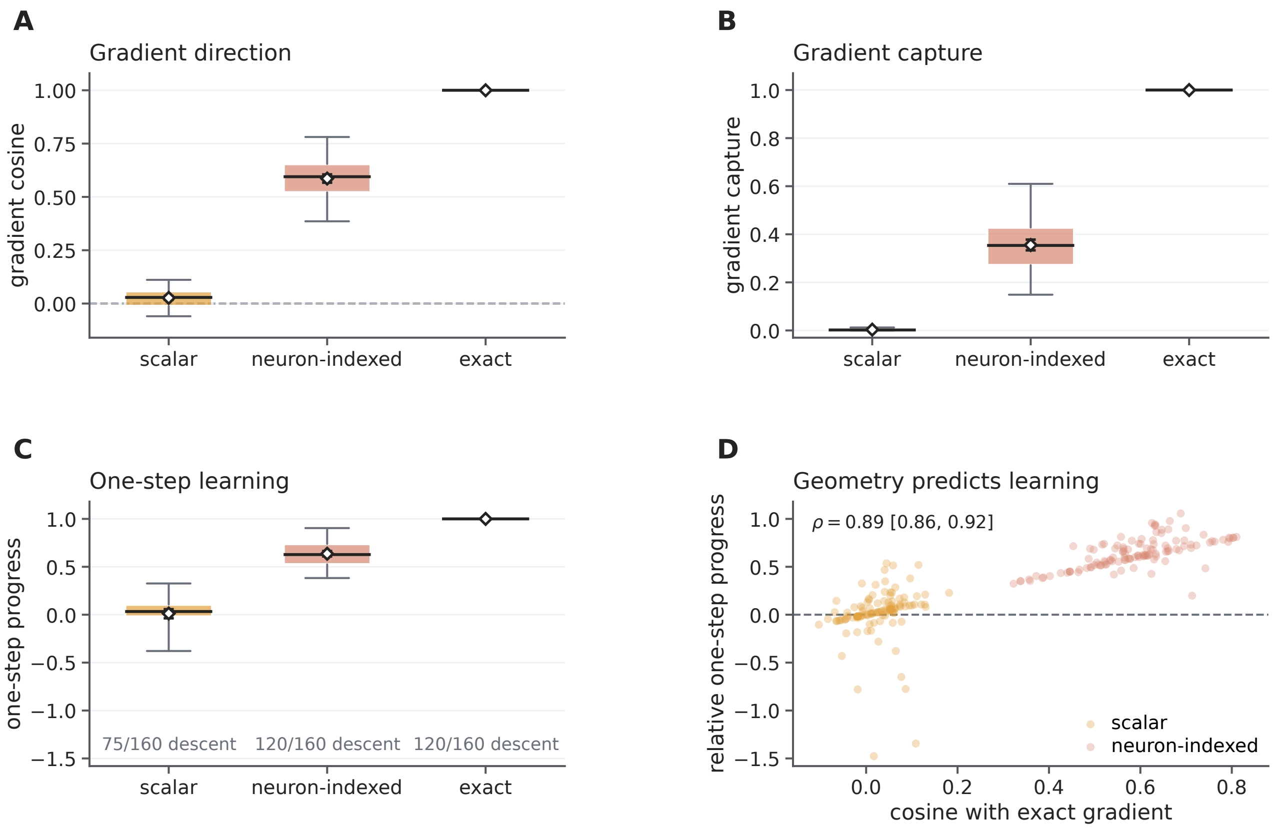

Figure S9: **Credit geometry predicts the loss change available to a local update.** **A**, Cosine between candidate and exact dendritic gradients at 120 input-valid matched backpropagation checkpoints. **B**, Eligibility-weighted gradient energy retained after optimal rescaling. **C**, Loss decrease after matching each candidate update to the exact-gradient norm; labels give the number of checkpoints in which the update decreased loss. **D**, Gradient cosine versus relative one-step progress for scalar and neuron-indexed fields; the interval for Spearman's ρ resamples checkpoints. In **A–C**, boxes show the interquartile range, whiskers the 5th–95th percentiles, and diamonds the mean with a 95% checkpoint-bootstrap interval. All panels use one frozen test batch per checkpoint and relative step 10^{-5}. Exact transport is an information oracle.

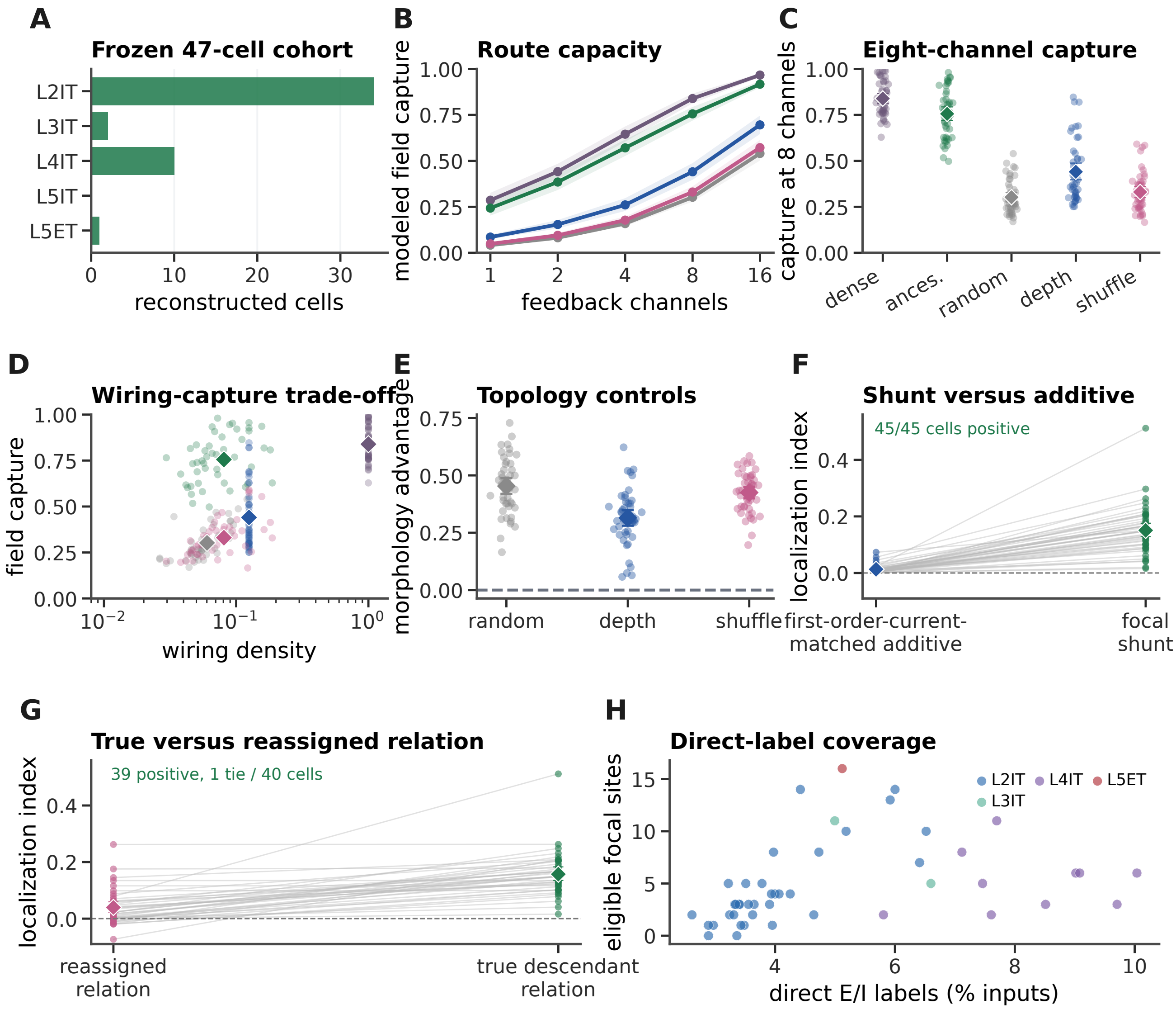

Figure S10: **Structural routing and focal shunting remain robust in a disjoint public-v661 cohort. A**, Composition of the 47-cell frozen analysis cohort by excitatory cell class. **B**, Cell-mean weighted modeled field-energy capture across feedback-channel budgets for a dense PCA oracle, morphology-defined paths, random paths, depth bins, and ancestry shuffles ($n = 47$ cells through eight channels; $n = 46$ at 16 channels). Twenty perturbation streams were averaged within cell; bands are 95% cell-bootstrap intervals. **C**, Cell-level capture at eight channels. **D**, Eight-channel wiring-capture trade-off; morphology paths used 8.1% of dense-feedback wiring. **E**, Within-cell morphology advantage over random paths, depth bins, and ancestry shuffles; all three contrasts were positive in 47/47 cells. **F**, Localization of exact-gradient changes after a unit-dose focal shunt and first-order-current-matched additive perturbation ($n = 45$ cells, 235 sites). **G**, Localization for true descendant relations and relation templates reassigned to other focal sites ($n = 40$ cells, 230 sites). **H**, Direct presynaptic E/I-label coverage and eligible focal sites per cell. Diamonds and intervals in **C,E–G** show means and 95% cell-bootstrap intervals. Cells, not sites or streams, are the inferential units. The cohort excludes the original eight nuclei but comes from the same mouse. It uses historical v661 reconstructions and direct E/I calls only (4.71% of incoming synapses). These are structural and passive-model tests, not measurements of learning in vivo.

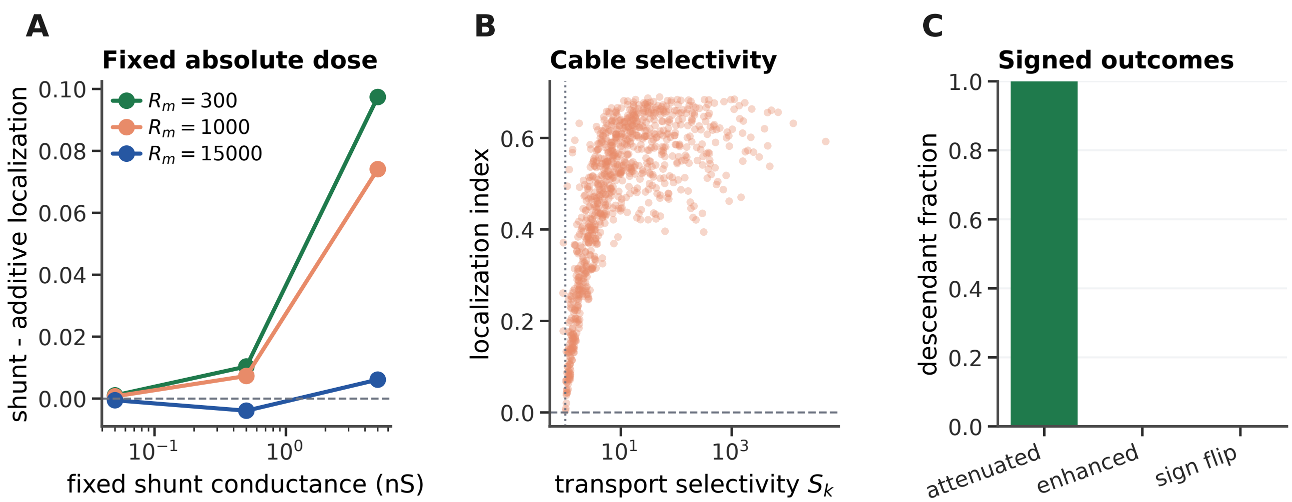

Figure S11: **Fixed-absolute dose and background conductance bound passive focal-shunt selectivity. A**, Shunt-minus-additive localization across fixed absolute conductance doses at three membrane-resistance settings with no added background conductance. Each curve averages focal sites within each of eight cells before the cell mean. **B**, Transport selectivity S_k versus full synaptic-gradient localization at unit input-conductance-normalized dose across 101 sites, three membrane resistances and three background levels. Points are site–regime values and are descriptive because sites and regimes are repeated within cell. The dotted line marks $S_k = 1$. **C**, Fraction of descendant gradients attenuated, enhanced or sign reversed at $R_m = 1{,}000\,\Omega\,\text{cm}^2$, background multiplier one and unit input-conductance dose. All tested passive focal-shunt conditions had the same qualitative signed outcome: attenuation without enhancement or sign reversal. Specific resistances and background levels are sensitivity parameters, not cell-specific fits.

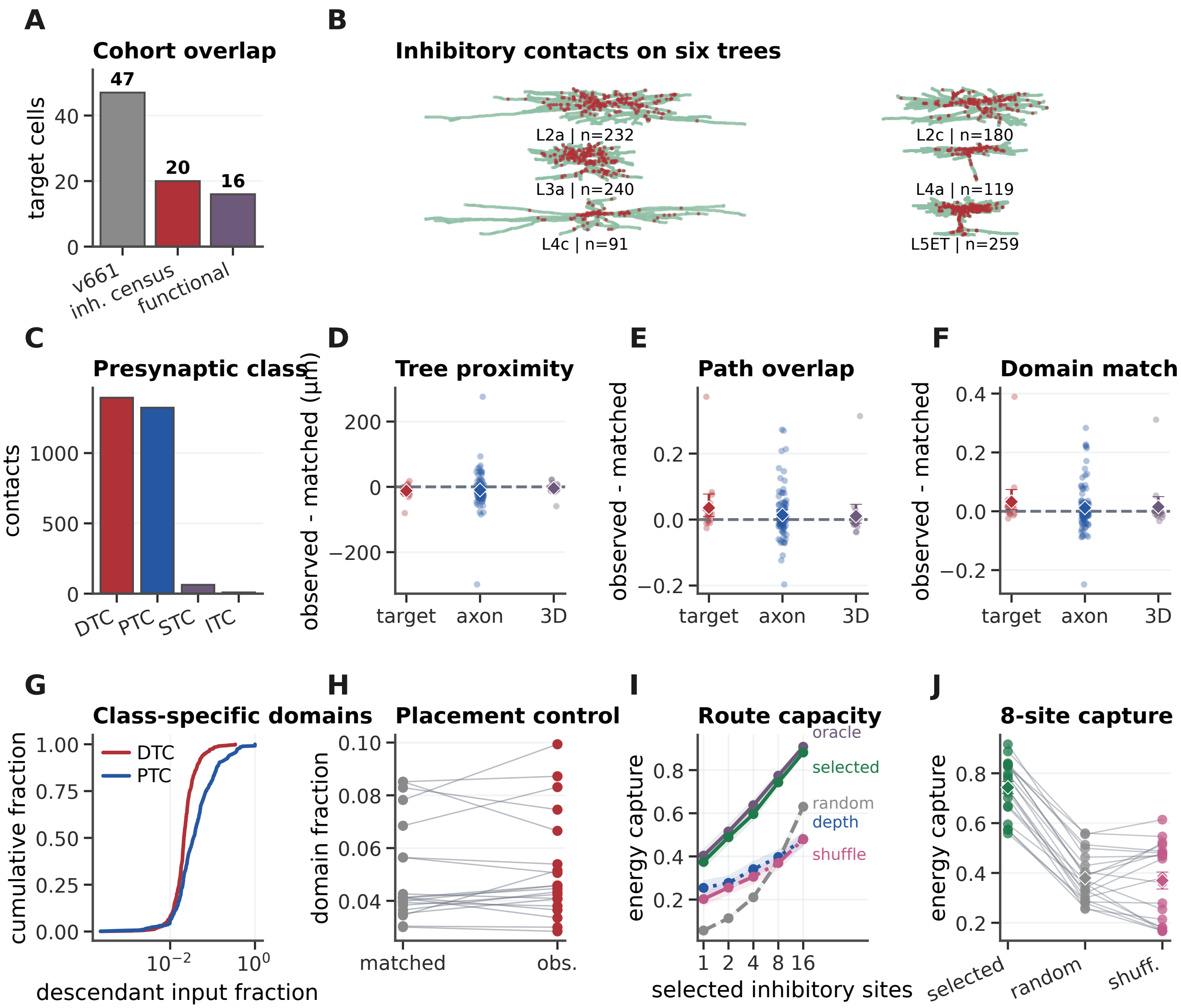

Figure S12: **Synapse-resolved inhibition reveals class-dependent dendritic scales.** **A**, Overlap between the 47-cell v661 sensitivity cohort, the v795 inhibitory census and available functional summaries. **B**, Six layer-diverse target reconstructions; red points are mapped known inhibitory contacts. **C**, Contacts in the overlap by the published inhibitory targeting class. DTC, distal dendrite-targeting; PTC, perisomatic-targeting; STC, sparsely targeting; ITC, inhibitory-targeting. **D–F**, Observed-minus-matched contrasts in tree distance, shared path and weighted descendant-domain overlap. Target denotes the primary 20-target analysis; axon first averages within each of 92 presynaptic axons; 3D jointly matches path distance and Euclidean proximity within target. Diamonds and intervals show unit means and bootstrap 95% intervals. Only tree distance persisted after axon aggregation, and no endpoint persisted after joint 3D matching. **G**, Descendant-domain size for DTC and PTC contacts, retaining one contact per axonal clump. **H**, Overall inhibitory-contact domain size versus matched locations; no reliable global localization advantage was detected. **I**, Weighted capture of the actual inhibitory-route dictionary as a function of site budget. **J**, Eight-site capture within each of 20 targets. Sites in **I,J** are selected by anatomical leverage; this is a structural capacity analysis, not evidence that the routes carried teaching signals in vivo. Target cells are nested within one mouse.

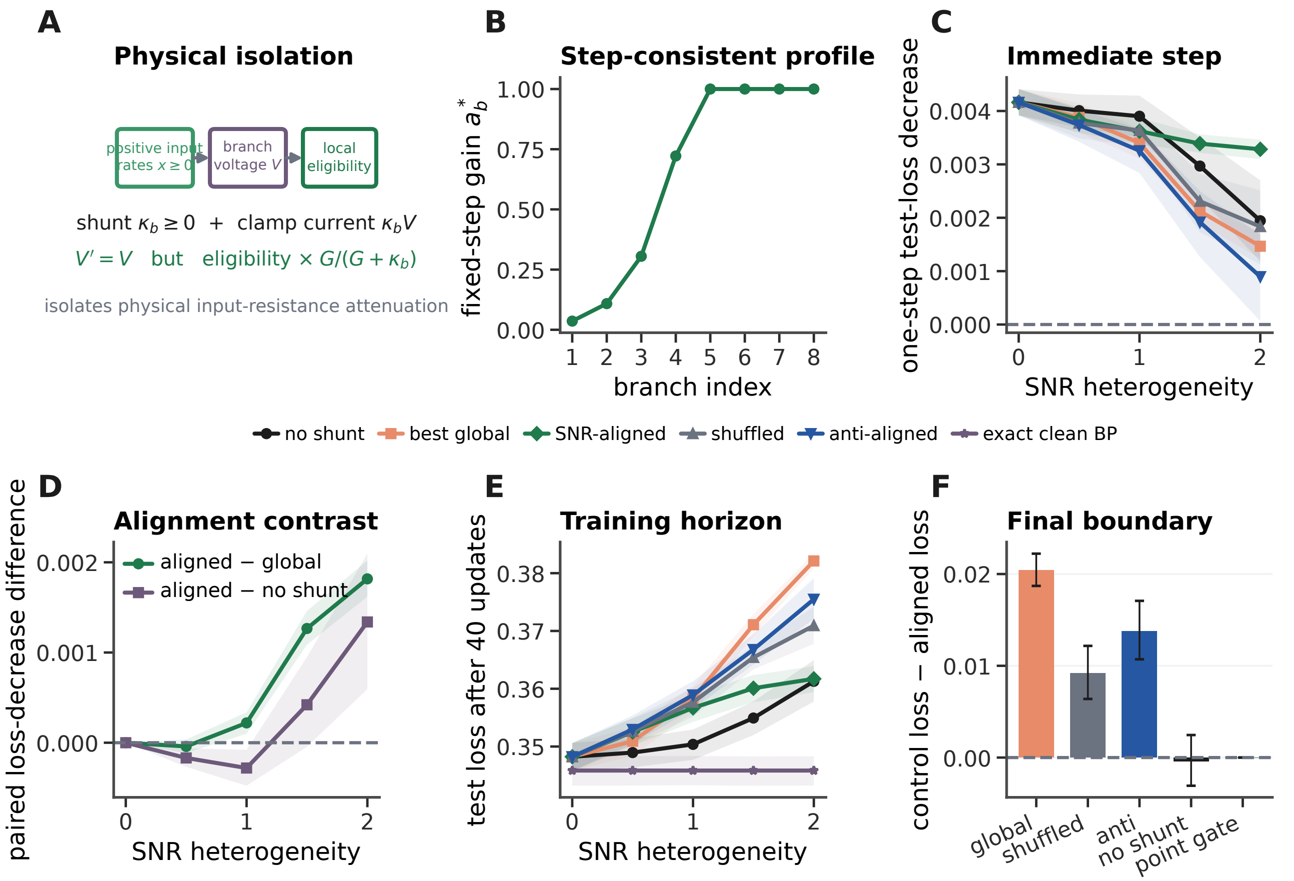

Figure S13: **A state-matched positive conductance approximates the predicted fixed-step branch-reliability factor, with a bounded training benefit.** **A**, A nonnegative shunt κ_b is paired with compensating current $\kappa_b V_b$, preserving branch voltage while reducing physical input resistance and local eligibility. **B**, Prescribed reliability profile $\min\{1, S_b/[c(S_b+N_b)]\}$ at maximum heterogeneity with $c = 0.5$. **C**, Mean one-step held-out loss decrease across 50 paired seeds. Lines and shading show seed means and 95% paired-seed bootstrap intervals. **D**, Aligned-minus-global and aligned-minus-no-shunt one-step contrasts. At heterogeneity two, the former is positive in 49/50 seeds; the latter has a positive mean interval, is positive in 33/50 seeds and has two-sided Wilcoxon $P = 0.0030$. **E**, Held-out loss after 40 updates. Exact clean backpropagation is the unattenuated optimization reference. **F**, Paired final-loss contrasts at maximum heterogeneity, displayed as control minus aligned so positive values favor aligned shunting. Aligned shunting beats the global, shuffled and anti-aligned shunts but has no reliable final difference from the unshunted noisy rule. The independently computed explicit point gate receiving the same sample-dependent factors agrees to numerical precision. All input rates and excitatory and shunt conductances are nonnegative. The oracle state clamp and fixed reliabilities isolate mechanism; they do not constitute autonomous shunt learning.

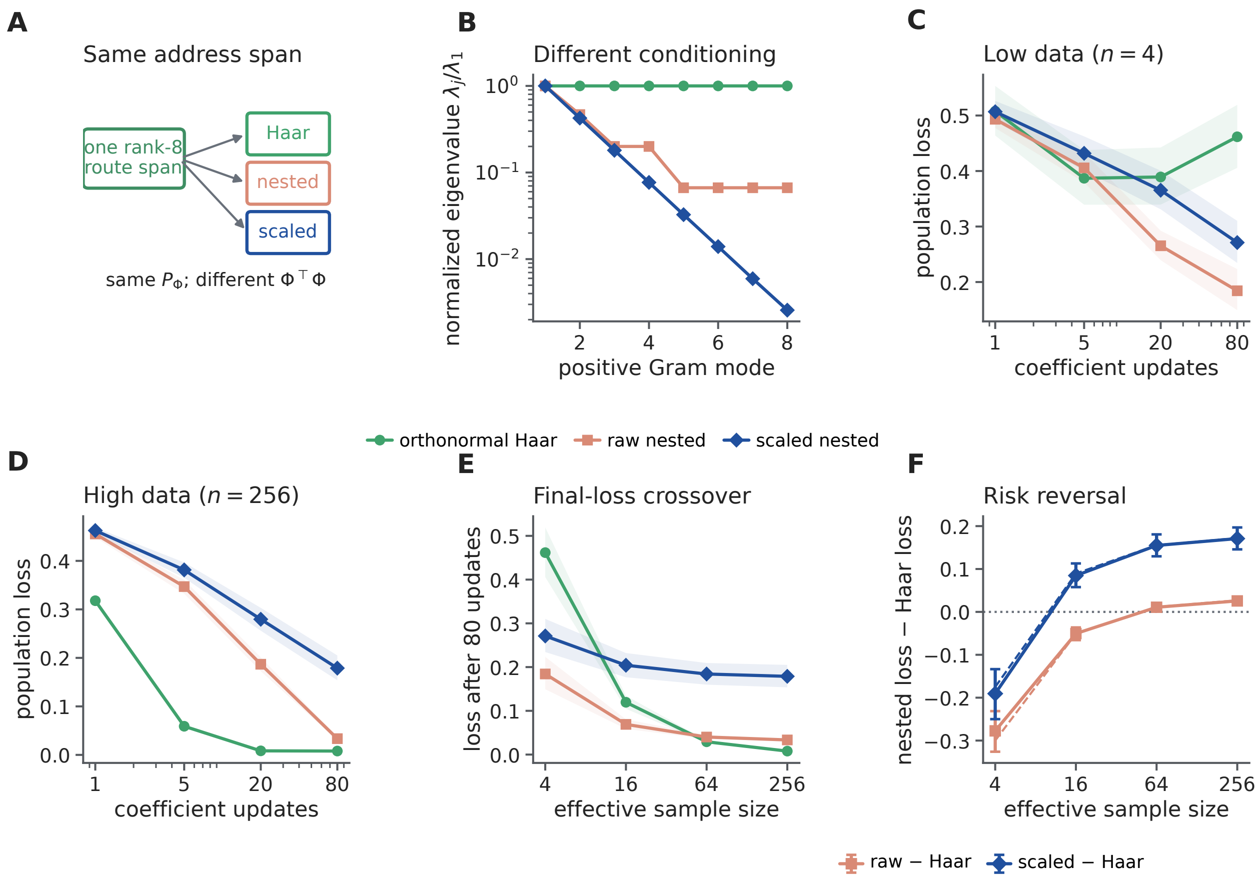

Figure S14: **Same address span produces a finite-data coefficient-learning crossover.** **A**, Orthonormal Haar, redundant nested and statically scaled nested dictionaries parameterize the identical rank-eight route span. **B**, Normalized positive eigenvalues of their field Gram operators; condition numbers are 1, 15 and 388.52. **C,D**, Mean population-loss trajectories under common noisy local target estimates at effective sample sizes 4 and 256. Shading shows 95% paired-seed bootstrap intervals across 50 seeds. At low data, slow nested modes filter variance; at high data, their retained bias dominates. **E**, Final loss across the sample-size series. **F**, Paired nested-minus-Haar contrasts. Solid points and bars are observed means and intervals; dashed curves are the exact finite-time risk from Eq. 53. The reversal is a bias–variance effect within one span. Gram-preconditioned controls (not shown) make all parameterizations agree to 1.16×10^{-15} and do not assert a biological implementation of the preconditioner.

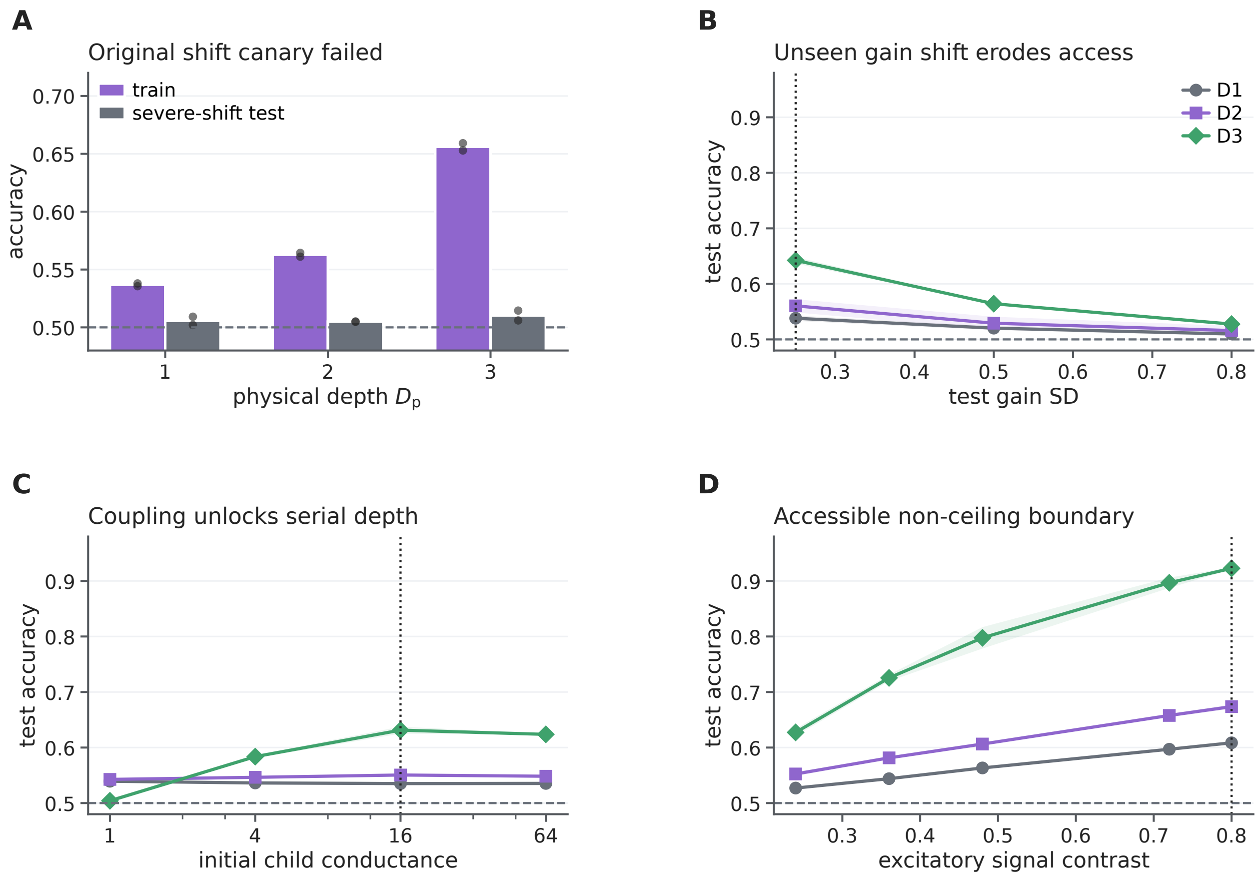

Figure S15: **Transparent calibration of the nonlinear physical-depth operating point. A**, The original aligned shunting-BP canary at signal contrast 0.24 fit the training distribution increasingly with depth but remained at chance under the prespecified severe gain shift (test gain SD 1.2). Bars are two-seed means and dots are the individual exploratory seeds. **B**, With training gain SD fixed at 0.25, increasing unseen test gain variation eroded accessibility at every depth. The dotted line marks the matched train/test condition. **C**, At matched gain and signal contrast 0.24, stronger child coupling selectively made serial D3 computation accessible; the dotted line marks child conductance 16 used subsequently. **D**, At matched gain and child conductance 16, increasing excitatory signal contrast moved D1–D3 away from chance without saturating D1 or D2. The separately run 0.80 boundary (dotted line) passed the frozen accessibility gate and selected the fresh-seed operating point. Lines show two-seed means; shading spans the two exploratory values and is not a confidence interval. All pilot, ladder and boundary seeds were excluded from the confirmatory contrasts.

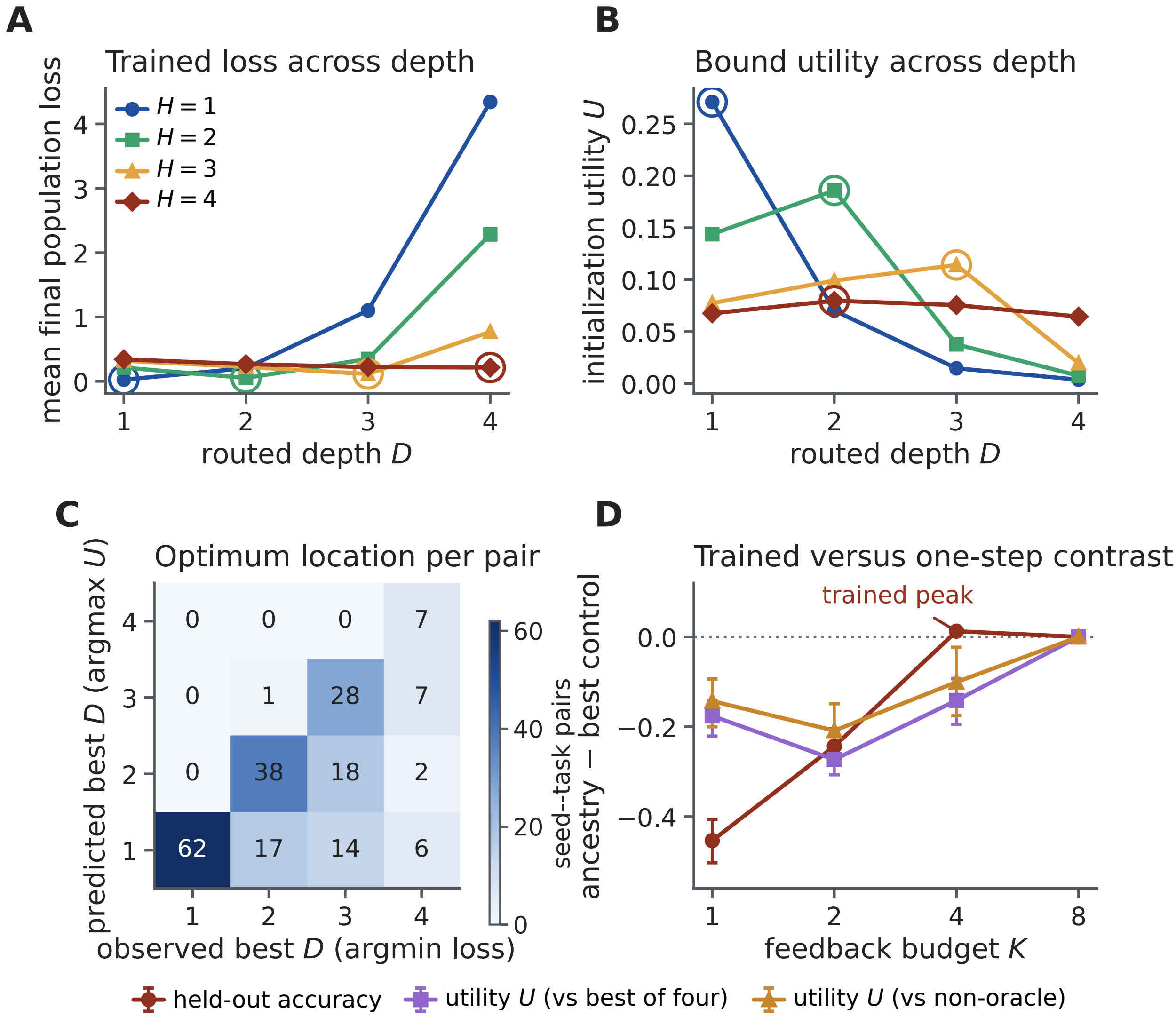

Figure S16: **The credit-operator bound locates the trained depth optimum but not the trained ancestry advantage.** **A**, Mean final population loss across routed depth D_r for task depths $H = 1$–4 in the frozen hierarchy-depth phase experiment (aligned tree, $n = 50$ seeds); open circles mark the loss minimum. **B**, Mean initialization utility (Supplementary Eq. 44) for the same conditions; open circles mark the utility maximum. The group-level argmax matches the trained optimum exactly at $H = 1, 2, 3$; at the full-rank boundary $H = 4$ the one-step bound is conservative and favours $D_r = 2$. **C**, Per-pair predicted (argmax utility) versus observed (argmin loss) best depth over all 200 seed–task pairs; 135 pairs lie on the diagonal. **D**, Ancestry-minus-best-control contrast across feedback budget K in the 2,700-fit factorial, for trained held-out accuracy and for initialization utility against the published four-control set and its three implementable (non-oracle) members. Points and bars are means and 95% seed-bootstrap intervals ($n = 20$). The utility reproduces the exact ties at $K = 8$, where all spans coincide, but not the trained $K = 4$ peak: the trained anatomical advantage accrues along the training trajectory rather than from one-step initialization geometry. Analysis: `scripts/analyze_interior_optimum.py`.

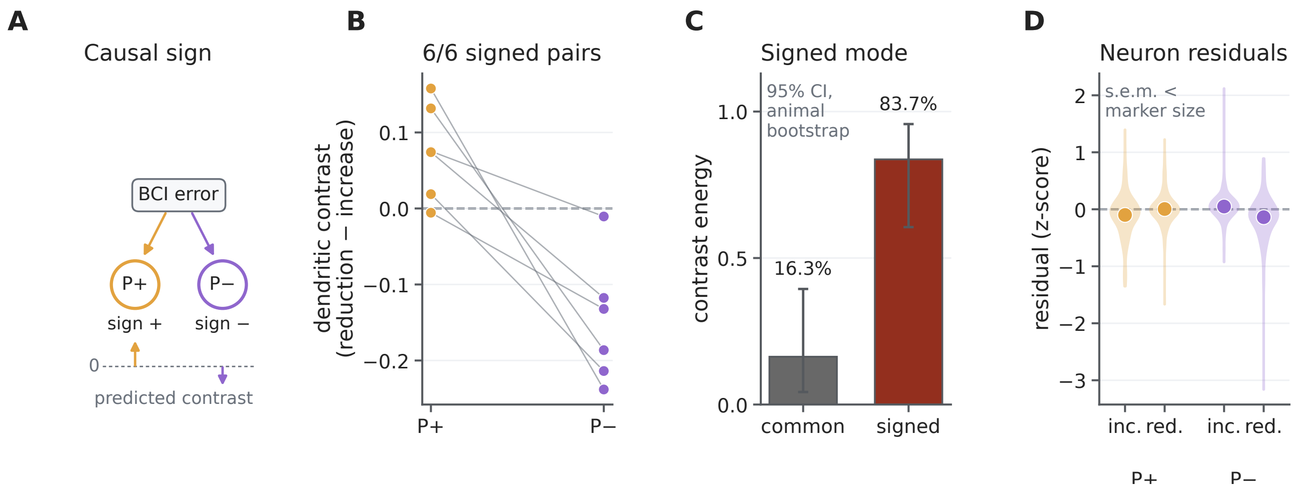

Figure S17: **Published animal data support signed neuron-specific teaching coordinates.** **A**, In the source brain–computer-interface experiment, the causal mapping assigned opposite signs to P+ and P- populations, predicting opposite dendritic contrasts. **B**, Error-reduction minus error-increase standardized dendritic residual for P+ and P- in six animals; lines pair the two populations within animal. **C**, Fraction of the two-coordinate contrast energy in a common scalar mode and the signed P+/P- mode; intervals are 95% animal-bootstrap intervals. **D**, Descriptive neuron-level residual distributions from the published source table; dots show means. Animal, not neuron, is the inferential unit. This external consistency test supports signed neuron identity but does not resolve within-tree transport or uniquely identify a conductance mechanism.

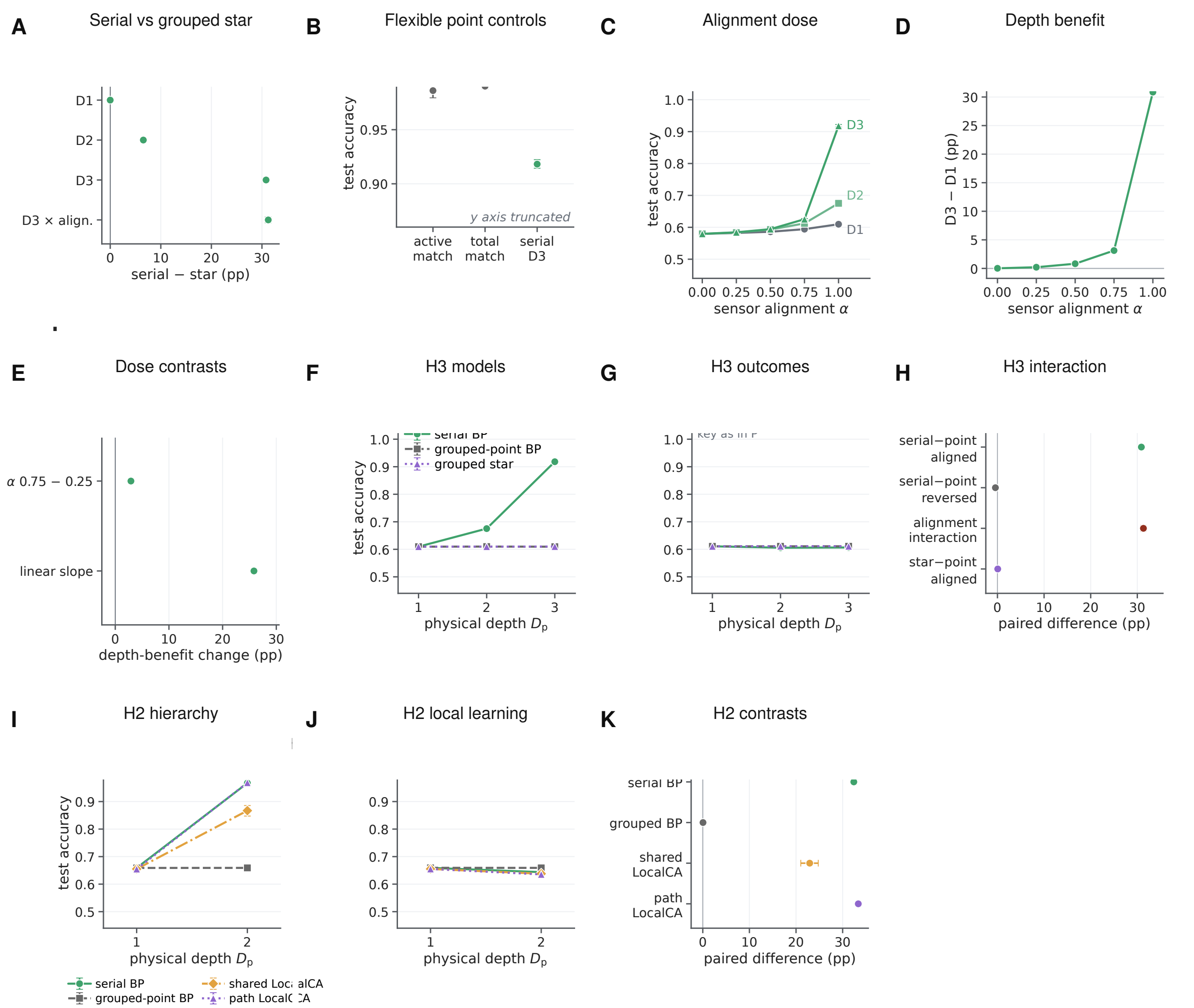

Figure S18: **Physical-depth controls, alignment interpolation and hierarchy replication.** **A**, Paired serial-minus-grouped-star effects and the D3 alignment interaction. **B**, Flexible point networks matched to active or total parameter count versus the serial D3 tree. **C–E**, Accuracy, D3-minus-D1 benefit and paired dose contrasts across sensor–task alignment α. Intermediate levels were specified after the endpoints and are an interpolation, not an independent replication. **F–H**, H3 serial tree, literal grouped-point and grouped-star controls under aligned and reversed placement. **I–K**, Independent H2 hierarchy with serial BP, grouped-point BP and shared- or path-transport LocalCA. Lines are paired-seed means and intervals are 95% seed-bootstrap intervals ($n = 10$). The two hierarchies belong to one calibrated synthetic task family.

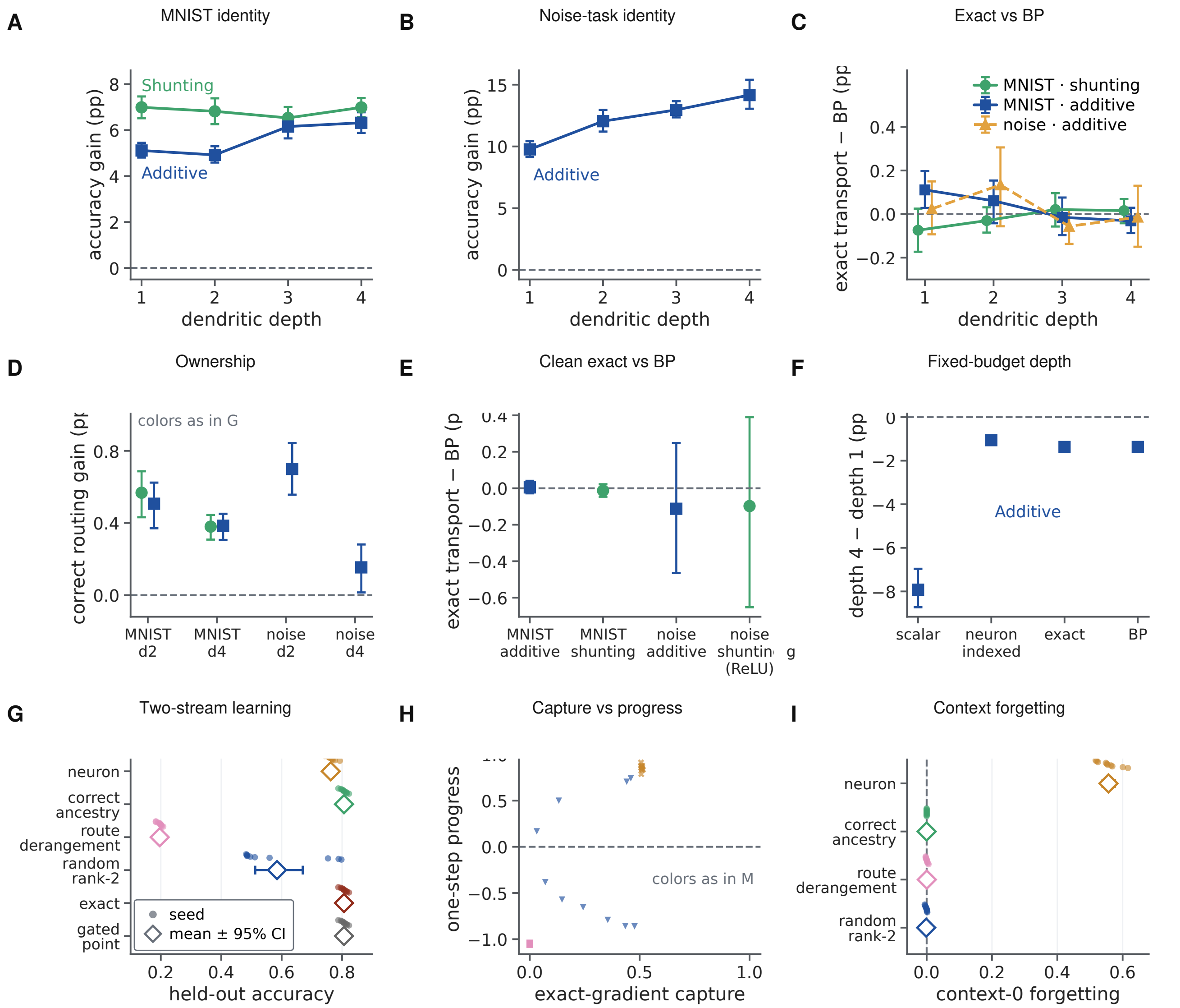

Figure S19: **Prospective identity, ownership and route diagnostics. A,B**, Neuron-indexed-minus-scalar accuracy across depth on MNIST and the input-valid additive noise task. **C**, Exact-transport-minus-BP accuracy in the retained historical conditions. **D**, Correct-minus- deranged neuron-to-tree ownership at matched bandwidth. **E**, Depth-averaged exact-minus-BP accuracy in the independent clean-source audit. **F**, Depth-four-minus-depth-one accuracy at fixed terminal branches and contacts. **G**, Held-out two-stream credit-reversal accuracy. **H**, Exact-gradient capture versus norm-matched one-step progress. **I**, Forgetting of the first context after adaptation to the second. Small points are paired seeds and larger symbols and bars are means and 95% seed-bootstrap intervals. Exact transport is an information ceiling.

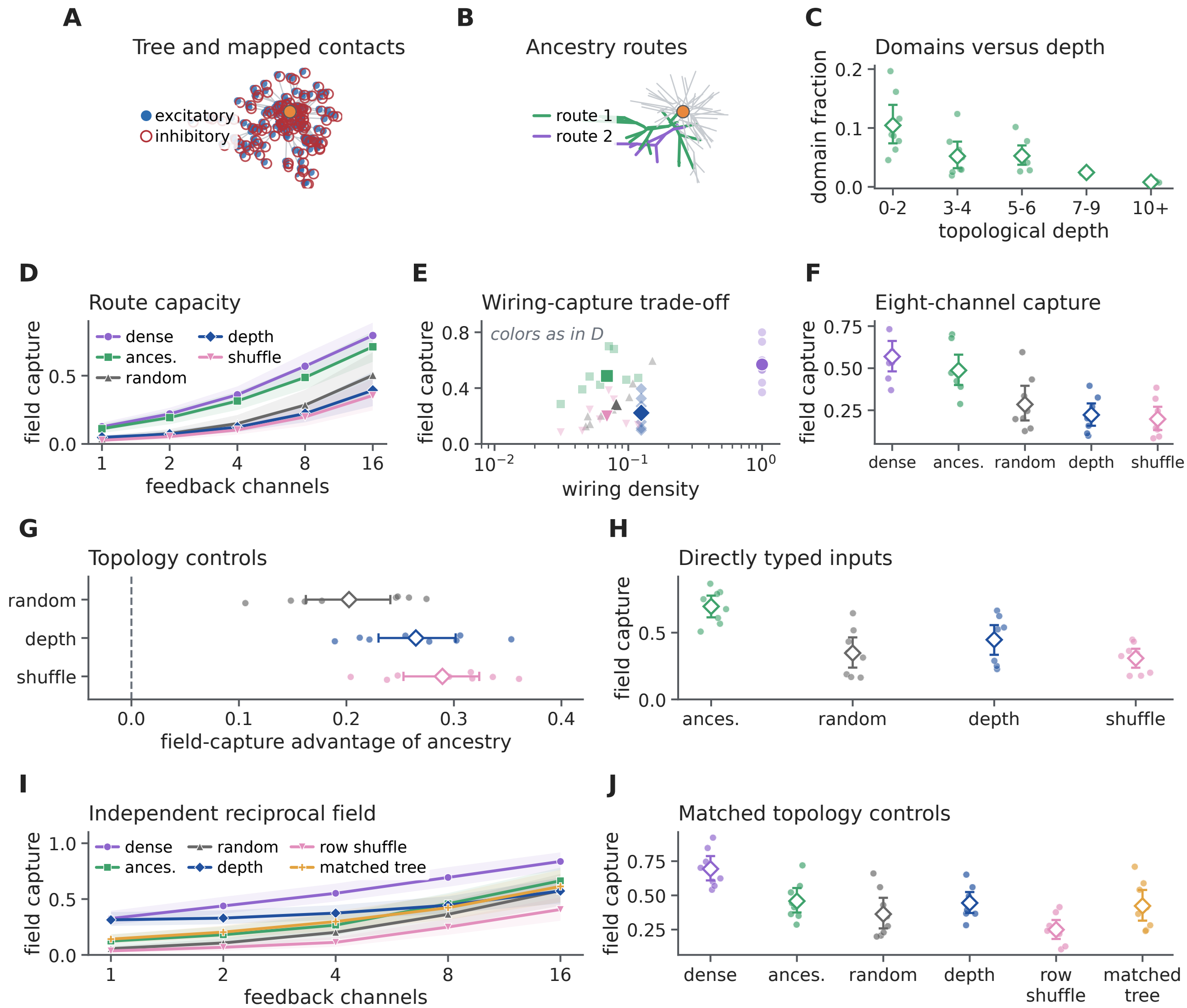

Figure S20: **Detailed morphology-route capacity and control analyses.** **A**, Reconstructed arbor with mapped excitatory and inhibitory contacts. **B**, Example ancestry-defined route domains. **C,D**, Route support and participation rank. **E,F**, Model-derived field capture across channel budget and the eight-channel comparison. **G**, Paired ancestry-minus-control contrasts. **H**, Eight-channel capture after restricting the analysis to directly typed presynaptic inputs. **I,J**, Independent reciprocal-cable field capture and its matched topology controls. Points are reconstructed-cell summaries; intervals are 95% cell-bootstrap intervals. This figure preserves the complete diagnostic display underlying the focused main Figure 6.

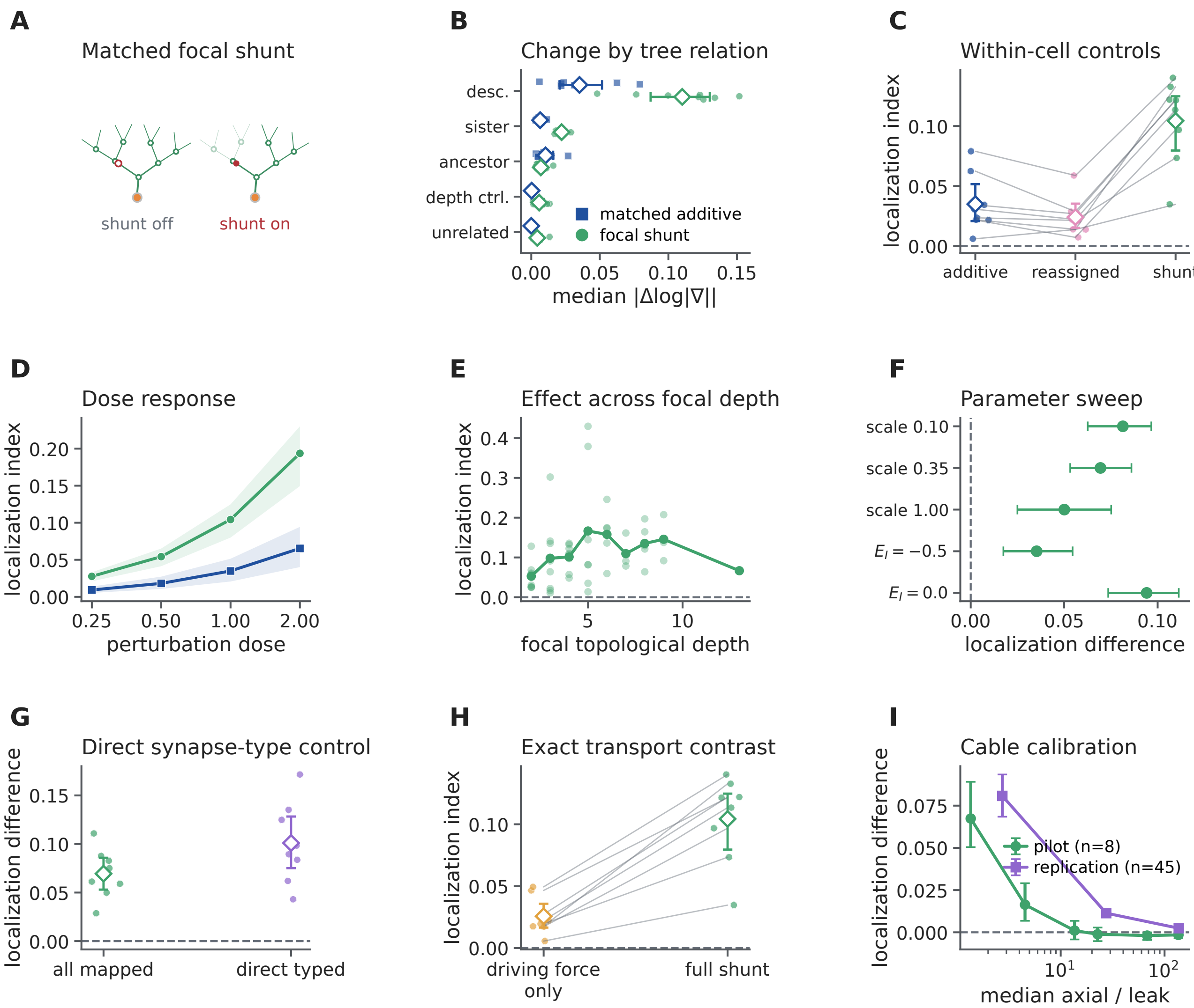

Figure S21: **Detailed focal-shunting controls and electrotonic boundary.** **A**, State-matched focal intervention. **B**, Exact-gradient change by topological relation. **C**, Within-cell shunt, additive and relation-reassignment controls. **D**, Perturbation-dose response. **E**, Effect across focal-site depth. **F**, Conductance-scale and inhibitory-reversal sensitivity. **G**, Direct presynaptic-type control. **H**, Exact factor-freeze contrast isolating transported error. **I**, Physical cable calibration across axial-to-leak regimes. Sites are nested within eight reconstructed cells; intervals are 95% cell-bootstrap intervals.

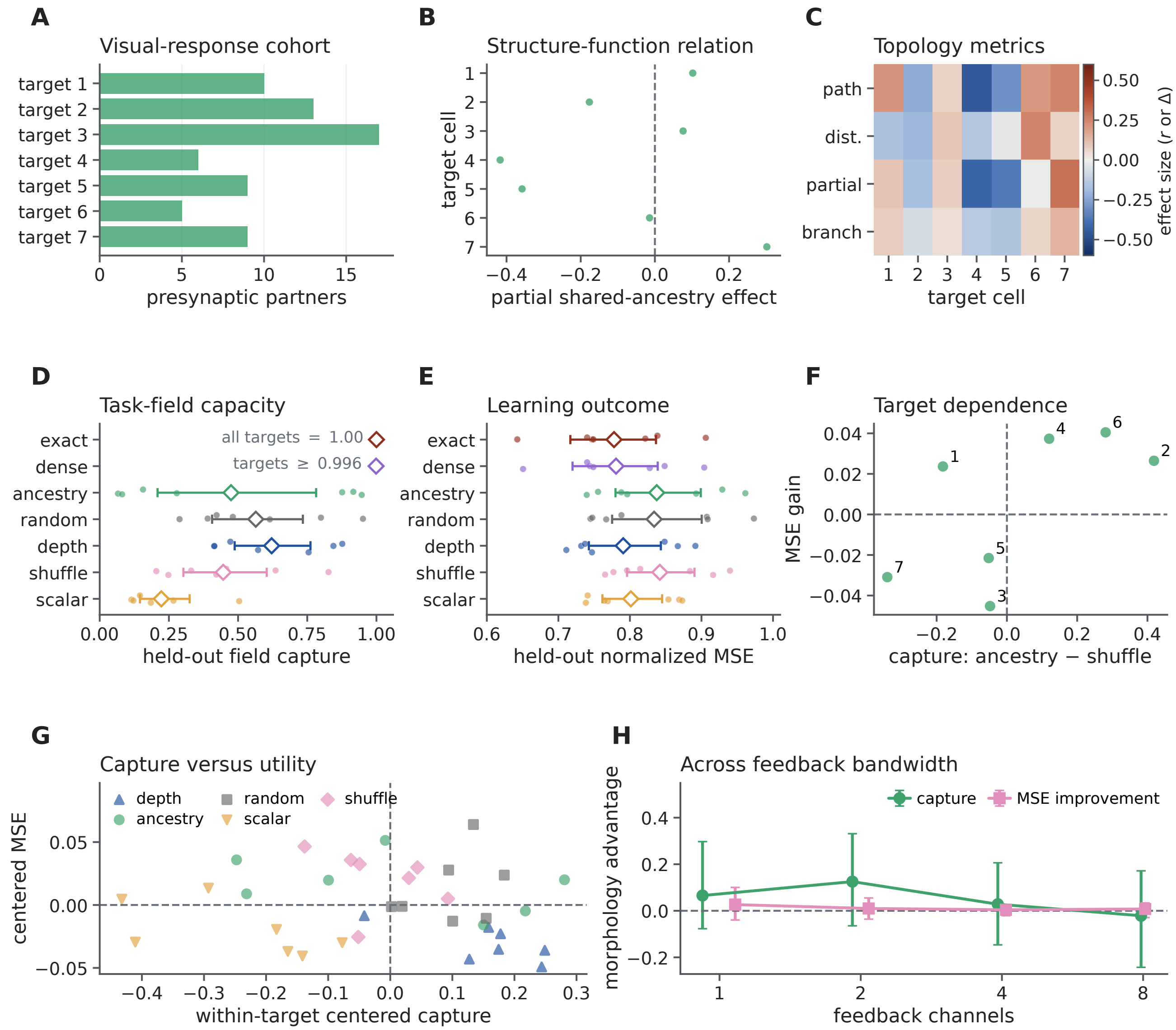

Figure S22: **Detailed measured-response topology and learning boundary.** **A**, Seven-target response cohort. **B**, Geometry-controlled structure–function relation. **C**, Target topology metrics. **D,E**, Four-channel task-field capture and held-out learning across exact, dense, ancestry and matched control dictionaries. **F**, Target-level dependence of morphology-minus-shuffle effects. **G**, Capture–utility association after accounting for target identity. **H**, Morphology-minus-shuffle contrasts across feedback bandwidth. Stimulus splits and scans are nested within seven target cells.

Supplementary Tables

Table S2: Confirmatory subtree-address factorial. Accuracy contrasts are paired across 20 seeds. The best matched non-anatomical control is selected within seed from route derangement, depth-interleaved, random-sparse and random-rank fields. Positive contrasts favor correct ancestry.

K	Correct accuracy	Versus deranged	Versus best control	Versus rewired tree
1	0.1866	0.0000	−0.3583	0.0000
2	0.4486	0.2639	−0.1540	0.2315
4	0.8010	0.6106	0.0127	0.0502
8	0.8100	0.5587	0.0000	0.0000

For the best-control contrast, 0/20, 5/20, 15/20 and 0/20 paired seeds favored ancestry at $K = 1,2,4,8$, respectively; the full-rank conditions are exactly tied by construction. At $K = 4$, the paired bootstrap 95% interval was 0.0059–0.0197 and the two-sided Wilcoxon value was $P = 0.0048$ (Benjamini–Hochberg-adjusted $P = 0.0064$ across $K = 1,2,4,8$). Correct-versus-rewired intervals were 0.210–0.254 at $K = 2$ and 0.042–0.060 at $K = 4$, with 20/20 seeds positive in both comparisons.

Table S3: Credit-operator phase experiment and frozen-factorial reanalysis. Phase contrasts use 50 paired confirmatory seeds. The depth contrast combines four task depths within seed; its positive reported magnitude favors matched depth. Utility correlations use 20 seed blocks from the existing factorial.

Endpoint	Effect	95% interval	Positive	Control / interpretation
Ancestry–random capture, $\rho = 1$, $K = 4$	0.5520	0.5353–0.5686	50/50	Dense rank-K upper bound; full rank tied
Matched versus best mismatched depth	0.0957	0.0782–0.1134	48/50 seeds	Four task-depth contrasts averaged within seed; $D_{\mathrm{r}} = 4$ equals full stochastic BP
Aligned versus best global reliability gain	1.2415	1.1160–1.3728	50/50	Explicit point gate matched exactly
Utility versus observed one-step progress, ρ	0.9374	0.9336–0.9412	—	Seed-block Spearman interval
Utility versus final accuracy, ρ	0.9157	0.9047–0.9292	—	Secondary initialization diagnostic

Two-sided Wilcoxon values for the spectral, depth and reliability contrasts were 1.78×10^{-15}, 3.00×10^{-13} and 1.78×10^{-15}, respectively. All 25 signal–noise condition means followed the analytical projection boundary. Static route-wise gains preserved projectors to 1.33×10^{-15}. The phase experiment is a fixed-state quadratic validation, not a trained positive-conductance network.

Table S4: Step-consistent state-matched positive-conductance reliability experiment at maximum branch-SNR heterogeneity. Effects are paired across 50 confirmatory seeds. Positive one-step effects favor aligned shunting; negative final-loss effects favor the left-hand aligned condition.

Endpoint	Effect	95% interval	Positive	Two-sided Wilcoxon P
Aligned–global, one-step loss decrease	0.001817	0.001618–0.002020	49/50	3.55×10^{-15}
Aligned–no shunt, one-step loss decrease	0.001339	0.000594–0.002096	33/50	0.0030
Aligned–global, final test loss	−0.020404	−0.022220––0.018705	0/50	1.78×10^{-15}
Aligned–shuffled, final test loss	−0.009204	−0.012171––0.006387	7/50	6.92×10^{-9}
Aligned–anti-aligned, final test loss	−0.013781	−0.017077––0.010708	3/50	9.52×10^{-13}
Aligned–no shunt, final test loss	0.000425	−0.002456–0.003071	33/50	0.154
Aligned–explicit point gate, final test loss	0	0–0	0/50; 50 ties	1

The independently computed state-matched unattenuated current control agreed with the unshunted noisy rule to numerical precision. Exact clean backpropagation had lower mean final loss than every attenuated condition. Shunts were fixed from initial oracle reliabilities and forward voltage was clamped exactly; excitatory conductance parameters, not shunts, were trained.

Table S5: Same-span noisy coefficient-learning contrasts after 80 updates. Effects are paired across 50 confirmatory target-and-noise seeds. Positive values mean greater loss than the orthonormal Haar coordinates.

Coordinate contrast	n_{eff}	Effect	95% interval	Positive	Two-sided Wilcoxon P
Raw nested–Haar	4	−0.2774	−0.3260––0.2313	0/50	1.78×10^{-15}
Scaled nested–Haar	4	−0.1906	−0.2500––0.1335	10/50	1.05×10^{-7}
Raw nested–Haar	256	0.02555	0.02171–0.02952	50/50	1.78×10^{-15}
Scaled nested–Haar	256	0.17091	0.14576–0.19686	50/50	1.78×10^{-15}

All dictionaries had the same rank-eight projector. Their Gram condition numbers were 1, 15 and 388.52. Median exact-risk crossover sample sizes were 38.53 for raw nested and 8.25 for scaled nested coordinates. Oracle Gram-preconditioned field trajectories agreed to 1.16×10^{-15}.

Table S6: Active steady-state focal-conductance ensemble. Values first average 64 accepted channel draws and focal sites within each of eight reconstructed cells.

Dose	Shunt localization	Additive localization	Shunt–additive	Shunt energy ratio
0.25	0.1587	0.0398	0.1189	0.7154
1	0.5287	0.1470	0.3817	0.3302
4	1.3387	0.4657	0.8730	0.0625

Dose is relative to local input conductance. Shunt-localization cell-bootstrap intervals were 0.1538–0.1628, 0.5130–0.5414 and 1.3066–1.3642. At every dose and in every cell, all modeled descendant gradients were attenuated without enhancement or sign reversal. Draws are nested sensitivity samples, not independent biological replicates.

Table S7: Functional topology across all eligible scans. Statistics were computed within 13 scans, averaged within each of seven target cells, and then summarized across targets.

Endpoint	Mean	95% target-bootstrap interval	Positive targets
Shared-path Spearman ρ	−0.0245	$[-0.2142, 0.1466]$	4/7
Negative tree-distance Spearman ρ	−0.0175	$[-0.1297, 0.1016]$	3/7
Partial shared-path ρ	−0.0485	$[-0.2237, 0.1183]$	3/7
Same-major-branch contrast	−0.0027	$[-0.0833, 0.0699]$	4/7

Two-sided exact Wilcoxon values were 1.0, 0.8125, 0.8125 and 1.0, respectively. Pairwise partner observations and multiple scans are nested within targets; all targets are from one mouse.

Table S8: Complete reconstructed-tree learning across all eligible scans. Values average ten held-out-stimulus splits within scan, then scans within each of seven target cells.

Teaching field	Normalized held-out MSE	Common-checkpoint update capture	Fits
Exact compartment error	0.7877	1.0000	130
Topology routes	0.8321	0.2322	130
Random routes	0.8333	0.3584	130
Site-shuffled routes	0.8391	0.4392	130

Positive MSE improvement favors topology. Topology improved over site shuffling by 0.0070 (95% interval −0.0207 to 0.0321) and over random routes by 0.0013 (−0.0090 to 0.0113). All restricted dictionaries had identical effective rank within run. None of 520 fits was excluded.

Table S9: Frozen architecture and data specification for the regular-tree MNIST experiments.

Component	Specification
Data	MNIST tensors; 784 flattened pixels; no normalization or augmentation; seed-controlled 48,000/12,000 train/validation split; official 10,000-image test set.
Modeled population	One layer of 128 excitatory modeled neurons; no separate inhibitory-neuron population.
Tree	$[3,3]$: nine distal, three proximal and one somatic compartment per neuron; no somatic input synapses.
Input contacts	Per dendritic compartment, separate TopK arrays selected 40 excitatory and 20 inhibitory inputs from 784 candidates per sign; 92,160 active contacts among 2,408,448 candidate synaptic parameters.
Parameterization	Nonnegative synaptic and child-coupling conductances obtained by softplus transformation; inactive local weights were not updated.
Reactivation	Learned $[\tanh(m(V-b))+1]/2$ at distal, proximal and somatic stages. Shunting used analytical initialization; raw additive used 0.1/0.9 occupancy-quantile calibration.
Readout	Linear map from 128 soma activities to 10 class scores.
Model size	2,414,602 trainable raw parameters in the archived feedback runs, of which 2,408,448 were candidate synaptic parameters.

Table S10: Training protocols for the regular-tree results. Each protocol used mini-batches of 256, categorical negative log likelihood, no weight decay, no mixed precision and restoration of the best validation checkpoint.

Protocol	Seeds; epochs	Core output	Rule and feedback	Adam learning rates
Feedback definition	42–56; 180	Identity	3F, local decoder; matched-width/scalar fallback or neuron-indexed sharing	TopK 1.5×10^{-3}; child coupling 7×10^{-4}; reactivation 5×10^{-4}; decoder 1.5×10^{-3}
Exact-transport factorial	42–46; 200	ReLU	3F or 5F crossed with local or backpropagated decoder; exact path-transported compartment error	TopK 1.5×10^{-3}; child coupling 7×10^{-4}; reactivation 5×10^{-4}; decoder 1.5×10^{-3}
Backpropagation reference	42–46; 200	ReLU	Standard backpropagation through the matched shunting tree and decoder	TopK and decoder 10^{-3}; child coupling and reactivation 5×10^{-4}

Adam used $(\beta_1,\beta_2)=(0.9,0.999)$ and $\varepsilon=10^{-8}$. The 3F local-gradient elementwise clip was 5. The exact-transport and backpropagation protocols shared architecture, seeds, epoch budget, and checkpoint-selection procedure, but used the method-specific learning rates listed above. Both were separate from the 180-epoch feedback-definition protocol.

Table S11: Archived provenance and execution records for the regular-tree results. "Not archived" means that the value was not recoverable and was not inferred from the current environment.

Cohort	Surviving run artifacts	Hardware record	Software record
Clean feedback cohort, 60 runs	All resolved configurations, checkpoints, logs, final metrics, frozen sweep manifests and SHA-256 hashes	NVIDIA H200; 4.006 summed GPU-hours across both arrays	Python 3.10.13; Git commit `4ba38bde`; tracked worktree recorded as dirty; hashes of generator and critical model sources frozen
Prospective depth-by-feedback cohort, 640 executions (480 retained)	All resolved configurations, checkpoints, learning curves, held-out metrics, resource records, frozen manifests, source hashes and the outcome-independent 1,840-run validity ledger	73.29 summed GPU-hours over heterogeneous NVIDIA accelerators	Frozen source identity plus a verified equivalence audit for two later changes that are inactive in every resolved configuration; signed-noise shunting cells are excluded from inference
Prospective checkpoint diagnostic, 3,200 historical rows (2,400 retained)	Five feedback-field labels and four step sizes at each of 160 checkpoints; publication analysis uses 120 input-valid independently trained checkpoints	Fixed-checkpoint analysis; no additional model training	Complete balance and finite-value checks; checkpoint-clustered bootstrap analysis and exclusion ledger frozen with the source tables
Clean exact-transport/backpropagation audit, 320 runs	Resolved configurations, seed records, checkpoints, learning summaries, final metrics, scheduler and training logs, frozen sweep manifests and SHA-256 hashes	NVIDIA H200 for every resource-corrected rerun; initial incomplete attempts are excluded and archived	Detached tracked-clean root commit `74792ca`; nested synthetic-data commit `4f3612a5`; signed transfer retained for additive noise, explicit ReLU transfer used for positive-conductance shunting noise
Exact-transport factorial, 20 runs	Resolved configurations, checkpoints, logs and grouped result table	NVIDIA A100-SXM4-40GB; 2.953 GPU-hours total	Exact commit, Python and package environment not archived
Backpropagation reference, 5 runs	Resolved configurations, checkpoints, logs and grouped result table	NVIDIA A100-SXM4-40GB; 0.474 GPU-hours total	Exact commit, Python and package environment not archived

Portable sweep files, all resolved configurations, source-data tables, source hashes and validation records for the clean 15-seed feedback cohort and the 320-run detached audit are supplied with the article. Hardware times are sums of Slurm elapsed time over GPU tasks, not wall-clock latency.

Table S12: Notation used in the theory and analyses.

Symbol	Definition
u, n, i	Indices for a neuron, compartment and synapse, respectively.
V_n, a_n	Conductance-stage voltage and transmitted activity of compartment n; $a_n = f_n(V_n)$.
g_i, $g^{\text{den}}_{c\to n}$	Scalar synaptic conductance and child-to-parent coupling conductance. These are distinct from bold gradient symbols.
E_i, x_i	Synaptic reversal potential and presynaptic activity.
g^{tot}_n, R^{tot}_n	Total compartment conductance and its reciprocal input resistance.
δ_u, $\delta_{u,k}$	Neuron-level teaching coordinate and its route-k refinement within neuron u.
$\delta_{0,u}$ (or δ^V_0), $\partial\mathscr{L}/\partial V_n$	Exact somatic error for neuron u (single-tree voltage-space notation in parentheses) and exact compartment voltage-space error.
α^{cond}_n, $\widetilde{\alpha}_n$	Conductance-only and effective path gains; the latter includes reactivation derivatives.
α, ρ	Sensor alignment in the physical-depth task and covariance alignment in the spectral phase experiment.
$\boldsymbol{w}$, $\boldsymbol{g}$, $\widehat{\boldsymbol{g}}$	Parameter vector, exact population gradient, and its unbiased stochastic estimate.
$\boldsymbol{q}$	Adjoint field; q_n is the loss sensitivity to a small current injected at compartment n.
M, Σ, $U(M)$	Credit operator, stochastic-gradient covariance, and the signal–noise phase utility.
K	Number of communicated feedback channels or route fields.
H, D_{r}	Task-hierarchy depth and routed address depth in the fixed-state phase experiment.
D_{p}; D1–D3	Physical dendritic stage count; D1, D2 and D3 mean $D_{\text{p}} = 1,2,3$. In $[b_1,\ldots,b_{D_{\text{p}}}]$, branching factors are ordered soma-to-distal.
A_T	Binary synapse-by-route ancestry matrix of tree T.
Γ_{ik}	Direct ancestry-route sensitivity from candidate route k to excitatory site i.
$\Phi_{T,K}$	K-channel feedback dictionary implemented by topology, route gains, channel allocation and weights.
W	Positive diagonal coordinate-weight matrix; summed excitatory synaptic area in the structural MICrONS analysis.
$\overline{P}_{\Phi}$	Ordinary Euclidean orthogonal projector onto the columns of $W^{1/2}\Phi_{T,K}$.
$\mathscr{C}_K$	Field-energy capture at feedback-channel budget K in the stated coordinates.
G, R_a, R_m	Passive conductance matrix, axial resistivity, and specific membrane resistance.
B	Mini-batch size when used in $\langle\cdot\rangle_B$.
S_K, C_g	Feedback-channel assignment matrix in Eq. 26 and gradient covariance, respectively.

Table S13: Feedback objects and the information each supplies.

Feedback field	Communicated values	Spatial assignment	Interpretation
Global scalar	Robust mean-or-signed-maximum reduction in Eq. 23	Same value at every receiving unit	Deliberately removes neuron and branch identity.
Matched-width/scalar fallback	Somatic vector at a same-width stage; otherwise the same robust scalar	Coordinate-wise only at the matched stage	Historical implementation; not ancestry sharing.
Neuron-indexed ancestry sharing	One coordinate per modeled neuron	Each coordinate repeated over all compartments descended from that soma	Tests neuron identity and correct neuron-to-tree ownership, not distinct within-tree addresses.
Correct sibling-subtree routing	K coefficients within one modeled neuron	Each coefficient assigned to its context-relevant subtree	Phase-1 trained test of within-neuron address support at $K = 2$.
Within-neuron derangement	The same K coefficients as correct routing	Fixed permutation among sibling routes inside the neuron	Alignment control at matched coordinate count and sparse nonzeros.
Rank-K random	K projected coordinates	Fixed random compartment mixing	Bandwidth control without anatomical routes.
Selected paths	K channel coefficients	Sparse real ancestry domains	Tests topology-matched route capacity.
Exact transport	Somatic error and exact sample-specific path derivative	Distinct exact value at every compartment	Information oracle; not proposed as an available biological signal.

Table S14: Principal regular-tree learning and gradient results. Values are mean $\pm$ s.d. over independent seeds. Accuracy uses checkpoints trained with the named feedback field; both gradient-cosine entries within an architecture are fixed-checkpoint diagnostics evaluated at its neuron-indexed-trained checkpoints.

Architecture	Feedback or learning condition	n	Test accuracy (%)	Dendritic-gradient cosine
Shunting	Matched-width/scalar fallback, 3F	15	90.54 ± 0.54	0.012 ± 0.059
Shunting	Neuron-indexed ancestry sharing, 3F	15	97.16 ± 0.11	0.708 ± 0.027
Additive	Matched-width/scalar fallback, 3F	15	91.76 ± 0.78	-0.001 ± 0.047
Additive	Neuron-indexed ancestry sharing, 3F	15	97.14 ± 0.11	0.625 ± 0.035
Shunting	Exact transport, 3F, local decoder	5	97.18 ± 0.12	Exact field
Matched reference	Backpropagation	5	97.03 ± 0.08	1 by definition

Table S15: Input-valid prospective depth-by-feedback results. Accuracy contrasts are ranges of condition means over one to four dendritic stages. MNIST retains both neuron models; the noise task retains the additive model. The exact- transport comparison reports the largest absolute difference between the three-factor local rule and matched backpropagation among the 12 retained task–model–depth conditions.

Task	Contrast	Accuracy difference	Paired-seed ordering
MNIST	Neuron indexed minus scalar	4.92–6.99 percentage points	Positive in 80/80 pairs
Noise resilience, additive	Neuron indexed minus scalar	9.76–14.16 percentage points	Positive in 40/40 pairs
Retained conditions	Exact 3F minus matched backpropagation	At most 0.14 percentage points in absolute mean difference	12 matched task–model–depth conditions

Each condition contains ten independent paired seeds. Exact transport is an information oracle rather than a proposed biological signal. Synthetic-noise shunting rows with a signed transfer output are excluded by an outcome-independent conductance-validity rule.

Table S16: Detached tracked-clean exact-transport/backpropagation audit. Within each task and core, accuracy was first averaged over depths one through four for each paired seed. Values are percentages; intervals are 20,000-draw paired- seed bootstrap intervals and *P* values are descriptive two-sided Wilcoxon tests.

Task	Core	Exact accuracy	BP accuracy	Exact − BP (95% interval)	Exact > BP	*P*
MNIST	Additive	97.013	97.008	0.005 (−0.028–0.041)	4/10	1.000
MNIST	Shunting	96.960	96.973	−0.013 (−0.047–0.022)	4/10	0.492
Noise resilience	Additive	33.378	33.490	−0.112 (−0.465–0.247)	4/10	0.625
Noise resilience	Shunting (ReLU)	36.690	36.788	−0.098 (−0.653–0.390)	7/10	1.000

The 320 runs comprise 160 seed-, task-, core- and depth-matched method pairs. One MNIST-additive depth-averaged seed was tied. Exact transport is an information-oracle implementation control; intervals containing zero do not constitute a formal equivalence test.

Table S17: Bandwidth-matched neuron-to-tree routing control. Values are correct minus fixed-deranged held-out accuracy in percentage points. Intervals are paired-seed bootstrap intervals; *P* values are exact two-sided Wilcoxon tests.

Task	Neuron model	Depth	Difference (95% interval)	Positive pairs	*P*
MNIST	Additive	2	0.51 (0.37–0.62)	10/10	0.0020
MNIST	Additive	4	0.39 (0.31–0.45)	10/10	0.0020
MNIST	Shunting	2	0.57 (0.43–0.69)	10/10	0.0020
MNIST	Shunting	4	0.38 (0.31–0.44)	10/10	0.0020
Noise task	Additive	2	0.70 (0.56–0.84)	10/10	0.0020
Noise task	Additive	4	0.15 (0.02–0.28)	8/10	0.1055

Five contrasts remained significant after Benjamini–Hochberg correction across the six tests; the additive depth-four noise contrast did not. Across conditions, 58/60 paired seeds favored correct assignment. The modest effect relative to the much larger scalar-to-neuron-indexed contrast separates a dominant coordinate-bandwidth contribution from a smaller assignment-specific routing contribution.

Table S18: Outcome-independent input-validity accounting for the prospective artificial-tree cohorts. "Retained" counts the units used in current figures and inference. The entire inhibitory-dose family is excluded because its prespecified endpoint compares against invalid signed-input shunting cells.

Cohort	Original units	Signed-shunting exclusions	Retained units
Depth by feedback	640 runs	160	480
Matched routing	160 runs	40	120
Inhibitory dose	400 runs	200	0
Spatial topology	320 runs	80	240
Fixed-budget depth	320 runs	160	160
Checkpoint diagnostic	160 checkpoints	40	120

The exclusion rule is task plus core plus resolved transfer activation; no accuracy or gradient outcome enters it. The complete 1,840-run historical ledger and all retained row tables are supplied as Source Data.

Table S19: Audit of fixed random and spatial sparse input maps. Each row summarizes 1,280 neuron-level maps: 128 neurons for each of ten topology seeds. Both maps contain 336 active contacts per neuron.

Connectivity property	Random map	Spatial map
Unique input coordinates, mean ± s.d.	276.2 ± 5.9	336.0 ± 0.0
Input coverage fraction	0.352	0.429
Repeated-contact fraction	0.178	0.000
Mean pairwise branch Jaccard overlap	0.0139	0.0000
Feature-degree coefficient of variation	1.507	1.155

Table S20: Prospective fixed-checkpoint credit-to-step audit at relative step 10^{-5}. Values are means over 120 input-valid independently trained backpropagation checkpoints. One-step progress is the loss decrease from a norm-matched candidate update divided by that from the exact update.

Feedback field	Gradient cosine	Gradient capture	Relative progress	Descent checkpoints
Global scalar	0.027	0.003	0.013	75/120
Neuron indexed	0.586	0.355	0.636	120/120
Exact transport	1.000	1.000	1.000	120/120

Across scalar and neuron-indexed fields, gradient cosine predicts relative one-step progress with Spearman $\rho = 0.893$ (checkpoint-clustered 95% bootstrap interval 0.860–0.923); gradient capture gives $\rho = 0.859$ (0.825–0.890). These are fixed-state optimization diagnostics, not convergence or generalization comparisons.

Table S21: Selected endpoints from the expanded regular-tree regime analyses. Values are test-accuracy means $\pm$ s.d. over five independent seeds.

Analysis	Contrast	Shunting or structured feedback	Additive or restricted feedback
Depth stress	One to four dendritic layers	$63.5 \pm 6.2\%$ to $57.4 \pm 4.2\%$	$54.9 \pm 5.4\%$ to $29.7 \pm 10.5\%$
Broadcast-noise stress	$\sigma = 0$ to $\sigma = 1$	0.617 ± 0.022 to 0.469 ± 0.062	0.465 ± 0.052 to 0.114 ± 0.009
Noise-task feedback	Scalar versus rank eight	0.662 ± 0.016 versus 0.795 ± 0.019	Exact transport: 0.847 ± 0.018
Flattened CIFAR-10	Scalar versus rank four	$29.7 \pm 1.3\%$ versus $35.0 \pm 1.0\%$	Exact transport: $47.2 \pm 0.6\%$; BP: $49.5 \pm 0.4\%$

The first two rows compare complete forward architectures and therefore establish robustness regimes rather than a causal backward effect of shunting. Exact transport is an information oracle. The flattened CIFAR-10 analysis is a harder-data mechanism control, not a state-of-the-art image classification comparison.

Table S22: Reconstructed-cell inputs to the structural analysis. Synapse counts are mapped and classified excitatory (E) or inhibitory (I) contacts.

Root ID	Type	Depth	Segments	E synapses	I synapses	Kernel rank	Map pass
864691135060651419	L2IT	11	146	4661	1378	38.36	0.983
864691135277186789	L3IT	10	104	3780	983	14.79	0.981
864691135324623260	L4IT	6	40	1534	506	16.66	0.967
864691135196576298	L5IT	5	22	1254	394	11.00	0.974
864691135409937097	L2IT	9	78	2870	744	21.15	0.984
864691135739661169	L5IT	6	34	970	446	11.96	0.982
864691135810666525	L2IT	14	128	4725	1934	31.29	0.986
864691136043380566	L2IT	9	64	1987	503	21.25	0.986
Total or mean		8.75	77.0	21,781	6,888	20.81	0.980

Table S23: Eight-channel routing capacity on model-derived perturbation fields in eight reconstructed cells. Intervals resample cells and Monte Carlo streams hierarchically.

Dictionary	Mean capture	95% interval	Wiring density	Ancestry advantage
Dense PCA oracle	0.569	0.481–0.665	1.000	–
Ancestry-selected paths	0.487	0.400–0.581	0.069	–
Random nonempty paths	0.285	0.191–0.398	0.081	0.202 (0.161–0.242)
Depth-only bins	0.222	0.157–0.292	0.125	0.265 (0.229–0.302)
Ancestry shuffle	0.198	0.132–0.272	0.069	0.289 (0.253–0.324)

The morphology advantage was positive in 8/8 cell means for each structural control; the two-sided exact paired Wilcoxon value was $P = 0.0078125$ in each exploratory comparison. The cohort was selected before this journal synthesis but was not a population-random sample.

Table S24: Eight-channel ancestry dictionaries evaluated on an independently generated reciprocal-cable shunt-response operator. Capture is the expected weighted response energy in the stated subspace.

Dictionary	Mean capture	Ancestry difference	Ordered cells
Dense SVD oracle	0.694	–	–
Ancestry-selected paths	0.458	–	–
Random real paths	0.363	0.095	8/8
Depth bins	0.445	0.014	5/8
Row-shuffled selected paths	0.249	0.209	8/8
Degree- and depth-matched surrogate tree	0.423	0.035	6/8

The target was the finite-difference response of the exact soma-state-matched excitatory log-gradient field to every focal shunt, rather than a field sampled from the tested ancestry kernel. Ancestry-minus-control cell-bootstrap intervals were 0.064–0.124 for random paths, −0.037–0.068 for depth bins, 0.179–0.244 for row shuffling, and 0.0069–0.0658 for matched surrogate trees. The paired two-sided Wilcoxon values were 0.0078, 0.844, 0.0078, and 0.078, respectively. The cell-bootstrap interval for the matched surrogate excluded zero whereas the prespecified exact paired test did not; we therefore treat that comparison as inconclusive. This remains a passive-model response on one mouse's reconstructed anatomy.

Table S25: Focal-perturbation sensitivity at unit dose. The primary condition is the first row. Scale multiplies normalized excitatory and inhibitory synaptic conductance; E_I is the inhibitory reversal potential in normalized units.

Scale	E_I	Shunt localization	Additive localization	Difference	Positive cells	P
0.35	−0.2	0.104	0.035	0.069	8/8	0.0078
0.10	−0.2	0.101	0.019	0.081	8/8	0.0078
1.00	−0.2	0.114	0.064	0.050	7/8	0.0156
0.35	0.0	0.115	0.021	0.094	8/8	0.0078
0.35	−0.5	0.089	0.054	0.035	7/8	0.0156

Values are means of site-averaged cell effects. P is the exact two-sided paired Wilcoxon value over reconstructed cells. Sensitivity rows reuse the same eight anatomical cells and are not independent replications.

Table S26: Physical passive-cable sensitivity of the focal shunt contrast. R_a is axial resistivity and R_m is specific membrane resistance.

Cohort	(R_a, R_m)	Median axial/leak	Difference	Positive cells	95% interval
Original eight	$(150, 300)$	1.36	0.0674	8/8	0.0502–0.0898
Original eight	$(150, 1000)$	4.53	0.0164	8/8	0.0068–0.0290
Original eight	$(150, 3000)$	13.60	0.0010	5/8	−0.0043–0.0068
Original eight	$(150, 15000)$	68.02	−0.0019	2/8	−0.0046–0.0000
v661 sensitivity	$(150, 300)$	2.74	0.0808	45/45	0.0687–0.0937
v661 sensitivity	$(150, 3000)$	27.41	0.0113	45/45	0.0090–0.0138
v661 sensitivity	$(150, 15000)$	137.07	0.0024	45/45	0.0018–0.0030

R_a is in Ωcm and R_m in $\Omega \mathrm{cm}^2$. Synaptic area was calibrated relative to the median physical leak and is not a measured conductance. Specific resistances are sensitivity settings rather than fitted cell-specific values. Complete R_a/R_m combinations are supplied as source data.

Table S27: Target-level inputs for the measured-response branch-model analysis. Every target contributed 464 trials, 280 unique stimulus identities and 136 repeated identities; all extracted response values were finite.

Nucleus ID	Root ID	Type	Session	Scan	DANDI asset prefix	Sites	Branches
256456	864691135810666525	L2IT	6	6	06c05eec	10	5
256550	864691135060651419	L2IT	9	4	ad27b3c6	13	5
258209	864691135277186789	L3IT	8	5	436f762f	17	8
264871	864691135739661169	L5IT	7	3	67384ebb	6	4
264888	864691135196576298	L5IT	7	3	67384ebb	9	5
264889	864691135324623260	L4IT	6	6	06c05eec	5	3
291111	864691135409937097	L2IT	6	4	dc68c26d	9	6

Full DANDI asset identifiers and NWB paths are included in the machine-readable target manifest. "Sites" are connected, functionally imaged presynaptic partners after automatic conservative matching; "Branches" are occupied first-level dendritic branches in the fitted model.

Table S28: Four-channel measured-response task analysis. Lower normalized MSE is better. Values average ten fixed stimulus splits within each of seven target cells before averaging targets.

Feedback field	Held-out field capture	Held-out normalized MSE
Exact backpropagation	1.000	0.778
Dense PCA oracle	0.999	0.781
Ancestry-selected paths	0.475	0.837
Random nonempty paths	0.564	0.834
Depth-only bins	0.621	0.791
Ancestry shuffle	0.446	0.842
Scalar broadcast	0.222	0.801

Ancestry routes minus ancestry-shuffle capture was 0.028 (target bootstrap 95% interval −0.146 to 0.206; 3/7 targets positive). The reduction in error from morphology relative to ancestry shuffle was 0.004 (interval −0.021 to 0.027; 4/7 targets improved). Neither contrast was reliable.

Table S29: Synapse-resolved inhibitory-census endpoints. Differences are first averaged within each of 20 postsynaptic targets. Confidence intervals bootstrap target cells. Directional one-sided Wilcoxon tests were frozen before the join for the first four endpoints and the three selected-site capacity comparisons; overall placement is two sided.

Endpoint	Mean difference	95% interval	Ordered targets	P
Same-axon tree distance (μm)	-12.003	$[-21.460, -4.151]$	15/20 lower	0.0032
Same-axon shared path	0.0357	$[0.0091, 0.0766]$	15/20 higher	0.0028
Same-axon domain overlap	0.0322	$[0.0060, 0.0743]$	14/20 higher	0.0053
DTC minus PTC median domain fraction	-0.0162	$[-0.0231, -0.0105]$	20/20 lower	9.5×10^{-7}
All contacts minus matched locations	0.0016	$[-0.0022, 0.0056]$	10/20 higher	0.546
Eight selected sites minus random sites	0.3638	$[0.3326, 0.3948]$	20/20 higher	9.5×10^{-7}
Eight selected sites minus depth bins	0.3462	$[0.3126, 0.3793]$	20/20 higher	9.5×10^{-7}
Eight selected sites minus ancestry shuffle	0.3738	$[0.3437, 0.4037]$	20/20 higher	9.5×10^{-7}

Table S30: Dependence and spatial sensitivity of the repeated inhibitory-contact analysis. Contrasts use the same endpoint sign as Table S29.

Control	Endpoint	Mean difference	Ordered units	P
Presynaptic-axon aggregation	Tree distance (μm)	-9.368	62/92 lower	0.0008
Presynaptic-axon aggregation	Shared path	0.0143	45/92 higher	0.104
Presynaptic-axon aggregation	Domain overlap	0.0117	41/92 higher	0.393
Joint 3D and path matching	Tree distance (μm)	-4.272	13/20 lower	0.108
Joint 3D and path matching	Shared path	0.0107	8/20 higher	0.826
Joint 3D and path matching	Domain overlap	0.0151	9/20 higher	0.54

The first control averages connection contrasts within each of 92 presynaptic inhibitory axons before testing. The second retains the target as the unit and draws matched sites by joint percentile rank of log path distance and three-dimensional Euclidean proximity. All units remain nested within one mouse. Only the coarse tree-distance contrast persisted after presynaptic-axon aggregation; no endpoint persisted after joint spatial matching.

Table S31: External six-animal signed-credit test from Francioni et al. source data. The contrast is error-reduction minus error-increase soma–dendrite residual. Directions follow the known causal mapping.

Contrast	Mean	95% interval	Ordered animals
P+	0.0753	$[0.0295, 0.1214]$	5/6 positive
P-	-0.1498	$[-0.2054, -0.0866]$	6/6 negative
P+ minus P-	0.2251	$[0.1416, 0.3108]$	6/6 positive

Exact one-sided random-sign permutation P values were 0.0313, 0.0156, and 0.0156. The signed and common modes contained 83.7% and 16.3% of pooled squared projection energy across the six animal contrast vectors, respectively. The signed fraction was 45.8–97.1% across individual animals, with an animal-bootstrap 95% interval of 60.6–95.7% for the pooled fraction and a leave-one-animal-out range of 76.5–90.2%. Animal-level values are provided in Source Data.

Table S32: Interpretive boundary for each evidence class.

Evidence	Supported conclusion	Not established
Exact factorization	Conductance-tree gradients separate local eligibility from transported compartment error.	A biological circuit computes exact gradients or has exact path transport.
Regular-tree feedback	Preserving neuron identity closes most of the scalar-feedback gap in both tested architectures; exact transport makes 3F sufficient; feedback quality orders norm-matched one-step progress.	Universal advantage of shunting, or superior accuracy, memory or latency relative to backpropagation.
Subtree-address factorial	Task-aligned nested ancestry has a small advantage over matched non-anatomical routes at intermediate bandwidth and over a rewired tree at intermediate budgets.	A uniquely dendritic advantage; matched gated point and flat implementations reproduce the same result, and unrestricted low-rank routes win at the smallest budgets.
Credit-operator phase	Retained signal, admitted noise and task hierarchy predict when a restricted route or additional address depth improves a stochastic update.	Superiority to exact full-batch backpropagation, a positive-conductance forward mechanism or learned branch reliability.
Physical-depth controls	Across $H = 2$ and $H = 3$ variants of one mechanism-matched task family, serial divisive composition beats resource-identical star and literal grouped-point controls only when task factors match physical order; shared and path LocalCA retain the aligned H2 effect. Flexible parameter-matched point MLPs beat the serial tree, and soma-broadcast controls localize the BP–local separation.	A dendrite-exclusive expressivity advantage, natural-task prevalence, biological availability of exact path transport or optimizer-independent equivalence of BP and LocalCA.
State-matched reliability conductance	Fixed positive shunts approximate the step-consistent oracle reliability ordering and outperform global, shuffled and anti-aligned shunts under heterogeneous branch noise.	Exact block-scalar attenuation, autonomous shunt learning, a dendrite-exclusive operation or sustained final-loss superiority over unshunted noisy learning.
Same-span coefficient learning	Coordinate conditioning creates the finite-data bias–variance crossover predicted by exact linear risk, despite identical route capacity.	A biological Gram preconditioner, unconditional benefit of orthogonal or ill-conditioned coordinates, or new address span.
Prospective input-validity audit	Identity, ownership, spatial and fixed-budget conclusions persist in the 1,000 retained artificial-tree runs; the inhibitory-dose family is excluded.	Any claim derived from signed synthetic inputs driving positive-conductance shunting cells.
Fixed spatial topology	Spatial partitioning improves forward learning and input coverage at fixed contact count.	Task-aligned dendritic credit routing; the effect persists under backpropagation and a randomly projected task.
MICrONS capacity	Real ancestry compresses model-matched fields and captures independently generated reciprocal-cable responses better than random or row-shuffled routes.	An advantage of fine topology over coarse depth and degree, task-gradient alignment, endogenous credit routing, or learning in the recorded mouse.
Inhibitory census	Inhibitory classes occupy different dendritic scales and repeated contacts are spatially co-targeted; actual-route dictionaries are internally compressible.	Geometry-independent descendant-route organization, inhibitory activity, a teaching signal, or biological selection of leverage-ranked routes; all targets come from one mouse.
Focal perturbation	In the normalized proxy, an exact factor-freeze control attributes descendant localization to changed adjoint transport; physical calibration specifies where the effect is large or small.	That the effect is universal across electrotonic regimes or that inhibitory synapses conveyed teaching signals during the animal's experience.
Active focal ensemble	Descendant-enriched attenuation survives exact local linearization at 512 accepted active-conductance steady states.	Channel kinetics, spikes, cell-specific channel fits or independent biological replication by the nested channel draws.
Measured responses	All 13 eligible scans and the 520-fit complete-tree model do not show reliable alignment between the tested anatomical routes and this visual-response learning objective.	Absence of topology–function organization in cortex or under other response windows, coverage and biophysical calibration.
Projection coefficients	Each dictionary's best attainable capacity can be compared under a fixed channel budget.	A locally realizable circuit for learning the oracle coefficients.
Alignment-controlled optimization	Rotating exact credit into a fixed reconstructed-tree basis changes both capture and loss reduction under matched quadratic objectives.	That the reconstructed cells acquired or used the imposed gradients in vivo.
Signed animal contrast	Known P+/P- causal sign predicts the dominant dendritic population contrast across six animals.	Within-tree routing or a unique conductance/shunting mechanism.
Interference identity	Route overlap and off-route leakage are sufficient to produce exact one-step forgetting in a static quadratic task.	Temporal neural dynamics or a causal explanation of the published motor-learning experiment.

Table S33: Alignment-controlled capture and one-step optimization. Values are means across eight reconstructed cells after averaging 40 nested Monte Carlo streams per cell. One-step progress is normalized by a norm-matched full-gradient update.

Aligned energy	Dictionary	Field capture	One-step progress
0.0	Ancestry-selected paths	0.000	0.000
0.0	Random paths	0.135	0.307
0.0	Depth bins	0.153	0.331
0.0	Ancestry shuffle	0.152	0.331
0.4	Ancestry-selected paths	0.400	0.615
0.4	Random paths	0.197	0.390
0.4	Depth bins	0.191	0.385
0.4	Ancestry shuffle	0.172	0.359
1.0	Ancestry-selected paths	1.000	1.000
1.0	Random paths	0.290	0.460
1.0	Depth bins	0.247	0.444
1.0	Ancestry shuffle	0.201	0.373

At 40% alignment, morphology capture exceeded the three controls by 0.203–0.228 and the difference was positive in 8/8 cell means. At zero alignment morphology lost to all controls in 8/8 cells; at complete alignment it exceeded all controls in 8/8 cells. Monte Carlo streams and gradient fields are nested sensitivity samples, not biological replicates.

References

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